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VALUE INVESTING

From Graham to Buffett and Beyond

BRUCE C. GREENWALD JUDD KAHN

ERIN BELLISSIMO MARK A. COOPER TANO SANTOS

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*To Diana Greenwald, without whose “gentle”
encouragement this book would not exist,
and Gabriel Kahn, wonderful son, marvelous father*

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New York that extended over two Premier League seasons. Even more crucial, she has been author, director, and producer of the videos that accompany this volume. If it weren't for her modesty, she would be listed as an author.

Introduction

In 1999, when we began to write the first edition of this book, value investing as conceived by Benjamin Graham and David Dodd and developed by their successors was in eclipse as a method of stock selection. Academic finance had for 30 years embraced the Efficient Markets Theory (EMT), which rejected the possibility of consistently successful active investing. The stock market boom produced by the first Internet bubble appeared to invalidate all the analytical principles on which value investing was based. Value practitioners, with the notable exception of Warren Buffett, were dismissed as old-fashioned and out of touch with contemporary economic reality. Fortunately, the collapse of tech and telecom stocks between 2000 and 2002, coupled with the superior performance by value investors, revived interest in the Graham and Dodd approach. At the same time, a mass of published academic evidence powerfully contradicted EMT. Statistically constructed value portfolios generally outperformed the stock market as a whole over almost all extended periods in almost all the national markets for which sufficient historical data were available.

Increasingly accepted academic studies in psychology, pioneered by Daniel Kahneman and Amos Tversky, spawned the field of behavioral finance and provided an explanation for the historical outperformance of these value portfolios, grounding it in deeply embedded human behavioral biases. As a result, a “value premium” in returns appeared likely to be a persistent feature of future financial markets. These studies and continuing innovation by value practitioners led to a more thorough understanding of Graham and Dodd principles and marked improvements in value investing practices, especially with respect to identifying and evaluating what are designated as “franchise” businesses.

The long bull market since the depths of the 2008–9 financial crisis has once again raised questions about the validity of a Graham and Dodd approach. During the years since 2009, many notable value

investors have significantly underperformed national and global market indices. Carefully constructed statistical value portfolios have seen the gap between value and overall market performance narrow significantly if not entirely disappear. A new generation of technology stocks have provided sustained returns that again appear to contradict established value principles. And value investors are once again being dismissed as old-fashioned and out of touch with current economic reality.

In part, a decline in the relative performance of value portfolios is a predictable result of the valuation excesses of the later phases of any long bull market. Value investors have historically performed relatively poorly in these periods, as they did in the late 1990s. However, other important factors appear to be at work. First the renewed success of Graham and Dodd investing in the years from 2000 through 2007 increased the popularity of a value approach. Especially in the United States, the proportion of value oriented investors rose significantly. An increased demand for value stocks may have compressed the spread between glamour and value stock valuation multiples, although the evidence is mixed. Second, economic developments have complicated the task of applying Graham and Dodd principles. The shift in economic activity from industry/manufacturing to services has increased the importance of intangible capital—customers, trained employees, product portfolios, and brand images—relative to tangible capital—inventories, accounts receivable, property, plant, and equipment that accountants have traditionally included on a business's balance sheet. Moreover, since investments in intangibles—advertising, hiring, training, and product development—are often counted as current expenses for accounting purposes, defining and measuring current earnings power has become more difficult. Technological developments have had a similar impact. Modern computer and Internet-based firms like Amazon, Google, Oracle, Facebook, Microsoft, and Netflix have relatively little physical capital. Much of their growth related investments are, for accounting purposes, buried in expenses, where they depress, perhaps excessively, reported earnings and raise some valuation multiples.

A further complicating factor is the increasing extent to which service and modern technology companies operate in local geographic markets or niche product markets. These local/niche markets are characterized by potential economies of scale and, through continuous customer interactions, high degrees of customer captivity. The result is an increased incidence of dominant local/niche market competitors who benefit from significant barriers to entry. In the language of value investing, “franchise” businesses with wide “moats” constitute

an increasingly large fraction of overall economic activity. For franchise businesses, net assets play a diminished role in determining profits and growth contributes significantly to overall value. The consequence is that equity valuations depend heavily on future cash flows, and often far distant future cash flows, whose values are difficult to measure using Graham and Dodd asset value/earnings power value methods. Also for franchise businesses, management performance, especially with respect to capital allocation, has an enhanced impact on firm valuations. Not surprisingly in this environment, many traditional balance sheet focused value investors have not done well.

A final newly important factor that plays a role in the valuation of franchise businesses is the heightened potential for disruptive change that may undermine a firm's franchise position. For competitive businesses without significant economies of scale, any decline in profitability should be roughly offset by fixed and working capital recoveries as the business contracts. For franchise businesses, where earnings power value exceeds asset value, disruptive decline has much more serious consequences. Loss of economies of scale undermines earnings without any compensating return of capital. High returns on capital mean that lost earnings are only slightly offset by any realized capital recoveries. Dying franchise businesses are far less valuable relative to their pre-disruption positions than dying "cigar butt" businesses. Any attempt to invest in undervalued franchise businesses requires a careful assessment of the consequences of disruption.

These changes mean that we have to revisit all aspects of the approach to value investing laid out in the first edition of this book. We have rethought the imperatives of searching for and then valuing potentially attractive opportunities once they have been identified. We have also carefully examined active research processes once a preliminary valuation has been made and have looked at the issue of risk management far more extensively than we did in the first edition. In this revision we have benefited from observing practicing value investors and noting the adaptations they have made to changing economic circumstances. In all these areas, we have explicitly measured the advantages of a modern Graham and Dodd approach against what is ultimately the fundamental challenge facing any active investor. While there is now overwhelming evidence that financial markets are not efficient in the academic sense, there is a fundamental and inescapable way in which markets are efficient. The average return to all investors in any asset class must be equal to the average return to all the assets in that asset class (i.e., the "market" return for that asset class). All the assets are owned by somebody and derivative

arrangements (e.g., uncovered short sales) net out since for every seller there is an offsetting buyer. Therefore, if one investor outperforms the market for a particular asset class, another investor must underperform by a compensating amount, weighted by the assets under management. Since this constraint applies to all asset classes, it applies to investments as a whole.

Graham and Dodd were fully aware of this efficiency constraint although they described it in slightly different terms. They understood that every time someone bought a security thinking it was likely to do well relative to alternative opportunities, someone else was selling that security because they thought it would underperform the relevant alternative opportunities. Depending on the outcome, one of these investors always had to be wrong. The essential characteristic of a well-conceived investment process is that at every step—search, valuation, research process, risk management—it should place an investor on the right side of the trade. The process must be superior to that of the investor on the other side of the trade. This criterion is what we have used to explicitly measure the modern value investing practices described in this second edition.

The search process involves not only a value orientation, a preference for non-glamorous, ugly, out-of-favor, and obscure stocks, but also some degree of specialization. If I, as a generalist, trade with an equally capable and highly disciplined specialist, the specialist will usually have superior understanding and information. He or she will therefore more often than not be on the right side of the trade. In this edition we have extended the search chapter to include a discussion of effective specialization strategies. Recent experience supports this point. Highly focused value investors tend to be unusually successful even compared to the value community as a whole. Successful but more broadly oriented value investors tend to perform better in industries and geographies where they have concentrated than in other areas. We have, therefore, added a number of specialized investors to our profile section. The Graham and Dodd concept of “circle of competence” applies not just to staying away from unfamiliar areas but actively defining focused fields of expertise.

We have over the years since the first edition learned a similar lesson with respect to valuation. Different assets with different return horizons present different valuation challenges. For liquidating assets or other short-term investments with catalysts, discounted cash flow (DCF) valuations are appropriate. The relevant cash flows are likely to be accurately estimated. Assets whose values are determined in competitive markets—real estate, natural resources, non-franchise businesses—are

generally ones for which growth does not create significant value. This limits the importance of far future cash flows. For those investments the Graham and Dodd asset values/earnings power value formula, which ignores growth, is better than a DCF or ratio valuation for reasons we discussed in the first edition of this book. It is the approach of choice for most value investors. However, it is not suitable for evaluating businesses for which growth creates significant value. In these cases, growth means that far distant cash flows are both difficult to estimate because small differences in growth rates compound into large differences in far future cash flows and constitute an important fraction of value. As a result, current intrinsic values for such businesses cannot be estimated with usable degrees of precision. Investment decisions are more usefully based on estimated future returns than estimated values. The major innovation in this edition of our book is the description through three extended chapters of a return-based approach to evaluating franchise businesses. The process we lay out may not turn out to be the last word in franchise business evaluation. However, we can say with confidence that a one-size-fits-all valuation approach is not likely to place investors consistently on the right side of the trade.

Close observations over the years since 1999 of the practices of successful value investors has led us to add a chapter on active research practices to the second edition. The rise of the importance of intangible investments and franchise earnings as global economies have evolved has meant the traditional financial statement analysis is no longer adequate to evaluate many businesses. The increasing importance of the quality of management stewardship of these assets to returns has also forced investors to look well beyond published financials in evaluating investments. These imperatives have created unprecedented demands on the investment analyst's time. Properly and efficiently focused active research processes are increasingly important to investment success. A better research process will put an investor more frequently on the right side of the trade. In this added chapter we describe what good active research processes look like.

Finally, the increasing importance of specialization and active research has created significant problems in risk management. Historically, individual investment managers have tended to build portfolios as if they managed most if not all of their clients' assets. They selected levels of diversification without reference to the fact that the wealth owners typically spread their holdings among many investment managers. For wealth owners it is the risks of their overall portfolios that matter, and these overall portfolios often include large illiquid positions or

businesses. In principle, wealth holders concerned with the risks of the complete portfolio should manage risks centrally. Decentralized risk management potentially leads individual investment managers to take offsetting positions—manager A embraces risks that manager B hedges—or positions whose returns are highly correlated with large illiquid wealth holdings. The result may be an expensive risk management process that does little to reduce overall portfolio exposures. With specialized investment managers, the benefits of diversification must be achieved at the wealth-holder level. Increasingly, therefore, risk management will be divorced from individual security selection. Centralized risk managers must receive sufficient information from their individual investment managers in order to do their jobs effectively. In light of these developments, we have included an extended discussion of risk management in the second edition. We carefully define risk from a Graham and Dodd perspective and describe effective practices for managing that risk.

The continuing vitality of the Graham and Dodd tradition has always depended on successful adaptation to changing economic and financial conditions by practicing value investors. We have revised the profile section of this book with this in mind. If investors we covered in the first edition have since passed away, we have included the original profiles without change to preserve the valuable insights they had to offer. The others we profiled have continued to innovate and improve their processes. Fortunately, most of them have spoken annually for many years in the MBA value investing course at Columbia. For them and other class speakers, we provide brief descriptions of where they stand in the broad spectrum of value investing practice and how their approaches have evolved over time. Taking advantage of the advances in technology, these summaries are supplemented with access to edited online videos of investors-classroom appearances. Most of the new investors we have added to the second edition—Tom Russo, Paul Hilal, and Andrew Weiss, all specialists to some degree—have spoken in the class many times and will be treated in the same way. Two other important investors, one seasoned, Warren Buffett, and one new, Jan Hummel, have not spoken in the class. For Hummel we have presentations that we have edited for online access and which we have supplemented with a brief description of his specialized investment approach. As before, Warren Buffett will receive his own written chapter based on extracts from his extensive public commentary.

When we began working on the first edition of this book in 1999, Bruce had taught the value investing MBA course about five

times, with some additional run-throughs in Executive MBA and 2 day Executive Ed versions. Twenty years later, even with a sabbatical now and then, he has taught it more or less continuously for around 25 years. In 2005, we published *Competition Demystified*, a detailed study of the factors that constitute sustainable competitive advantages and what distinguishes franchise businesses, firms protected by barrier to entry, from companies subject to competitive pressures. And during those years, both of us have had direct investment experience, working in a large global mutual fund and three smaller hedge funds. There is no doubt that all the additional teaching, thanks to the students and the guest investors who generously contributed their time and expertise to developing the Graham and Dodd tradition, has expanded our own understanding. At least as important has been our time in the field, so to speak. As the historian Edward Gibbon wrote in his Memoirs about his service in the Seven Years' War, "The discipline and evolutions of a modern battalion gave me a clearer notion of the phalanx and the legion; and the captain of the Hampshire Grenadiers (the reader may smile) has not been useless to the historian of the Roman Empire."

Bruce Greenwald and Judd Kahn, New York, 2020

1

Value Investing: Definitions, Distinctions, Results, Risks, Principles

What Value Investing Is

Value investing is an approach to investing originally identified in the 1920s and 1930s by Benjamin Graham and David Dodd. Since then, the approach has developed and flourished in the hands of a notable but relatively small group of investors, the most famous of whom is Warren Buffett, who was their student in the early 1950s. As initially defined by Graham and Dodd, value investing rests on three key characteristics of financial markets:

1. The prices of financial securities are subject to significant and capricious movements. Mr. Market, Graham's famous personification of the impersonal forces that determine the price of securities at any moment, shows up every day to buy or sell any financial asset. He is a strange fellow, subject to all sorts of unpredictable mood swings that affect the price at which he is willing to do business.
2. Despite these gyrations in the market prices of financial assets, many of these assets do have underlying or fundamental economic values that are relatively stable and that can be measured with reasonable accuracy by a diligent and disciplined investor. In other words, the

intrinsic value of the security is one thing; the current price at which it is trading is something else. Though value and price may on any given day be identical, they often diverge.

3. A strategy of buying securities only when their market prices are significantly below the calculated intrinsic value will produce superior returns in the long run. Graham referred to this gap between value and price as “the margin of safety;” ideally the gap should amount to about one-half, and not be less than one-third, of the fundamental value. He wanted to buy a dollar for 50 cents; the eventual gain would be large and, more important, secure.

Starting with these three assumptions, the central process of value investing is disarmingly simple. A value investor estimates the fundamental value of a financial security and compares that value to the current price at which Mr. Market is offering it. If price is lower than value by a sufficient margin of safety, the value investor buys the security. We can think of this formula as the master recipe of Graham and Dodd value investing. Where their legitimate descendants differ from one another—where each may add his or her unique flavor—is in the precise way they handle some of the steps involved in the process:

- Selecting securities for valuation;
- Estimating their fundamental values;
- Calculating the appropriate margin of safety required for each security;
- Deciding how much of each security to buy, which encompasses the construction of a portfolio and includes a choice about the amount of diversification the investor desires;
- Deciding when to sell securities.

These are not trivial decisions. To search for securities selling below their intrinsic value is one thing, to find them quite another. It is because the Graham and Dodd descendants have devised a variety of approaches to those tasks that value investing has remained a vital discipline through all market conditions in the more than eight decades since Graham and Dodd first published *Securities Analysis*.

What Value Investing Isn't

A common and brief summary of value investing is that value investors search for and buy only “bargains,” securities selling for less than their

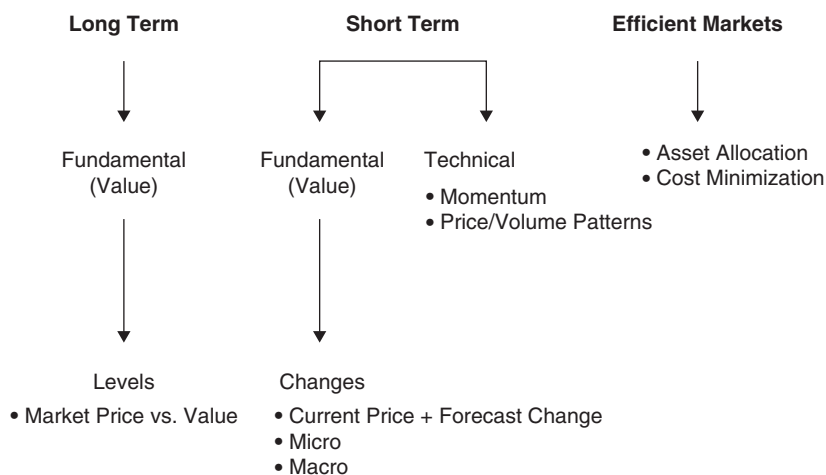


Figure 1.1 Approaches to Investing

true or intrinsic values. There is a problem with this simple definition. No rational investor admits to searching for securities selling for more than their underlying value. Everyone is looking to buy low and sell high.¹ We need to be clear about what differentiates real value investors from all the others who trade in the securities markets (see Figure 1.1).

One large class of investors who obviously do not qualify are “technical” analysts, or technicians. Technicians avoid fundamental analysis of any kind. They pay no attention to a company’s line of business, its balance sheet or income statement, the nature of its product markets, or anything else that might concern a fundamental investor of any stripe. They care nothing for economic value. Instead they focus on trading data, that is, the price movements and volume figures for any security. They believe that the history of these movements, reflecting the supply and demand for that security over time, traces patterns that they can analyze to infer future price movement. They construct charts to represent this information, and they scrutinize them for signs that will predict how prices will move next and thus allow them to make a profitable trade. For example, momentum investors extrapolate the current price trend,

¹ We are going to confine our general discussion throughout this book to the “long” position side of investing and ignore those investors who short (sell without owning) securities that they think are priced at more than their fundamental value. At certain points in his career, Graham used short sales to hedge other positions he had taken, and there may be bona fide value investors today who make active use of shorting securities. We discuss the pros and cons of short sales as an approach to risk management later in this book. In the main, however, value investing is identified with uncovering fundamental value and buying it at bargain price.

buying securities whose prices are rising in the expectation that they will continue to go up. Sometimes they compare the day's price for the security to a trend line made up of a moving average of the last 30, 90, 200, or some other number of days' prices. Crossing that trend line, up or down, can indicate a change in direction. Surely they intend to buy low and sell high, but *low* and *high* here refer to the previous and future prices of the security, unconnected to its fundamental value. For technical investors, Mr. Market is the only game in town. It is also a game that lends itself to trading—buying and selling over a short term. Very few traders ignore technical information. Today's chartists are much more likely to use sophisticated computerized algorithms to detect patterns, and to search for those patterns among different security prices rather than focus on the price history of a single security. But like most technicians, they are at best marginally interested in the fundamental economic value of the businesses underlying the securities.

Even when we turn back to people who legitimately see themselves as fundamental investors, concerned with the real economics of the companies whose securities they buy, Graham and Dodd value investors are distinct.

We can divide the class of fundamental investors into those who focus on macroeconomic issues and those who concentrate on the microeconomics of specific securities. Macrofundamentalists are concerned with broad economic factors that affect the universe of securities as a whole, or at least in large groups: inflation rates, interest rates, exchange rates, unemployment rates, and the rate of economic growth at the national or even international level. They closely monitor the actions of policy makers, like the Federal Reserve Board, and aggregate investor and consumer sentiment. They use their information to forecast broad economic trends, and they then use the forecasts to decide which groups of securities (or even individual issues) are likely to be most affected by the changes they predict. Their approach is often referred to as *top-down*, starting with the overall economy and working down to specific industries and securities. Like every other investor, they intend to buy low and sell high, using what they hope are their superior predictions to trade before the market as a whole recognizes what is happening. They do not, as a rule, do direct calculations of the value of individual securities or particular classes of securities, though such calculations could be consistent with a macro-fundamentalists approach. Although there are some famous and successful macro value investors, value investors in the Graham and Dodd tradition are basically microfundamentalists.

Within the society of microfundamentalists—those who analyze the economic fundamentals of companies and look at securities one by one—value investors in the Graham and Dodd tradition are still a minority. A more common approach to microfundamentalist investing takes the current price of a stock or other security as the point of departure. These investors study the history of this security, noting how the price has moved in response to changes in those economic factors that are thought to influence it: earnings, industry conditions, new product introductions, improvements in production technology, management shake-ups, growth in demand, shifts in financial leverage, new plant and equipment investments, acquisitions of other companies and divestitures of lines of business, and so on. There is more than enough to examine. They then try to anticipate how the critical variables on this list are likely to change, relying in large measure on company and industry sources as well as their more general knowledge.

Most forecasts focus on company earnings. Security prices incorporate the market's aggregate prediction about future earnings. If these investors find that their estimates of future earnings and other important variables exceed the market's expectations, then they purchase the securities. They assume that when new information about earnings and the other matters are released, their predictions will be validated and the market will drive up the price of the securities. They have bought low, based on a superior estimate of the future, and they intend to sell high.

Though this approach shares with value investing a concentration on economic fundamentals and specific securities, there are major differences. First, it focuses on prior and anticipated *changes* in prices, not on the *level* of prices relative to underlying values. One could apply this analysis equally well to a stock trading at 10, 20, or 50 times forecast earnings. A value investor would not regard these situations as equivalent. Second, this approach does not incorporate an identifiable margin of safety to safeguard the investment from Mr. Market's capricious behavior, which, after all, has been known to sink the price of shares in response to good news. So while Graham and Dodd value investing is most frequently a microfundamentalist approach, not all, or even most, microfundamentalists are value investors.

Each of these alternatives to value investing can lead to a successful investment record, provided it is carefully and diligently pursued. Statistical studies increasingly suggest that security prices and volumes do trace consistent and recognizable patterns; there are positive serial correlations in the short run and reversion to the mean over the longer term. There are successful technical investors. Macroeconomic variables can

be forecast with some accuracy and will affect securities markets in systematic and identifiable ways. There are successful macrofundamentalist investors. Analysts who energetically pursue information from company and industry sources, ferreting out trends ahead of the pack, should in theory and sometimes do in practice obtain above-average investment returns.

Another approach to investing outside the value tradition rejects all these possibilities. It arose from Modern Portfolio Theory and its sibling the Efficient Market Hypothesis, which were developed in academic finance departments beginning in the 1960s. The underlying premise of the theory is that current prices for securities, which are set by the collective perceptions of all market participants, accurately incorporate all the legally accessible information about future prices and values. Misperceptions and non-rational decisions are assumed to be essentially random. Excessive optimism, for example, on the part of some investors would be offset by excessive pessimism on the part of others. Correct perception, by contrast, being shared by many energetic and intelligent investors, would determine market prices. These prices would reflect the best forecasts of future developments affecting the companies' value. As a result, future price movements would depend either on random investor behavior or relevant new information that could not have been anticipated. Given these assumptions, future price changes would be unpredictable and current prices would be the best predictor of average future prices. All attempts by individual investors to outperform a portfolio based on current prices would be futile, since all changes would be random.

Given this view of security markets, the wise investor should focus on minimizing transaction costs and managing risk. The first task of risk management is to fully diversify away the idiosyncratic risks of individual investments, much as a risk-averse oil wildcatter drills a lot of holes to minimize the prospects of coming up dry by drilling only the few that look the most promising. For securities, this approach means buying a proportionate share of all the available risky securities to obtain the "market" return.

Second, the remaining risk, the systematic risk—which cannot be diversified away because it is the risk of the entire market—can be managed by combining market portfolios with investments in a risk-free asset, normally short-term government debt. By adjusting the proportion of overall wealth allocated to the risk-free asset, investors can obtain the exposure, from all in risk-free to all in the market portfolios, that suits their appetite for risk. Should some individual investments offer a better risk-return trade-off than the market portfolios, well-informed investors

would crowd into those opportunities. By driving up their prices and thereby lowering future returns, this activity would bring all prices back into the risk-return alignment that the Efficient Market Theory posits as the normal condition. In the end, for investors who believe in market efficiency, the investment process consists of decisions about asset allocation—the right combination of risk-free and market—and minimizing transaction costs with index mutual funds and Exchange Traded Funds (ETFs).

Extensive empirical evidence has largely discredited the strong form of the Efficient Market Hypothesis. Some individual managers and firms have outperformed market portfolios over long periods of time. In addition, portfolios selected by simple statistical rules, such as those with lowest book to market value, have significantly outperformed market portfolios with no apparent increase in risk over extended sub-periods—15 years or so—going back to the 1920s in the United States. The same results hold true, with a few exceptions, for overseas markets throughout the periods for which adequate stock market return data has been available.

Nevertheless, there is one important sense in which markets are inescapably efficient. The average return earned by all investors, weighted by assets owned, in any asset category must equal the average return on the assets that make up that category. For example, the total of the returns that investors in the US stocks earn in any given period must be equal to the total return produced by US stocks in that period. All these stocks are owned by somebody, and any related derivatives net out (e.g. for each short seller, who does not own the shares in question, there is a corresponding buyer of those non-existent shares). Since the total value of stocks at any moment must also equal the total value of investors' stock holdings, the average percentage return earned by investors in US stocks must equal the average percentage returns produced by the US stock market as a whole—the market return. This means that if some investors outperform the market in question, other investors must underperform by an equal amount. Before transaction costs, security trading is a zero-sum game; with transaction costs included it becomes negative-sum.

A way to think about this constraint, one that should focus attention on its implications, is to recognize that every time you buy a security because you expect an above-average return, someone is selling it to you because they think the return will be below average. At least one of you will always be wrong; if the security does nothing, you will both lose the transaction costs and one of you the foregone return on the risk-free

asset. It should not come as a surprise, therefore, that when we take the effort and expenses into account, 80–90% of active fund managers have done worse than the efficient markets strategy of buying low-fee index instruments.²

This inescapable reality is the most important feature of active investing, one that an active investor should never forget. An active investor must be able to identify convincing reasons why he or she will be on the right side of the trade more often than not—why, in other words, you will earn above-average returns. If you can't make that case, then investments in appropriate index funds make sense. We can acknowledge the effectiveness of index funds—known as *passive* investments because they seek only to mimic the market, not beat it, and make no investment decisions other than to be invested or not—without subscribing either to the idea that the price Mr. Market offers for a security is always the best measure of its fundamental value or that no investment approaches will outperform a passive approach over time.

Does Value Investing Work?

The case for value investing must confront this zero-sum constraint. The argument has both theoretical and empirical dimensions. We develop the theory in our detailed discussion of the procedures of modern value investing. Empirically, the historical record confirms that value investing strategies have worked; over extended periods, they have produced better returns than both the leading alternatives and the market as a whole.

Three distinct sources provide evidence of this superiority in practice. The first comes from a battery of mechanical selection tests. A researcher typically sorts all the stocks in the universe they have defined on a measure of value, such as market price to the book value of the equity or market price to earnings. They group the sorted stocks into buckets, frequently deciles, from the cheapest (value) to the most expensive (glamour). They record the total returns for each bucket over a defined period, usually one year. They then repeat the process for a number of years. At the end they have the returns over a long period of time—some go back 90 years—and can see how a mechanical value strategy has done relative both to the glamour stocks and to the market as a whole.

² See Footnote 4 below for details.

Many studies have been conducted employing different versions of this approach.³ The results demonstrate almost invariably that the value portfolios produce better than average returns—average here meaning returns on the entire market—in almost all periods and all kinds of markets. Using Kenneth French’s market to book data, the return of a portfolio that was long the cheapest 30% of stocks and short the most expensive 30% (a zero-cost portfolio before transaction fees) had a compound annual return of 3.35% from 1927 through 2018 on the gross amount invested in each bucket for an overall portfolio requiring no net investment; the outperformance has diminished since the publication of the article. Low price-to-earnings portfolios have had a similar success. Portfolios constructed of highly priced stocks, measured by market-to-book and price-to-earnings ratios, have done poorly. They are highly priced mainly because the companies have experienced rapid sales and earnings growth in the recent past; hence the label “glamour.” Unfortunately, all of that success and expectations that it will continue have already been incorporated into the stock price by the time the portfolios are constructed.

These mechanical selections of stocks produce portfolios that look very much like those that a diligent value investor, analyzing stocks one by one, would construct, especially as value investing was practiced in its early period. But value investing is not the same thing as a mechanical approach—a computer program—that selects stocks on the basis of a statistical measure that indicates which ones are cheap. Calculations of intrinsic value are usually more intricate and require more detailed

³ Some of the most important papers have been written by Eugene Fama, an early and principal proponent of the efficient market theory, for which he was awarded a Nobel Prize in economics, and his co-author Kenneth French. The original article is Fama, Eugene F. and French, Kenneth R. (1992). “The Cross-section of Expected Stock Returns,” *Journal of Finance* 47: 427–465. They have published many articles since then, and Professor French, currently at Dartmouth, has kept the original data and many other approaches up to date—and extended it back in time—on his website http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. All investors—active, passive, or any variant thereof—are deeply in his debt. See also the papers produced by Fama’s student Clifford Asness at his company’s website, <https://www.aqr.com/Insights/Research/White-Papers>, and additional research published by Wesley Gray, another Fama student, on his website <https://alphaarchitect.com/alpha-architect-white-papers/>. Both Asness and Gray run investment firms that try to capitalize on market inefficiencies. The Internet has made research more available and more fluid, meaning it changes all the time. So far, value as a factor has withstood all the attention, although it has underperformed in the recovery period starting in 2009.

knowledge of company and industry economics than are disclosed by simple financial ratios.

Nevertheless, the striking historical success of these value portfolios produced by mechanical selection should remind us of the high standards that an active value investing strategy must meet. According to Standard & Poor's, more than 80% percent of active fund managers underperformed their market benchmark over a 5-, 10-, or 15-year period ending in June 2018.⁴ In that 15-year period, the S&P 500 returned 9.3% a year; the cheapest quintile of stocks sorted by earnings to price returned 11.8%.⁵ It is reassuring, therefore, that investment management institutions that have adopted systematic value strategies in the Graham and Dodd fashion have return records that outperform the market as a whole.⁶ The performance of these institutions is our second source of support for the argument that value investing produces superior returns. Unlike the mechanical studies, which are "backtests" of selection rules applied to historical data, these institutions have generated real returns for real clients. Value investing works in the world as well as the lab.⁷

Finally, among those notable investors who have earned returns well above market indices over long periods of time, value investors who trace their intellectual origins back to Graham and Dodd are heavily over-represented. The most famous of these is Warren Buffett. The list also includes many of the investors profiled in this book, those identified by Buffett himself in his 1984 article "The Superinvestors of Graham

⁴ Standard & Poor's maintains a scorecard to assess the performance of active fund managers against the specific market indices (benchmarks) to which they should be compared. It is called the SPIVA US Scorecard and is updated every 6 months. For the period ending June 30, 2018, fewer than 10% of active managers had beaten their S&P benchmark, whether large, mid, or small cap, over the previous 15 years. <https://us.spindices.com/documents/spiva/spiva-us-mid-year-2018.pdf>

⁵ This data is from Kenneth French's website.

⁶ For example, The Tweedy Browne Value Fund, inception 1993, and The Sequoia Fund, inception 1970, have outperformed the SP 500 since they began despite the underperformance since the recovery in 2009. The Dodge and Cox Stock Fund has also outperformed, even over the last 20 years.

⁷ We should be clear that, like card-counting in blackjack, value investing does not work all the time. If it never underperformed the market, then everyone would become a value investor, and the advantage would be, in the terminology of the discipline, "arbitraged away." The statistical value portfolios typically underperform the market from 3 to 4 years out of 10. But over periods of 10 years or longer, value portfolios almost invariably outperform the market. The same applies to the value-oriented individuals and institutions. They may underperform significantly for extended periods, as in the later 1990s and in the period from March 2009 through this writing, but measured over significantly long windows, say 10 years or more, they almost always have outperformed.

and Doddsville.”⁸ and others who have pursued value strategies without necessarily publicly embracing the Graham and Dodd tradition.

The theoretical case for value investing starts with the zero-sum nature of investing. A consistently superior approach must succeed at the expense of investors who underperform the overall market. This success should rest on advantages at every stage of an active investment process, whether value oriented or not. The process consists of four basic steps (see Figure 1.2):

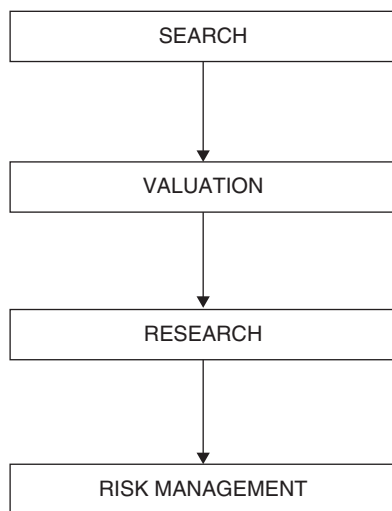


Figure 1.2 The Sequence of Investment Steps

The first step is to craft a search strategy. No investor can look carefully at all of the thousands of investments available in the world today. They must identify an intelligently selected subset of the entire group to examine. Ideally, this subset will contain a disproportionate number of opportunities that will more often than not put our investor on the right side of the trade. Some search strategies are determined by the nature of the investment firm. For example, a firm or fund that invests only in US listed convertible bonds has the origins of a search strategy built into its investment mandate. An investor who seeks out US equities with attractive growth prospects, either with statistical screens or less formally by reading the appropriate literature, going to conferences, or talking with like-minded colleagues, has a search strategy. At a minimum, a good

⁸ Originally published in *Hermes*, reprinted in Benjamin Graham, *The Intelligent Investor*, 4th and subsequent editions.

search strategy should answer the questions of why you are looking at a particular investment and why, given the opportunities identified by the search, you are likely to be on the right side of the trade it identifies.

After the search strategy identifies a security for detailed investigation, the second step of the process is to decide what it is actually worth. This step is valuation. Whether valuation is done explicitly or implicitly, for example by picking stocks whose values are expected to increase, any investment decision must be based on the judgment that the future value of the security exceeds the price at which it can be purchased today. The various methods of valuation, like the alternative ways of conducting search strategies, define the investor's underlying approach. Technical investors assess future values based on prior price movements and trading patterns. Short-term fundamentalists adjust today's price for their estimate of near-term future developments at either the macroeconomic or individual security level. Efficient market investors default to today's price as the best measure of value. The point here is that everyone has a valuation rule. They are not all equally successful. The test of quality is the returns the investor earns by taking one side of the trade against someone whose valuation measure produces different, less accurate estimates.

Most useful valuation methods will identify important uncertainties related to the value of the security under examination. Professional investors will employ a systematic process of active research to resolve these uncertainties. The first characteristic of a well-conceived research process is that it is focused on those uncertain variables that most importantly affect valuation. Meeting this criterion requires a sound valuation approach that identifies these key uncertainties. Research efforts that are mechanically focused on a predetermined list of variables, unrelated to their relevance to the particular investment in question, will not be as efficient as those driven by a superior valuation methodology. Second, there is always significant indirect information available that may either confirm or challenge the initial valuation estimate. For example, has the chief financial officer recently sold all of his or her shares? Are other well-informed and disciplined investors buying or dumping the shares? What are the commonly held beliefs that underlie the current price of the shares, and why do you think your valuation is better than the consensus? Know yourself. How have you acted in similar situations in the past? Have you made decisions that look like this one, and how have they worked out? Are you, in other words, a sucker for certain kinds of stories? An effective research process will collect and analyze this relevant indirect evidence to an extent that is

at least as comprehensive and efficient—within the law—as that of the people on the other side of the trade.

Finally, having searched, evaluated, examined additional evidence to check the initial valuation, and made a decision, every investor needs a process for managing the risk that this investment adds to (or reduces in) the portfolio. What is the appropriate size of the position within the portfolio? Does it move in tandem with other holdings, or does it have a low or even negative correlation with them? Will it provide insurance in a crisis as gold, cash, and derivatives may do? Unlike the trading of securities, risk management is not a zero-sum game, especially when different investors face different risks. But even at this stage of the work, it is useful to think in competitive terms: are your definitions of risks and the processes you use to manage it as least as good as widely employed and available alternatives?

Our contention is that the approaches developed by Graham, Dodd, and their talented successors are at each stage of the process generally superior to the methods commonly employed by those outside the value tradition. We aim to convince the reader that not only does historical evidence—statistical studies, firm performance, and the success of individuals—support a value approach to investing but that until others develop alternative processes at least as effective as those of the Graham and Dodd community, value investing will continue to prevail in the future.

The Rest of the Book

Part II discusses valuation in depth, with chapters on valuing assets, earnings power, and growth. Two detailed analyses are incorporated, Hudson General as an example of an asset value approach, and Magna International for an earnings power valuation. Part III focuses on franchise businesses and a new approach to valuing growth in these firms. Again, there are two detailed examples, this time for franchise stocks: WD-40 and Intel. Part IV discusses research strategies and risk management.

In addition to the text between the covers of this book, we have been able to make available online some of the presentations delivered over the years in the value investing course at Columbia Business School, taught for a quarter century by Bruce Greenwald and now by Tano Santos. Some of the most extraordinary value investors have devoted time and effort to make these presentations, in some cases virtually annually.

We are grateful beyond measure to them and the contributions they have made to the evolving discipline of value investing.

Appendix: Is Extra Return the Reward for Extra Risk?

One final issue regarding the factual evidence in favor of a value approach must be addressed. It is certainly possible that the higher returns achieved by value investing from each of these three sources—mechanically selected portfolios, value-oriented institutions, and individual Graham and Dodd investors—arise only because these portfolios are riskier than the market as a whole. If that were so, then their superior returns would be nothing more than an appropriate reward for bearing this increased risk. Many academic financial experts have been emphatic in arguing not only that higher return is the reward for higher risk but also that there is no way to beat the market's average return other than by assuming additional risk and that the best way to add risk is to leverage the market portfolio, since no stock selection process will outperform the market.

The problem with this argument is that when standard academic measures of risk—either annual return variability or betas as defined by modern finance theory—have been calculated for value portfolios, they have generally been no higher than the same risk measures applied to the market as a whole. In addition, value portfolios have proven to be less risky than the market as a whole when tested by other measures of risk, such as how much a stock drops in reaction to bad news about the company, the extent of price declines during bear markets, or simply the level of maximum loss experienced. These measures are closer to our commonsense understanding of risk and more appropriate for value investors, who regard price fluctuations as opportunities to buy or sell, not as accurate estimates of the intrinsic worth of the security.

For our mechanically selected value portfolios, which have been subjected to the most thorough statistical scrutiny, their average 1-year returns have been higher, their average 3-year holding period returns have been higher, their average 5-year holding period returns have been higher, they have provided superior returns during recessions, and they have outperformed glamour portfolios during the worst months for the stock market as whole. The value approach, even in its mechanical application, is no fair-weather friend. Looking at maximum losses

(drawdowns), in the 30 years from 1989 to 2018, the S&P 500 had two major bear markets:

- August 30, 2000–September 30, 2002 (45%)
- October 21, 2007–February 28, 2009 (51%).

As a comparison, using Kenneth French's cheapest quintile as measured by earnings-to-price ratio, the two largest were:

- June 30, 2001–September 30, 2002 (19%)
- March 31, 2007–February 28, 2009 (51%).

In one case, the value portfolio lost considerably less money; in the other, they were equivalent. The compound annual returns for the entire 30-year period were 9.2% for the S&P and 12.9% for the cheapest quintile. Using drawdowns as a measure, it is impossible to say that the value portfolio outperformed by assuming more risk.

As another alternative approach to risk, we refer to Warren Buffett's classic account of how he came to buy a large chunk of the shares of The Washington Post Company. The date was late 1973. It was a miserable time for the economy, the stock market, the national temperament, and, naturally, a great moment for value investors. The market capitalization of The Washington Post Company had dropped to \$80 million. At that moment, the whole company could have been sold to any of 10 buyers for at least \$400 million. Clearly Mr. Market was in a dreadful mood. Now, Buffett asked, had the market value of the stock declined again, from \$80 million down to \$40 million, would that have made a purchase of the shares more risky? According to modern investment theory, yes, because it would have increased the volatility of the prices. According to Buffett, not at all, because it would have increased an already ample margin of safety and lowered whatever risk—he thinks there was none to begin with—existed in the purchase. As a calculation of risk, the margin of safety has nothing in common with the volatility of a security's price. In order to use it, you have to acknowledge the existence of an intrinsic value and feel confident about your ability to estimate it.

2

Searching for Value: Finding the Right Side of the Trade

The investment world offers so many choices that any serious investor needs to carve out a limited part of that world before beginning his or her analysis. Either by design or by practice, investors must specialize. Specialization is the first, most important yet frequently overlooked factor in an effective search strategy. As an extreme case, consider a situation in which one investor has spent 20 years studying and investing only in South Texas Gulf Coast onshore oil leases. A second investor flies in from Europe and, based solely on the belief that in politically stable areas such as the United States oil is a good investment, buys an oil lease from our local expert. Sometimes that investment will work out. Oil prices may rise more than expected or the leased property may have output capacity higher than anticipated. But there should be little doubt that most of the time the local expert will come out ahead on a trade with a counterparty whose focus is on the price of a global commodity. Specialization, particularly in markets that are not intensely followed by a large number of other investors, is an obvious way to improve the odds of being on the right side of the trade.

We will take our own advice in the discussion here and specialize on only a portion of the larger investment universe. We will not address areas

like government debt, exchange rates, broad market-level instruments like index mutual funds and ETFs, or commodities like oil, wheat, or copper in which many investors are making decisions based on macroeconomic factors. In avoiding these markets we mirror the practices of most Graham and Dodd investors. We will also ignore very simple investment choices, such as bank savings accounts and certificates of deposit, where there is limited scope for a professional investor to add value. Finally, we will not discuss areas like collectables or intricate derivatives, where specialization is essential but the knowledge required is very narrow, specific, and outside the interests of most investors, whether value or otherwise. We will concentrate on corporate securities, primarily equities, where the central value questions are related to the fundamentals of the underlying business. Benjamin Graham and David Dodd started here, and most value investors have followed their lead.

Specialization

A well-established principle of Graham and Dodd investing is to remain within one's "circle of competence." No investor is capable of understanding every business in every industry. Warren Buffett has often stated that he has no understanding of technology firms; when he invested in Apple, it was an extremely successful consumer goods company. Since they are outside his "circle of competence," investments he might make in those industries are unlikely to land him on the right side of the trade. A circle of competence delineates one's areas of specialization. Investors should always stay within the circle and regard the regions beyond as unprofitable territory. The key difference between establishing a specialization and identifying a circle of competence is that specialization works from the inside out, starting from a particular industry or geography or other narrow area of interest. Once that area has been mastered, an investor may move on to adjacent areas of interest, mastering them one at a time. A circle of competence is generally described from the outside in, beginning with a full set of investment opportunities and eliminating those beyond one's current understanding. Formally, the two may ultimately be equivalent, especially for experienced investors. But particularly for beginning investors, a process of specialization is likely to be a more efficient way to define and develop fields of investing advantage. Thus, in what follows we will focus our search discussion on specialization rather than more traditional circles of competence.

This description actually understates the power of intense specialization. High productivity through the division of labor that Adam

Smith described in a pin factory has been a cornerstone of economic prosperity for more than two centuries. Individuals specialize in professions such as law, medicine, accounting, journalism, education and, almost invariably, on a relatively narrow segment area within the profession, like patent law, labor law, family law, and white-collar criminal law. Surgeons don't treat infectious diseases, and heart surgeons do not replace hips. Managers usually spend the greater part of their careers in one industry. Professional athletes who excel in one sport, even Michael Jordan in basketball, are generally out of their league when they try another, like baseball. Firms that succeed in particular industries and sometimes in specific geographies have historically outperformed firms that try to do many things—conglomerates—or operate on a global basis. Walmart could not replicate its success when it expanded beyond North America. Even within a well-defined circle of competence, investors have generally performed better starting from a very narrow focus and working outward. Warren Buffett has been more successful investing in insurance, banking, old media, and consumer non-durables than in other industries.

Within the investment profession, the most common form of specialization is by industry. Analysts on both the buy and sell side of the market are almost always industry specialists. They commonly cover industries like autos, banks, retailers, or software companies on an increasingly global basis. The challenge for both investors and analysts is that there may be long periods during which there are no attractive opportunities within any particular industry. Fortunately a viable circle of competence can extend beyond a single industry. An experienced investor or analyst should be able to master several. But those who attempt to become experts in many industries are likely to be masters of none, which puts them at a disadvantage when they trade against investors more focused in each area. This view of specialization is considerably more restrictive than what is traditionally considered a circle of competence.

Geographical specialization is another useful constraint. Focusing on firms within a particular region is an advantage particularly for those who invest in small- to medium-sized companies. First, for service businesses that operate locally, a regional focus allows the investor to monitor customers, suppliers, competitors, and local business communities at a level of detail that cannot be matched by investors who attempt to cover the industry everywhere around the globe. Second, a local presence eases continuous access to management. Visits can be more frequent within a restricted territory, and the investor can see more of the business's operations. Compared to those who rely on annual fly-ins for

meetings at company headquarters, the locally focused investor is much better informed about the current state of affairs and better able to assess where the business is headed. Third, many regional economies have an industry specialization—medical devices in Minnesota, pharmaceuticals in New Jersey, information technology in Silicon Valley, biotechnology in greater Boston—so that regional and industry specializations may overlap. For smaller companies with less analyst coverage, these advantages may be particularly potent. When Warren Buffett was running the Buffett Partnership in the 1950s and 1960s, many of his most successful investments were in companies based in and around Omaha (often in insurance or related areas), not otherwise known as a hub of innovation or profitable investment opportunities.

Specialization at the level of security selection need not conflict with the value of diversification at the portfolio level. The portfolio manager of a diversified fund can combine the choices of a group of analysts to run a product that has the benefits of both detailed industry or geographic knowledge and risk control features that come with diversification. At a higher level, the overall holdings of an institution or household can be more diversified, across asset classes, and yet be composed of securities selected by specialists operating within narrowly circumscribed circles of competence. Although reaching this goal by choosing the right investment managers is not an easy task, it is possible. We will address the challenge directly when we discuss risk management in Chapter 10. Most investment managers have resisted specialization in the manner we have described here, even though specialization represents the element of a search strategy that is most likely to put the investor on the right side of the trade.

The Value Advantage

In the first chapter, we mentioned a number of studies that employ a mechanical approach to stock selection to test various investment styles. These studies use one or several variables to rank the universe of stocks at a given moment, group these ranked lists into equally sized buckets, and compare the investment returns of these groups, now considered portfolios, over future periods. Overall, these studies demonstrate convincingly that portfolios of stocks ranked high on value outperformed the market as a whole and trounced portfolios ranked high on “glamour,” meaning they were expensive by the measure chosen.

A few noteworthy examples will illustrate the nature of this research. As we noted in Chapter 1, in 1992 Eugene Fama and Kenneth French published a seminal paper intended to reinforce the Efficient Market Hypothesis.¹ Using data on US common equities from 1963 to 1990, they sorted stocks into 10 groups based on book to market values from the lowest (glamour stocks) to the highest (value stocks) at the end of each year. They also sorted the same stocks based on size as measured by market capitalization. They then measured the monthly performance of each basket of stocks for the period July 1963 through December 1990. The average monthly return for the highest book to market decile was 1.63%; for the lowest book to market it was 0.64 %. The value decile outperformed the glamour decile by around 12% a year over the entire period. A similar if slightly smaller return difference emerged between small and large stocks; the smallest basket averaged 1.77% per month compared to 0.95% for the largest. Combining the 2 sorts to create 100 buckets ranking from largest and most expensive to smallest and cheapest added little to the picture. The smallest and cheapest averaged monthly returns of 1.92%; the largest and most expensive 0.89%.

Two findings of their work stand out. First, it is difficult to improve on a simple value strategy by adding additional variables. The cheapest decile outperformed the most expensive by 0.99% per month. Adding size, the smallest and cheapest yielded a premium of 1.03% per month, an economically insignificant gain of 0.04%, and one difficult to capture in practice because of the liquidity constraints imposed by small company shares. Second, most of the premium for value lies at the extremes. Going from the third cheapest decile to the first added far more to the return than moving from the eighth to the sixth. The same pattern held at the glamour end; moving from the third most expensive decile to the top detracted far more than moving from the middle to the third. In searching for value and avoiding glamour, it is the cheapest of the cheap you want to embrace and the most expensive you want to avoid.

A second notable paper, published in 1994 by Joseph Lakonishok, Andrei Shleifer, and Robert Vishny, replicated the Fama and French study over a slightly different period. It used earnings to price and cash flow to price as alternative measures of value. Its findings are similar: the most extreme value portfolios outperformed the most extreme glamour portfolios by around 9% annually regardless of the value metric employed.

¹ Fama, Eugene F. and French, Kenneth R. (1992). "The Cross-Section of Expected Stock Returns," *Journal of Finance* 47 (2): 427-465.

Adding a second dimension to the analysis, the authors sorted their universe of stocks into three groups based on price to cash flow and three based on sales growth over the prior five years. They found that the value stocks with the slowest rate of sales growth returned 21% a year on average whereas the glamour stocks with the highest sales growth returned 11%.²

A third key study published in 1989 by Werner De Bondt and Richard Thaler examined the impact of past stock market performance on returns going forward. At the beginning of each year, they sorted the universe by total market return over the previous 12 months. They focused on two buckets, one containing the stocks in the lowest decile of returns (worst performers) and the other with the stocks in the highest decile (best). They then tracked the difference between the returns of the two portfolios over the subsequent 30 months. Over that entire period, the two portfolio returns were approximately equal, although in the first 12 months the best prior performers continued to outperform (this is a well-established “momentum” effect, about which more below). They then repeated the process using the previous 24 months of returns to sort for the worst and the best performers. The most disappointing stocks over the prior two year period outperformed the high return stocks by about 12% over the next 30 months. Finally, they formed portfolios from the stocks that had done the worst and the best over the prior 36 months. These long-term worst stocks earned returns on average of 18% above those of the best stocks over the subsequent 30 months. Again, the more extreme—in this case the most extended the period of underperformance—the greater the outperformance going forward.³

An extensive group of studies examining equity returns in the United States and overseas over longer time periods confirm these general results.⁴ Portfolios of ugly, disappointing, obscure (small), and boring (e.g., low growth) stocks repeatedly generated higher returns than both the market as a whole and, more strikingly, portfolios of attractive, highly profitable, well-known (big), and glamorous

² Lakonishok, Josef, Shleifer, Andrei, and Vishnu, Robert W. (1994). “Contrarian Investment, Extrapolation, and Risk,” *Journal of Finance* 49 (5): 1541–1578.

³ Werner, F., De Bondt, M., and Thaler, Richard H. (1989). “Anomalies: A Mean-Reverting Walk Down Wall Street,” *Journal of Economic Perspectives* 3 (1): 1898–202.

⁴ The literature is abundant and grows larger every month, if not every day. We have mentioned 2 of Fama’s students, Clifford Asness and Wesley Gray, as maintaining websites that include both their firms’ ongoing work and in Gray’s case, the work of others. For Asness’s firm, see <https://www.aqr.com/>; for Gray’s, see <https://alphaarchitect.com>. One outstanding article deserves mention: Asness, Clifford, Moskowitz, Tobias, and Pedersen, Lasse. (2013). “Value and Momentum Everywhere,” *Journal of Finance* 48 (3).

(e.g., fast-growth) stocks. Their low prices more than compensated for their unappealing characteristics. Moreover, as these studies demonstrate, especially that by De Bondt and Thaler, it is the extremely unappealing portfolios—three years of disappointing returns—that produce the best results. The future returns of portfolios organized by market-to-book ratios increase steadily with the degree of cheapness. The cheapest decile bucket of stocks often see two-thirds of the companies go bankrupt. But the ones that do survive do so well relative to their very low starting prices that the ugliest portfolios are the best performing among all the buckets.

In addition to these basic value/glamour measures, other variables have been studied to assess their impact on portfolio performance. They come in three varieties: (1) purely fundamental; (2) purely technical; (3) a combination that relates market price to a fundamental factor, like price to book value or price to earnings. Those that use only fundamental variables refer only to a company's business performance, not the relationship between that performance and its share price. Studies have sorted stocks using returns on equity or on total capital invested, growth in earnings per share, growth in assets—as opposed to sales growth—and various measures of profit margins. Companies with high marks on these variables are successful firms whose shares are inherently attractive to investors. However, consistent with the studies we discussed above, it is often the firms that ranked lowest on these measures—low returns on capital or narrow profit margins—that have tended to generate the highest future market returns.⁵ At worst, these studies demonstrated little or no positive relationship between future returns and the quality of current firm performance. Michelle Clayman's telling phrase, "in search of disaster" has been a better investment guide than "in search of excellence."⁶ A supportive example of this phenomenon emerged from the widely read and much admired book *Good to Great* by Jim Collins.⁷ Collins selected 11 companies that transformed themselves from good to great. His selection was based on hefty stock market outperformance over the prior 15 years. He and his researchers identified the common

⁵ In double sorts—sorting by return on capital and on market to book value, for example—the business quality measures tend to add to returns only among the middle stock by value. For the extreme low value stocks—the truly cheap—the quality measures at best add nothing to returns.

⁶ Clayman, Michelle. (1987). "In Search of Excellence: The Investor's Viewpoint," *Financial Analysts Journal* 43 (3).

⁷ Collins, Jim. (2001). *Good to Great: Why Some Companies Make the Leap ... and Others Don't*. New York: HarperCollins, 2001.

characteristics that distinguished the greats from peers who did not make the transformation. The book was published in 2001. In the next few years, two of the “great” firms, Circuit City and Fannie Mae, managed to go bankrupt; in the case of Fannie Mae, in a spectacular fashion. For these two, Good to Great became Good to Great to Gone.⁸

A second type of study examines stock market returns and trading volumes without reference to any fundamental data and looks for patterns like those in the De Bondt and Thaler paper. This is technical investing, most commonly focused on price momentum. A frequently used metric is relative strength, which is simply the price performance of the stock relative to all other stocks in the pool. Choosing a portfolio of last year’s best performing stocks is a bet that winners will continue to excel, and the bet has paid off. For the 40 years from 1979 to 2018, a portfolio that was long the prior year’s best 30% performers and short its worst 30% returned 4.2% annually. In the period since the recovery in 2009, when bull market returns belied traditional value measures, this momentum approach returned 2.8% per year.⁹

But momentum plays out fairly quickly. The French portfolios are rebalanced every month. And as De Bondt and Thaler discovered, when stocks are selected on the basis of price change over three prior years rather than last year alone, the results are reversed—the worst do better. The high flyers fall back to earth, the downtrodden arise. We witness a phenomenon found so frequently in nature and culture—reversion to the mean. The children of tall parents do not, on average, exceed their parents in height; if they did, centers in the NBA would not command the salaries they do. And if prior winners in the stock market continued to outperform over sustained periods, their market capitalizations would rise to the sky, leaving behind any fundamental value that might support the share price. There are manias in the history of investment; they always collapse.

The third type uses both financial and price variables to relate the price of the shares to some fundamental company information: share price to earnings, share price to cash flow, share price to book value,

⁸ The remaining nine did poorly in the stock market through 2010. As a group, they had negative returns. <https://memnorthwestern.wordpress.com/2013/01/02/not-so-good-to-great-companies-and-a-heuristic-thinking-strategy/>

⁹ The data is from the momentum factor on Kenneth French’s website. We should note that long-short portfolios constructed like this are “zero cost,” meaning that minimal money has to be invested in them, and the return on the investment is astronomically high—provided you don’t go bust in any year.

share price to sales, share price to dividends, and more. The Fama-French paper is in this camp, as is that by Lakonishok and co-authors discussed above. In all of these studies, the value stocks, those with low share prices relative to each of these other variables, outperform the glamour stocks. These ratios offer slightly different snapshots of the same underlying picture: how much investors are willing to pay for the future success of the company, whether measured by earnings, cash flow, sales, net assets, or dividends. The less investors want to pay, the better the prospects of the shares, if not the company itself.

Overall the statistical evidence both for the United States and overseas markets paints a remarkably consistent picture. Embracing stocks that are ugly, boring, obscure, disappointing, and therefore cheap, has historically been the best way to buy stocks, in which you are likely to be on the right side of the trade. These are the stocks that have outperformed the market as a whole. The uglier, more boring, more obscure, more disappointing, and therefore usually the cheaper the stock, the better the returns have been. That, at any rate, is the historical record and that history, thanks in large part to studies we have cited and more we have not, is now widely recognized. The next and obvious question is whether this situation will persist.

The argument in favor of persistence is the fact that the value premium, the return advantage of value relative to glamour stocks, which was first identified by Benjamin Graham in the 1930s and systematically quantified since the 1970s, appears not to have diminished significantly over time. Nevertheless, it is crucial to understand why value search strategies worked so well in the past and whether the forces responsible will continue to operate in the future.¹⁰

The Anomaly of Value and the Nature of Human Behavior

Superior investment returns from value stocks over time constitute an anomaly. Once investors know that cheap stocks outperform expensive stocks, they should bid up the price of the cheap stocks and eliminate

¹⁰ As we write, in 2020, the value premium has gone missing essentially since the current bull market began in 2009, and particularly since 2016. There have been prior periods in which value trailed glamour and the market as a whole. If reversion to the mean is no longer operative, the economic world of the future will be a very different place than it has been until now.

the superior performance. Yet the value premium has endured. There have been two distinct responses to the evidence that cheap stocks provide superior performance. One seeks to explain it, the other to explain it away.

The explanation for why the value anomaly exists in the first place, has been durable over time, and is likely to persist into the future, lies in certain deeply embedded features of human behavior. These behaviors lead people to make decisions that violate rational economic calculations. They are apparent in activities like the ubiquitous popularity of lotteries that produce negative returns, a fact that virtually everyone knows even as they buy their tickets. They have also been identified and investigated in the field of cognitive psychology, most notably by Daniel Kahneman and Amos Tversky.¹¹ The field of behavioral finance is a direct outgrowth of this work.¹²

Three of these behaviors are particularly important for understanding the existence of the value premium. First, human beings have always been prone to overpay for the dream of getting rich quickly. Lotteries, as we mentioned, have been successful ventures in almost every society and in every period of human history for which we have a record. Yet lotteries are and have always been astonishingly poor investments. Lottery operators, whether governments or criminal associations, have taken between 30 and 70% of the cash paid in before distributing the rest to the winners. People who buy lottery tickets know that the odds are seriously stacked against them, but that knowledge has not made lottery tickets harder to sell. They have persisted even when they have been made illegal. A broad cross-section of the population buys tickets; the appeal of instant riches is not confined to the poor or less educated.

In the more sober world of equity investing, glamour stocks are the lottery tickets that hold out the promise of spectacularly high returns. Because of this appeal, some investors are willing to pay irrationally high prices relative to their fundamental values. This pattern came into sharp focus, at least after the fact, during the technology boom of the late 1990s. Established companies such as Intel, Microsoft, AOL, and Cisco, plus a large group of immature start-ups, commanded what were in retrospect absurd prices. But even during less frothy times, “glamour” valuations are a regular component of equity markets. Sometimes it is

¹¹ See Daniel Kahneman, *Thinking, Fast and Slow*, New York: Farrar, Strauss and Giroux, 2011, for a summary of much of the research.

¹² Richard Thaler, whom we cited above, has been a pioneer in the field and a winner of the Nobel Prize for his efforts, as has Daniel Kahneman, even though his discipline has been psychology.

national markets, such as Japan in the late 1980s or the BRICs in the 2000s, sometimes it is specific industries, such as telecommunications or biotechnology, sometimes specific companies such as Apple or Cisco, that trade at prices inflated by their “lottery ticket” appeal. Investors who succumb to this widely shared attraction will, as a group, underperform the market as a whole. Given the zero-sum nature of investing, value investors who take the other side of these trades will outperform. As long as lotteries and their stock market equivalents retain their attraction, this source of the value anomaly is not likely to disappear.

The second major behavioral feature contributing to the value anomaly is an inverse of the preference for lottery tickets. In investing as in life, humans avoid situations that look ugly, troublesome, and appear to offer a high probability of loss (and lack the perverse highs associated with more exotic forms of risk-taking, such as skydiving).¹³ In the field of behavioral finance, this trait of investors is referred to as loss aversion. For most people, the pain of an anticipated loss outweighs the pleasure of an equivalent gain.¹⁴ As a consequence, people will embrace highly risky choices to avoid certain losses that they would never countenance in exchange for foregoing certain gains. As a consequence, if offered a portfolio of stocks two-thirds of which may go bankrupt, most investors will recoil in horror before considering the potentially enormous gains from the one-third that do survive, gains large enough to make the whole investment highly profitable. It should come as no surprise that these investors avoid ugly and disappointing stocks in the first instance and, for stocks that do turn sour when already owned, dump them reflexively on the market without careful analysis of the events that have given rise to the ugliness and disappointment. The natural consequence is that these stocks will be chronically undervalued. Being on the other side from this behavior should be comparably rewarding. Again this kind of loss/ugliness aversion has been a part of human behavior for millennia and is unlikely to change anytime soon.

The third psychological factor underlying the value anomaly is so deeply embedded in human behavior that it dominates discourse without widespread awareness of its impact. Humans have an innate

¹³ The foundational paper is Kahneman, Daniel, and Tversky, Amos. (1979). “Prospect Theory: An Analysis of Decision under Risk,” *Econometrica* 47 (2): 263–291.

¹⁴ Loss aversion is different from risk aversion in several ways. First, risk aversion applies to overall *levels* of wealth compared to needs. Loss aversion applies to changes in wealth around current levels. Thus, increasing wealth should lower risk aversion but has no impact on loss aversion. Second, even in situations of certainty, sure losses hurt much more than sure gains, even if the changes in overall wealth are relatively small.

tendency toward overconfidence; we embrace certainty and ignore alternative possibilities. In discussions of public policy issues, such as the presence of weapons of mass destruction in Iraq in 2002–03, few talked of probabilities. The discussion was couched in the language of competing certainties. One side insisted that the weapons were there for sure; the other side insisted with equal certainty that they were not. Virtually no one ever addressed the issue in terms of probabilities, such as a probability of one quarter or one third or one half or two thirds. Framing the issue as a set of alternative probabilities was simply not part of the debate. After the fact, of course, they were either there or not there. But beforehand no one knew for certain whether the weapons were there or not. Rational opinions should have taken the form of these kinds of probability statements. Both rhetorically and in their own minds, each side of the debate was radically overconfident in the likely accuracy of their judgments. Such overconfidence is a systematic human tendency.

Overconfidence dominates investment thinking and discussion in the same way. When talking about “good” stocks with “great” prospects, advocates never say that they are likely to outperform average market returns two-thirds of the time. An investor who achieved that degree of accuracy over any significant period of time would be in the top 10th of 1% of all investors. They tout the stocks with absolute assurance and are very surprised when they subsequently underperform the market. Those who recommend avoiding other stocks do so with the same degree of certainty. They talk of poor investments as “dogs,” not as stocks with at least some possibility of substantial gains.

The effect of this overconfidence is to intensify the overvaluation of the glamour stocks, since the possibility of an unanticipated adverse consequence is implicitly minimized. At the same time, overconfidence amplifies the undervaluation of ugly and disappointing stocks since their potential recoveries are similarly minimized or ignored. Overall, overconfidence works to magnify the return distortions between glamour and value stocks. Until evolution works to eliminate it as a human characteristic, the value anomaly is likely to persist.

Overconfidence also adds an important temporal dimension to the value anomaly. In good times, overconfidence tends to lead investors as a whole to minimize potential negative future developments much more than a rational calculation of probabilities would suggest. Investors extrapolate present favorable conditions into the indefinite future. Evidence of this practice appears in those recurring radical overvaluations in

market prices that Robert Shiller has labelled “irrational exuberance.”¹⁵ In these periods estimates of market volatility, which are implicit in the prices of derivative instruments, typically fall to historic and unrealistically low levels.¹⁶ In good times, option prices tell us that investors typically expect smooth sailing when the historical record shows that on average subsequent market movements have been exactly the opposite. In bad times, when prices have fallen and the economic outlook is gloomy, investors are doubtful that good times will ever return. They are overconfident in their pessimism. Then demand for option protection is correspondingly high and options are expensive. In a very real sense, the neurotic extremes of Benjamin Graham’s Mr. Market are a tribute to the power of overconfidence. Unless Mr. Market somehow settles down, either through finding the right bipolar medication or from a change in human behavior, intertemporal (and thus cross-sectional) value anomalies will continue to exist.

Psychological experiments indicate how difficult it is to escape this overconfidence. In one well-known experiment, subjects looking through eyeholes into a darkened room were asked to estimate the distance to a luminous square. Since they were not told the size of the square—squares of different sizes were selected at random and placed at random distances—they could not possibly give an accurate estimate of this distance. Yet subjects invariably gave a precise distance estimate in feet and when asked to judge the likely accuracy of this estimate gave error brackets that were small (e.g., plus or minus 1 foot). Faced with a situation of fundamental uncertainty, the subject reflexively embraced a high degree of certainty. In fact, their estimates of distance were completely unrelated to the actual positions of the squares, and their actual average errors far exceeded their error estimates. The experimenters then showed their subjects the squares of different sizes and explained that squares would be chosen at random. The experiment was rerun and subjects continued to give precise estimates of the square locations. Their error estimates did increase (e.g., from 1 to 2 feet) but remained well below the actual errors. Overconfidence appears to be highly persistent, even in the face of contrary evidence that seems to be discounted via a mechanism known in the behavioral literature as “hindsight bias”—the “I knew that already” impulse. A large number

¹⁵ Shiller, Robert, *Irrational Exuberance*, Princeton, 2000.

¹⁶ If investors feel the future course of securities prices are only slightly uncertain, then the cost of options to buy or sell at fixed prices will be low since the prospect of large gains are considered unlikely.

of other experiments have been run with both experts and lay subjects who are asked to estimate present or future quantities; both groups persistently overestimate the accuracy of their judgments. And, if anything, the experts are worse than the lay subjects.¹⁷

A further issue regarding the past and future of the value anomaly concerns the organizational structure within which investment decisions are made. Individual investors may well continue to be subject to lottery preference, loss aversion, and persistent overconfidence. However, as most investment dollars are now in the hands of investment organizations, these organizations should in theory be able to put rules and structures in place that control aberrant or counter-productive individual behavior. Still, it is individuals working for organizations that make and implement these processes. These people have their own interests and agendas, some of which may not be in line with the interests of the organizations for which they work. They also have their own psychologies over which they may have little control.

Organizational policy will always be in tension with the instinctual behaviors and incentives of both their own employees and their retail and institutional clients. Therefore, it makes sense to think about both organizational imperatives themselves and the behavior of managers within the organizations. When we do that it will become clear that organizations are as likely to amplify individual behavioral biases as to reduce them.

For organizations, investment biases are generally a consequence either of policy or size. We'll start with policy. Many investment organizations are prevented, whether by charter, by stated investment policy, or by legislative intervention, from owning certain kinds of stock. Shares of companies that engage in businesses deemed socially irresponsible, whether for environmental, health, or regime related reasons, are off-limits. If many funds adopt policies that compel them to avoid the same companies, the normal demand for these shares may be substantially reduced. Unless there are enough funds with "socially irresponsible" investment mandates—buy only stocks of tobacco companies, manufactures of defective infant car seats, or polluters—shares of the "dirty companies" may be permanently undervalued as measured by current earnings or growth prospects. It will take a change in investment policy, a change in corporate social behavior, or a reorganization of the

¹⁷ See the work of Philip Tetlock, especially *Expert Political Judgment: How Good Is It? How Can We Know?* Princeton, NJ: Princeton University Press, 2005.

business to eliminate this prohibition and allow the shares to be revalued upward. So long as the prohibition is stable, the stocks may remain permanently depressed.

The issue of size bias is more important and more interesting. Many funds cannot invest in small companies, either because their mandates do not allow it or, more frequently, because they have too much money to manage and small companies just can't absorb enough of it to make it worthwhile. If a diversified investment company—most mutual funds—with \$10 billion to invest wants to own stock in 100 companies, it needs on average to buy \$100 million in each. Since the fund does not want and is often not allowed to own more than 10% of any company's stock, this limits the universe of investment choices to firms with market capitalization of \$1 billion or more. The specific sizes of funds, the market capitalization of suitable firms, and the percentage of ownership will vary, but the impact of company size persists. Many funds simply cannot buy shares in small companies. For similar reasons, investment analysts within institutions will also concentrate on large capitalization opportunities. An insight into a \$40 billion company that can absorb a \$2 billion investment will be worth far more than an equivalent insight into a \$400 million firm that can absorb only \$20 million. The consequence is that coverage of and demand for small company shares will tend, all other things being equal, to be below that of large company shares, and small company stocks should be cheaper.¹⁸ Hence, the shares in small companies are cheaper, all other things being equal, than shares in large companies. Among the "all other things" that need to be held constant, growth prospects are the most important. Small companies typically have the opportunity to grow faster than large ones, which already control major segments of the markets in industries that are likely to be well developed.

The only thing that small companies need do to qualify for large funds is grow. There are plenty of other investors out there to raise the value of the company's shares as its revenues and earnings increase. At some point, a small company, with a market capitalization of \$200 million, has grown into a mid-cap firm, with a capitalization of \$1 billion. As it matures, it becomes an eligible purchase for more funds; its shares

¹⁸ The share of small companies may be chronically undervalued, but higher earnings returns should result in higher distributions and/or, if retained earnings are effectively invested, higher rates of growth in value. Small firms can produce attractive returns even if prices never fully converge to value. Also, years of profitable growth transform small firms into larger ones, with more coverage, allowing them to shed the small firm discount.

are priced at less of a discount. It ceases to be an opportunity and emerges as a success. This cycle is one of perpetual renewal, and new small firms spring up to take its place in the shade. The shares of companies too small for big funds are always available on sale.

A revealing example of the size bias as it affects stock performance concerns the record of corporate spin-offs. Sometimes a company rids itself of a division or other unwanted business unit by issuing stock in this new corporation and distributing them to existing shareholders. The shares in this spun-off company now trade on their own. In most cases, the new firm is small, especially when compared with the giant from which it has just been separated. Funds that owned stock in Giant now find themselves holding shares in New Micro Enterprises, with a market capitalization of \$100 million. They may know little about the new business but are certain of one thing: it isn't big enough for them to spend much time learning about it since it is too small for them to buy. So they sell—dump—the shares on the market and pocket the money. Spin-offs are a wonderful opportunity for investors who are not constrained by questions of market capitalization. Since many of the shares are sold for reasons unrelated to the company's prospects, there are bound to be gems tossed away by the large funds for whom a small company stock, though perhaps a potential star, is still a nuisance. Spin-offs intensify the small company bias. The ordinary small company has its shares ignored by the large funds; the spin-off has them actively sold off. This is only an extreme illustration of the organizational bias against the shares of small companies that has made them, over time, a fertile field for value investors.¹⁹

Client demands are a second influence on organizational behavior. Investment company managers, whose clients are individuals, must respond to the irrational biases of those individuals. Organizations who invest for institutions such as charitable endowments, pension funds, and family offices must answer to the non-investment professionals who often run those institutions, human beings with the usual individual behavioral biases. In marketing to clients, investment managers spend

¹⁹ Noted value investor Joel Greenblatt let the cat out of the bag when he published *You Can Be a Stock Market Genius Even if You're Not Too Smart*, New York: Simon and Schuster, 1997. Since then, spin-off investing became more popular. New funds were organized to take advantage of the opportunities, and existing funds devoted more attention. The returns have been uneven and less spectacular than those Greenblatt reported. Given the famously adaptable nature of capitalism, similar adjustments to other organizational constraints probably mean that these restrictions are a less durable source of investment opportunities than human nature.

more time discussing their big winners—investing in Apple at \$10 and seeing it subsequently go to \$700—and devote less attention to their long-term average returns and to the investments that did not work out. The need for great stories to tell induces the organizational managers to overinvest in the same glamour stocks as individuals. When an organization discloses its portfolio holdings, clients are likely to judge them, at least partially, on the basis of their own loss aversion. Portfolios of troubled, disappointing stocks are a hard sell to clients. As a result organizations are prone to “window dressing” by purchasing shares of respectable, high-growth, high-return stocks just before any disclosures are due. This behavior reinforces rather than offsets the tendency for such stocks to be overvalued and amplifies the undervaluation of the diseased and disappointing stocks that they avoid or sell before they disclose holdings. Finally, there is substantial evidence that institutional clients tend to remain with their existing investment managers unless these managers significantly underperform their relevant market benchmarks. The risk-minimizing path for organizations is to buy what everyone else is buying. This herding instinct reinforces the tendencies to buy stocks that are glamorous and have recently been successful and shun the value stocks that have disappointed.

In the mid-1960s, these forces led organizational investors to focus largely on the “nifty 50” stocks. These were large stable companies—IBM, AT&T, General Motors, Xerox, Morgan Guaranty—that both managers and clients saw as having high return potential with little risk. Between 1965 and the end of 1974, a “nifty 50” portfolio typically lost more than 80% of its value. But the lessons of investment history have a short half-life; the cast of managers and their clients changes rapidly. In 2015, all of the gains in the S&P 500 came from four stocks: Facebook, Amazon, Netflix, and Google. The Nifty 50 had become the FANG 4. Microsoft and Apple joined the select group in the next several years. Any management firm that did not join the chorus almost certainly suffered by comparison. Organizational imperatives, unless they change radically, appear unlikely to undermine the value anomaly in the future.

The herding impulse is also felt by the individuals within investment companies who make the actual investment decisions. Money managers are employees, hired to produce results by following prescribed investment policies. Though there may be rewards for original thinking that leads to extraordinary success, the safest path is to look pretty much like everyone else investing with the same mandate. The now antiquated

adage of data processing managers, that nobody was ever fired for buying computers from IBM, applies as well to money managers. Nobody loses a job for average performance or for holding the same securities as the rest of the managers. If you stray and buy stock in a company that nobody else wants, the payoffs are skewed. Should the company recover and the stock rise, the praise is real but short-lived. If the company does poorly and the stock falls, everyone remembers that you chose that “dog.” The situation becomes most extreme toward the end of a reporting period, when managers, as we have noted, “window-dress” their portfolios, dumping the stocks that have fallen in price and loading up on the past year’s (or quarter’s) successes. This pruning has the effect of driving up the price of currently successful stocks and depressing even further stocks that are already downtrodden. The pressures on individual managers within organizations largely mirror the pressures on the organizations themselves. They tend to reinforce rather than offset the value anomaly.

Taken as a whole, these elements of human psychology and organizational behavior provide an explanation for the empirical data that has demonstrated the superiority of a value-oriented search strategy: look for opportunities among ugly, disappointing, boring, obscure (small) companies. As long as they continue to play their historical role going forward, the value strategy will continue to outperform, even after evidence about its exceptional performance has been widely disseminated. Thus, in a world where there are colossal rewards for uncovering discrepancies between the current price of a stock and its true, intrinsic, or ultimate worth, some discrepancies persist. These discrepancies outline the types of opportunities in which a searcher for fundamental value is more likely to be on the right side of the trade.

Explaining Away the Value Anomaly

The behavioral and organizational features that we have described are at the center of our efforts to explain the existence and persistence of the value anomaly. A second approach, to which we alluded earlier, is to try to explain it away. This approach assumes that the value portfolios are riskier than the market as a whole, which, in turn, is riskier than glamour

portfolios.²⁰ In a rational market, greater risk should be associated with greater reward. It is higher risks that explain the “anomaly” of above market returns for value portfolios and lower risks that explain the below market returns of glamour portfolios. Essentially, this explanation is a defense of the Efficient Market Hypothesis. But if the pattern of relative risk explains return variations, the risks at issue should be observable and measurable. Otherwise, how would investors know which stocks should command higher returns? By “risk” here we are referring to volatility, accepting for the moment the meaning of the term as defined in the Efficient Market Hypothesis. Unfortunately for the Efficient Market Theory, historical statistical evidence suggests that value portfolios have been no riskier than the market as a whole and that glamour portfolios have, if anything, been riskier than the market. These results persist whether we use variance as the volatility measure or beta, comparing the responsiveness of the portfolio to changes in the market as a whole. To repeat, if volatility is our definition of risk, value portfolios have not been riskier, and glamour portfolios have been less risky than the overall market.²¹ A disciplined value investor, employing sound value approaches to risk management, should not hesitate to look for opportunities to produce higher than average returns in exchange for taking on theoretical, albeit undetectable and perhaps non-existent, higher risks.

Search Strategies in Practice

In principle, specialization and value-based search strategies are complements and not alternatives. In practice, their relative weights depend on the particular mandate under which an investment manager operates. Managers with broad mandates—large cap global equity, United States fixed income, or responsibility for the greater part of the wealth of

²⁰ This assumption is not limited to the value anomaly but is attributed to all other statistical anomalies as well. If there are portfolios with inexplicably higher returns, they are all assumed to entail higher risks, even if those risks have never been detected.

²¹ See the Appendix to Chapter 1 for data on value portfolios performance during bear markets. The paper by Lakonishok et al. cited in Footnote 2 has a useful discussion of value portfolios performance during down markets.

their investors—will necessarily have well-diversified value-based search strategies. Even under these conditions, however, the benefits of specialization apply. Developing industry or geographic expertise will still enhance the chances of being on the right side of particular transactions. For investors with more restricted mandates, expected to hold portfolios with fewer and larger positions, specialization will be a necessary point of departure. But whether in industries or geographies, value principles still apply. Looking initially at small, boring, ugly, and disappointing firms within the parameters of a specialization will have all the advantages that we described above. Indeed, the advantage of specialized knowledge is likely to be even more significant with respect to value stocks when one is considering wounded companies, and the critical and difficult challenge is to distinguish those that are terminally ill from those that are only temporarily impaired.

Specialized knowledge develops slowly over time. Older, more experienced investors will be capable of managing a greater range of specializations than newly minted ones. The experience of successful specialists suggests that at a minimum it takes a concerted effort of 1000 hours or more to develop a decent working knowledge of an industry. Given the other demands on an investor's or analyst's time, that translates into at least a full year to achieve an adequate level of mastery.

The knowledge and experience that constitute specialized expertise must be constantly nurtured and extended, especially in times of transition and uncertainty. For example, retailing has historically rewarded superior merchandising talent, such as the ability to create positive store experiences and select the right products. Successful retailers have generally been outstanding merchants. The rise of Walmart was built on the increasing importance of efficiency in local store operations and especially logistics. Local market share, thanks to the density of stores, with the associated benefits of scale in local storage management, distribution, and advertising, all of which allowed for lower prices, became far more important than innovative merchandising, which is much easier for competitors to copy. The increasing importance of web-retailing diminished the importance of store operations and elevated distribution logistics and order management into significant sources of retail success. Understanding the nature and pace of these transformations is essential for evaluating both the strengths of competitive positions and the competencies of individual retail managers, upon which successful retail investments depend. Only by continuously deepening one's retail specialization will an investor be able to keep abreast of such developments.

A second benefit of continuous industry or geographic engagement is the cultivation of an improved network of industry contacts. Attempting to do everything oneself is not usually an efficient approach to investing or indeed any undertaking. Being industry focused means following and repeatedly interacting with operating managers in a particular industry, which can produce a sharper understanding of their strengths and weaknesses and a better calibrated judgment of their opinions regarding industry developments. Maintaining productive relationships with other capable industry experts also has value, especially when these other experts have geographic focus different from one's own.

Value-based search strategies come in two varieties: quantitative and qualitative. At one extreme, quantitative strategies seek to reproduce or improve on the laboratory portfolios whose outperformance constitutes an important part of the case for a value approach. These strategies use statistical measures of cheapness, size, price momentum, and industry/company level accounting information to build portfolios from the stocks within the value buckets that these variables have defined. Often these portfolios are modified to limit or overweight concentration in particular industries or geographies in order to control overall portfolio risk. Some quantitative managers adjust the underlying models generating these portfolios to respond to putative evolution in market conditions. These adjustments are not always successful. Some managers run portfolios for which these search strategies are not just the beginning but also the end of the investment process. These "quant" value managers may have been successful, but the ubiquity of computing power and statistical expertise have allowed others to quickly detect the specific strategies and, by crowding in, curtail if not eliminate any "quant" advantages beyond a simple value approach.

Other managers begin by using software and financial databases to generate lists of potentially desirable investments from which they select certain companies for further detailed investigation. Basic measures of size, cheapness, and disappointment are used to screen for potential investment. This practice has limitations. On the one hand, given the large number of potential securities, even screens that employ three variables (e.g., small size, low market-to-book, low prior returns over the previous two years) to limit the list will produce a long, highly varied set of companies across many industries and geographies. Such a list may be dauntingly large. Also, it changes relatively slowly so that repeating the process at regular intervals does not add much. In practice, therefore, more useful screens are things like the "new lows" lists on various markets that can be quickly and conveniently reviewed, since

opportunities to buy at value prices usually arise from price declines. On the other hand, if screening is limited, say by industry or to large capitalization stocks, in order to produce a more manageable set of potential opportunities, experienced investors often find that they are already familiar with most of the names. Quantitative screening tends to be valuable more for novice investors than for veteran ones.

Beyond specialization, most value search strategies are commonly implemented informally. Start with industries in trouble; ones that appear disappointing, dispiriting, downtrodden. These are likely to trade at low multiples of value. Within industries, look for the same characteristics in firms, especially among smaller firms that are inaccessible to large investment funds and hence lack coverage by security analysts who need to get paid for their work. The company that has been doing the same thing for years, growing slowly and profiting modestly, is not going to spend its funds courting attention from analysts. A change in the fortunes of this kind of firm is more likely to go unnoticed than if it were doing something with a lot more flash.

Undesirability has other signs. Companies in bankruptcy or suffering from severe financial distress are clearly undesirable, except to the knowledgeable investor who sees the real value of the assets and the business that may emerge after reorganization. Companies in industries that are suffering from overcapacity, a sudden increase in imports, general decline, or the threat of legislative or regulatory impairment, may also be undesirable. Lawsuits, both current and potential, may make companies unattractive. And few things are more depressing than protracted underperformance. We are not referring here to a stock whose price drops by 50% in a week, or even a day, but to one that has substantially lagged the market for two or three years. These indicators of undesirability identify potential areas of opportunity; as investors flee from bad news or poor performance, they discard stock at prices that may exaggerate the company's distress. Not always, of course. Sometimes things are worse than even the gloomiest analyst imagines, and the current low price of the stock is actually too high. But overreaction is frequent enough so that the informed and diligent investor may find bargains in the trash.

There are securities that are mispriced because of institutional constraints or mandates and other temporary aberrations. When the Resolution Trust Company in the late 1980s and early 1990s disposed of assets it had acquired in taking over failed savings and loan companies, its aim was to get itself out of business and get these assets back onto the tax rolls. So investors who had the expertise and made the effort to value these assets, whether real estate, junk bonds, or the savings institutions

themselves, were able to purchase them at sale prices. Though opportunities such as these are not everyday events, they happen with sufficient frequency to keep value investors attentive to the next opportunity.

Finally, there are companies with divisions performing so poorly that the record of the whole company suffers. If the stock price reflects the earnings—often losses—of the whole company, then the only thing management need do to turn things around and boost the share price is to kill the division. Most of these situations do not escape notice from the sharp eyes of Wall Street analysts, but there are always a few with situations that are novel or complicated enough to avoid detection. They await the value investor with the knowledge and time to disaggregate the company's results and spot the earnings potential. They require as well some catalyst to encourage the company's executives to rid themselves of the albatross and let the true value emerge. It doesn't always work out that way.

We want to emphasize that all of this work is a starting point. The purpose of the search effort is to reduce the investment universe to a manageable size so that valuation analysis in depth can begin. The actual work of valuation and investment starts after the candidate opportunities have been selected. But a well-conceived search strategy plays an additional and essential role in the investment process. If, after careful analysis, an investment appears to be available at a large discount to intrinsic value, you always have to ask why, in a world of energetic, intelligent investors, this opportunity is being presented to you in particular. Why are you more likely to be on the right side of the trade? In the case of specialization, it is because you are particularly qualified to identify value. In the case of a broad value approach, it is because you have inoculated yourself against the behavioral and institutional biases that affect other investors. In both situations, the search strategy provides an important part of the case for any investment.

3

Valuation in Principle, Valuation in Practice

Assets, Earnings Power, and Growth, Maybe

For a Graham and Dodd investor, valuation is naturally at the heart of the investment process. The same standard applies to a valuation process as to a search strategy: is it better than the valuation practices being used by those on the other side of the trade? If Graham and Dodd valuations produce more accurate, better understood estimates of what you are getting for your money than widely used alternatives, then valuation is a second key step in the investment process in which value investors will have an advantage.

Part of this advantage is rooted in the recognition that securities have intrinsic values determined by the fundamentals of the underlying businesses. In Graham and Dodd's time, much market activity was driven by speculation about how the trades of knowledgeable "inside" investors would influence future prices. Fundamental analysis of company operations was rare and applied largely to fixed-income securities. Essentially value was defined as tomorrow's security price, and market dynamics—technical analysis of a primitive type—were assumed to be the dominant factor in determining that price. As a prime mover behind the professionalization of security analysis, Benjamin Graham naturally assumed that his own attention to business fundamentals would give him a large advantage over the speculators on the other side of the trade.

Even today market level technical and macro strategies, neither of which pays attention to specific company financials, still play a significant role in approaches to valuation and security selection. However, Graham's success has meant that attention to underlying business fundamentals is the rule rather than the exception. Yet there remain critical differences between fundamental analysis in the value tradition and much of what is being done today by people outside that tradition. One important difference is rooted in the time horizons of investors. Investors with short horizons, whether months, weeks, or even days, typically focus on changes in values, not on the level of those values. A large number of professional security analysts concentrate on forecasting changes in those business operating variables, such as earnings, revenues, returns on investment, and product market shares, which are assumed to determine changes in the prices of securities. Because these are short-term forecasts, they generally take the existing levels of securities' values as given by their market prices. If the analysts' forecasts are more optimistic than the market consensus, they anticipate that prices will rise in the short term and they buy the securities in question. If they are more pessimistic, they sell.

From a Graham and Dodd perspective, this approach ignores vast amounts of pertinent information that would allow an assessment of intrinsic value independent of the current market price, especially information related to a company's long-term prospects. If Mr. Market has temporarily inflated the price of a security, this will be neither recognized nor taken into account by short-term, forecast-driven valuations. For overvalued stocks, even optimistic forecasts that turn out to be accurate may produce disappointing returns if the starting price is too high. The opposite often holds true for undervalued stocks and pessimistic forecasts; low purchase prices can produce positive returns even if the company performance has not improved. Overlooking important information about the level of intrinsic value will put short-term investors at a significant disadvantage.

Adherents to value investing as a discipline focus on measuring the intrinsic value of a security through careful analysis of business fundamentals from a long-term perspective. They believe opportunities for profitable investments are available when the current market price of a security deviates significantly from its intrinsic value. The essential task of the value analyst is to determine intrinsic value with enough accuracy to take advantage of the market's mispricing and to have the patience to wait for the market or some event, such as a take-over, to close the gap between price and value.

This basic approach of comparing price to a fundamentally determined intrinsic value is far more widely practiced by investors than it

was in Graham's time, including many who would not put themselves in the value camp. Competition for buying mispriced securities is significantly greater than in the past. But there are methods in abundance for estimating an asset's true value. We contend in this chapter that the Graham and Dodd approach, as it has been refined over the years by value practitioners such as Warren Buffett and others, remains superior to commonly used alternatives. The historical and likely future successes of a value approach arise from a superior valuation methodology as well as a better search strategy.

Common Approaches to Fundamental Value Measurement – Multiple Valuations

The single most widely used approach to determining the value of a business's operation, and thereby the value of its securities, is a multiple valuation. A multiple valuation has two components: (1) some measure of the distributable cash flow that is generated today by a business or security; (2) a ratio of value per dollar of this cash flow—the multiple—that an investor should pay to buy this cash flow from today forward. The product of the cash flow estimate and the multiple is a current estimate of the value of the business or security. The heart of a multiple approach is the selection of a measure of the distributable cash flow and the determination of the proper multiple to apply to that cash flow.

In selecting a cash flow variable, analysts who have confidence in the accuracy of financial accounting take future net income to be an appropriate measure. Those concerned with the distorting impact of leverage and taxes use operating earnings—earnings before interest and taxes, or EBIT. Others who recognize that amortization of goodwill is often an accounting rather than a real cost add back amortization, to produce EBITA. Often analysts will apply the same logic, sometimes inappropriately, to depreciation and look at EBITDA, operating earnings plus depreciation and amortization. Analysts concerned that adding back depreciation ignores the investments required to maintain a business's capital assets and to support growth, money that is not available for distribution to owners, prefer free cash flow, usually defined as cash flows from operations minus investment spending.¹

The multiple applied to any chosen measure of cash flow is generally determined by (1) defining a set of comparison companies for

¹ Sometimes even measures such as sales or gross profits that reflect potential and not necessarily current cash flows are used in multiple analyses.

whom the relationship of that cash flow to value should be comparable to the relationship for the target company being valued and (2) calculating an average ratio of market value to the cash flow variable in question for this set of comparables over an appropriately determined time period. The characteristics of the comparables will never exactly match those of the target company, so attempts are often made to adjust the multiple to account for any significant differences. Determination of an appropriate multiple will depend on the nature of a business's cash flows. Safe, stable, predictable cash flows should be more valuable than risky, unstable ones and should command higher multiples. Since stability and safety are impaired by high levels of debt, multiples should be lower for highly levered firms, compared to those with little leverage. Better businesses with higher margins should have relatively higher multiples both because they are likely to be safer and because they have better future business prospects. More generally, businesses whose cash flows are expected to improve in the future should have multiples above those of stagnant or declining businesses. This principle applies as well to businesses whose current cash flows are temporarily depressed by industry- or economy-wide cycles; their multiples may in fact appear elevated when they merely reflect investors' recognition that the cash flows will recover. Conversely, the cash flows of businesses operating at or near cyclical peaks should command lower multiples than they may currently exhibit. In all of these instances, the direction of changes in multiples are clear.

Other business characteristics may have a more ambiguous impact on multiples. Good managements, for example, should enhance value per unit of cash flow. They will operate more carefully, lowering risk, and allocate retained earnings more effectively, enhancing future returns. Still, overall future business performance will depend on the quality of management in the future, and how that compares to the people presently in charge. Here careful reasoning produces counterintuitive conclusions. Firms with good management see that quality embedded in current cash flow measures. They are unlikely to get better. Firms with poor management, by contrast, have substantial room for improvement. Given these future prospects, it is bad managements that should command higher multiples, however obtuse that may initially seem.

Making qualitative judgments about appropriate multiples in situations like those described above is only the initial step. Value estimates are necessarily quantitative, and multiples also must be quantitatively determined. A crucial part of this quantitative determination is the selection of the time period over which the average levels of market value to cash

flow are calculated. Especially when market multiples are inflated by excess optimism or lowered by excess pessimism, current market multiples may be highly inaccurate. An alternative is to estimate average multiples using extended historical periods (e.g., periods encompassing 2 complete business cycles). But in changing business environments, these long-term averages may no longer be appropriate, and the resulting value estimates may be just as inaccurate as those based on current multiples. In practice, uncertainties in cash flow estimates and valuation multiples mean that plausibly computed multiple valuations may differ by as much as 100%.

The Net Present Value of Current and Future Cash Flows

Given the shortcomings of multiple valuations the most common alternative is to use a net present value/discounted cash flow approach. In theory the intrinsic value of any asset, whether an office building, a gold mine, a company selling groceries at the corner or over the Internet, a government bond, or a share of Amazon stock, is determined by the present value of the disposable cash flows that the asset supplies to its owner. Net present value is properly calculated as the sum of present and future cash flows, both outlays and receipts, with each dollar of future cash flow appropriately discounted to take into account the time value of money. The term *Discounted Cash Flow* (DCF) is used to describe the calculation. (See Appendix to this chapter.)

Graham and Dodd disciples accept the concept and the calculation of net present value, as do all other fundamental investors. The techniques are taught at every undergraduate and graduate school of business. Investment bankers and corporate financial officers use them. Governments depend on them to evaluate the returns from potential capital projects and other investments. Calculators are programmed to produce NPV figures, and electronic spreadsheets have financial functions that will do the work. NPV analysis is inescapable. But what is true in theory need not provide an appropriate model for measuring intrinsic value in actual practice. We can say, only partially in jest, that the practical value of NPV analysis should be discounted.

The standard way of calculating NPVs, and hence intrinsic value, is to begin by estimating the relevant cash flows for the current and future years out to a reasonable date, perhaps 10 years in the future. Then one selects—estimates—a rate for the cost of capital that is appropriate to the

riskiness of the asset in question. The cost of capital determines the separate value per dollar of each future period's cash flow. Multiplying the estimated cash flows in each future year by these discount factors produces the values today—the present value—of those future cash flows. The sum of the individual present values for the first ten years is the present value of our foreseeable cash flows.

The customary practice for dealing with the cash flows in the distant future is to come up with what is called a terminal value. The terminal value is invariably calculated by assuming that beyond year 10—or whatever year is the last for which we have done annual cash flow calculations—cash flow grows perpetually at a constant proportional rate. Under this assumption, the value of those cash flows, looking forward from the end of year 10, will be the projected cash flow for year 11 times a multiple. In theory this multiple is equal to one divided by the difference between the cost of capital and the perpetual growth rate. For example, if we project a cost of capital of 10% and a growth rate of 5% a year, then the multiple is $1/(10\%-5\%) = 20$. In practice, any of the methods that we identified above for ratio valuations may be used to provide a multiple as either a supplement to or a substitute for this theoretical measure. Since we won't see this terminal value until we look forward from the end of year 10, we need to discount the terminal value back to the present. We add that to the NPV of our first 10 years of cash flows to get an intrinsic value for the current and all future cash flows.

There is a serious inconsistency here between the precision of the algebra and the uncertainties infecting the variables that drive the model. We estimate rates of growth for the first ten years and then another growth rate from the end of year ten to forever. This is a heroic, not to say foolhardy, exercise. Suppose that in two or three years, the company faces more competition, technological challenges, a spike in its costs of materials that it cannot pass on to customers, or any of a host of reasonable possibilities that will curtail, and may even eliminate, the growth of its cash flow. Imagine how accurate our estimates are likely to be even for a stable company like Proctor and Gamble or Pfizer, much less for dynamic firms like Netflix or Google. We also assume that our company will have access to long-term financing at a predictable cost of capital on an ongoing basis. Yet who knows today what lenders will demand in five years, or how much potential equity investors will be willing to pay for new shares? Profit margins and required investment levels, which are the foundations for cash flow estimates, are equally hard to project accurately into the far distant future.

Compounding the problem, valuations vary significantly if the underlying assumptions are off by only small amounts. Consider the terminal value and the cash flow multiple. If future perpetual growth is 4% and the future cost of capital is 8%, then the terminal value multiple is 25 ($1/(8\%-4\%) = 25$). If our estimate is wrong by only 1% in either direction for the cost of capital and the growth rate, the terminal value multiple can vary from a high of 50 (7% cost of capital minus 5% growth rate) to a low of 16.7 (9% cost of capital less 3% growth rate). This is a range greater than 3-to-1. And in many—probably most—valuations, the terminal value is the dominant component of the total NPV.

Investors are certainly aware of these difficulties, and there are ways of attempting to deal with them. One method, which certainly saves time and effort, is to simplify the valuation process by relying on multiple-based value calculations. This amounts to moving the terminal value calculation from a future year to the immediate present, a change that can hardly be expected to add precision and has all the problems of selecting the appropriate multiple we discussed above.

Another widely employed approach to dealing with the uncertainties of NPV is to perform an exhaustive number of sensitivity analyses. Here the analyst varies the projected operating parameters of a company that determine the future cash flows—growth rates in sales, profit margins, investments required per dollar of sales, the cost of capital—and then looks at the corresponding variations in the company's valuation. The purpose is to capture the full range of valuation possibilities. The problem here is that the range is usually large. Since the underlying parameters are linked together in complicated ways, it is not clear which of the many possible valuations is the one likely to materialize. Sensitivity analysis has the virtue of making explicit the unreliability of NPV estimates, but pointing out the problem is not the same as solving it.

There are two things that can be said in favor of a net present value/discounted cash flow approach to valuation. First, it is theoretically correct. If we knew the expected future cash flows without error and the exact cost of capital at which to discount them, an NPV analysis would yield the proper intrinsic value. Second, it is in theory superior to ratio valuations since, as we pointed out above, ratio valuations can be thought of as just abbreviated NPV calculations. To the extent that there are practical problems with the NPV approach, they apply equally, if not more seriously, to ratio valuations. However, in practice, there are three fundamental problems with using NPV to calculate intrinsic value that make it a poor candidate for consistently producing valuation

estimates that will prove superior to those made by the investors on the other side of the trade.

The Problems with Net Present Value Estimates

The first and most obvious problem with the NPV approach is that it largely ignores an important source of valuation information—the current balance sheet. Future cash flows are derived essentially from estimates of future income and cash flow statements. They track future operational flows. They pay little or no attention to the contemporaneous values of assets and liabilities on the balance sheet. Sometimes for firms in distress, an analyst will calculate a liquidation value using the current balance sheet to supplement the estimate of the net present value of future cash flows. But for going concerns, these calculations have little relevance. Yet the balance sheet is one of the best and often the most reliable sources of information concerning the value of any business enterprise. Balance sheet items can in general be examined directly at the time of the analysis. And useful collateral information can often be collected to calculate their current values. In contrast, future cash flow numbers are estimates whose accuracy can only be judged as actual operations unfold over time. This relative precision should be a major inducement to use current balance sheet information.

Although the balance sheet may not always be decisive in making good investments decisions, it will be critical in a significant number of cases. In these instances, an investment analyst looking carefully at the balance sheet will have a distinct advantage over an investor who has ignored it. Investors who rely solely on NPV analysis and overlook balance sheets are more likely to find themselves on the wrong side of the trade than those who do pay attention to the balance sheet.

The second major shortcoming of an NPV approach is that more often than not the terminal value number dominates the overall value and thus becomes central to the investment decision. As we described above, terminal values are measured with large degrees of error. This should not be surprising since terminal values depend on cash flows in the far distant future, which are inherently difficult to predict. This problem is not the fault of the terminal value formula but stems from the fundamental unpredictability of the cash flows that far into the future. The shortcoming of the NPV approach arises from this increasing uncertainty. The NPV is the sum of the present values of each of the individual future cash flows. Once the time value of money has been

taken into account by the discounting process, all those individual cash flows are given equal weight. However, the near-term cash flows are measured with far more precision than those in the distant future. By weighting all cash flow present values equally, the NPV takes very good information—the near-term cash flow estimates—and adds it to very poor information—the distant future cash flow present value estimates. As any engineer will attest, the inclusion of bad information corrupts the whole. Large estimation errors in the bad information completely overwhelm the small estimation errors in the good information. By failing to segregate good information from bad, the NPV approach wastes the good information in forming its valuation estimate. This is hardly a prescription for landing on the right side of the trade.

Those who promote the use of DCF analysis have argued that this problem is significantly reduced by the effect of the discounting itself. The poorly estimated future cash flows are reduced more by greater discount factors than are the accurately estimated near-term cash flows, lessening their impact on the NPV. Although true, the effect of this greater discounting is limited for two reasons. First, there are many more distant cash flows—in theory infinitely many more—than there are near-term ones. Second, for growing companies or other assets whose returns grow over time, distant cash flows will be larger than near-term ones and will have more weight in the NPV calculation. The practical importance of these two factors is confirmed by the fact that terminal values, representing distant future flows, typically dominate NPV valuations.

A final fundamental problem with the NPV approach is a little more subtle. In the abstract, a valuation process is a machine into which an analyst puts current variables and assumptions about the future. The analyst turns the crank, and the machine spits out an estimate or a range of estimates of the value of the asset in question. The quality of these estimates depends critically on the quality of the assumptions that are fed into the machine. Assumptions that are by nature subject to significant errors are rarely going to produce estimated values that are not equally subject to large errors. The assumptions that go into an NPV evaluation machine are all particular numerical variables: the level of estimated cash flow in each future year is usually calculated based on assumptions about numerical revenue growth rates, gross profit margins, operating profit margins, effective tax rates on corporate or partnership earnings, capital investments requirements needed to support growth—driven by numerical estimates of fixed and working capital intensities—and predictions of the cost of capital that determine discount factors. These individual

parameters are difficult to estimate with precision, especially for distant future periods. Moreover, because these numbers are highly correlated over time, these errors tend to accumulate rather than cancel one another out when computing an NPV. Given the inherently low quality of these estimates, calculated NPVs are subject to large margins of error.

Of course, if this is the best that investment analyst can do, then however poor, NPV estimates would be no worse on this account than other alternatives. However there are things we know about the future of a business with far greater confidence than we can know the particular numerical values that go into an NPV machine. We may not know the future of Ford's sales, margins, capital intensities, or cost of capital 10, 20, or 30 years from now. But we do know with some confidence that there will be a viable global automobile industry. In the late 1960s a major consulting firm was asked, in connection with the establishment of the Urban Mass Transportation Administration, to forecast the long-term future of transportation in the United States and the world. The report concluded that in the years 2010–20, transportation would consist largely of “individually directed” vehicles operating on “multiuse rights-of-way”; in other words, cars, trucks, and buses operating on roads. Fifty years later that prediction has been borne out. The future today looks very similar going forward, even with self-driven/electric/shared vehicles.

Also, there will almost certainly be a viable North American automobile production industry. The advantage of off-shore production has rested on lower labor costs. The offsetting disadvantage has been in higher transportation cost and being further removed from final customers. As productivity in auto manufacturing, indeed in all manufacturing, continues to improve rapidly, required labor inputs will fall. The net impact can be dramatic. In the early 1980s when General Motors initially got into trouble, it had 360,000 workers in North America. Today, producing roughly half as many vehicles, it is heading toward 20,000. As labor input per car shrinks, the benefit of cheaper labor becomes less significant. Transport costs and the benefits of being close to the consumer are changing at a far slower rate. As a result, manufacturing—but not manufacturing employment—is returning and will continue to return to developed economic areas such as North America. Ford, as a North America automobile producer, will continue to be an economically viable operation in the future.

A final aspect of the Ford Motor Company's future that seems predictable is the degree of competition it will face. For the past 30-plus years, the market for automobiles has been global and highly competitive.

None of the large auto companies, including Ford, have had technology, operating scale, or access to customers that have not been available on essentially equal terms to other large companies. In this market, returns on invested capital have on average been at or near these companies' cost of capital. Given this history, it seems unlikely that Ford will be able to establish sustainable competitive advantages in the future. It is equally unlikely that Ford will have to operate at a long-run competitive disadvantage.²

These are the strategic judgments regarding industry and company viability and competitive positions that we can reliably make about an extended future. An effective valuation machine would be able to integrate such assumptions directly and clearly into its estimated values. Unfortunately, there is no simple and obvious way to accomplish this feat in the context of the NPV framework. A superior alternative approach would be one that (1) integrated all the significant value information, especially from the balance sheet; (2) organized the components of value by the reliability with which they can be estimated, ideally from most to least reliable; and (3) made clear the implications of broad strategic assumptions for estimated levels and components of value.

The Graham and Dodd approach to valuation avoids the problems inherent in the NPV approach as currently practiced. It examines all the accounting information, especially a firm's balance sheet. It segregates information by reliability grade, so that good information is not contaminated by poor information. Finally, it directly integrates broad strategic judgments about competitive positions into the valuation process.

This does not however mean that DCF valuations are never appropriate. When an investment project involves future returns that are well-defined and likely to be realized within a short time period, then an NPV approach will often be the best available since the problems we have identified will be limited. Much event-based investing in takeovers, corporate reorganization, bankruptcies, and the like will meet these criteria and is an area where an NPV approach is appropriate. The larger lesson here is that a one-size-fits-all valuation approach (which

² There is potential concern that developments of electric and self-driving vehicles may undermine the traditional auto producers. Two factors, both broadly strategic, should limit the impact of these changes. First, batteries and "smart" driving systems are just two of many components of a typical vehicle. The established companies have historically adapted well to integrating the products of independent outside suppliers into their final products. Second, for a competitive industry, the effect on profits of a shrinkage in demand should in principle be fully offset by capital recoveries. (On this latter point, see Chapter 7.)

DCF often purports to be) is not a sensible way to measure how much value you are getting for your money.

A Three-Element Approach to Valuation: Assets, Earnings Power, Profitable Growth

The skepticism with which Graham and Dodd investors regard NPV calculations might be nothing more than a worldly cynicism toward all systematic efforts at valuation if these investors did not offer an alternative approach that avoids the pitfalls of the NPV approach. Fortunately they have developed a valuation method that meets our three standards. It is based on a thorough grasp of the economic situation in which a company operates. It puts more emphasis on information about the firm that is solid and certain, and it values the company's future prospects with more realism and less optimism than is customary on Wall Street. It refuses to pay anything for even the rosiest prediction that has no current or historical foundation. In the late 1990s, Charlie Munger of Berkshire Hathaway said that if he were giving a test calling for an analyst to value a new dot-com Internet company, he would fail anyone who answered the question. Some dot-coms did grow into their enormously high valuations, but most of them disappeared. To quote Wittgenstein, "Whereof one cannot speak, thereof one must be silent."

Of what then can we speak? Let's go back and look at Ford. It would be rash to predict its cash flow in 2030, but there are some things we can state with confidence.

Element #1: The Value of the Assets

The natural place to start is with the most reliable information we possess, information about the present state of the company. This state is presented on the company's balance sheet. Following Graham and Dodd, we will speak first about the net asset value of the firm as revealed in the balance sheet at the end of the most recent operating period (or periods if seasonal variations are significant) as determined by the company's accountants. We know that these accounting values are going to be more accurate for some assets than others. Thus, as we work down the balance sheet, we accept or adjust the stated numbers as experience and analysis dictate. Because of accounting principles and tax regulations, certain assets—a product portfolio, trained workers, customer relationships, or

local business franchises—do not even appear on the balance sheet. They are nevertheless important parts of the assets of a firm, and they must be valued. To estimate them with adequate precision requires the development of techniques often based on detailed industry expertise. The value analyst who has mastered these skills is collecting important information that is being ignored by most other analysts, raising the chances of being on the right side of the trade.

We do the same for the liabilities side of the balance sheet, including appropriate adjustments to accounting entries and estimates of unlisted but real liabilities. At the end of the process, we subtract liabilities from assets to obtain a current net asset value, essentially the book value of the equity, properly adjusted. There is no need for us to forecast the future. The assets and liabilities exist today. Many of them are tangible or quasi-tangible, such as money in the bank account as confirmed by the bank, and these can be valued directly with great precision.

Starting at the top of the balance sheet has another advantage. As we work down the asset accounts from cash at the top, whose value is unambiguous, to various intangible assets, whose value is often highly uncertain, we recognize the decreasing reliability of our estimated values. Graham himself preferred to rely entirely on current assets that could be realized within a year and whose accounting values did not vary far from the actual cash that could be obtained by liquidating them. He subtracted all the firm's liabilities from these current assets to arrive at his famous "net-net working capital" figure for the value of the company.

The results of an asset valuation depends on the principles employed to assign or adjust values to each asset and liability type. Here strategic judgment directly and significantly enters the process. If an entire industry does not appear economically viable in the future or if a specific firm is situated in a part of the industry where it is unlikely to survive, then assets and liabilities must be valued at what they are likely to bring in liquidation. This situation will arise whenever an industry or a firm is in terminal decline. Under these circumstances, generic assets—cash, accounts receivable, raw material inventory, some multi-use buildings and equipment—can be more or less accepted at full financial statement value. More specialized assets—finished goods inventory, specialized plant and equipment—are likely to be disposed of for cash amounts well below the figures on the balance sheet. In the worst case, they will be valued at zero. Intangibles will often be worth little or nothing in liquidation unless they are associated with independent and profitable operating divisions. Graham and Dodd, seared by their formative experiences in the Great Depression, favored liquidation value as

the safest approach to asset valuation. In the post-Depression years, they were able to find many opportunities to invest in companies selling at less than net-net working capital, having assigned a liquidation value of zero to all non-current assets.

Fortunately, depression conditions have turned out to be the exception rather than the rule.³ Almost all companies are in economically viable situations almost all the time. Under these conditions, the strategic assumption that will most often underlie our asset value procedure is one of future economic viability for which liquidation values are largely irrelevant. For example, as we noted above in connection with Ford, the North American automobile industry is not going away. If it is to continue doing business, it must invest in the assets—working capital, fixed capital, and intangibles—that are essential to its business operations. As these assets wear out or become obsolete, they must be replaced, presumably at the lowest possible cost given the existing state of technology. Otherwise the industry will not continue operating and will not supply the goods required of it. We will refer to the sums that must be spent to maintain business operations as the reproduction cost of the assets. The asset values of a firm in a viable industry should be equal to this reproduction cost. If they were available for less, competitors could buy a firm's assets more cheaply than it would cost to reproduce them. Bidding among competitors for the firm would then drive the value of the assets up to reproduction cost. At the same time, the assets should never command more than the cost of replacing their function in the most efficient way possible, since no intelligent investor would pay such a price.

The process of actually estimating asset reproduction values involves working down the balance sheet from working capital to fixed capital to intangibles, whether listed or not, on the asset side and taking a comprehensive view of liabilities. Again, as in the case of liquidation values, the cost of cash, accounts receivable, and inventory is relatively easy to calculate and close to accounting book value. The farther down the list, the more adjustments may be required. But there are appraisers who make a living by valuing plant and equipment, so we are still dealing with a more solid value than estimates of earnings growth rates 10 years into the future. Most investment analysts do not look carefully at asset reproduction values. This is an area in which industry expertise and

³ In the Great Depression, real output fell by about one-third from 1929 to 1932 and prices fell by comparable amounts. The combination lowered the dollar value of economic activity by roughly 50%. Net farm income, which supported one-third of the US population, fell by 80%. The comparable decline in the Great Recession of 2008–09 was about 4½%.

relationships with industry sources who have detailed knowledge of the net cost of reproducing the functions of existing plants, replacing depleted natural resource reserves, or acquiring new customers to compensate for those lost to natural attrition will be especially valuable. Having the creativity and the energy necessary to develop techniques for estimating reproduction values is likely to lead to sustainable investing advantages.⁴

Element #2: Earnings Power Value

The second most reliable measure of a firm's intrinsic value is the next calculation made by Graham and Dodd, namely the value of its current earnings, properly adjusted. These can be estimated with more certainty than future earnings or cash flows, and they are more relevant to today's values than earnings in the past, even though prior earnings will play a part in this estimate. To transform current earnings into an intrinsic value for the firm requires us to make assumptions both about the relationship between present and future earnings and about the cost of capital. Because we need to rely on these assumptions, intrinsic value estimates based on earnings are inherently less reliable than estimates based on assets.

The traditional Graham and Dodd earnings assumptions are that earnings levels and the return investors require to justify an investment in those earnings—the cost of capital—are constant for the indefinite future. They assume that a business neither grows nor shrinks. Constant size means no net investment in either working or fixed capital. Thus, if the depreciation levels used in calculating earnings equal maintenance capital expense,⁵ estimated net income—earnings—should equal distributable cash flows. Graham and Dodd referred to the average sustainable level of current earnings as *earnings power*. *Earnings Power Value* (EPV) then equals the earnings power capitalized at the average future cost of capital. The formal equation for determining EPV is

$$\text{EPV} = \text{Earnings Power} \times 1/R$$

⁴ We discuss some of these approaches and the issues involved in Chapter 4 on Asset Valuation.

⁵ Maintenance capital expense is the investment necessary to restore a company's physical assets at the end of the year to the condition and capacity they were in at the start of the year.

where R is the currently measured constant future cost of capital. Given these assumptions, the EPV of a business is equal to the NPV of its distributable cash flows.

Estimation of earnings power requires significant adjustments to currently reported accounting earnings. These adjustments, which we will discuss in greater detail in Chapter 5, include:

1. Rectifying accounting misrepresentations, such as frequent “one-time” charges that are supposedly unconnected to normal operations; the adjustment consists of finding the average annual ratio that these charges bear to reported earnings before adjustments and reducing the current year’s reported earnings before adjustment proportionally.
2. Resolving discrepancies between depreciation and amortization as reported by the accountants and the actual amount of reinvestment the company needs to make in order to restore a firm’s assets at the end of the year to their level at the start of the year—maintenance capital expense; the adjustment adds or subtracts this difference.
3. Taking into account the impact on earnings of the current position in the business cycle and other transient effects; this adjustment estimates average earnings to eliminate temporary variations.
4. Considering other modifications we will discuss in Chapter 5.

The goal is to arrive at an accurate estimate of the current disposable cash flow of the company by starting with reported earnings data and refining them. We assume, to repeat, that this level of cash flow can be sustained and that it is not growing. Although the resulting EPV is somewhat less reliable than the pure asset based valuation, it is considerably more dependable than a full-blown net present value calculation that assumes a rate of growth and a cost of capital many years in the future. And while the equation for EPV looks like other multiple-based valuations we have previously faulted, it has the advantage of being based entirely on currently available information and is uncontaminated by more uncertain conjectures about the future.

Part of the benefit of this Graham and Dodd definition of earnings power is that by deferring issues of growth, it greatly simplifies calculation of the value of today’s earnings. We will address the value of growth separately since that value is both highly uncertain and depends critically on the strategic position of a business.

One further set of adjustments must be made to arrive at a final EPV. The value of a firm may consist of more than just the perpetuity value

of the sustainable level of operating earnings. Many firms have assets that are unnecessary for the operations of their ongoing businesses. They may have excess cash, portfolios of securities, partial ownership of other unrelated ventures, excess real estate, and other assets. To the extent that these extraneous assets can be disposed of without affecting the ongoing functioning of the core businesses as covered by the income statement, they represent a source of value in addition to the value of sustainable earnings. This figure must be added to the initial EPV estimate to arrive at an overall earnings power value. At the same time, companies often have inherited liabilities that are also unrelated to the ongoing operations of the business. Historically underfunded pension liabilities, environmental liabilities beyond those created in the “normal” course of business, and other event-based liabilities such as from past litigation fall into this category. To mirror the extraneous assets described above, these liabilities must be subtracted from our initial EPV estimate to arrive at a final EPV.⁶

Element #3: The Value of Growth

We have isolated the growth issue for two reasons. First, this third and last element of value is the most difficult to estimate, especially if we are trying to project it for an extended period into the future. Uncertainty regarding future growth is usually the main reason why value estimations based on net present value calculations are so prone to error. By isolating this element, we can keep it from infecting the more reliable information incorporated into the asset and earnings power valuations.

Second, in many circumstances, growth in sales and even growth in earnings add nothing to a firm's intrinsic value. This claim seems to contradict a common article of faith about a company's sales and profits—growth is good. In fact, these conditions arise only in a limited range of economic environments. These growth-friendly environments are readily identifiable, based on a strategic assessment of industry economics. In most instances, a strategic analysis will indicate upfront whether growth is likely to create value. Where it does not, our analysis can ignore growth dynamics in the process of estimating value. Graham and Dodd investors certainly recognize that in some important situations, growth does create value and constitutes a third element in

⁶ Both the extraneous assets and extraneous liabilities appear on a firm's balance sheet, so they are already accounted for in the firm's net asset value.

valuation, along with assets and earnings power. The distinctiveness of the Graham and Dodd approach to valuation is to address these three elements of value separately, not to integrate them into a single ratio or NPV calculation.

Addressing Strategic Dimensions of Valuation within the Graham and Dodd Framework

The elements of the Graham and Dodd approach to valuation are summarized graphically in Figure 3.1. The first column represents the asset value. It is the most reliably estimated element of value. For a viable business, it is the net reproduction value of the assets—what it would cost someone to replicate the assets necessary to operate this business. For a non-viable business, it is the liquidation value.

The second column represents the earnings power value.

It is the second most reliably calculable element of value. The data from which the EPV is calculated—historical earnings and cash flows, industry conditions, and financial market variables that determine the cost of capital—are quite different from the data underlying the calculation of asset value. Thus, asset value and EPV provide two distinct estimates of value that can be triangulated to obtain a more complete picture of any business whose securities we are considering buying. The strategic assumptions here are that the business is viable and that earnings are sustainable but not growing. If the industry itself is not viable, earnings power is only temporary and unlikely to add anything to the liquidation value of the assets.⁷ The earnings power value estimates may exceed or fall below the asset value. In Figure 3.1 we have drawn earnings power value in excess of asset value, but as we shall see, that is not always the case.

The third column captures the total value of the firm. This differs from earnings power value by incorporating the effects of growth, which may either add to value, as in the figure, or subtract from it. We have so far said nothing about how this total value might be arrived at, except to caution that it is difficult to estimate accurately. To its credit, the NPV model makes an attempt to account for the value of growth. We have

⁷ In the case of a non-viable business that dies over a few years, an orderly liquidation may lead to a relatively predictable series of near-term cash flows. In this instance, an NPV evaluation may be appropriate, but it is unlikely to differ greatly from the asset liquidation value, especially if management wastes resources trying to survive.

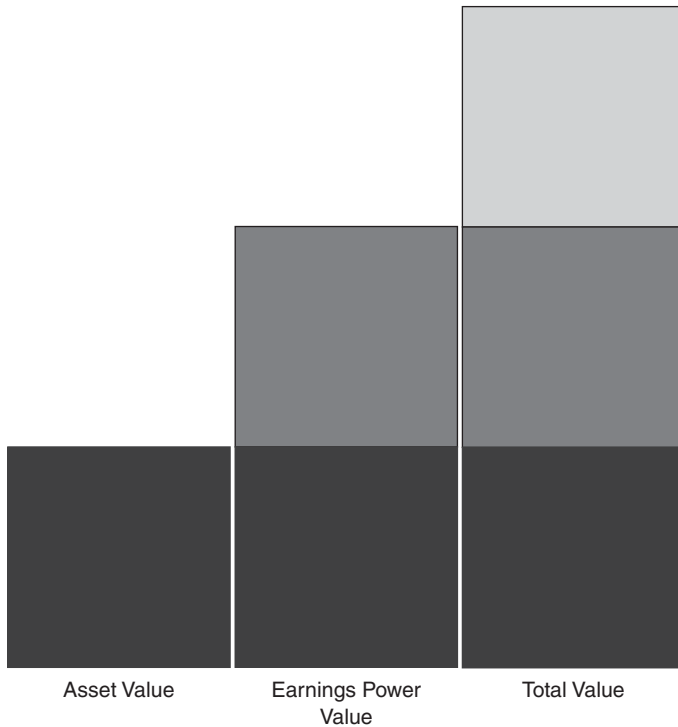


Figure 3.1 Three Elements of Value

already identified what we consider the shortcomings of that effort. In later chapters, we make the case that, for practical purposes, a Graham and Dodd approach to valuing growth does a superior job and is more likely to put an investor on the right side of the trade.

For a value investor, the valuation process typically begins with a careful estimate of asset value and then calculates earnings power value including extraneous assets and liabilities. When these independent estimates are compared to one another, there are only three possibilities, each represented in Figure 3.2: the asset values exceed the EPV (Case A); they are approximately equal (Case B); the EPV is greater than the asset value (Case C).

In Case A, where the asset value exceeds the EPV, there is the possibility that we have inadvertently overestimated the asset value by using a reproduction value where a liquidation value is more appropriate. This is a question of economic viability that we ought to be able to answer by looking at the long-term prospects of the company and the industry, just as we did earlier for Ford. If we conclude that the company is viable, two alternative explanations remain. First, the condition may be

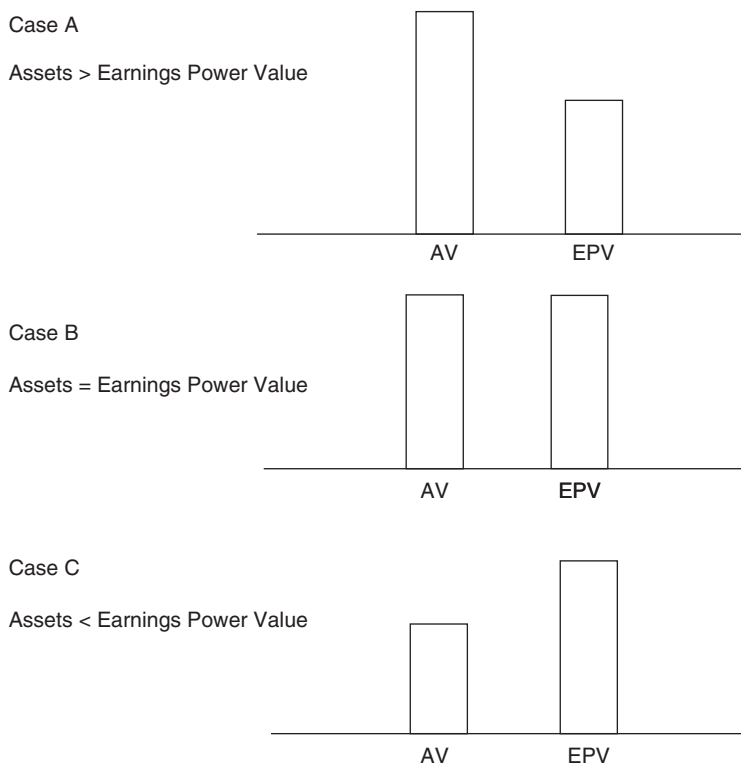


Figure 3.2 The Possibilities When Comparing Asset Value to Earnings Power Value

temporary, and we have used an inappropriately short-term measure of earnings in computing EPV. An industry cannot be viable in the long term if the value of sustainable earnings is below the reproduction cost of the required operating assets. Suppose, for example, that the asset value is \$10 billion and the EPV is \$5 billion (earnings power of \$500 million divided by a cost of capital of 10%). Then the return on those assets is only 5% (\$500 million divided by \$10 billion), which is not enough to attract capital voluntarily. In this case, investments necessary to sustain the company in the future will not be made, undermining the viability of the company.

The first response to this situation should be to re-examine the estimate of sustainable earnings power or to rethink the most efficient way of reproducing the assets. If this analysis fails to account for the discrepancy, the second possibility is that the assets are in the hands of an inferior management. This is a common situation when we find the relationship depicted in Case A. It can be confirmed by looking at the

historical operating and strategic performance of management. Assets of \$10 billion in the hands of a competent management should be worth their reproduction value; if they are not, current management needs to be improved or replaced. Should the inferior management persist, the business will be worth the \$5 billion earnings power value, at best. A more likely and discouraging scenario is that as current management reinvests a significant fraction of earnings and continues to earn returns below the 10% cost of capital, the company's value deteriorates to substantially less than \$5 billion.

This situation is a classic "value trap." The asset value is there, but investors will never see it unless the inferior management is replaced. The critical factor in the success of the investment is the ability to replace or refocus management. Until there is a clear prospect of such an event, the only reason to consider an investment in a Case A company is if it can be acquired at a deep discount not only to asset value but to EPV. This clear insight is a feature of the Graham and Dodd approach and would be difficult to extract from a welter of NPV sensitivity analyses.⁸ This strategic perspective, and the ability to measure valuation estimates against alternative strategic judgments, is a feature of the Graham and Dodd approach. It is difficult or impossible to extract these insights from an NPV valuation.

Turning to Case B, if the properly calculated asset value and EPV are approximately equal, within 25% of each other, this indicates a very different strategic situation for the business in question. It is what one would expect to see for firms with capable managements operating in competitive markets: in economic terms, no barriers to entry; in value investing terms, no moats. Why?

Consider a company with \$2 billion in earnings and a 10% cost of capital. Its EPV will be \$20 billion. If the net asset value were only \$10 billion, an entrant could reproduce the business for that net cost. In a market without structural barriers to entry, potential competitors will seize such an opportunity aggressively. Competition in the business will intensify and profits will inevitably be driven down, even if the industry is not perfectly competitive. It can happen in stages. Suppose the first wave of new entrants reduces the incumbent company's earnings to \$1.5 billion and the EPV to \$15 billion. That will still exceed the required \$10 billion investment. Only when the discrepancy between EPV—opportunity—and asset value—cost—is eliminated will the

⁸ The potential upside in Case A is also apparent. Good replacement management will usually be able to create value equal to or slightly above the asset value.

process of entry stop.⁹ At that point, asset value and earnings power value will be approximately equal.

On the other hand, if the company, in the hands of capable management, earned \$500 million with a 10% cost of capital, the EPV of \$5 billion would be significantly below the asset reproduction value of \$10 billion. Other companies in the industry with capable managements would earn comparable amounts. Average industry returns of \$500 million represent a 5% return on capital, well below the 10% return required by investors. Confronting this reality, managements would no longer find it profitable to replace depleting assets, industry capacity would fall, some companies would leave, and competition would diminish. After a time, profits for the surviving companies would rise. And, like the entry/expansion process, this contraction, even though very likely slower than the expansion, would not end until the EPV equaled or exceeded the asset value. Again, economic forces in a competitive market drive asset and EPV toward one another.

As with Case A, the implications of Case B can be tested against broad strategic judgments about the nature of the company and the industry. In the instance of Ford, the unlikelihood of sustained company competitive advantages would lead us to expect the relationship depicted in Case B, an approximate equality of assets and EPV. If we do not find a rough equality, we will have to reconsider our asset and earnings power value estimates. If the asset value estimate is accurate and the market is competitive—historically there have been no significant barriers to entry—then the estimated level of earnings will not be sustainable. If EPV is below asset value and management is capable, earnings will rise until EPV approaches the asset value. Thus, an investment should be profitable so long as there is a margin of safety between price and asset value. If EPV is above asset value, it is unlikely that the level of earnings is sustainable, and careful investors will want to see a market price below the value of the assets.

Again, the Graham and Dodd approach leads to a clear investment decision based on current asset and earnings power values and a strategic assessment of the nature of the market. An NPV analysis, even with extensive sensitivity trials, is unlikely to produce equal clarity.

A second benefit of the Graham and Dodd approach arises because in Case B our two estimates of value are derived from distinct types of information. Asset values begin with the figures on the balance sheet and look

⁹ We describe the process in more detail in Chapter 7.

at evidence related to what they would cost to reproduce with current technology. The firm's earnings power valuation begins with historical income and cash flow statements and looks at evidence from the annual operating performance of the firm as well as its likely cost of attracting capital. If we have confidence in our broad strategic judgment that automobiles or any other industry will continue to be viable, with little possibility for individual firms to enjoy sustainable competitive advantages, then as we noted above, we have two independent estimates of the company's value that can be triangulated to obtain the most accurate valuation possible.

One approach to triangulating between asset and EPV in Case B would be to average the two. But we can do better. For some companies, earnings are highly unstable and difficult to predict. Commodity producers and other heavily cyclical businesses fall into this category. But asset reproduction values—the cost of plant capacity, working capital, the cost of replacing depleted reserves—are often quite stable. In these situations, asset values will be a more reliable measure of value than earnings power. Other industries—service businesses, branded consumer products—have stable earnings, but many of the assets are intangible and difficult to measure. For these companies, EPV without growth will be the more reliable starting point for valuation. In either situation, the estimated values are mutually supporting. When you begin with EPV, the asset value still tells the extent to which these earnings are protected from competition by asset requirements. When you begin with assets, the EPV reveals the extent to which foreseeable earnings levels will support these asset values. Nothing like this is available from an NPV approach.

Finally, for Case C, the EPV substantially exceeds the net reproduction value of the assets. A number of companies fit this picture. The average value of Coca-Cola's sustainable earnings in 2015–16 of \$50–60 billion (\$4–5 billion of earnings power at a cost of capital of roughly 8%) was far greater than any reasonable cost of reproducing its assets. The book value was \$30 billion, and even if we included the cost of replicating the secret Coca-Cola formula, the total would still be far less than \$50 billion in EPV. The same disparity applies to other well-known companies such as Walmart, Microsoft, Google, Nestlé, and American Express. Case C is the inverse of Case A, where poor management often produces earnings that do not take full advantage of the value of the assets. Some of the excess earnings may therefore be attributable to superior management. But firms like Coca-Cola have had superior earnings for decades while the quality of management has varied widely over this period. Consistently superior earnings power is largely attributable to industry

conditions. As Warren Buffett has pointed out, when an industry with a “bad reputation” meets a management with a “good reputation,” it is almost always the reputation of the industry that survives. The opposite may also be the case; a firm in an industry with a “good reputation” can survive sub-par management, at least for a while.

The necessary industry conditions can be inferred from the previous discussion of Case B. As long as there are no structural barriers to competitive entry into a company’s market, EPV in excess of the reproduction value of the assets will ultimately be competed away. Thus, from a strategic viewpoint, Coca-Cola’s earning must be protected by significant and sustainable barriers to entry. In the language of Warren Buffett, Coca-Cola’s markets must be defended by wide moats.

In business strategy analysis, barriers to entry exist if and only if an incumbent firm has competitive advantages that a new entrant cannot match. Competitive advantages stem from (1) privileged access to customers, (2) proprietary technology, or (3) economies of scale.¹⁰ These advantages challenge potential entrants with the prospect of being forced to compete at a disadvantage; in most cases they deter entry. Thus, despite the potential attractions of Case C, EPV higher than asset costs, would-be entrants are never going to reap those benefits. The firm within the moat continues to achieve earnings far in excess of the reproduction cost of its assets. The term commonly applied to this excess EPV is “franchise value.” The critical strategic questions associated with Case C are the strength and sustainability of this franchise. An NPV valuation pays little if any attention to the importance of barriers to entry. In the case of Ford, with no sustainable competitive advantage in the foreseeable future, we would distrust valuations that found sustainable EPV greatly exceeding the reproduction costs of the assets. In Chapter 6 we discuss in detail how such strategic judgments can be made.

Growth: A Brief Preview

We have thus far ignored the value created by growth, and here we will only preview how growth considerations are integrated into a Graham

¹⁰ Entrant competitive advantages, for example an entrant network that is newer and better than that of established competitors, lead only to relentless competition. In these cases, once the entrant has wiped out incumbents and has frozen its network technology, the next entrant with the next new technology wipes out the original entrant who has become the incumbent. Earnings power in excess of asset reproduction value will never be sustained.

and Dodd approach to valuation. While there are positive value attributes to growth—everyone knows a growing income stream is more valuable than a static one—there are negative value consequences as well. Not all growth creates value. Growth requires investment, and the cash that must be committed to the business reduces the cash that a company can distribute today. A growing income stream will therefore be associated with smaller percentage distributions at each point in time. The net value of growth depends on which of these two factors—more rapid growth versus a lower starting point—has the greater impact.

Fortunately, if we examine this question from the investment perspective, the answer is immediately apparent. Suppose a company were to invest \$100 million in growth. Assume further that the cost of capital for that investment—what it would have to pay to attract investors voluntarily¹¹—was 10%. Then, an annual charge of \$10 million would have to be covered out of the returns on the “growth” investment before any value would be created. If, because of poor management, the investment yields only \$5 million, or 5%, then even though there will be earnings “growth” of \$5 million, existing business owners will be \$5 million worse off after paying the new investors. Growth investments that earn less than the cost of capital destroy value. Growth investments in competitive markets will earn average returns in the long run of 10%. Higher returns will be competed away. Lower returns will be driven up by lack of new investment. In this case, existing shareholders see earnings increase by \$10 million but pay the full amount to the new investors. Investments for growth in competitive markets, where they earn the cost of capital, neither create nor destroy value and should not influence valuation. To repeat, only growth investments that earn above the cost of capital will produce profits in excess of the returns required by new investors. For these returns to be sustainable, the investments and the growth must be protected from competition by barriers to entry. Only franchise growth creates value.

These results can be summarized by reference to three cases in Figure 3.2. Growth in Case A—poor management—will almost certainly earn less than the cost of capital and destroy value. Here growth is your enemy, and growing companies of this kind are worse investments than static or shrinking ones. The need for management change is amplified, not alleviated, by growth.

¹¹ This cost of capital is also what the firm should earn for its own investors, whose earnings it has the capacity to reinvest involuntarily in growth by not distributing it in dividends.

In Case B—competitive markets with capable management—growth is irrelevant to valuation. It generally neither creates nor destroys value. Ignoring growth will not impair the quality of valuations. The fact that in NPV models, growth, even in competitive markets, usually produces an increase in estimated value is further evidence of the superiority of the Graham and Dodd approach.

It is only in Case C, franchise businesses growing within their areas of competitive advantage, that growth creates significant value. It is in these situations that we must take account of growth if we are to develop accurate valuations. Doing this work well requires a complete shift of valuation perspective, one we will discuss in Chapters 7 and 8. Here we will note that a modern Graham and Dodd valuation approach begins with asset and earnings power value, incorporates strategic assessments of management, industry viability, and competitive advantages, and recognizes that for growing franchise businesses, Case C, valuation requires an entirely new perspective. Once this new method is included, the modern Graham and Dodd approach is uniformly superior to NPV valuation, even when growth must be taken into account.

We have described a situation in which one approach to valuation—NPV calculations—is correct in theory, precise, and is generally applied to any asset that produces a flow of income or cash to its owner. Unfortunately, it has three defects: (1) it ignores balance sheet information; (2) it lumps together estimates based on good information with those based on very uncertain assumptions, and the combination of the two taints the lot; (3) it relies on making accurate estimates of company operating parameters in the distant future even as it ignores strategic judgments that are more reliably made and more important for valuation.

The value alternative puts more emphasis on current information and on fundamental competitive conditions. It depends on specific knowledge about particular industries and assets, and it places much less faith in projections of rosy futures unless substantiated by current hard data. This is the discipline of value investing in the Graham and Dodd tradition. It applies to those investments with long-term future returns whose levels are determined by future business conditions. The investments in question are quite different from short-term investments with well-defined returns for which a DCF approach is appropriate. The larger lesson here is again that valuation approaches must be

tailored to investment realities. One-size-fits-all valuation approaches are a bad idea.

Appendix: The Present Value of Future Cash Flows

The process by which money to be received at various dates in the future can be equated with money in hand today is called *discounting*. The term *discount* refers to the fact that we prefer to have a dollar in hand today than the promise—even the iron-clad guarantee—of a dollar at some time in the future. The bank will pay us *interest* if we give them the dollar, as will other collectors of funds. In one year's time, at 8% simple interest, the bank will return \$1.08. At 8% simple interest, \$1.08 in one year is the equivalent of \$1.00 today; put differently, \$1.00 is the *present value* of \$1.08 one year hence, *discounted* at 8%. The algebra is rudimentary: \$1.00 times $(1+.08) = \$1.08$; \$1.08 times $(1/(1+.08)) = \$1.00$. The expression $(1/(1+.08))$ is the discount factor. We can think of the discount rate as the equivalent of the interest rate in reverse, the rate at which the future money is reduced to determine its present value. As with the interest rate, part of the discount is to compensate the investor for inflation, the rest for risk and the willingness to part with the money.

At 8% interest, compounded annually, in two years a dollar deposited today will be worth \$1.00 times $(1 + .08)$ times $(1+.08)$, or \$1.164. Conversely, the present value of a dollar guaranteed to us two years in the future, at a discount rate of 8%, has a present value of \$1.00 times $1/(1.08)$ times $1/(1.08)$, or \$.857. The present value of the future cash flow is reduced more if we have to wait longer for it. The expression that captures this relationship is *the time value of money*. Combined with the right algebra, the concept allows us to transform a whole series of future values into their value today. The two variables we need are time, which is almost always stated in years, and that other expression we have called both *interest* and the *discount rate*. Both terms refer to the rate at which people will voluntarily commit funds to acquire the asset in question. Other phrases for this concept that are more or less equivalent are *rate of return* (which is how much the investor demands) and *cost of capital* (which is how much the user of the funds has to pay for them.) The general equation for the present value of a dollar received in the future is

$$PV = \$1 \times 1/(1 + R)^T,$$

where R is the cost of capital (or rate of return) per year, expressed as a percentage, and T is the number of years until payment, with the current time as 0.

Let's see how this works out with a bond. We buy a 10-year government bond with a face value of \$1,000 that, for simplicity sake, pays us \$80 at the end of each year. At the end of year 10, it also repays the principal amount. What is the present value of the bond, given this stream of payments? If our rate of return is 8%, then the present value exactly equals the face value of \$1,000. Here are the calculations:

$$\text{Discount Factor} = 1/(1 + R)^T; R = 8\%.$$

Year	Payment	Discount Factor	Present Value
1	\$ 80.00	0.93	\$ 74.07
2	\$ 80.00	0.86	\$ 68.59
3	\$ 80.00	0.79	\$ 63.51
4	\$ 80.00	0.74	\$ 58.80
5	\$ 80.00	0.68	\$ 54.45
6	\$ 80.00	0.63	\$ 50.41
7	\$ 80.00	0.58	\$ 46.68
8	\$ 80.00	0.54	\$ 43.22
9	\$ 80.00	0.50	\$ 40.02
10	\$ 1,080.00	0.46	\$ 500.25
		Sum =	\$1,000.00

Everything looks perfect because the bond's stipulated interest payments, at 8%, are identical to the rate of return we require or can earn on an equivalent investment elsewhere. But suppose that, after the bond is issued, interest rates on this kind of investment increase to 9%. That becomes our required rate of return; we don't want to take less than everyone else. What happens to the present value of the bond? The only change is that R now equals 9%, but as a consequence, the present value of the bond drops by \$64.

$$\text{Discount Factor} = 1/(1 + R)^T; R = 9\%.$$

Year	Payment	Discount Factor	Present Value
1	\$ 80.00	0.92	\$ 73.39
2	\$ 80.00	0.84	\$ 67.33
3	\$ 80.00	0.77	\$ 61.77
4	\$ 80.00	0.71	\$ 56.67
5	\$ 80.00	0.65	\$ 51.99
6	\$ 80.00	0.60	\$ 47.70
7	\$ 80.00	0.55	\$ 43.76
8	\$ 80.00	0.50	\$ 40.15
9	\$ 80.00	0.46	\$ 36.83
10	\$ 1,080.00	0.42	\$ 456.20
		Sum =	\$ 935.82

The only difference between the term *Present Value* and the almost identical expression *Net Present Value* is that *NPV* includes the initial flow, which is usually the money laid out in year zero. In the first example, we pay \$1,000 for the bond, which is a negative flow unreduced by a discount factor because it happens today. The Net Present Value of the cash flows would be zero, which only means that we receive back the present value of our current outlay. If interest rates go to 9% and we still insist on paying \$1,000 for the note with the \$80 coupon, then the NPV of the investment turns out to be negative, since we only receive back funds with a present value of \$935.82. One of the cardinal rules of investing is not to make investments that have an NPV of less than zero.

4

Valuing the Assets: From Book Value to Replacement Costs

The first step in a Graham and Dodd valuation is to calculate the asset value of a company. For many traditional value investors, this has been essentially the only step. The proper approach depends, as we noted in Chapter 3, on the strategic prospects of the company. If it is not economically viable because, for example, its industry is in terminal decline, then the value of the company's assets should be based on what they will bring in liquidation. If the company is viable, that means its assets will need to be reproduced as they wear out. In that case, they should be valued at the reproduction cost.

The reliability of these estimates diminishes the further down the balance sheet they are located. At one extreme, Benjamin Graham began by considering just the current assets: cash, accounts receivable, inventory, and whatever else particular firms carried as current assets. Because they would be realized within a year their values could be estimated with only minor errors on either a liquidation or reproduction cost basis. He then subtracted the full book value of the company's liabilities; what was left was a "net-net" asset value. He sought to buy companies with market prices below two-thirds of this "net-net" figure. In the early part of Graham's career, he was able to find investments meeting this stringent standard with reasonable frequency. Today these opportunities

have largely disappeared. As a result, contemporary value investors have had to move further down the balance sheet to take into account plant, property and equipment, and even intangible assets such as product portfolios, customer relationships, brand images, and trained employees. The error band here is wider, and valuing these assets, especially the intangibles, requires skill, ingenuity, and industry expertise.¹

Analysts who pay close attention to the balance sheet should have a leg up on those who don't. Looking at all assets carefully, even those that are difficult to value, expands this advantage. Specialization matters in this process. One's understanding of the cost of reproducing assets or the money that can be realized from liquidating them is an area in which capabilities improve with experience. It is also importantly related to industry knowledge. An example of the usefulness of asset values beyond their immediate valuation implication arises in judging business quality. Cash flow-oriented analysts of value place great stress on the quality of a business, especially its competitive advantages. They usually judge business quality by the returns on capital or equity as reported on the balance sheet, using book capital or book equity. A consistent 25% return on book equity is regarded as proof of a competitive advantage, a significant value-enhancing characteristic. But book equity often ignores important intangible assets. If these assets are included in equity, as they should be because they represent genuine investments, then actual returns on total equity employed may be much lower and thoroughly prosaic. The only systematic way to perform these adjustments is by a comprehensive asset valuation. Asset values are always worth estimating, and if they are worth estimating, they are worth estimating well.

Values in Liquidation

In the Red, Inc., is not anyone's idea of a wonderful business (Table 4.1). Between 2017 and 2018, the company's retained earnings fell by almost \$4 million, putting its net worth below zero. Perhaps this was a temporary setback and the firm will be able to convince its lenders to extend it more credit to meet its interest obligations. We can't tell from this fragment of information, and we really don't care.² All we want to be able to do is to estimate the value of the assets if the firm is to be liquidated.

¹ This effort is worthwhile only for economically viable firms. Intangibles in liquidation have little or no value.

² How it got into this situation is a question related to earnings power and EPV, not assets.

Table 4.1 Balance sheet of In the Red, Inc. (values in millions).

In the Red, Inc.	2017	2018
Assets		
Current assets		
Cash	\$ 195	\$ 150
Marketable securities	\$ 100	\$ 25
Accounts receivable	\$ 1,595	\$ 1,667
Inventories	\$ 2,250	\$ 2,328
Total current assets	\$ 4,140	\$ 4,170
Property plant and equipment, net	\$ 7,750	\$ 7,500
Goodwill	\$ 2,400	\$ 2,250
Deferred taxes	\$ 155	\$ 150
Total assets	\$ 14,445	\$ 14,070
Liabilities and equity		
Current liabilities		
Notes payable	\$ -	\$ 2,200
Accounts payable	\$ 850	\$ 1,300
Accrued expenses	\$ 725	\$ 1,275
Current portion of long-term debt	\$ 500	\$ 520
Total current liabilities	\$ 2,075	\$ 5,295
Long-term debt	\$ 9,250	\$ 9,500
Deferred taxes	\$ 150	\$ 125
Preferred stock	\$ 350	\$ 350
Paid-in capital	\$ 850	\$ 850
Retained earnings	\$ 1,770	\$ (2,050)
Total liabilities and equity	\$ 14,445	\$ 14,070

For cash and marketable securities, there should be no discount from the amount as stated on the company's books, provided the securities are short term or have been marked to market. Accounts receivable will probably not be recovered in full, but it is trade debt, and there are plenty of specialists who know how to collect it. We will estimate that 85% of book value (which already includes an allowance for bad debt) is recoverable in liquidation. What the inventory will bring depends on what it is. For a manufacturing firm, the more commodity-like the inventory, the less the discount necessary to sell it. It is yesterday's fashions that have to be marked down, not the cotton yarn. On the other hand, if the inventory consists of cartons of last season's unsalable toys, then it may be necessary to pay someone to cart it away. We will estimate in this case that we can realize 50% on the inventory; if the inventory is highly specialized, then the valuation would have to be substantially lower. In those situations in which the value of the inventory is critical to the overall valuation, an expert appraiser can be called in to offer a more precise figure than our back-of-the-envelope estimate (Table 4.2).

Table 4.2 Assets and liquidation value of fictional firm.

Assets	2018	% realized	Value
Current assets			
Cash	\$ 150	100%	\$ 150
Marketable securities	\$ 25	100%	\$ 25
Accounts receivable	\$ 1,667	85%	\$ 1,417
Inventories	\$ 2,328	50%	\$ 1,164
Total current assets	\$ 4,170		\$ 2,756
Property plant and equipment	\$ 7,500	45%	\$ 3,375
Goodwill	\$ 2,250	0%	\$ -
Deferred taxes	\$ 150	0%	\$ -
Total assets	\$ 14,070		\$ 6,131

The same holds true for property, plant, and equipment. Detailed knowledge of the real estate and the equipment is necessary to come up with an accurate estimate. Certain broad principles apply. Generic assets such as office buildings will be worth far more, relative to their book values, than specialized structures such as chemical plants. We have put down 45% in our example as another quick and dirty valuation; if this entry is critical, we can hire another expert to do the appraisal. We ascribe no value to the goodwill; it merely represents the excess over book value that the firm paid in making those acquisitions that may have gotten it in trouble.³ Deferred tax assets, the refunds the company can expect over time from the IRS, are offset against deferred taxes owed. Putting all these figures together, we come up with a value of \$2.756 million for current assets and \$3.375 million for property, plant, and equipment, for a total of slightly more than \$6 million.⁴

Who might want to invest in this company's securities? Certainly not a traditional equity purchaser, no matter how value oriented. But there is room for profit here, provided one is a specialist in buying up distressed debt. Though it looks fairly certain that if the company is liquidated there will not be enough money left to pay anything to owners

³ In theory, the accounting entry for goodwill represents the cost of acquiring intangible assets such as products, customers, and market positions that are "real" but not represented on the balance sheet of the company being acquired. The natural way to proceed in putting a value on goodwill is to identify the underlying tangible measures—product lines, number of customers, trained workers, repeat customers, cable population density—and to value each of them separately. But for a company in a non-viable industry, these highly specialized assets are not likely to have any significant value. Thus, our bottom line for goodwill is zero.

⁴ We ignore here any tax benefit of liquidation.

of either common or preferred shares, there probably will be funds for the owners of the debt. Accounts payable and accrued expenses amount to only \$2.575 million. Even if they receive priority over the debt and dollar-for-dollar reimbursement, there would still be \$3.556 million left for the debt holders. Everything else can flow to the holders of the debt. Debt on the books comes to \$12.220 million, but given the condition of the company, the bonds would certainly have been available for substantially less. If the discount is steep enough, and there is enough value in the property, plant, and equipment, this might be a lucrative opportunity for an expert in distressed debt securities and liquidation values.

Orderly Liquidations

Our balance sheet-based liquidation value is the asset value for businesses at the extreme end of the non-viability scale. It applies to companies clearly worth more dead than alive, and usually the faster they are put out of their misery, the better the return. Graham, being naturally conservative, chose this kind of valuation as a worst-case possibility against which to measure his purchase price. More often companies in distress will have some kind of future with positive operating cash flows, albeit only a short-lived one. The term applied by value investors to this kind of company is “cigar butt.” It has a few puffs of tobacco left and then only the value of the remaining husk. For these companies, an orderly liquidation over, say, 3 to 5 years may yield more value than an immediate interment.

In these cases, because the valuation horizon is short, an NPV will be appropriate. A plan should be developed to continue to operate the firm over this period while essentially making no new investments. Accounting profits may well be negative, but cash flows, including depreciation recoveries, should be positive. As sales decline, accounts receivable and inventories can be worked off and effectively recovered at near full value, adding to interim cash flows. Salable general assets such as land and general-purpose buildings can be sold off at non-fire-sale prices. Then, at the end of this period of positive albeit declining cash flows, the current balance sheet can be evaluated on an immediate liquidation basis. The net present value of the projected interim cash flows plus the liquidation recovery can then be compared to the proceeds of immediate liquidation. If they are more valuable, the orderly liquidation over time will make sense. And because the time period over which this cash flow is

realized is strictly limited, the NPV calculation should be sufficiently accurate. There is always the danger that a management caught up in this process will try to extend the life of the firm beyond its useful term. For orderly liquidation to work, this temptation must be resisted.

Assets for a Going Concern: How Much to Get into the Business?

Liquidation value is one conservative approach to calculating what the assets of a firm are worth in the worst possible circumstances. Our principal purpose in valuing a firm based on its assets is to discover whether the economic value of the assets is accurately reflected in the price at which the firm's securities are being traded. Opportunities lie in the gap between value and price. We have already made the case that for a firm in a viable industry, the economic value of the assets is their reproduction costs—what a potential competitor would have to spend to get into this business at the scale of our firm. How do we make these estimates? Table 4.2 presents the balance sheet for another fictional company.

What adjustments do we need to make here to get at reproduction costs (Table 4.3)?

Cash is cash and nothing is required. For marketable securities, we have to find the current market prices. This may be difficult if the securities are not liquid, but generally this category is used only for securities that are actively traded. In principle, the book value of these securities should be based on their actual or potential market values. The serious work starts with accounts receivable; from here on, the book value should be modified, up or down, to get a more realistic reproduction cost. A firm's accounts receivable, as reported in the financial statement, generally contains an allowance for bills that will never be collected. A new firm starting out is even more likely to get stuck by customers who for some reason or another do not pay their bills, so the cost of reproducing an existing firm's accounts receivables is probably more than the book amount. Many financial statements will specify how much has been deducted for bad debt to arrive at this net figure. That amount can be added back, or an average of similar firms can be used.

Valuing the inventory is more complicated. The stated number may be too high or too low by a substantial amount. Our attention should be drawn to an inventory that has been piling up. Say it equals 150 days' worth of COGS (cost of goods sold) in the current year, whereas previously it had averaged only 100 days. The additional 50 days may represent

Table 4.3 Adjustments to asset for fictional firm (values in thousands).

Assets	Book value	Adjustment to arrive at reproduction cost
Current assets		
Cash	\$ 2,250	none
Marketable securities	\$ 6,750	none
Accounts receivable, net	\$ 31,250	add bad debt allowance; adjust for collections
Inventories	\$ 25,000	add LIFO reserve, if any; adjust for turnover
Prepaid expenses	\$ 5,900	none
Deferred taxes	\$ 4,250	discount to present value
Total current assets	\$ 75,400	
Property plant and equipment, net	\$ 54,000	original cost plus adjustment
Goodwill	<u>\$ 26,250</u>	related to product portfolio and research and development
Total assets	\$ 159,900	

items that will never sell, or sell only at closeout prices. In this instance we would be justified in reducing the reproduction cost downward. By contrast, if the company uses a LIFO (last in, first out) method for keeping track of inventory costs, and if the prices of the items it sells have been rising, then the reproduction cost of the inventory is higher than the published figures indicate. This difference is the LIFO reserve, the amount by which the current cost of any item exceeds the old, recorded cost. The new entrant can't build this year's inventory at last year's prices so will have to pay more to reproduce it.

Prepaid expenses, such as rent or insurance, are what they are, small and, if the accountants have done their job, properly measured by book value. They should require no adjustment.

Deferred taxes as an asset are the value of the future deductions or refunds the company will get from the government. Since we are interested in the value of the assets today, we ought to get the timing of the reductions or payments and calculate their present value. In our example, deferred taxes are listed as a current asset; this firm expects to cash them in within the year, so any discount based adjustment will be minor. But they might just as easily have been non-current, in which case the present value analysis has more significance.

The adjustments we end up making to the book value of current assets are not in most cases going to be large enough to matter. The assets are current because we expect them to be turned into cash within a year, so there has been little time for great disparities between recorded

costs and reproduction costs to build up. The situation changes when we examine non-current, or fixed assets. The land that cost \$2,000 an acre in 1985, purchased because it was cheap, plentiful, and close enough to an adequate labor market, even though a little remote from a decent restaurant, may now sit 200 yards from a new laboratory incubator developed as a private-public partnership. Our company owns 500 acres. Suppose that last month property similarly situated sold for \$10,000 an acre. Either the land is worth that much to our company because of its proximity to the new facility, or else we should sell it and move elsewhere, pocketing the difference. In either case, the gap between the book value and either the costs of reproducing it or the net gains from an outright sale is large enough to catch our eye.

Establishing the reproduction cost for plant and equipment is more complicated, less precise, and requires a higher level of specialized industry knowledge than adjusting current assets. Consider, for example, the reproduction value of an oil refinery capable of producing 60,000 barrels per day of crude oil. Suppose that the cost of a new facility on that scale is \$10,000 per barrel, an estimate obtained by consulting engineering firms that manage the construction of oil refineries. This suggests a reproduction cost of \$600 million. The calculation is simple enough, but there are three significant problems with this figure. First, a new facility will almost certainly have lower operating costs than the refinery on the balance sheet. The present value of the savings over the life of the existing facility can be calculated with reasonable accuracy, as they are of finite duration and should not grow rapidly and ought to be well defined. The value of the savings must be subtracted from the reproduction costs of the refinery.

Second, the new facility will have a longer life than the existing one. The present value of the difference between the residual value of the new facility and the scrap value of the existing one must also be subtracted from the \$600 million cost of the new facility.

Third, and often most important, a new facility may not be the most efficient way to reproduce the function of the old facility. No new oil refineries have been built in the United States for decades even as refining capacity has expanded steadily. The added capacity has come from incremental improvements to existing "brownfield" refineries, and not from newly constructed "greenfield" refineries. The cost of adding brownfield capacity has been far less than the cost of greenfield activity, around \$2,000 per barrel compared to \$6,000 per barrel, even after making the first two adjustments we described above. The most efficient way of replacing our 60,000 barrels of refining capacity—our goal is increased

oil production, not increased refinery construction—is by extending existing refineries at a total cost of \$120 million. This is the properly calculated reproduction cost of the facility.⁵ In competitive situations, where companies compare the costs of building either new construction or brownfield extensions to the cost of purchasing capacity in secondary markets, these prices should be roughly equivalent. Still, it is helpful to perform the calculations and have two distinct sources of information for valuation.

This kind of direct assessment of plant value is one way to estimate its reproduction cost. Another way is to start with the financial information provided by a company's balance sheet, including the associated footnotes. This generally reports original acquisition costs separately for land, plant and structures, construction in progress, equipment, and furnishings and fixtures.⁶ Accumulated accounting depreciation is then subtracted, usually as a simple aggregate, to arrive at book value. This depreciated net value of PPE will always differ significantly from reproduction cost. Land does not depreciate and may appreciate significantly due both to general inflation and economic growth. So too may plants and structures: factories, oil refineries, office buildings, motels, stores, restaurants, and facilities such as long-lived fiber optic cable linking together large cities both at home and overseas. The potential disparity between book value and reproduction cost may be enormous for two reasons. First, although plant and structures, unlike land, do wear out, they often do so more gradually than the rates assumed by accounting standards. For example, our fiber optic cables wear out slowly, if at all. But accountants depreciate them over 20 years or so. Second, inflation and economic growth positively affect the values of plant and structures just as they do land. For land, plant, and structures we can estimate the percentage impact of growth and inflation by looking at the rate of appreciation of comparable facilities that are sold at publicly recorded prices. We then have a basis for adjusting original costs upward. We can also make an offsetting downward adjustment to capture a realistic rate of wearing out. We put the 2 together to arrive at a reported balance sheet reproduction value for plant and equipment. These figures can

⁵ This also assumes that our viable refining industry is not in a situation of overcapacity that will only be eliminated several years in the future. If such overcapacity exists, then the incremental value of capacity today must be discounted from the point at which it actually comes online.

⁶ Some firms include software in PPE, but we will deal with that issue when we address the valuation of intangibles.

be compared with the direct reproduction value estimates for PPE described above.⁷

Construction in progress is easy to value; these are structures in the current process of reproduction so that the original book value should equal the reproduction cost unless the facilities in question are over cost or inappropriate.

Equipment, furnishings, and fixtures wear out more rapidly than structures or fixed facilities. Depreciation plays a greater role in determining value and must be carefully considered. However, for a firm which is neither growing nor shrinking, equipment, furnishings, and fixtures should be roughly half-way through their useful lives. Even for the firm growing at 5% per year, whose equipment will be newer on average than that of the static firm, the difference from the one-half of life is only on the order of 10%. Thus, for a first approximation, we can assume that equipment, furnishing, and fixtures are halfway through their useful lives. This original estimate can then be modified by trends in equipment prices, usually adjusted for changes in quality. The United States Department of Commerce tracks these prices, which have been falling relatively rapidly. To illustrate this adjustment, assume that a company's equipment has an average life of 8 years, making the average age of the existing equipment 4 years. If equipment prices have been falling at about 4% per year, the price of equipment to replace four-year-old equipment will be about 85% of its original cost. Thus we can estimate reproduction cost for existing equipment as one-half of the original cost times 85%. This figure can then be added to the estimated reproduction cost of land, building, and construction in progress to obtain a quick estimate to compare with book value after depreciation.⁸

We turn now to goodwill and other intangibles. The goodwill that appears on the company's balance sheet usually represents the residual impact of past acquisitions. When Company A acquires Company B, the difference between the price paid and the "fair" net value of company B's assets appears on company A's balance sheet as goodwill. Over time this excess purchase cost, which covers intangibles that are not represented on the balance sheet, things such as product portfolios and customer relationships, is reduced—amortized—to account for the gradual decay

⁷ We go into more detail in the next chapter about comparing accounting depreciation to the economic costs of restoring fixed assets to their value at the start of the year.

⁸ We have already calculated depreciation from original costs when we multiply by one-half so that including accounting depreciation here would be double counting.

in their value. At any given moment, the value of goodwill is the current amortized level of the acquisition premium the company paid.

Recently, companies have begun to capitalize certain costs such as research and development, software creation, and customer acquisitions that have benefits over many future periods. Traditionally, firms deducted the full amount of such costs against current revenues when calculating profits. When capitalized, the cost of such “investments” is not charged against current revenues. Instead they are accumulated in a balance sheet item labeled “intangibles.” This asset is then depreciated over time, the depreciation being charged against revenues in calculating profits.⁹

Both intangibles and goodwill are determined by accounting conventions and historical choices that may not reflect the reproduction value of the assets in question. A company may simply pay too much to acquire another firm, either because of perceived competitive threats or grandiose plans that don’t work out, or it just makes a mistake. This acquisition premium is included in goodwill, but it may not represent any economic value. A new entrant does not need to reproduce anything here in order to compete. This portion of goodwill may represent a prior blunder, and we are justified in ignoring it entirely in coming up with our asset-based valuation. The value of goodwill depends on its source, and to determine that we need information and industry knowledge. Because of all these uncertainties, we think the best course is to set the value of these accounting entries to zero and estimate the reproduction value of identifiable intangibles, including those that are part of goodwill, from first principles.

We begin with a list of the relevant intangibles. This will not be limited to the “intangibles” entry on the balance sheet. For most companies this will consist of a product portfolio with development and/or brand value, a current customer base, and a trained labor force operating within an effective organizational structure. For each of these items we can think through several alternative processes of reproduction to get a composite estimate of their reproduction value. Multiple approaches are especially useful for intangibles that are hard to value.

Let’s start with a company’s product portfolio. This portfolio has been created by research and development activities whose purpose is to invent, design, and produce products and services for sale. Many firms

⁹ This is obviously a less conservative approach to calculating profits and one more subject to manipulation. We will discuss the impact on current cash flows when we look at calculations of earnings power values. Here we are concerned only with handling these items for the purpose of calculating asset values.

will have no significant R&D expenses, and the reproduction cost of their product portfolio will be low or zero. Others will spend only modest amounts that don't warrant their own entry in the income statement but may be specified in the footnotes. However, many companies have highly significant levels of R&D expense. As a general rule, the higher the technology, the more R&D is embodied in a company's products and the greater the asset value of its product portfolio. Table 4.4 presents R&D spending of some leading non-financial corporations as a percentage of sales. (Those companies with "NA" in the column did not break out R&D spending in their financial statements.)

Converting R&D spending into a valuation of a company's product portfolio is most simply done by multiplying annual spending by the number of years of R&D embodied in the company's product line. If most current products were developed over the last six years, then six times current R&D spending is a useful starting point for valuing the firm's product portfolio. A new entrant would presumably have to spend a roughly equivalent amount to offer a comparable range of products. In cases where R&D output is episodic and highly significant, we need to get into more detail. For a company such as Boeing, average R&D may be relatively low as a fraction of sales in particular years. However, R&D and sales are driven by new product cycles. A new model, such as the

Table 4.4 R&D spending of major US firms as a percentage of revenue.

	2017	2018
Apple	5%	5%
Microsoft	13%	13%
Amazon	13%	12%
Facebook	19%	18%
Alphabet	15%	16%
Johnson & Johnson	15%	13%
Exxon Mobile	5%	4%
Walmart	NA	NA
Procter & Gamble	NA	0%
Intel	21%	19%
Cisco System	13%	13%
Verizon Communications	22%	18%
AT&T	NA	NA
Pfizer	15%	15%
Coca-Cola	NA	NA
Boeing	3%	3%
Ford	5%	5%

787 Dreamliner, will have an estimated development cost of \$10–\$20 billion with uncertain sales levels, since it is still early in its production cycle. A competitor will have to develop a comparable aircraft, although perhaps without the technological missteps and therefore at lower cost than the 787. Competitors will also have to offer alternatives to the full range of mature Boeing models, the 737, 747, 757, and 777, at current costs which, because of inflation, higher performance standards, and less experience than Boeing, will almost certainly exceed Boeing's historical costs. In this instance, it will be necessary to refer to expert industry sources who can more accurately estimate the current cost of developing a comprehensive Boeing-compatible product line. Merely multiplying Boeing's R&D levels and "average" product life, which may range from 15 to 40 years, will usually not be precise enough.

For research-centered industries such as pharmaceutical, things are even more complicated. Existing drugs that have been tested and satisfied international regulatory standards are not the focus of current R&D spending. As established products, they have associated revenue and production cost streams that are usually highly predictable and largely independent of historical R&D spending. The best way to value these is by calculating the NPVs of these earnings streams, which tend to be of relatively limited duration before the drug in question goes off-patent. The cost of reproducing these drugs, which are typically not going to be reproduced since they are likely to be replaced by lower profit generics or improved substitutes, are irrelevant. R&D spending in pharmaceuticals typically drives the value of a pipeline of future drugs, not existing products. Since the pipeline extends over many years, most of the future product is visible today although its value may be highly uncertain. For this part of the product portfolio, there are experts in evaluating drug pipelines in particular therapeutic categories, like oncology, cardiology, and gastroenterology. Non-specialists in pharmaceuticals are at an obvious disadvantage. The overall lesson here is that where intangibles such as product portfolios are especially important, industry specialization is critical.

A second important intangible asset is embodied in the company's book of business, that is, its revenue base. The customer relationships that underlie this revenue cost money to develop and should be regarded as an important intangible asset of the firm. Again, the best way to estimate the reproduction value of this asset is to think through the process of acquiring a particular level of revenue. For example, companies are constantly bringing new brands to market. A marketing budget and a set of revenue targets are associated with each new brand introduction. At the

start of the process, costs will exceed revenues, and the cash flow will be negative. Once the brand reaches its target sales level, there should be net positive cash flows over its future life. The cost part of this plan is usually estimated reasonably well. Budgets are determined for the new brand introduction, and overrunning those budgets can have consequences for management. The sum of outlays during the investment phase can then be divided by the target for mature sales, all discounted by the timing of the cash flows, to obtain a reproduction cost per dollar of sales. However, only one-third to one-half of new product introductions actually succeed, so this theoretical reproduction cost must be multiplied by a factor of 2 to 3 to calculate the reproduction cost of a successful brand that becomes part of a company's long-term revenue base. Data on these costs and success rates are usually obtainable from industry experts, often retired executives who have experience of many actual new brand introductions. A typical adult woman's clothing brand in the late 1990s cost between \$.20 and \$.30 per dollar of mature sales to develop. With a success rate of between one-third to one-half, the average cost per dollar of successful brand sales was between \$.40 (2 times \$.20) and \$.90 (3 times \$.30), or roughly \$.65 per dollar of sales.

An alternative method of sales acquisition is to outsource the process. In many industries, there are independent sales agencies that will undertake to sell a company's product for a commission, usually set as a percentage of first-year sales. These commissions vary by industry and product type. For industrial products they range from 5% to 15%. This information is relatively easy to obtain since these prices are widely quoted and well known to industry participants. They provide a useful check on the costs of developing the revenues internally as estimated in the business plan.

A second way to outsource revenue acquisition is to buy businesses directly from competitors. Major women's fashion firms buy product lines from other firms all the time either by acquiring specific brands or taking over smaller firms. The amount that they pay, after subtracting the value of other collateral assets acquired, can be divided by average acquired future revenues to get a cost per dollar of sales. The average cost per dollar of sales acquired for adult women's mature clothing brands in the late 1990s was between \$.60 and \$.70, in line with our direct estimate cited above. The outsourced cost can be used as a check on internal business case estimates of the cost of revenue generation. Keeping records of revenue acquisition costs in an industry is an important part of valuing intangible assets.

A third kind of intangible is embodied in a trained labor force. The cost of acquiring this staff can be estimated either from past experience or by looking at the fees charged by personnel search firms and agencies that provide trained workers of various types.

Finally, there are the costs to create the necessary business structures, which include human resource management, information technology, and other unglamorous but essential functions. These do not simply spring, like Athena, fully armed from the head of Zeus. After we have estimated the reproduction values of a company's product portfolio, its current customer base/book of business, and its trained labor force, we must still account for what it takes to get the business off the ground and running smoothly. Usually this will be some multiple of annual administrative costs. The multiple in question will depend on the time that it takes to ramp up operations in a particular industry, but typically it will fall between one and three years. If administrative costs are not reported separately but are buried in the SGA line on the income statement, then we need to identify the approximate share of this total attributable to administration. If we have not been able to break out the reproduction values of the product portfolio, book of business, and trained labor force, then we can take a multiple of total annual SGA as a less precise but still useful estimate of the intangible assets embedded in a functioning business operation.

A further approach to calculating the value of intangibles is to look, as in the case of brands and revenues, at purchase costs in the market for similar businesses as a whole. This is particularly important in situations where intangibles are highly specialized, such as broadcast rights for radio and television, local cable, or utility franchises; licenses to run casinos or sell alcoholic beverages in markets with a limited number of such licenses; mineral rights, oil and gas reserves; long-term local service contracts for transportation facilities; local distribution franchises for global companies such as Coca-Cola or AB-Inbev; and professional sports franchises. These assets may be difficult or impossible to reproduce at historical costs. However, both discrete assets and operating business incorporating these assets are frequently bought and sold among industry participants. Where the transaction between knowledgeable buyers and sellers takes place for cash, as opposed to stock or other kinds of less certain considerations, these private market values are important indicators of the current reproduction values of these assets. The well-prepared analyst keeps track of these transactions; the values they establish may be an important source of information in similar situations.

Care must be taken when using such private market values. First, the transactions taking place will necessarily differ widely in size. A local cable system with 120,000 subscribers and \$180 million in revenue is not the same as one with 6,000 subscribers and \$6 million in revenue. It is always possible to translate the overall price paid into a “per” figure: price paid per subscriber, per regional population number, per dollar of revenue, or per dollar of operating profit. But different measures may tell different stories. Suppose, in our simple example above, the big cable system is 20 times the size of the small one measured by subscribers but, due to economies of scale, 30 times the size in income. Careful consideration rooted in detailed industry knowledge is needed to decide on the most applicable metric.

Second, transactions take place at widely varying times, and prices often reflect changing perceptions of future industry profitability rather than asset reproduction costs. Thus, asset values in these industries will not be fully independent of the earnings power value. In some cases, the operations in question are protected by barriers to entry so that earnings power value including the value of growth is the appropriate measure. For these dealings, the lack of transaction-based asset values that are independent of earnings potential is not a problem since asset values will be of secondary importance, and private market value will still provide a worthwhile check on the reasonableness of earnings-based valuations. But in cases where markets are competitive, such as buying oil reserves, private market value estimates should be calculated as averages over time to eliminate extreme periods of optimism and pessimism over a commodity cycle. Also, in these latter cases, unstable private market values introduce a considerable element of risk. Value investors who rely heavily on these private market values typically look for catalysts, such as takeovers, that will cause these private market values to be realized in the relatively near future.¹⁰

The same approach can be used in valuing a subsidiary business within a company. Private market purchases of similar businesses provide a basis for determining the worth of the subsidiary. Instead of using the price paid per subscriber or other operating figure, the standard practice is to use a multiple of a cash flow measure, like EBITDA. Say, for example, that a property and casualty insurance company has bought or built a subsidiary that provides online information to insurance adjusters on the price and availability of replacement auto parts.

¹⁰ Mario Gabelli pioneered this “private-market-value-plus-a-catalyst” approach.

This information is available to all insurance firms, who pay for it by a combination of subscription fees and usage charges. Although it serves the insurance industry, the subsidiary is really a different business. Its success has nothing to do with careful underwriting or skillful investing of premium revenue. It is an information services provider. In arriving at a value for the whole insurance company, it makes sense to break out the earnings and assets of the information subsidiary and to look at the price at which comparable information services companies have sold in the private market. Perhaps the insurance company will take advantage of the higher P/E multiples assigned to Web-based businesses and sell this subsidiary, or spin it off and retain some of the shares. Whatever it decides to do, if its value is heavily affected by ownership of this other business, the other businesses deserves a separate valuation.

Liabilities and Net Asset Value

So far we have only examined the asset side of the company's balance sheet. An investor in the company's stock who buys a proportionate share of its assets acquires an equally proportionate share of the liabilities that encumber these assets. Investors should properly be concerned with the net value of what they have bought. To arrive at a net asset value, we must subtract the negative value—cost—of the company's liabilities from the positive value of its assets.

A company's liabilities consist of debt, both long- and short-term, current non-debt liabilities, and long-term non-debt liabilities. We will address each of these liability categories separately.

Current non-debt liabilities are those due for payment within one year. They consist chiefly of accounts payable to suppliers, accrued liabilities largely due to workers, and also include taxes owed to various governments. Unlike current assets such as accounts receivable, there are no allowances for non-payment of these items. Failure to pay will lead to default judgments and bankruptcy. Thus, the book values of these liabilities are the amounts a company actually has to pay, which, in turn, is their intrinsic value, which turns out to be negative. We can depend on book values in estimating the costs of these current non-debt liabilities.

Long-term liabilities other than debt are either amounts incurred in the course of normal operations that come due more than a year in the future or long-term future obligations that have arisen over the course of historical operations. Long-term normal operational liabilities are usually owed to the same groups as current non-debt liabilities, including dealers

who supply marketing and support services, workers, and governments. Like current liabilities, they must ultimately be paid if a company is to continue to operate. Because payments may be deferred more than a year, the book values of these liabilities could be discounted by a cost of capital to arrive at their current intrinsic cost. But, except for inflationary, high cost-of-capital environments, the effect of discounting should be small, and book values should be close enough to actual values. In high cost-of-capital environments, average periods until payment should be estimated and book values discounted accordingly.

Inherited long-term liabilities consist chiefly of accumulated legal and environmental costs, unfunded future pension and benefit obligations, and the cost of deferred executive compensation. In theory, a company's auditors should estimate the present value of these obligations fully and accurately. The book values of these liabilities should, therefore, be proper measures of their actual negative values. However, if an item is large and the company in question has a reputation for accounting manipulation, then an independent estimate of the liability must be made. The information necessary is usually available in footnotes to a company's reported balance sheet. Available footnotes normally present detailed calculations with clearly identified assumptions for unfunded pension and benefit liabilities and for deferred executive compensation, including stock options. Major potential legal and environmental costs are also detailed in either footnotes to a company's balance sheet or a separate part of its annual report. This information can then be used to calculate alternative estimates of long-term non-debt liabilities.¹¹

Debt liabilities appear on company balance sheets at face value. At the time the debt is issued, interest rates are generally set so that its cost to the company—the present value of the associated payments of interest and principal—is at or near this face value. If interest rates subsequently rise, then the cost of these future payments will fall below face value. If rates fall, the cost will rise above face value. For short-term debt, except in extreme inflationary environments, these effects will be small and book value will be an appropriate measure of the cost of the debt. For longer term debt, the difference may be material. But for these debts, the true cost per dollar can be measured by looking at the average price, as a percentage of face value, at which debt of comparable duration and

¹¹ There are separate, often complicated calculations involved for each type of liability. Several good books on forensic accounting describe these calculations, including Cain, Michael, Hopwood, William, Pacini, Carl, and Young, George. (2016). *Essentials of Forensic Accounting*, Hoboken: John Wiley and Sons.

risk trades in public markets. Applying this adjustment factor to the book value of a company's outstanding long-term debt yields a usable measure of the true cost of that debt.

A second situation in which actual debt costs would deviate from book value arises when a company is in financial distress, facing a high probability of bankruptcy. But under these circumstances, the asset value of the company's equity will be zero. Existing asset values will be divided up among the liability holders in an order of priority determined by a bankruptcy proceeding. For viable businesses, debt should be measured at book value, adjusted where necessary to account for the impact of changing interest rates.¹²

Summing Up

In the value tradition, there are three alternative approaches to determining the net asset value of a company. The first, and least demanding, is to trust the accountants and use the book value of equity, the assets minus the liabilities. The second is Graham's "net-net" approach—current assets minus all liabilities—which is highly conservative. The third is to compute carefully a liquidation value for non-viable businesses and a reproduction value for viable ones. The broad characteristics of each of these approaches are outlined in Table 4.5. Finding the reproduction cost of a firm's assets and liabilities takes more work, more knowledge, and more everything than relying on book value or net-net figures. It is, however, the most frequently applicable approach and the one most likely to put an investor on the right side of any investment opportunity. Still, in practice portfolios of net-nets, each purchased at prices of two-thirds or less of net-net value, have generally outperformed broad stock market indices. The same applies to low price-to-book value

¹² The treatment of liabilities that we have described above preserves the role of net asset value as a measure of necessary investment for a potential competitor. For assets, we have already belabored this point. For entrants to compete on equal terms with existing firms, they need to reproduce the functional capacities of their assets. The cost of doing so is the definition of reproduction value. Liabilities represent sources of financing that offset part of the asset reproduction costs. If an entrant has access to non-interest bearing sources of funds that arise in the normal course of business (e.g., accounts payable) equal to that of an incumbent firm, then the off-setting liabilities for the entrant should be the same as the level of the incumbent's liabilities. The incumbent's net asset value should, therefore, equal the net investment required of the entrant. This argument clearly applies to debt and current and long-term operational liabilities. Legacy liabilities are another matter.

Table 4.5 Various approaches to calculate asset value.

Approach	Asset Value Approaches		
	Graham and Dodd	Book value	Reproduction costs
Opportunities	none	limited	more extensive
Value in practice	yes	yes	yes
Industry knowledge required	none	none	substantial
Stability/reliability	high	low	intermediate
Value of goodwill	0	book	reproduction costs
Value of debt	book (low debt)	book	market

ratios. Indeed, consistently outperforming these mechanically selected portfolios is a feat achieved by very few investors. Those who rely on detailed analysis of reproduction values must do it well if the effort is to pay off.

Example One: Hudson General

In our discussion of valuation, we noted that although asset and earnings power values are generally mutually supporting, there are instances when they diverge. In those cases, there are some occasions in which asset values are more useful and others where earnings power values provide better information. The choice of an approach depends on the nature of a business, the impact of management, and other future developments. It is necessary, before producing any detailed numerical analysis, to develop a clear picture of each of these elements of a business. This requirement is especially the case when the business is a complex combination of distinct operations that may obscure the determinants and level of intrinsic value.¹ To illustrate the kind of preliminary analysis required, we will return to an example we analyzed in the first edition of this book, that of Hudson General in the summer of 1998.²

A glance at the company's income statements from 1996 to 1998 suggests that something out of the ordinary had happened during this period. (See Table Ex1.1.) Reported revenues declined by 96% while net income fell by only one half. At the same time, the income statement shows a strange mix of operations: a business providing ground support, fueling, and other services to airlines at major airports (Hudson General

¹ This very complexity often leads these businesses to have hidden sources of value, so that they represent fruitful targets for careful analysis.

² Mario Gabelli introduced Hudson General as a company deserving analysis in Bruce Greenwald's value investing course at Columbia Graduate School of Business.

Table Ex1.1 Hudson General income statement (000s).

	1996	1997	1998
Revenue	\$ 157,100	\$ 5,064	\$ 5,783
Operating income	\$ 19,436	\$ (3,755)	\$ (2,724)
Equity in earnings of Hudson General	\$ 855	\$ 11,955	\$ 9,426
Equity in loss of Kohala joint venture	\$ (3,021)	\$ (11,292)	\$ (2,822)
Interest income	\$ 379	\$ 3,985	\$ 4,156
Earnings before taxes	\$ 14,949	\$ 866	\$ 8,036
Provision for taxes	\$ 7,183	\$ 391	\$ 2,780
Net earnings	\$ 10,466	\$ 475	\$ 5,256

Source: Data from Securities and Exchange Commission Form 10-K, Annual Report

LLC) and a joint venture for real estate development in Hawaii (Kohala). The Hudson General balance sheet also shows substantial cash, securities, and other assets held by the parent company that are part of neither enterprise. (See Ex1.2.) To develop a clear understanding of Hudson General's value, we need to begin by examining each of these separate sources of value, the assets and liabilities of Hudson General LLC, Kohala, and the parent company. Then we can consider how they fit together to deliver value to the stockholders in Hudson General (see Table Ex1.2).

A first preliminary step is to decide whether Hudson General is a suitable object of analysis for disciplined value investors, whether it would be identified as a good investment opportunity by an appropriate value-oriented search strategy. There were some promising characteristics of Hudson General at the July 31, 1998, price of \$48 per share. The company was small and obscure. Its market capitalization was only \$85 million. It was not covered by any sell-side analysts. And it had a total of only 182 shareholders. While not cheap, Hudson General's stock by late 1990s standards was certainly not expensive. With average earnings of roughly \$3 per share, it was selling at 16 times earnings, below the average market price-to-earnings multiple of 20. Despite a recent increase in price, its performance had trailed the market for a number of years. On the other hand, the dividend return was only 2% (\$1 per share divided by \$48), and the stock traded at 1.25 times book value. All things considered, it was worth a look although we would not have been surprised if a more detailed analysis proved it to be an unattractive investment.

Hudson General Parent

The balance sheet in Ex1.2 includes a 74% interest in Hudson General LLC (HLLC), a 50% interest in the Kohala land development venture,

Table Ex1.2 Reproduction cost of Hudson General assets (000s).

	Book value 1998	Reproduction
Assets		
Cash	\$ 19,001	\$ 19,001
Marketable securities	\$ 19,002	\$ 19,002
Receivables	\$ 563	\$ 563
Advances to HLLC	\$ 2,057	\$ 2,057
Prepaid expenses	\$ 56	\$ 56
Total current assets	\$ 40,679	\$ 40,679
PPE net	\$ 2,389	\$ 2,389
Investment in HLLC	\$ 22,306	\$
Investment in Kohala joint venture—net	\$ 4,962	\$
Notes receivable from Hudson General LLC	\$ 3,130	\$ 3,130
Total assets	\$ 73,466	\$ 46,198
Liabilities		
Account payable	\$ 200	\$ 200
Other current liabilities	\$ 2,628	\$ 2,628
Total current liabilities	\$ 2,828	\$ 2,828
Default income taxes	2197	NPV ≈ \$ 2,000
Total liabilities	5025	» \$ 5,000
Net worth		» \$ 41,000

Source: Data from Securities and Exchange Commission Form 10-K, Annual Report

and a number of assets and liabilities at the parent company level, most prominently \$38 million in cash and securities. The mystery of the 1996–97 decline in revenues is reflected in the first of these items. In that period, Hudson General received \$23 million from a branch of Lufthansa Airlines in exchange for 26% of the HLLC business plus an option to buy an additional 23%. After the sale, Hudson General deconsolidated the operations of HLLC in its financial statements, and HLLC's revenues disappeared from the income statement.

The remaining activities at the parent company were very limited. It provided support services to HLLC, for which it received revenues of \$5.1 million in 1997 and \$5.8 million in 1998. The parent also invested the excess cash and supervised the Kohala joint venture. An obvious question is whether the parent company operations provided any value at all. If HLLC were to be sold entirely to a company like Lufthansa, the buyer could almost certainly perform the parent company's functions within its existing infrastructure at relatively little incremental cost.³

³ Given the decentralized nature of HLLC operations across many separate airports, any benefit provided by the central parent company would have had to be very limited and hence should be easily performed by any existing overhead operations such as Lufthansa's.

Given the historical performance of the Kohala joint venture, Hudson General's real estate development prowess looked minimal at best and thus dispensable. If the residual parent company's assets were liquidated and distributed to shareholders either as dividends or through stock buy-backs, there would be no need to manage them, a cost savings of roughly \$3 million. The operations of Hudson General were almost certainly worth more dead than alive. The best way to value the extraneous parent company assets was liquidation value. The earnings power, and therefore the earnings power value, of the parent was less than zero.

Calculating a liquidation value for the LLC parent is straightforward. The cash and securities of \$38 million would be worth \$38 million. Money owed it by HLLC, receivables of \$563,000, advances of \$2,057 thousand, and the note receivable of \$3,130 thousand would all certainly have to be fully paid by whoever purchased HLLC. Like the cash and securities, they should yield book value of \$5.75 million in liquidation. This would go first to settle the current liabilities of \$2.83 million, leaving roughly \$2.9 million in net value. Recoveries from prepaid expenses and property, plant, and equipment would be less than book value. At a 50% recovery, they would yield about \$1.2 million. This resulting loss would offset some of the deferred tax liability, lowering it from \$2.2 million to roughly \$2.0 million at most. As a whole, the non-cash and securities assets of the parent should yield roughly \$2.1 million (\$2.9 plus \$1.2 minus \$2.0). These adjustments bring the total liquidation value of the Hudson General parent to approximately \$40 million (\$38 million in cash and securities plus \$2 million in other net assets).

Hudson General LLC

Hudson General's 74% interest in HLLC was a very different asset than the largely cash holdings of the parent. HLLC was a viable ongoing business for which Hudson General published a separate income statement (Table Ex1.3) and balance sheet (Table Ex1.4).

This part of the company should be valued using a full asset value/earnings power value approach. The resulting value estimate could be carried up to the parent level in either of two ways. First, it could simply be multiplied by the 74% of the business that Hudson General owned. However, Lufthansa had announced its intention to exercise its option to buy a further 23% of HLLC at a price of \$29.6 million. At the completion of this transaction, Hudson General would have an additional \$29.6 million in cash and own 51% of HLLC. Thus,

Table Ex1.3 Hudson General LLC income statement.

June 30	1996	1997	1998
Revenue	\$ 168,811	\$ 167,729	\$ 168,947
Operating expenses	\$ 130,696	\$ 136,259	\$ 139,880
Overhead 1	\$ 13,052	\$ 13,625	\$ 14,459
Operating income	\$ 25,063	\$ 17,845	\$ 14,608
Provision for taxes	NA 2	\$ 2,085	\$ 1,748
Net income	NA 2	\$ 15,939	\$ 12,738

¹Including service fee to Hudson General.

²NA for 1996, when Hudson General LLC operations were fully consolidated in the Hudson General income statement.

Source: Data from Securities and Exchange Commission Form 10-K, Annual Report

Table Ex1.4 Hudson General LLC balance sheet.

Hudson General LLC balance sheet (000)

June 30, 1999

Cash	\$ 3,393
Accounts, notes receivable	\$ 16,886
Other current assets	\$ 6,391
Total current assets	\$ 26,670
Net property, plant, and equipment	\$ 45,639
Other assets	\$ 643
Total assets	\$ 72,952
Accounts payable	\$ 17,336
Accrued expenses, other liabilities	\$ 19,045
Advances from Hudson General	\$ 2,057
Total current liabilities	\$ 38,428
Deferred income taxes	\$ 339
Note payable to Hudson General	\$ 3,130
Equity	\$ 31,055
	\$ 72,952

Source: Data from Securities and Exchange Commission Form 10-K, Annual Report

a second, more accurate alternative would be to add \$26.9 million cash plus 51% of the estimated value of HLLC to the \$40 million of parent company value.

Earnings power value is the best way to begin assessing the value of HLLC. HLLC was a relatively stable service business with lots of intangible assets: long-term airport contracts and relationships, trained

local labor forces, and established local management teams. Evaluating these intangible assets would be far more difficult and imprecise than forecasting and valuing sustainable future earnings. Revenues for HLLC over the period 1996–98 had been stable at roughly \$168 million per year. However, operating earnings declined steadily from \$25 million in 1996 to \$14.6 million in 1998. Hudson General attributed this decline to “increased competitive pressure on pricing of aviation services” as a result of “alliances between major commercial airline carriers.” But the company had seen this trend stabilize during 1998, a development noted by other airport service companies. Thus, a reasonable starting point for estimating earnings power value would be the \$14.6 million of operating earnings reported for 1998.⁴ To this we have to add back the \$5.7 million paid for unnecessary services by HLLC to Hudson General’s parent. This leaves us with a sustainable earnings power estimate for HLLC of \$20.4 million before tax. The tax level for HLLC reported in Table Ex1.3, about 12%, is clearly unrealistic for evaluating HLLC as an independent entity. It results from its special status as a subsidiary operation of its taxable parents, Hudson General and Lufthansa. An appropriate tax rate in 1998, including allowance for US federal, state, and local taxes, and Canadian taxes, would have been around 37%. This implies taxes on HLLC operating earnings of \$7.5 million (\$20.4 times 37%) or after-tax earnings of \$12.9 million. Replacement CapEx and accounting depreciation and amortization for HLLC appear to have been about equal so that no further significant adjustments to earnings power seem necessary.

The year 1998 had been preceded by a long-term bull market, and expectations of future equity returns were relatively high. Because HLLC had almost no debt, an equity cost of capital of 10% seems appropriate for even a relatively stable company such as HLLC. Thus, our estimate of \$12.9 million in earnings power should have a value of \$129 million (\$12.9 million divided by 10%). From this we have to subtract HLLC’s roughly \$2 million in net debt (\$5.2 million owed to Hudson General less \$3.4 million in cash) to arrive at a final equity earnings power value of \$127 million.

Since the reproduction and book values of HLLC’s property, plant, and equipment look roughly similar, and adjustments to net working capital are likely to be small, a natural starting point for estimating the

⁴ A longer history, including at least one recession, would be desirable here. In this case, that history would produce a number near the \$14.6 million above.

asset value of HLLC is the book value of \$31 million.⁵ To this we must add the reproduction value of the LLCs intangible assets, which are substantial. Its contracts last from 5 to 10 years and only 10 to 15% of them turn over every year. Building local labor forces and infrastructure might take even longer. Thus, developing intangibles equivalent to those underpinning HLLC's business might involve as much as seven years of overhead and marketing expense or \$100 million (7 times \$14 million of annual HLLC overhead). But this estimate will be subject to significant potential error.

Fortunately, we have in this case a private market estimate of the value of HLLC assets from a knowledgeable buyer willing to pay cash. That buyer was Lufthansa's ground service subsidiary. In 1996, Lufthansa paid \$23 million for 26% of HLLC, establishing a value of about \$88.5 million on HLLC net assets (\$23 million divided by 26%). In 1998, it was willing to pay \$29.6 million for another 23% of LLC, valuing all of those assets at \$128.7 million (\$29.6 million divided by 23%). As this asset value is very close to our estimated earnings power value of \$127 million, the Lufthansa bid was by no means an unreasonably optimistic one. And Lufthansa was not paying for a control position, which, by eliminating HLLC operating inefficiencies beyond the unnecessary fees paid to the Hudson General parent, would further enhance HLLC's value.

These calculations indicate that HLLC as a whole was worth at least \$128 million. For Hudson General parent, HLLC added \$29.6 million in cash and \$65 million of value for its residual 51% of HLLC. Together with a parent company liquidation value of \$40 million, mostly in cash, this brings the total value to roughly \$70 million in cash and \$65 million in its share of HLLC, compared to a Hudson General market value of \$85 million (\$48 per share times 1.77 million share equivalent outstanding).⁶

Kohala Development

The last piece of Hudson General was its 50% interest in the Kohala land development on the island of Hawaii. The balance sheet for Kohala is shown in Table Ex1.5. The income statement (not presented) was a sad tale of continuous losses: \$5.6 million in 1998 after a \$22.6 million

⁵ This conclusion follows from the analysis that replacement capital expense and accounting depreciation and amortization are approximately equal.

⁶ This figure is arrived at by multiplying \$48 times the total number of outstanding shares and options and subtracting the cost to the option holders of exercising their options.

loss in 1997, which included a \$17 million write-down in the value of Kohala's assets. After having been halted by the filing of an environmental lawsuit, a recently favorable court decision would allow land sales for residential construction to resume. But the long-term reality was that Hudson General had effectively put \$29 million into Kohala and would recover only a fraction of that amount. Hudson General's⁷ experience as a Hawaiian land developer had been a flop. As a result, the Kohala venture would be liquidated. Therefore, like the Hudson General parent, Kohala should be assessed at liquidation value.

Table Ex1.5 Balance sheet of Kohala.

Kohala balance sheet (000)

June 30, 1998

Cash	\$ 355
Land	\$ 9,210
Mortgages, notes	\$ 2,137
Foreclosed property	\$ 2,186
Other current assets	\$ 1,549
Total assets	\$ 15,437
Account payable	\$ 860
Net worth	14577
	\$ 15,437

Source: Data from SECURITIES AND EXCHANGE
COMMISSION FORM 10-K, Annual Report

We will assume that recoveries from the non-land assets of \$6.2 million would be 100% for the cash, 80% for the mortgages and notes, and 50% for the foreclosed property and other assets. These total \$3.9 million. From this amount, the payables of \$0.9 million must be subtracted, leaving \$3.0 million net or \$1.5 million for Hudson General's half interest in Kohala. These are crude calculations based on common rules of thumb that may not apply to Kohala, but as the amount involved is relatively small, a more detailed analysis is unwarranted.

⁷ The difference between 50% of the value of Kohala in Table Ex1.5 and the value of Kohala on the Hudson General balance sheet has to do with the treatment of accrued interest on advances to Kohala from its partners. The accrued interest is treated as being paid in to the partnership, but it is not recognized as such at the Hudson General level as part of the investment in Kohala. Because it is not being paid, it is treated as a future liability by Kohala but not as an asset of Hudson General. Since the amount involved is shared equally by the two partners, the difference is immaterial to a liquidation analysis.

The remaining Kohala asset was 1,820 acres of partially developed and undeveloped land. Of the original acquisition of 4,000 acres, 2,100 acres had been sold in phases one and two of the development, which took place before the lawsuit, and another 80 acres that were sold as part of phase three. Hudson General's share of this remaining portion was 910 acres. The standard way of valuing real estate assets is to rely on local assessors.⁸ In this case, the assessment produced an estimate of between \$15,000 and \$20,000 per acre, or from \$27 million to \$36 million for the entire 1,820 acres. This was well above the book value of \$9.2 million but far less than the \$55 million net land investment by the partners (\$58 million paid less \$3 million non-land net assets). If we use the lower end of this range, Hudson General's share of the land value was \$13.5 million, from which roughly 6% in sales cost would have to be deducted. This adjustment leaves the total liquidation return for Kohala at about \$14 million (\$13.5 times 94% plus \$1.5 million for the non-land assets).

Calculated as a sum of its various parts, Hudson General should be worth about \$150 million:

- \$70 million in cash from the parent liquidation;
- \$65 million in residual 51% LLC holding;
- \$14 million from the Kohala liquidation.

Compared to the market price of \$85 million, this total includes a margin of safety of \$65 million, or 43% of our estimated value.⁹

Hudson General as a Whole

Unfortunately that \$150 million of value was “potential” value, what an investor would realize if Hudson General's management were to do the right thing, dismember the company and eliminate their own jobs. As an ongoing enterprise in the hands of current management, Hudson General was considerably less valuable. As of June 30, 1998, Hudson General generated about \$5.3 million in annual earnings (see Table Ex1.3) and

⁸ The costs of this generally amounts to only several hundred dollars.

⁹ Notice that we have ignored future growth prospects. This is not just because Hudson General has not been growing recently—if anything, earnings are shrinking. It is also because Hudson General's business is not protected by barriers to entry, and growth typically does not generate value for such businesses, as we discussed in Chapters 6, 7, and 8.

paid out about \$1.8 million in annual dividends. In fiscal 1997, the company did buy back roughly \$9 million worth of stock, but the next year it was a net stock issuer. And its record of capital allocation to ventures like Kohala did not auger well for value creation in the use of its \$40 million cash hoard.

If Hudson General management were to pay out the full \$5.3 million in earnings, the company might be worth \$66 million even without growth. The dividend return would be about 8%, typical for a stable dividend-paying utility. If it destroyed only 50% of the value of its cash hoard, then Hudson General's value would be about \$85 million (\$66 plus \$20), which was equal to its current market value. Hudson General is a classic Case A company according to our taxonomy at the end of Chapter 3, for which the potential asset value of \$150 million far exceeds its current earnings power value of \$85 million. The critical issue here is one of management behavior. If that behavior remained unchanged, Hudson General was likely to be an unsatisfactory investment. Even if management stopped destroying value, the best an investor could earn without a sale of the assets would be a cash distribution that did not match the value of assets trapped within the firm.

Situations like this require a "catalyst" to replace existing management. Hudson General was potentially an exemplary "value trap," with asset value in excess of market price. Without management change, the full value of these assets would never be realized.

Given the need for a change in management to affect either a sale or an improved return, Hudson General might not have looked like a promising investment in the summer of 1998. Four of its eight board members were company officers—chairman, vice chairman, president, and executive vice president. Another was an investment banker who had earned significant fees from the Lufthansa transaction. Without a clear path to management change, investing in a company like Hudson General is a perilous undertaking. In this instance, however, that change was in fact imminent. A noted value investor with an activist orientation owned a near majority of the stock and was clearly interested in forcing management change. Under these circumstances, management did not need much prompting. In November 1998, it offered to purchase the company for \$100 million. Fortunately for the shareholders, the managers' move just initiated the bidding. Between then and February 1999, when Globeground, the Lufthansa unit that by then owned 49% of HLLC, put in its winning bid of \$133 million, there had been a sweetened offer from management at \$106 million and several bids from other companies in the aviation services industry. All this attention kept

management from enriching itself at the expense of outside shareholders, though who knows what door prizes they were able to cart away. Did Globeground pay top dollar? From our analysis we know that they got both the Hawaiian acreage and control of HLLC for nothing. But since they were the logical purchasers from a strategic sense, it is probably unreasonable to expect anyone to have bid more. For a company whose market price was \$85 million only months before, a sale at \$133 generated a return of more than 50% in virtually no time at all, a rich reward for someone who had the skill and patience to work their way through the morass of Hudson General.

5

Earnings Power Value

The second step in a traditional Graham and Dodd valuation is calculation of a company's earnings power value (EPV), which is based on sustainable current earnings. Information related to current earnings is second only to assets as the most reliable basis for determining what a company is worth. By current earnings power we mean the average distributable earnings of a company as it exists today. This level of earnings is assumed to be constant over an indefinite future. The value of such a stream, the EPV, is calculated as this current earnings power divided by the company's cost of capital. The cost of capital, in turn, is the expected annual return that the company must offer in order to attract willing investors. If a firm must offer an expected return of 8% and has constant future annual earnings of \$10 million, then investors will properly value the firm at \$125 million (\$10 million divided by 8%).

Because there is a forecast embedded in the assumption of a constant level of future earnings, the EPV is subject to greater potential errors of measurement than the asset value. But by concentrating on sustainable earnings as the company exists today, this estimate of future earnings is firmly based on the observable past history of the company, its management, and its industry. The EPV calculation does not attempt to anticipate future changes in a company's operations. Security analysts who incorporate developments such as future growth from new products in their forecasts have a poor record subject to both random and systematic error. By isolating the EPV from these more speculative future forecasts, Graham and Dodd estimates of sources of value move in a disciplined way from the most to the least reliable measure of value. They

do not ignore the potential effects of new developments on a company's value but defer consideration of their impact until after calculating a more basic and reliable valuation. We deal with future developments when we discuss the valuation of growth.

Earnings Power

In the definition of earnings power, sustainable distributable earnings are the average earnings that a company would generate in the future if its underlying operations were to continue without change. We include "average" in this definition to smooth out business cycles and other temporary fluctuations, positive and negative, to which short-term earnings are subject. Sustainable earnings are earnings measured after making the investments necessary to return a company's operations at the end of the year to their condition at the beginning of the year. This calculation involves correcting accounting mismeasurements of non-cash expenses such as depreciation. The distributable part of the definition is intended to deal with a second set of accounting issues. Many growing firms incur expenses for research and development, advertising, marketing, and the hiring and training of new personnel that exceed the level required to maintain the firm's current economic position. A proper sustainable earnings calculation should eliminate these "excess" growth-related expenditures from costs before calculating adjusted accounting earnings. Shrinking firms may underinvest in similar expenditures (e.g., cutting R&D below what would be necessary to sustain earnings or raising prices to increase current revenues at the expense of future market position). For these firms, the estimated shortfalls must be subtracted from adjusted accounting earnings to arrive at a sustainable earnings power.

Companies often record one-time events, both positive and negative, as adjustments to earnings. Usually these are related to gains from operations sold off or losses on prior investment mistakes. Theoretically, since they are unrelated to the ongoing operations of the firms, they should be ignored in calculating sustainable earnings. But for some companies, these supposedly isolated incidents are frequent, even annual, recurrences, used to conceal ongoing costs that should in fact be deducted from sustainable earnings. In such cases, some fraction of the average level of one-time charges should be treated as actual operating costs for the purposes of calculating sustainable earnings power. Even when these regular charges are legitimately independent of ongoing operations, they represent a reduction in the value of distributable earnings. To the extent

that they are a regular feature of management's performance, their average level should be subtracted in calculating earnings power.

Revenues tend to be far more stable than profit margins over business cycles and are subject to smaller random fluctuations. Therefore, except for highly cyclical businesses such as natural resources where revenues do fluctuate substantially, the customary way of calculating sustainable earnings is to begin with the current year's revenue and multiply that by an average level of margins over a period that ideally encompasses at least two business cycles. A simple arithmetic mean is the natural place to start. But calculation of average margins may in practice be more complicated. Often margins exhibit secular trends, increases or decreases, that must be taken into account. A common temptation will be to extrapolate these trends by using a sustainable margin figure well below the current margin when profitability is declining or, conversely, by selecting a higher margin when profits margins are growing.

This temptation should be resisted. Graham and Dodd understood the dangers of forecasting. Forecasts, especially reflexive trend extrapolations, should not be allowed to contaminate basic value estimates. There are two ways of dealing with secular margin trends in the spirit of Graham and Dodd. The first is simply to ignore them and use the arithmetic mean. This is likely to be the best way to proceed when there is no observable economic or strategic reason that can account for the trend in margin. If, on the other hand, there is an obvious source for changes in margin, such as continuous adverse technological developments that intensify competition and impair industry margins or improve operating leverage as revenues grow, then using the most recent margins, adjusted as necessary to account for other cyclical conditions, is appropriate. Any further future margin improvement or deterioration should be dealt with as part of the value of growth.

A second question is which margin figure to use. Gross margins, operating margins, pre-tax margins, and net income margins are all potential candidates.¹ The argument for using gross margin, revenue minus direct product costs, is that it is the hardest to manipulate. The

¹ As everyone familiar with basic accounting knows, and therefore virtually every reader of this book knows, margins are percentages, the numerator being a figure such as gross profit (revenue minus direct cost of production) and the denominator being revenue. Operating margins are revenues (direct production costs + SGA and other operating expenses) divided by revenue. To keep the narrative flowing, we will assume that the reader understands that we are referring to these margins as percentages even when we do not describe the complete calculation.

overhead expenses that represent the difference between gross margins and operating margins, including research and development, marketing, and general and overhead expenses, can be increased or decreased by management to make current operating margins look better than they really should be. For example, by cutting research and development, management can raise current operating earnings with relatively little impact on current sales. However, there may be a substantial cost in lost future opportunities. But even gross margin is not a pristine figure. Accountants have considerable discretion in allocating expenses between direct production costs and overhead. Moreover, they can usually do this on paper rather than by distorting company operations so that the economic costs of fiddling with gross margins will generally be lower, and therefore a greater temptation, than the cost of manipulating operating margins. Moreover, overhead costs are real costs. Ignoring them will always lead to a mistaken estimate of distributable earnings. Our preference here is to focus on operating earnings but to adjust them for inexplicable changes in R&D, marketing, and other overhead expenses that are likely to be due to management earnings manipulation.

There are two significant differences between operating income, often referred to as EBIT (earnings before interest and taxes), and pre-tax income, or EBT (earnings before taxes). The first and usually the more important is net interest, which is the difference between interest earned on cash and other financial assets and interest paid on a company's debt. The amount of net interest generally depends on the degree to which the company relies on debt financing. There are two ways to account for the extent of such leverage. The first is from the income statement. We can simply base our earnings power estimate on post-interest, pre-tax earnings. This approach assumes that the burden of leverage is constant over an indefinite future. But debt levels, especially increasing debt levels, are a matter of financial policy that may be changed relatively quickly at the discretion of management. And interest rates will fluctuate. When interest rates are unusually low, current pre-tax earnings will be at levels that are not sustainable when interest rates rise. The same problem in its inverse applies when interest rates are unusually high. They are likely to come down, and pre-tax margins will increase. A firm's underlying business, by contrast, is usually more stable and changes only slowly over time. If it does not have these characteristics, it is going to be difficult to value the firm at all. Operating income, therefore, is a better place to begin thinking about sustainable earnings power than pre-tax income, even when interest is a significant expense (Table 5.1).

Table 5.1 Alternative earnings margins.

	2019	Margins
	(000s)	
Revenues	\$ 1,000,000	
Cost of revenue (direct cost of production)	\$ 600,000	
Gross profit	\$ 400,000	
Gross profit margin		40.0%
Marketing	\$ 100,000	
R&D	\$ 60,000	
General and administrative	\$ 70,000	
S,G, and A	\$ 230,000	
Operating income (earnings before interest and taxes, EBIT)	\$ 170,000	
EBIT Margin		17.0%
Net interest expense	\$ 75,000	7.5%
Earnings before taxes	\$ 95,000	
EBT margin		9.5%
Taxes	\$ 20,900	
Net income	\$ 74,100	
Net income margin		7.4%

A second way of dealing with leverage is to look at the level of net debt rather than current interest payments. We can calculate an *enterprise earnings value* for a firm's operating business ignoring net interest payments and then adjust that value to get an equity value by subtracting the estimated value of net debt. As a first estimate we can use the latest book values of debt minus cash and marketable securities. If necessary, these figures can be adjusted to approximate market value by looking at current market prices of the firm's debt securities, both owned and

owed. We prefer the net debt approach because the balance sheet levels of debt, cash, and marketable securities are measurable with considerable accuracy, whether based on book values or market prices. On the cash and securities side of the equation, a company's accounts should generally already have marked the book values of cash and investments to best estimates of the market value.

EPV is the value of a company's operating business. Some of this value goes to holders of the debt, leaving less for the owners of the equity. To calculate the EPV for the equity holders, we must subtract this amount, which should in theory be the value of the outstanding debt. At the same time, the portion of the cash and financial investments that are not required to support the basic operations of a firm represent value to the equity owners above the basic earnings power value of the company. Conventionally, all the cash and financial investments are assumed to be extraneous to operations; in practice a firm needs some of these assets, perhaps 0.25 to 0.5% of revenues, for its smooth functioning. These amounts should be subtracted from overall cash and investments before adding them to the EPV of the company.² Thus there is no compelling need to add the errors inherent in projecting future interest rates to the calculated earnings power.³

A second major difference between operating earnings and pre-tax income is the earnings of unconsolidated subsidiaries, joint ventures, and minority interest in consolidated operations. As in the case of debt, these items appear on both a firm's balance sheet and its income statement. If they are peripheral to a firm's principal business, as often happens, there is a strong case to be made for handling these factors as adjustments to operating income. These assets and liabilities rarely trade in public markets. Private market values are often difficult to obtain unless full details for the relevant operations are included in footnotes to a firm's financial statements. Book values are based on original equity investments, adjusted for historical gains and losses that may bear little relationship to current economic values. For example, an unconsolidated venture may be near the end of the period of losses incurred in building the business,

² Similar treatment should be applied to other non-essential assets such as excess real estate whose returns are not included in operating earnings. There are also legacy liabilities that are equivalent to debt but not counted as debt for accounting purposes (e.g., unfunded pension liabilities); these should be included as part of debt. Later in this chapter we discuss in detail the differences between equity and enterprise approaches to valuation.

³ Graham and Dodd generally avoided investing in firms with high levels of debt for reasons of risk management, which we will discuss later in this book. Therefore, differences between operating and pre-tax earnings for the firms they were interested in were usually small.

but may have significant current profitability. Actual value will be more accurately reflected in current profitability. If there is a significant discrepancy between the value of unconsolidated operations based on current earnings power and the values recorded on the balance sheet, then operating income should include these items before calculating average sustainable distributable income. If there is no such obvious discrepancy, then balance sheet adjustments similar to those used in accounting for debt can be used. In either case, the basic starting point for estimating earnings power will be operating margins and operating earnings.

Taxes are real and inevitable reductions of earnings power, and operating earnings must be adjusted to take them into account. Again, operating earnings are the best place to begin. Average sustainable operating income, properly calculated, reflects the underlying economics of a company's business. For firms that can be reliably valued, this figure must be fairly stable, even taking into account average cyclical fluctuations. Taxes depend on government tax policies that are subject to periodic change. Average operating margins, including at least one or ideally two business cycles, may be calculated over a period of seven to fifteen years. Current average tax rates are likely to be stable over no more than five years. There is an obvious mismatch between the two time periods. The purpose of this entire exercise is to calculate the average distributable, meaning after-tax, earnings as the firm sits there today. If there has been a major recent tax law change, then average tax rates since that change will be the relevant ones.

It is common to make a distinction between cash taxes actually paid and tax liabilities calculated for income accounting purposes. For NPV calculations, the annual cash flow impact of taxes actually paid is the proper measure of the tax burden. But earnings power calculations are designed to capture average tax levels assuming no growth. Since long-term discrepancies between cash taxes and accrued taxes depend largely on growing tax credits, the no-growth assumption argues for using accrued taxes. In the long run these accrued taxes must be paid. They are, therefore, the proper adjustment to earnings power.

We can now lay out the steps in a basic earnings power calculation before accounting and other adjustments.

1. Calculate an average operating margin over an appropriate period.
2. Multiply this margin by current sales or an estimate of average sustainable current sales. The product is an estimate of operating earnings (EBIT).

3. Adjust this estimate for the impact of the earnings of unconsolidated operations.
4. Calculate after-tax operating income by multiplying the pre-tax figure by one minus an appropriate average tax rate. (Adjusted EBIT $\times [1 - \text{average tax rate}]$.)
5. The result is an estimate of sustainable, distributable net operating profit after taxes (NOPAT).

Adjusting for Non-cash Costs: Depreciation and Amortization

Depreciation is the most significant area where there may be major differences between accounting measurements of non-current costs—the gradual wearing out of capital equipment—and economic reality. Accounting depreciation is calculated by spreading the historical acquisition costs of capital investments over an estimated useful life of each of the various types of investments involved. In the estimation of earnings power, actual depreciation is the cost of restoring the condition of the company's fixed capital at the end of the year to its state at the beginning of the year. It is the capital expense required to replace the lost capability of the firm's capital stock over the course of the year, or replacement CapEx (capital expenditures). Since replacement occurs in the present, the appropriate measure is the current cost of capital equipment, plants, and structures, rather than some arbitrary portion of the historical cost.

There are two obvious differences between these accounting depreciation figures and replacement CapEx requirements. First, if there are trends in capital goods prices, current costs will differ substantially from historical costs. Inflation-driven trends of increasing costs, in railroad construction for example, will cause replacement prices to exceed historical ones, and accounting depreciation will understate true depreciation. Technologically driven cost reductions, as in information technology equipment, especially when improved machine performance is taken into account, will lead to replacement costs below historical costs, and accounting depreciation will overstate them. In either case, direct estimates of replacement capital expense should be used in place of accounting depreciation.

A useful method of estimating replacement CapEx is to start with overall capital expense and then subtract an estimate of the capital

expense devoted to supporting growth. Estimation of growth CapEx can be done in at least two ways.

One way to segregate the two is to look for the periods of little or no growth in the history of the company or of comparable competitors. During this time, total capital expenditures, reported on the cash flow statement, should consist almost entirely of replacement capital costs since growth-related capital costs should be negligible. Any discrepancy between total capital expenditures and accounting depreciation should capture the necessary adjustment that needs to be made to actual earnings. A second approach is to estimate growth capital costs by multiplying growth in sales by an estimate of the firm's capital intensity, which is the ratio of fixed capital to sales calculated over an appropriately recent historical period. For example, if each dollar of sales appears to require \$.30 of capital investment, then a year in which sales grow by \$80 million should require roughly \$24 million in growth-related capital expenditures. This growth CapEx can then be subtracted from total CapEx to obtain an estimate of what is being spent purely to restore the fixed assets to their level at the start of the year. This estimate can be used in place of accounting depreciation as a more accurate figure of actual replacement capital costs.

Some companies such as railroads will report their own estimates of replacement directly. Others report the details of their capital expenditures that can then be roughly divided into growth and replacement amounts. Detailed industry knowledge may enable an analyst to make these adjustments with reasonable accuracy even if only aggregate investment and facilities growth numbers are known. Those who make a diligent effort to improve on reported accounting depreciation in the calculation of earnings power will have an advantage over those who simply rely on the official financial statements.

One shortcut should always be avoided. Analysts will often use earnings before interest, taxes, depreciation, and amortization (EBITDA) in their measure of the company's basic earnings power. The unstated assumption is that replacement costs for both tangible (depreciation) and intangible (amortization) capital are zero. This is hardly ever the case. Buildings and other structures will often have accounting lives that are far shorter than their real economic lives. Their replacement values may actually increase rather than decrease over time. Adjustments to accounting depreciation of these assets must certainly be made. However, even buildings eventually wear out, and like equipment, they must be replaced. Assuming zero depreciation is inappropriate even in these instances. The same is true for amortization. A product

portfolio, customer base, or a trained labor force will deteriorate over time and need to be replenished. In the case of intangibles, however, the necessary replacement expenses may in fact be included in overall operating costs. Under these circumstances, including amortization as a cost will involve double counting and adding back amortization to EBIT will be appropriate. In general, however, there is always a danger that relying on EBITDA will overstate earnings and lead to overvaluation and therefore overpayment for securities.⁴ A carefully calculated earnings power with proper adjustments to depreciation, expensed investments, and other accounting distortions will be a far superior measure of sustainable distributable earnings.

EPV and Estimating the Cost of Capital

The final step in computing a company's earnings power value is to divide its estimated earnings power by another estimate, this one of its cost of capital. This process converts the flow of earnings power into a level of value. Ultimately, we want to know what a firm is worth, based on an estimate of reliably identifiable future earnings, for comparison to the market price we must pay. Value is what an intelligent, well-informed investor is willing to pay for the firm at that level of sustainable earnings. Suppose the firm generates \$25 million as a constant average annual future level of after-tax distributable earnings. The amount an investor would be willing to pay depends on the return he or she requires in order to voluntarily commit funds to acquiring the firm. If this figure is 10%, the investor will pay \$250 million; \$25 million represents an annual return of 10% on a \$250 million investment. If the investor requires 12.5%, he or she will pay only \$200 million, since \$25 million represents a 12.5% return on \$200 million. At an 8% required return, the firm will be worth \$312.5 million since the \$25 million is 8% of \$312.5 million. In each case we obtain what the investor is willing

⁴ An alternative to using EBITDA is to begin with net operating cash flows from the cash flow statement and subtract gross investment, also from the cash flow statement. The resulting figure is often referred to as free cash flow, supposedly a direct measure of distributable earnings. There are two shortcomings with this approach. First, some real costs, including options, deferred taxes, and accrued retirement liabilities, are not counted as costs in net operating cash flows because they are not associated with current cash outlays. Second, for growing companies, much working and fixed capital expense included in this calculation applies to future investments, not current cost. It should be included in distributable future earnings.

to pay by dividing the earnings power by the investor's required return. Thus the earnings based value of the firm is

$$\text{EPV} = \text{EP} / \text{Required Return}$$

This required return is just the definition of a firm's cost of capital, the returns it must pay to attract investment voluntarily. The lower the required rate of return, the higher the value of the company.

Estimation of the cost of capital, therefore, means understanding how investors determine their required returns and how these individual required returns translate into an appropriate required return for the firm. If we were asking about the average cost of labor per hour, there would be no mystery about how to proceed. We would identify the different types of labor to be hired, from highly skilled professionals to unskilled workers. We would estimate the fraction of the labor force in each category and the level of wages necessary to attract sufficient employees in each group. In competitive labor markets we would look at what other firms are offering for similar jobs. Average wages would be calculated by multiplying the required wages for each class of workers by the fraction of such workers in our labor force and summing to get the total.

We use an equivalent procedure to estimate the weighted average cost of capital. Firms basically employ two types of capital, debt and equity. The first step is to decide on the appropriate mix of debt and equity used in financing a firm's operations. Looking at the historical financial structure will be the best way to begin, just as we would look at the historical mix of labor types employed to estimate our hiring mix. However, there is one important difference when we consider the cost of capital. Debt financing imposes fixed payment requirements on a company that equity financing does not. Consider a firm that relies heavily on debt. As a rule, after-tax interest costs per dollar of financing are lower than equity costs, if only because interest payments are tax-deductible for firms. A highly leveraged firm, one with a high fraction of debt financing, will have a low calculated average cost of capital. However, this debt imposes a hidden cost. In times of economic stress, high debt, because of its payout obligations, may significantly impair the efficiency of a firm's operations; in a worst case it will force the firm into bankruptcy. These distress costs are hard to quantify but still need to be taken into account in calculating the cost of capital. They are not part of the cost of equity because equity financing does not interfere with company operations. Using a weighted average observed cost of debt and equity may grossly understate the true cost of capital for a company with excessive leverage.

One way to handle this problem is to use a weighted average based on historical levels of debt and equity, but to set the weight of the debt at a conservative level below which it is unlikely to significantly impair a company's operations. For example, suppose that a company's recent debt level has been 70% of overall financing, but the upper limit of a "safe" level of debt is about 30%. Above that level there is a significant probability of operational impairment. In that case, we would use 30%, not 70%, as the weight of debt in calculating the cost of capital. Graham and Dodd solved this problem by not investing in companies with high leverage, but we want to add a level of precision and open up opportunities.

The way to estimate the cost of debt is analogous to estimating the cost of labor: look at the yields available on publicly traded debt with comparable risk characteristics to the debt of our company. If that debt trades at yields of 5%, then our company will have to pay 5% to attract debt funding. This is the pre-tax cost of debt. If interest on debt is subtracted from profits before paying corporate income tax, then the net after-tax cost of the debt will be reduced by the tax saved. If corporate tax rates average 30%, that makes the after-tax cost of the company not 5% but 3.5%, thanks to a corporate tax saving of 1.5% (30% of 5%) on debt financing.⁵ It is in this adjustment to the cost of capital that we account for any tax savings associated with debt financing. That is why we apply the corporate tax rate to all of operating earnings in calculating NOPAT.

The cost of equity is more difficult to estimate than the cost of debt. Debt instruments provide well-defined returns whose yields, in the absence of default, are easy to calculate. Stocks produce both dividends, which may vary due to business conditions and are at the discretion of management, and capital gains, which depend not only on the performance of the company but on overall financial market circumstances. For equity investments we cannot simply identify comparably risky stocks and look at a stated yield on an investment in those stocks. Despite these hurdles, it is possible to make sensible estimates of the cost of equity capital.

We begin by recognizing that because debt receives a contractual payment schedule and has a prior claim on a company's assets and earnings, debt investments are inherently less risky than equity investments.

⁵ As we write (2020), the current corporate tax rate is 21%. We are interested in long-term rates and are leaving the 30% rate here because we think the rate may change again, and the 30% is a likely average rate.

Thus, if investment grade (at least BAA rated) bonds with low probabilities of default yield returns of 5%, a reasonable lower bound on the cost of attracting higher risk equity financing is going to be 6%. At the other end of the risk spectrum, venture capital is widely regarded as the riskiest form of equity investment. It involves buying the shares of promising companies without either established economic positions or documented operating histories. A high proportion of these investments fail. Venture capital firms that make such investments regularly provide two kinds of information on the returns they must offer to attract capital. They report the levels of returns on their past funds and they respond to industry surveys asking directly for the level of expected returns necessary to market their latest venture funds. Not surprisingly, these figures tend to cluster together. Historically, these venture costs of capital have been as high as 18 to 20%. More recently they have been in the range of 13 to 14% as overall investment returns have fallen with interest rates. Required equity returns for established firms without dangerous levels of leverage should be below this venture capital level, at around 13% or less. Thus the cost of equity capital in 2020 is likely to fall in a range between roughly 6% and 13%.

If we are able to make qualitative judgments of the risks associated with investments in different firms, we may make our estimate more precise. We can divide company equities into low risk (e.g., utilities and stable consumer non-durable companies such as Coca-Cola), moderate risk (e.g., general non-financial service businesses such as United Parcel), and high risk (e.g., cyclical industrial and commodity companies such as Freeport-MacMoRan). Low-risk equities typically have cost of capital of 6–8% or roughly 7%, moderate risk equities need to offer 8 to 10%, or about 9%, and high-risk companies would face a cost of equity capital of 11%–13%, an average of about 12%.⁶ By deciding into which risk category a particular company should be assigned, we can narrow the overall 6%–13% range for establishing its cost of equity.

As an example, we have a company in the moderate risk category that is financed with 20% debt and 80% equity. The debt has a pre-tax cost of 4.5%, which translates into 3% after taxes. The equity costs 9%. Putting the two together, we have a calculated cost of capital of 7.8% (Table 5.2). In practice, this level of precision in estimating the overall cost of capital is unrealistic. We should use 8% for computing our earnings power

⁶ Cost of equity capital of highly leveraged companies will often fall outside this range, but individual investors could still ask themselves (and others) what its expected return levels and equity would have to be for them to be comfortable holding such investments.

Table 5.2 Weighted average cost of capital.

Capital source	Percent of total capital	Cost (after-tax)	Weighted cost
Debt	20.0%	3.0%	0.6%
Equity	80.0%	9.0%	7.2%
Total weighted average cost			7.8%

value. Unless we had grossly misjudged the company's perceived risk and therefore its cost of attracting capital, we are not likely to be off in this estimate by more than around 1% or so in either direction, close enough for financial work.

An alternative approach to estimating the cost of equity financing is provided by Modern Portfolio Theory (MPT). One of the tenets of that theory is that the relevant measure of risk for any security is the incremental variability that it adds to a fully diversified portfolio, which is defined as the whole relevant market. The more variability the stock adds to the portfolio, which is designated *beta* (β) in MPT, the riskier the investment, at least as "risk" is defined here. The market itself has by definition a *beta* of 1: a 1% change in the market equals a 1% change in the market. If a stock rises or falls by only 0.5% for each 1% move in the overall portfolio, adding more of that stock will lower the overall variability of the portfolio. This is the sense in which a stock like this, which has a *beta* of 0.5, represents a low risk investment. In contrast, if a second stock rises or falls by 1.5 % for each 1% move in the market portfolio, adding that stock to the market portfolio will increase the overall market variability. This second stock, which has a *beta* of 1.5, is a high risk investment. The higher the *beta*, the riskier the stock.

According to MPT, investors should require higher rates of return from stocks with higher *betas*, which are riskier by this definition. The amount of that required return can be calculated by looking at the average historical difference between the return on the market portfolio, with its *beta* of 1, and the return on a risk-free asset, usually assumed to be the short-term debt of a stable government's securities, such as three month Treasury Bills. Estimates of this extra return—the market risk premium—per 1 unit of *beta* risk run from 3 to 7%, depending on the relevant estimation period used. Another estimate of the overall cost of capital for the equity of a company with an arbitrary *beta* is the risk-free rate plus the risk premium per unit of *beta* multiplied by the estimated *beta* of the company's equity:

$$\text{Cost of equity capital} = \text{Risk free rate} + \text{beta} \times (\text{market risk premium.})$$

Betas are estimated from the slope of the historical plots of returns on the equity security in question against contemporary market returns on either a daily, weekly, or monthly basis. The *beta* of an average stock is 1. Errors in estimating *betas* are on the order of plus or minus $\frac{1}{2}$. Thus, the *beta*-based estimate of the cost of capital for a moderate risk stock at a time when the risk-free return is about 1% runs from 2.5% (1% plus a risk premium of 3% times a *beta* of 0.5) to 11.5% (1% plus a risk premium of 7% times a *beta* of 1.5). In practice, for all the elegance of the theory, this range of estimated values is far too wide to be useful. The simple qualitative process described above will usually do far better.

Earnings Power Value, Equity Value, and Enterprise Value

Once we have estimated an earnings power, including appropriate corrections for accounting depreciation and other adjustments, and a cost of capital, we can calculate the Earnings Power Value (EPV). This is an earnings-based valuation of a company's basic operations as described in its financial statements. However, there is a final set of adjustments that must be made before the EPV can properly be compared to a company's asset value and its market price. Some of these adjustments relate to earnings. As we have noted, there may be peripheral, unconsolidated operations that are not included in operating income. Other adjustments are found on the balance sheet. There are valuable, non-operating assets that could be disposed of without impairing operations, such as excess cash, marketable securities, and non-core real property holdings. On the liability side, there are obligations that a company has incurred that may exist independently of its operating business and that do not involve explicit interest payments. Such legacy liabilities include unfunded pension obligations, unfunded retiree medical benefits, litigation cost overhangs, environmental remediation responsibilities, and similar balance sheet entries.

There are a few key distinctions we need to keep in mind as we proceed; between earnings power value and asset value; between the enterprise as a whole and the equity portion of it; between intrinsic value and market price. This last distinction is at the core of value investing, but it needs increased attention here because the term "enterprise value" can be used to refer to both an intrinsic value and a market price.

We will start with the distinction between an equity approach and an enterprise value approach. Table 5.3 summarizes the key points of difference, which we discuss in more detail below.

Table 5.3 Distinctions between equity value and enterprise value.

	Equity	Enterprise
What you buy and pay	All the equity at market price	All the equity at market price plus all the debt at either book value or market price; pay off unfunded liabilities
What you sell	Nothing	Assets not essential for the business, including excess cash, property, and non-core business operations. This sale lowers the cost of buying the business.
What you own	Everything	Everything minus what you have sold. <i>You are left with the core business enterprise.</i>
What you owe	Debt and debt-like obligations (not including spontaneous liabilities)	Nothing (not including spontaneous liabilities)
What you earn	Net income after interest and tax payments including income from non-core businesses	Net operating income from core business after tax

The earnings power value—the earnings power divided by a company’s cost of capital—represents the value of the company’s ongoing core operations. It differs from the value of the company’s equity in two important ways. First, the company will often own assets that have significant value but are not required for sustaining ongoing operations. The value of these extraneous assets must be added to the core enterprise value in order to arrive at the equity value of a company. Just like the earnings of the enterprise, these assets belong to the firm’s shareholders. Second, there are claims on the firm’s operations and assets that must be paid before the shareholders receive their due. Outstanding debt is the most obvious of these obligations, but there are other debt-like liabilities that are not intrinsic to the firm’s operations, which also must be satisfied. These include those pension obligations and other legacy liabilities, such as environmental charges and other legal settlements. The total of these claims, which we will refer to simply as debt, must be subtracted from the sum of core earnings power value plus extraneous assets to arrive at the value of the shareholder’s stake in the company. Spontaneous liabilities

such as accounts payable and accrued taxes arise in the normal course of business operations and need not be repaid (on average) as long as a business sustains its current level of operations. And they do not usually require explicit interest payments. These spontaneous liabilities should not therefore be subtracted from the enterprise value in arriving at an equity value.

To recap, the core earnings power value (EP/cost of capital) is an enterprise value. The value of a firm's equity is this enterprise value plus the value of extraneous assets minus debt obligations broadly defined (but not including spontaneous liabilities). By contrast, net asset values, which are calculated by adjustments to the book value of a firm's equity, are equity values from the start. Extraneous assets have been included and the accountant's estimate of debt obligations has been subtracted to arrive at net worth, which is shareholder equity.

Since we are primarily looking at equity investments in this book, it is natural to think of a margin of safety as the difference between the estimated value of a firm's equity (triangulating between earnings power and asset values) and the firm's market capitalization, which is the cost of buying the company's outstanding stock at its current market price.

But we could equally well look at a margin of safety in buying the firm's ongoing enterprise. The value of that enterprise is measured by its earnings power value (sustainable earnings divided by cost of capital) on the one hand and the net asset value associated with the core enterprise on the other. In this case, we must make adjustments to the traditionally calculated asset value of the equity. First we must subtract the value of extraneous assets since they do not support core operations. Second, we must add back the value of the debt liabilities since we are calculating the firm's full enterprise value, not just the part of that value that accrues to stockholders. As with the equity valuation, spontaneous liabilities should not be added back since they are intrinsic to the enterprise and do not require payments out of enterprise earnings. With these adjustments, we can triangulate between the earning power value and the asset value of the ongoing core enterprise to arrive at an intrinsic enterprise value.

The margin of safety in buying this core enterprise is the difference between its intrinsic value and what we would have to pay to acquire it in the financial marketplace, including the debt and deducting what we could get from selling off assets extraneous to the core operating business. The result of subtracting the value of the extraneous assets from a firm's market cap and then adding the value of the outstanding debt is usually referred to as the firm's enterprise market value. This figure can be subtracted from our estimated enterprise intrinsic value to arrive

at the margin of safety that applies to purchasing a firm's core ongoing operations.

We can consider investments as either equity or enterprise purchases provided we are careful to compare equity values to equity market caps and intrinsic enterprise values to market enterprise values—market cap minus extraneous assets plus debt. Even though the simplest way to proceed is to look at equity investments in terms of equity values, there are at least two advantages to taking an enterprise value approach.

When we come to address the question of growth, we emphasize that growth rates typically apply to a company's core operating businesses. Non-core assets, legacy liabilities, and debt are driven by very different factors for which business growth is likely to carry limited weight. In assessing the contribution of growth to value, it clarifies matters considerably to focus on the enterprise and not overall values.

In addition, where a company is highly leveraged, the extent to which its equity appears to be a bargain may be dangerously misleading. Suppose we estimate a company's basic EPV to be \$800 million. Assume it has \$50 million in cash and securities and debt of \$700 million, or \$650 million of net debt. The total equity value of the firm will be \$800 million minus \$650 million = \$150 million. If the market value of the firm's common stock is \$75 million, it looks as if we have found a bargain with a large margin of safety. We can buy the firm for 50% of its equity value. However, if the accuracy of our estimated enterprise value (EPV and AV combined) is plus or minus 15%, which is in practice a high level of precision, then the enterprise value may easily be as low as \$680 million (\$800 million – \$120 million). At this level, the value of the equity will be only \$30 million (\$680 million minus net debt of \$650 million), well below the \$75 million market price of the equity. In enterprise terms, we are paying \$725 million (\$75 million for the stock, \$700 million for the debt, less the \$50 million in cash we collect) for a company operations with a value of \$800 million, which leaves a margin of safety of less than 10%. Since value miscalculations are related to the size and uncertainty of the enterprise value, the enterprise margin of safety is usually a more appropriate basis for investment decisions than the equity margin of safety.

One last point should be stressed. Valuations may readily be calculated on either an enterprise or an equity basis. Assets values are naturally calculated on an equity basis, but can easily be adjusted by adding debt and legacy liabilities and subtracting cash, securities, and non-core assets to obtain an enterprise value. In contrast, earnings power values are fundamentally calculated on an enterprise basis, but can easily be adjusted

to obtain equity valuations by deducting interest payments. In either case, there must be consistency. Equity asset values must be compared to equity earnings power values and the costs of acquiring a firm's equity. Enterprise earnings power values must be compared to enterprise asset values and enterprise acquisition costs.⁷ This admonition is obvious, but even seasoned investors have been confused or ignored it altogether.

⁷ Enterprise values for some companies, chiefly financials, cannot be calculated easily since the definition of debt is not clear cut. Bank deposits, for example, are debt but they are an organic part of a bank's business, not peripheral to it. For these kinds of companies, equity values will be more meaningful, but the equity valuation techniques employed will often be customized for these industries. Investing in these stocks, even more than is normally the case, is an area best left to specialists.

Example Two: Magna International

Warren Buffett's classic advice—to be fearful when others are greedy and greedy when others are fearful—applied with special force in March 2009. Following the Lehman Brothers bankruptcy on September 15, 2008, global equity markets collapsed. Driven by fears of the second Great Depression, the S&P 500 index fell from 1400 in May 2008 to 683 on March 6, 2009. A bottom appeared to be nowhere in sight, although that 683 print turned out to be the nadir. An environment so redolent of fear has historically presented extraordinary opportunities to produce stellar returns by buying companies at very low prices. Taking maximum advantage of these opportunities means looking at the most damaged parts of the stock market for companies that are unlikely to go bankrupt or suffer severe impairment to their operations.

In early March 2009, those financial institutions that survived, often with government help, represented the scariest part of the stock market. But their safety and potential for survival was very difficult to assess. The second scariest part of the market was the automobile industry. In January, Chrysler had announced a potentially life-saving alliance with Italy's Fiat, but without major government support the deal could not be done, and without the deal Chrysler would have been forced to declare bankruptcy. On February 26, 2009, General Motors announced that it had lost \$30.9 billion in 2008, its cash reserves were down to \$14 billion, and without substantial government assistance it too would be forced into bankruptcy. These were not idle threats. On April 30, Chrysler filed for bankruptcy; in June General Motors followed suit.

Independent parts suppliers to the major auto manufacturers were in even worse shape. Delphi, the leading parts supplier that had been spun out of GM in 1999, had filed for bankruptcy in 2005. In May 2008, it had tried and failed to put together a plan to emerge from Chapter 11. Visteon, spun out of Ford in 2000, was delisted from the New York Stock Exchange in March 2009 as its stock price had fallen from \$0.07 to \$0.02. Visteon's UK subsidiary subsequently filed for bankruptcy on March 31, 2009, and Visteon itself followed on May 28. Lear Corporation, a major producer of automotive interiors and electrical systems since 1917, sought to save itself by spinning off its money-losing interior systems division in 2009. Lear had to pay the buyer to take it, but it was still near bankruptcy in March 2009. Ultimately, it filed for Chapter 11 protection in July. The imminent failure of their major customers—Chrysler, GM, and perhaps even Ford—threatened to deal a final deathblow to the already weak parts suppliers, who would be forced to write down the value of their accounts receivable and see their future business evaporate. To make things even chancier, many of these suppliers operated with large debt loads that heightened their vulnerability.

Magna International, the one parts supplier unburdened by leverage, had other concerns. Incorporated in Canada, it was controlled via a dual share class by its controversial founder Frank Stronach. Stronach held the majority of Magna's B shares, each of which had 500 votes compared to 1 vote each for the A shares. As a result, even though he owned only about 20% of the economic interest in Magna, Stronach had absolute control of the board selection process and hence the company itself. He exercised this control without restraint. In 2008, he was paid a relatively small salary from the company, but a consulting firm of which Stronach was the sole employee had a long-term contract providing management services to Magna International. In the 13 years from 1996 through 2008, Stronach received more than \$350 million from this arrangement, almost \$30 million per year.

That deal was not the only way that Stronach used his control of Magna International to enrich himself and support his personal interests. Passionately involved with thoroughbred horse racing, he was one of the largest racehorse owners in North America. In 1998 Magna, a car parts supplier, began to invest heavily in racetracks and auxiliary racetrack betting enterprises. These investments were organized into a separate subsidiary, Magna Entertainment Corporation (MEC). In 1999, MEC was spun off as a distinct entity and listed on US and Canadian stock exchanges, even though 90% of the stock remained in Magna International's hands.

As a separate entity, MEC consisted of the horse racing properties, a greyhound track, two golf courses, and additional real estate to help ensure its economic viability. By December 31, 1999, Magna had invested \$550 million in MEC. Profits were minimal. In 2000 and 2001, MEC had pre-tax earnings of \$5.4 million and \$2.4 million, respectively. In April 2002, MEC completed a secondary offering of 23 million shares, raising \$142 million in additional capital. In 2002, the company had a pre-tax loss of \$25.2 million, despite additional racetrack acquisitions for stock. At that point, Magna decided to place its residual 59% interest in MEC, together with its remaining portfolio of real estate (almost entirely leased to Magna International), into a new entity, MI Development (MID), which was spun-off to stockholders on September 2, 2003. Shareholdings and share classes in MID mirrored those of Magna International so that Stronach continued to have absolute control of the company.

MID's subsequent history was as troubled as that of MEC. Its MEC subsidiary continued to suffer large losses; from 2003 to 2007, they totaled \$412 million. These losses required substantial capital inflows in order to sustain MEC as an operating enterprise. From 2004 to 2008, contributions from MID came to \$341 million. Despite the additional cash, MEC filed for bankruptcy on March 5, 2009. The shareholders of MID were dissatisfied with these practices, and MID was the subject of significant shareholder activism and lawsuits. Ultimately, in an effort to repair some of the reputational damage, MID changed its name to Granite Properties. Overall, Stronach's interest in horse racing may have cost Magna's shareholders as much as \$1 billion.

A second troublesome Stronach intervention in the operations of Magna International was the installation of his daughter Belinda as CEO in February 2001. Belinda Stronach had attended one year of college and then left to begin working at Magna International. In 1988, at the age of 22, she became a director of the company. By 1997, she was a corporate vice president and head of its Divisia Operating Group. In 1998, she was promoted to executive vice president and within three years was CEO. She resigned from that position in January 2004 to pursue an interest in Canadian politics, where she had a brief but eye-catching career. She ran for the leadership of the Conservative Party against two experienced politicians and, having refused to join any of the public debates, finished a distant third. She later won election as a Conservative MP and then crossed the floor to the Liberal Party and provided the single vote margin that the Liberals needed to survive in a parliamentary vote of no-confidence. As a Liberal she received a junior ministerial position,

but her controversial social relationship with a senior Liberal minister was a factor that contributed to the demise of the Liberal government. As a result of that and health issues, Belinda Stronach left politics. By 2007, she was back at Magna International as a senior executive in charge of special projects, for which she received \$3.8 million as compensation in 2008.

One last aspect of Magna's situation in March 2009 merits attention. In 2007, Stronach had entered into a joint venture with Oleg Deripaska, one of the more controversial Russian oligarchs, to produce automobiles in Russia. As part of his contribution to this venture, Deripaska purchased shares in Magna International at slightly below prevailing market prices. In the subsequent financial crisis, commodity prices dropped sharply, including the price of aluminum. Deripaska's fortune was based on his control of Russia's leading aluminum producer. His core enterprise under pressure, Deripaska was forced to sell his recently accumulated Magna shares. This added selling pressure helped contribute to a decline in Magna's share price from the peak of \$97 per share in October 2007 to slightly under \$20 per share at the beginning of March 2009.

The Magna situation looked grim. It was operating in an industry under long-run economic pressure with two of its major customers facing imminent bankruptcy. It was controlled by a manager with a checkered history of using the company as his personal piggy bank. The broader financial climate was dominated by fear and uncertainty. Magna's stock price had fallen by roughly 80%. By conventional measures, Magna was cheap. At \$20 per share, the market value of Magna's equity was \$2.24 billion, just 30% of its equity book value. The dividend yield was 6.3%. Although the P/E ratio for 2008 was 31.5 times, that was due to an unusually large temporary tax bill and depressed 2008 earnings. With a more normal tax rate of 33% and still depressed earnings at the 2007 level, the P/E ratio would have been 3.5 times. Magna International in early March 2009 was potentially an extremely appealing investment opportunity. However, given the many associated problems—which are what created the opportunity—an investor would need to do a painstakingly careful analysis before committing any funds.

The first step in that analysis was to determine the extent to which Magna was subject to the risk of bankruptcy, the ultimate permanent impairment of equity capital. The December 31, 2008, balance sheet for Magna is presented in Table Ex2.1. It shows \$2,757 million in cash, \$194 million in long-term financial assets (auction rate notes with little default risk that had to be reclassified as long-term only because the auction market rate selling mechanism was temporarily impaired), and

Table Ex2.1 Magna International, Inc., balance sheet (millions).

December 31, 2008	
Cash equivalents	\$ 2,757
Accounts receivable	\$ 2,821
Inventories	\$ 1,647
Other current assets	\$ 126
Total current assets	\$ 7,351
Fixed assets (net)	\$ 3,701
Goodwill	\$ 1,160
Investments	\$ 194
Other assets ¹	\$ 783
Total assets	\$ 13,189
Current debt	\$ 1,066
Accounts payable	\$ 2,744
Accrued liabilities	\$ 1,283
Current liabilities	\$ 5,093
Long-term debt	\$ 143
Deferred taxes	\$ 136
Other liabilities ²	\$ 454
Total liabilities	\$ 5,826
Shareholder equity	\$ 7,363
Total	\$ 13,189

Source: Data from Magna Annual report 2008

\$1,209 million in debt. Most other auto parts suppliers were highly leveraged; Magna had \$1,742 million in net cash at the end of 2008. Cash flow from operations minus investment was \$315 million (\$1,054 million less \$739 million in investment). In prior recession years, cash flow from operations had been only slightly negative as decreases in investment to less than depreciation and amortization and reductions in working capital offset net losses. Unless conditions in the auto industry deteriorated severely for a very long time, Magna International was not going bankrupt.

The next question was the long-run future of North America and European auto industries (roughly half of Magnus business was in Europe) and of independent parts suppliers such as Magna. In the long run, it appeared highly unlikely that either the industry or the independent parts suppliers were going to disappear. In 1967, McKinsey

¹ Future tax assets \$182, LT receivables \$67, patents, licenses, \$54, prepaid tooling, \$230, wage buy down \$52, other \$198.

² Pensions, retiree med \$146, hedges \$89, termination payments \$88 (NPV), deferred revenue \$31.

had provided a report to the newly established United States Urban Mass Transportation Administration on the future of US transportation. McKinsey forecast that the future would involve “independently guided vehicles” operating on “shared-use rights-of-way”; translated from consultant-speak, it predicted there would be cars, trucks, and buses operating on roads. As late as 2009, McKinsey had been proven right. Nothing about transportation technology in the intervening years seemed to indicate that the advantages of “independently guided” vehicles and “shared-use rights-of-way” would vanish.

North American and European automobile production was protected by another significant long-term trend. Automation in manufacturing was rapidly reducing the labor input required in car production, which, in turn, reduced the benefits of manufacturing in low-cost labor markets outside Europe and North America. Reductions in transportation costs were much less dramatic. Seaborne transport already used very little labor, and the benefits of being close to end markets were, if anything, increasing. As a result, by the mid-2000s, high-end manufacturing production such as automobiles, but not manufacturing employment, was returning to developed economies from the less developed ones. This trend was likely to continue.

Finally, continued unionization and/or high wage levels at the big auto producers in Europe and North America meant that the cost advantages of independent parts manufacturers, like a Magna, were not going to erode. Having confidence that the immediate bankruptcy risk of Magna was low and its long-term economic future was not in serious jeopardy, an investor should be prepared to expend the effort to develop a detailed evaluation for Magna International.

The way to begin that valuation was by looking at Magna’s asset and earnings power values in early 2009. The value of growth was likely to be minimal. Magna’s customers were large automobile manufacturers—General Motors, Ford, Daimler-Benz, BMW—with large professional purchasing departments, not about to let themselves be captive to any particular parts supplier. Economies of scale were also likely to be relatively insignificant. There were many companies like Magna, each of which had multiple manufacturing facilities. Dominant competitors in any auto parts market were rare. Without barriers to entry, the value of growth for Magna, as for any company without a sustainable competitive advantage, was likely to be minimal. Historical returns on equity (see Table Ex2.4) confirm the reality of a competitive market.

Asset Value

In Chapter 4, we computed the reproduction value of assets of several fictitious firms on a line-by-line basis going down the asset side of the balance sheet. We then did the same for a company's liabilities. In those cases, which we constructed to be representative, the estimated reproduction value of many assets and liabilities differed only minimally from their book values. This applied to most liabilities and most current assets. In practice, therefore, it is usually more efficient to begin with the net book value of assets and then add or subtract amounts that reflect differences between reproduction and book values only for those assets and liabilities for which these differences are significant. For Magna, the important differences are (1) between the book value of goodwill and the reproduction values of the intangible assets giving rise to that goodwill, and (2) between the book value and reproduction costs of fixed assets.

The book value of equity for Magna at the end of 2008 was \$7,336 million (see Table Ex2.1). The various elements of fixed assets are presented in Table Ex2.2. As we did in the case with Hudson General, for land and buildings we called local assessors in the area where Magna's properties were located. They estimated that on average the properties had appreciated at least 25% in value since they were acquired (acquisition dates are roughly available from historical annual reports). Adding 25% to original costs yields estimated reproduction values for land and buildings of \$265 million and \$1,070 million, respectively. Construction in progress represents assets in the process of reproduction so that original and reproduction costs for them should be the same. Magna's machinery had an estimated average life of eight years. For a stable company, neither growing nor shrinking, machinery in place should be on average halfway through its useful life. For growing companies, it will be less than halfway, for shrinking companies more than halfway. Magna had been growing steadily, but we will make the conservative assumption that its machinery was halfway through its useful life and retained half of its original value. Next we recognize that the cost of acquiring new automotive manufacturing equipment had been declining over time at about 4.7% per year.³ Machinery that was four years old on average should have been replaceable at 20% below its original cost (a decline of 4.7% per year compounded for four years). Dividing the \$7,700 million dollars first in half and then by 1.2 yields a reproduction value for machinery

³ This figure comes from the US Commerce Department.

Table Ex2.2 Magna International, Inc., fixed assets values (millions).

	Original cost	Reproduction value (est.)
Asset		
Land	\$ 212	\$ 265
Buildings	\$ 856	\$ 1,070
Construction in progress	\$ 313	\$ 313
	\$ 1,381	\$ 1,648
Machinery	\$ 7,700	\$ 3,208
	\$ 9,081	\$ 4,856
Depreciation	\$ 5,380	\$ 0
Book value	\$ 3,701	\$ 4,856

Source: Data from Magna Annual Report 2008

of \$3,208 million. That brings the total estimated reproduction value of all of Magna's fixed assets to \$4,856 million (\$1,648 million for land, buildings, and CIP and \$3,208 million for machinery). This figure is \$1,115 above Magna's year-end fixed asset book value of \$3,701 million. In computing overall reproduction value for Magna's net assets at the end of 2008 in Table Ex2.3, we begin by adding this \$1,115 million to the equity book value of \$7,363 million.

The book value of intangibles on Magna's balance sheet at the end of 2008 was represented by the goodwill item of \$1,160 million (see Table Ex2.1). But this figure was the result of following accounting rules to value Magna's many past acquisitions. It was not based on a careful consideration of Magna's actual intangible assets. To clean this up, we will subtract this artificial figure from Magna's book value and replace it with a detailed valuation of Magna's genuine intangible assets. Magna had significant value for its intangibles in three areas—its book of annual business, its product portfolio, and its trained labor force. For other companies the value of an established organizational structure could be added to this list. But Magna had a highly decentralized organization that had arisen organically from its everyday activities. The expense associated with reproducing that organization was likely to be minimal.

The sustainable size of Magna's customer base, its book of business, was about \$24 billion. Revenue of \$26 billion in 2007 represented a cyclical peak; the subsequent drop to \$23.7 billion in 2008 was the beginnings of movement toward a cyclical trough. The measure of reproduction value per dollar of revenue is the cost of acquiring those customers in the most efficient way possible. One way is through an

Table Ex2.3 Magna International reproduction value of assets.

December 31, 2008	
	Value
Book value of equity	\$ 7,363
Adjustment for excess	
Reproduction value of fixed assets	\$ 1,155
Subtraction of accounting goodwill	\$ (1,160)
Reproduction value of product portfolio ($5 \times \$450$)	\$ 2,250
Reproduction value of customer business ($0.075 \times \$24,000$)	\$ 1,800
Reproduction value of trained labor force ($10 \times \$75$)	\$ 750
Total reproduction value of net assets (equity)	\$ 12,158
Less net cash	\$ (1,742)
Net reproduction value of net assets (enterprise)	\$ 10,416

Source: Data from Magna Annual report 2008

independent sales agency. These typically charged 5–10% of first year sales as commission for obtaining sales contracts from customers like Magna's. If we use a midpoint estimate of 7.5% in fees per dollar of sales, then Magna's \$24 billion book of business would have a reproduction cost of \$1.8 billion (7.5% of \$24 billion).

Magna's labor force of 75,000 in early 2009 was made up of very different kinds of workers. About one quarter were unionized. The net asset value of such workers to Magna, given their wage levels, was near zero. A second quarter were skilled technicians, engineers, and managers with average salaries of roughly \$60,000. Human resource recruiting firms typically received about one third of a year's salary for successfully providing these employees. At that rate, their reproduction cost value would have been roughly \$20,000 each (one third of \$60,000). The remaining half of the trained labor force would have an acquisition cost/value about halfway between that of the unionized workers and the skilled professionals or roughly \$10,000 each. The overall reproduction cost/value of Magna's labor force was 75,000 times the average acquisition costs per worker of \$10,000, or \$750 million.

Magna's product portfolio consisted of between 4,000 and 6,000 individual items manufactured for the major automobile companies.

Some of the cost of developing and preparing to produce these items was specifically reimbursed by the big automakers. In 2008, the value of these reimbursements for one year's new products was \$92 million. This appears to have represented about one fifth of the full cost of product design, manufacturing, development, and tooling. A product portfolio roughly five years or less represented a Magna investment of about \$2,250 million (\$450 times 5). This comes to about \$0.5 million for each of Magna's 5000 or so products, excluding assembly operations, not an unreasonably high figure.

Magna's intangibles had a total value of roughly \$4.8 billion (\$1.8 billion plus \$750 million plus \$2.25 billion), or \$3.7 billion above book value. These calculations are set out in Table Ex2.3. The total figure of \$3.7 billion of non-acquisition-related intangible assets represents about three years' worth of Magna's approximately \$1.2 billion of SGA. Given the size and complexity of the company's operations, this seems an appropriate level for the reproduction cost that an entrant would incur in order to reproduce these intangibles.

The resulting net asset value on an equity basis for Magna in early 2009 was about \$12.2 billion; the asset value of its ongoing operations was about \$10.4 billion with the remaining \$1.7 billion representing net cash.

Earnings Power Value

A 10-year operating history for Magna's automobile business is presented in Table Ex2.4. It includes two down cycle years for the industry, 2001 and 2008. In general, the period from 1999 through 2008 was a difficult one for the auto companies, especially in North America, where rising gas prices starting in 2003 undermined sales of relatively high margin SUVs and light trucks.

Average sustainable revenues for Magna in 2008 were probably between the \$26.0 billion of 2007 and the \$23.7 billion of 2008. We used \$24 billion in calculating the value of Magna's book of business, which allows for the effects of the recession once every eight or so years. Sustainable operating margins are more difficult to estimate for two reasons. First, there was a steady downward trend in margins from 7.8% in 2000 to 4.1% 2007 and then to 1.0% in the recession year of 2008. If we merely extrapolated this trend, we might estimate sustainable margins of between 2% and zero. Second, the whole period after 2003 was one of poor demand conditions for Magna and the

Table Ex2.4 Magna International, Inc., operating history (millions).

Dec 31 FY	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
Revenue	\$ 9,260	\$ 10,513	\$ 11,026	\$ 12,971	\$ 15,345	\$ 20,653	\$ 22,811	\$ 24,180	\$ 26,067	\$ 23,707
Operating income	\$ 666	\$ 810	\$ 856	\$ 888	\$ 1,011	\$ 1,137	\$ 940	\$ 765	\$ 1,079	\$ 247
Operating margin	7.2%	7.7%	7.8%	6.8%	6.6%	5.5%	4.1%	3.2%	4.1%	1.0%
Tax rate					36.1%	34.4%	31.0%	33.3%	33.0%	78.4%
Net Income	\$ 438	\$ 622	\$ 627	\$ 611	\$ 661	\$ 738	\$ 650	\$ 528	\$ 663	\$ 71
Net income margin	4.7%	5.9%	5.7%	4.7%	4.3%	3.6%	2.8%	2.2%	2.5%	0.3%
Dividend	\$ 90	\$ 103	\$ 108	\$ 133	\$ 163	\$ 164	\$ 167	\$ 163	\$ 131	\$ 140
Buybacks ⁴	\$ (275)	\$ (4)	\$ -	\$ (225)	\$ (58)	\$ (51)	\$ (21)	\$ (28)	\$ (226)	\$ 247
Dep. and amort.	\$ 332	\$ 372	\$ 399	\$ 428	\$ 505	\$ 598	\$ 711	\$ 790	\$ 872	\$ 873
CapEx	\$ 859	\$ 653	\$ 525	\$ 898	\$ 801	\$ 859	\$ 848	\$ 793	\$ 741	\$ 739
Equity	\$ 3,933	\$ 4,202	\$ 4,482	\$ 5,421	\$ 4,930	\$ 5,335	\$ 6,565	\$ 7,157	\$ 8,642	\$ 7,363
Return on equity	11.1%	14.8%	14.0%	11.3%	13.4%	13.8%	9.9%	7.4%	7.7%	1.0%
Shares (millions)	78.6	78.6	83.3	95.6	96.4	96.9	109.3	109.9	116.1	112.6

Source: Data from Magna Annual report 2008

⁴ Negative denotes shares sold.

automobile industry as a whole, reflecting a slow adaptation to rising energy prices. Simply taking the 10-year average operating margin of 5.4% may actually understate future margins as the industry adjusted to lower demand. At margins below 4%, it is not certain that returns on capital would be sufficient to attract the investment necessary to sustain the output of what we have already identified as an economically viable business. Returns on equity for Magna during this period, with operating margins below 4%, were consistently less than 10%, which would be an appropriate cost of capital in early 2009 for a cyclical industry such as autos and auto parts. We will use an initial sustainable operating margin estimate of 5% and then look at the reasonableness of the resulting earnings power value.

With a sustainable margin of 5% and revenues of \$24 billion, sustainable operating earnings would be \$1.2 billion, close to the operating earnings of 2007. An average tax rate of 33% lies between a US rate, including state and local taxes, of 39% and lower rates in Europe—where half of Magna’s operations were located—and Canada. Sustainable taxes of \$0.4 billion leave net operating profit after taxes of \$0.8 billion, which is our estimate of the earnings power of Magna’s operations.

The principal adjustment that must be made to this estimate involves depreciation. As we noted in computing Magna’s asset value, accounting depreciation overstates actual depreciation for two reasons. First for accounting purposes, building values are depreciated over time, typically at 2–3% of original cost per year. In practice, the value of these buildings, if properly maintained, actually increases. Thus, accounting depreciation overstates economic depreciation for these components of Magna’s fixed assets. Second, the cost of automotive production machinery between 2000 and 2008 was declining relatively rapidly. Economic depreciation for this equipment would be the cost of replacing it and maintaining an unimpaired capital stock. Falling machinery prices would make replacements lower than the original cost of the machines on which accounting depreciation was based. In this area as well, accounting depreciation would overstate actual depreciation.

The amount of excess depreciation can be roughly estimated by looking at the years 2007 and 2008. Over this period, Magna’s revenues declined slightly so that the fraction of capital expenditure to accommodate growth should have been minimal. If we assume that for these years all capital outlays were maintenance capital expense, then “true” depreciation averaged \$740 million (\$739 million in 2008 and \$741 million in 2007; see Table Ex2.4). The average accounting depreciation and amortization was about \$873 million (\$873 million in 2008 and \$872 million

2007). The difference of \$133 million is a useful estimate of the excess accounting depreciation that must be added to accounting earnings to calculate actual earnings.⁵ In 2008, this extra \$133 million in earnings was protected from taxation by the depreciation tax shield—it simply did not appear as part of taxable profits. However, as lower maintenance capital expense leads to lower subsequent depreciation, the tax shield will gradually disappear. Once the tax shield is gone, the excess depreciation part of profits will be fully taxed at the 33% rate. On average this portion of profits would be taxed at almost 10% (in present value terms). The average excess depreciation after tax was thus about \$120 million (133 million minus \$13 million average tax). These adjustments bring the earnings power of Magna's core business to \$920 million per year in early 2009 (\$800 million plus \$120 million in excess depreciation).

In early 2009, B-rated corporate bonds were trading at yields of roughly 7%. At the same time, returns required to attract investors to new venture capital funds were around 14%. Thus, the range of possible costs of capital for Magna lay between 7 and 14%. A relatively risky firm such as Magna, but with excess cash, would have an equity cost of capital equal to that of an average firm. Within the 7–14% range, the low-risk firm would have had an 8% cost of equity, an average risk firm would have had a 10% cost, and a high risk firm would have had a cost of 12–13%. Thus, an appropriate cost of capital for Magna from early 2009 going forward would have been about 10%. The earnings power of Magna's core business at that time was \$920 million (\$800 million plus \$120 in excess depreciation.) Capitalizing the \$920 million in earnings power at this rate (1 divided by 10% or 10 times) yields an earnings power value for Magna's core operating enterprise of \$9.2 billion. The earnings

⁵ An alternative approach to measuring the gap between maintenance CapEx and depreciation and amortization is to estimate an average difference between the two for the 10 years from 1999 through the end of 2008 and then extrapolate a 2008 adjustment from the average. From 1999 to 2008, total depreciation and amortization was \$5.87 billion. Total capital expense was \$7.7 billion. Sales grew by \$17.3 billion. The ratio of fixed assets to sales averaged about 16%. If each dollar of revenue growth required \$0.16 of capital investment, then the \$17.3 billion revenue increase would have required \$2.76 billion of growth capital expense (\$0.16 times \$17.3 billion). Subtracting this estimated growth investment from the total investment of \$7.7 billion leaves an estimated \$4.95 billion of maintenance capital expense. The difference between this measure of real depreciation and accounting depreciation of \$5.9 billion was \$920 million or an average of \$92 million per year. But by 2008, Magna was 50% larger than it was in an average year between 1999 and 2008. Adding this 50% growth to the \$92 million average level of excess depreciation yields an estimated 2008 excess depreciation level of \$138 million (\$92 million times 1.5). This figure is very close to the \$133 million estimate obtained using only the 2007 and 2008 data.

power value of Magna's equity includes this \$9.2 billion plus \$1.7 billion in net cash, a total of about \$10.9 billion.

Final Valuation

The earnings power value, asset value, and market price for Magna International in March 2009 are summarized in Table Ex2.5. The first point about these numbers is that the asset and earnings power value estimates are essentially equal, within 12% of one another. In terms of three possibilities for comparisons between asset value and earnings power value, this is a Case B (see Chapter 4): a firm without significant competitive advantages but with a capable management. The lack of a clear competitive advantage for Magna is also apparent from the return on equity levels in Table Ex2.4. The Ten-year average return on equity of 10.4% almost exactly equals our 10% estimated cost of equity capital for Magna. Qualitative considerations further argue against the existence of sustainable competitive advantages. First, as we noted, its customers were large auto companies with professional purchasing departments. They were unlikely to be captive to any individual supplier for very long. Second, Magna's technologies were well known and seasoned. Proprietary techniques and patents did not play a significant role in its operations. Third, given the large number of decentralized manufacturing, marketing, and other operational facilities, it is unlikely that the economies of scale were significant. Without these barriers to entry associated with sustainable competitive advantages, growth for Magna would not create value (see Chapter 8). Thus, for valuation purposes, we need not concern ourselves with future growth rates in demand and/or cost reducing technological change at the industry level.

The fact that earnings power value is so close to asset value is somewhat surprising, given the historically questionable behavior of

Table Ex2.5 Magna International valuations (millions).

March 2009	Equity value	Enterprise value
Asset value	\$ 12,158	\$ 10,416
Earnings power value	\$ 10,940	\$ 9,200
Market price	\$ 2,240	\$ 500

Source: Data from Magna Annual Report 2009

management. But it is greatly reassuring in two respects. First, the asset value of \$12.2 billion assures that in the long run, the negative trends in margins could not continue. An earnings power value significantly below \$12.2 billion would not provide a return on assets sufficient to elicit the investment necessary for the continued viability of the North American auto and auto parts industries. Since the qualitative economic factors that we considered at the beginning of our analysis of Magna argue strongly for their continuation, future operating margins must be at or near the 5% level that we assumed in calculating our earnings power value. The triangulation between asset and earnings power values strengthens confidence in our earnings projections, confidence that was necessary to make an investment in Magna in the face of the troubling circumstances surrounding the company in March 2009.

Second, despite the difficult industry conditions facing Magna between 1999 and 2008, the company's management achieved operating margins, provided you believe the accounting numbers, that produced a cost-of-capital return on assets. This was accomplished despite the misuse of funds for racetrack investments and the consequences of nepotism in executive appointments. The contrast with Hudson General here is instructive. Hudson General's management destroyed substantial value, as was evident from the large gap between asset value and continuing earnings power value. No such gap is apparent for Magna International.

Given these considerations, the margin of safety from the valuation analysis summarized in Table Ex2.5 is comforting. In equity terms, Magna in March 2009 was selling at just 20% of its roughly \$11.6 billion intrinsic value. In enterprise terms, after deducting the net cash of \$1.74 billion from Magna's \$2.24 billion market price, Magna's operating business was selling for about 5% of its \$9.9 billion intrinsic value. Unless we have overlooked some important negative factors, Magna, as might be expected under the then prevailing economic and market circumstances, was a highly attractive investment.

Potential Issues

The final step in our investment analysis is to consider problems, like the continuation of management's depredations at Hudson General, that might undermine the profitability of an investment in March 2009 in Magna International. Three potential issues are worth examining in detail. They are (1) the possibility that the auto industry was on a path

of terminal collapse, perhaps along with the rest of the global economy; (2) the potential that the costs of mismanagement were not being fully taken into account; and (3) the chance that the accounting numbers on which we relied for our valuations were fraudulent. We consider each of these possibilities in turn.

We addressed the issue of North American auto industry failure at the beginning of this analysis. Barring a global economic collapse, it was not going to happen. It is worth remembering that much of the world had begun to emerge from the Great Depression before the onset of World War II. Moreover, once the war ended, and contrary to the opinion of most contemporary economists, the Depression did not return. Modern industrial economies have proved to be very resilient. Since the end of WWII, the scope of active government support for private economic activity in times of stress has greatly increased, further reducing the already small chance of total economic meltdown. Complete economic collapse, and with it the complete collapse of the auto industry, does not seem to have been a realistic economic prospect facing Magna in early 2009.

The potential damage from self-serving management behavior also appeared to be limited. First, it is important to remember that the operational and balance sheet numbers on which we base our analysis were recorded after taking account of management expropriations. Thus, only if management abuses were to increase after March 2009 would our calculated valuations be too optimistic. And the recent trend was in the opposite direction, toward a lower cost of management misbehavior. The racetrack folly was largely resolved with the spin-off of MID in 2003. More recent transactions with MEC involved golf, not horseracing, which was a much less costly undertaking. In 2006, Magna bought two golf courses in Ontario, Canada, and Austria from MEC for a total of \$84 million including assumed debt of \$21 million. In 2007, Magna bought additional Austrian land for \$29 million from MEC. Finally in the first quarter of 2009, Magna arranged to purchase additional Austrian land from MEC for \$5.7 million. The trend in the cost of Magna's interactions with MEC was clearly downward.

The danger that significant company operations would come under the control of Belinda Stronach, which had never been high given the decentralized structure of divisional management, had essentially vanished by 2008. Payments to the Stronach family in salary and consulting fees seemed to be shrinking. Finally the threat from any Deripaska joint venture had largely receded due to his financial difficulties. Overall, the cost of family control was diminishing.

On the other hand, if Magna's accounting statements were fraudulent, our investment would be in genuine danger. Here, however, we were paradoxically protected by Frank Stronach's dodgy reputation. Magna's accounts in 2008 were audited by Ernst and Young, a Big Four accounting firm. In the wake of the Enron and WorldCom scandals in 2002–3, which led to the demise of Arthur Andersen, large accounting firms were not likely to give the benefit of any auditing doubt to someone like Stronach. One reason that WorldCom and Enron were able to escape careful scrutiny was that they were regarded as leading-edge innovative firms run by visionary managements. In the event of an auditing dispute, a management with even a slightly questionable history was not likely to be able to manipulate a well-established auditor. The danger of accepting management accounting misrepresentation was more apt to come from firms such as Amazon, Tesla, Ali Baba, Google, or Facebook than from Magna International.

The credibility of Magna's profit numbers nevertheless deserves careful scrutiny. One way is to examine financial statements over an extended period of five years or more. Short-run accounting manipulations are far easier to sustain than long-run misstatements. From 2004 through 2008, Magna reported cumulative net earnings of \$2,650 million (see Table Ex2.4). It also reported sales growth of \$8,359 million. The profit figure might look inflated, but tampering with the sales growth numbers would be difficult. Magna's customers, who were the counterparties for its reported revenues, were large US and European automobile manufacturers. In an audit, they would be contacted to verify reported sales, and they were unlikely to support any material revenue misrepresentations by Magna's management.

Of the \$2,650 million of reported profits, \$686 million were distributed to shareholders as dividends and net stock buybacks. These payments leave a clear paper trail about which there is little definitional ambiguity. The \$686 million of reported net distributions would have been almost impossible to fake. Next, from December 31, 2003, to December 31, 2008, Magna's audited balance sheets showed an increase in net cash of \$512 million, from \$1,230 million to \$1,742 million. Outside of Italy before the Parmalat scandal, in which Parmalat claimed to have \$5 billion invested in hedge funds that were non-existent, cash balances and debt obligations are essentially impossible to misrepresent. The transactions involved leave clear paper trails, and the auditor sends confirmation letters to the institutions holding the funds. Thus, of the total \$2,650 million in reported profits over the five years, \$1,198 million could reliably be accounted for by net cash accumulation and net

distributions to shareholders. The remaining \$1,452 million of earnings had to support revenue growth of \$8,359 million, an investment of roughly \$0.17 for each incremental dollar of sales.

This is a very low level of net investment, especially compared to long-run levels of Magna and industry operating margins of \$0.05 per dollar of revenue. If Magna were inflating its earnings by 10%, then true earnings would have been \$265 million lower than the reported five-year total of \$2,650 million. As a result, true investment would have only been \$1,187 million (\$1,452 million minus \$265 million). This would mean that Magna was investing only about \$0.14 per incremental dollar of sales, a remarkably productive use of capital. The actual 10-year figure was \$0.16. Similar calculations for Enron in 2002 would have shown investment of more than \$3 per \$1 of incremental revenue. For WorldCom, the number would have been essentially infinite, since WorldCom was reporting huge implied investment levels while revenues were falling. Given the implied efficiencies of capital allocation implicit in Magna's numbers, it is very unlikely that Magna's reported long-run profits had been exaggerated. Indeed it is clear that for all of Stronach's dubious behavior, Magna was a highly efficient capital allocator and operator. This is also evident from Magna's long-term record of solid growth and its ability to satisfy its demanding automobile manufacturing customers. The likelihood that Magna's reported results involved a significant degree of misrepresentation was very small.

As a result, an investment in Magna International equity at \$20 per share around March 1, 2009, looked like an attractive opportunity. In contrast, a DCF valuation based on continuing projected declines in Magna's income would not have identified the asset protection that effectively ruled out such a scenario. Thus, investors approaching Magna from this non-Graham-and-Dodd perspective generally missed the opportunity. Ten year later, at the end of February 2019, the share price was \$52 after a 4-for-1 split, an annual return on price appreciation of 26%.⁶ In retrospect, the Magna investment seems so obvious, given the asset protection and the deep margin of safety. But in retrospect, we are all geniuses. It takes courage to be greedy when others are fearful, and one path to courage is to have confidence that you are equipped with the analytical tools appropriate for the job.

⁶ Auto stocks as a whole experienced comparable increases, but almost all of these other automotive stocks were highly leveraged and therefore risky.

6

Growth

Asset value and earnings power value are the first two components of a complete valuation description. We now look at the third element, growth. It has generally been taken as axiomatic that the value approach, with its preference for tangible assets and current earnings power, discourages its adherents from investing in high technology companies and other “growth” stocks. Though there is some truth in this idea, especially as it applies to more traditional Graham and Dodd investors, it ignores some significant innovations in valuation that modern value investors have devised. Warren Buffett and a number of other prominent value investors favor the shares of “good” companies, by which they mean firms like Coca-Cola—companies that increase steadily in value—over those that are merely “cheap.” In this context, a good company—sometimes elevated to a “great” company—is one that can pay out cash to its investors even as it funds its growth. Today these companies, “franchise” businesses in Buffett’s terms, constitute a large and growing fraction of the market value of publicly traded securities. At the same time, hard asset-based businesses are becoming less important.

The trend toward franchise-based value is evident in the high profitability and sustained trillion dollar valuations of companies such as Microsoft, Google, and Apple. It is also clear from the way in which Walmart and McDonald’s have created hundreds of billions in market value for traditionally atomized service industries. Health care, now a major sector, was formerly represented in securities markets almost exclusively by the major pharmaceutical companies. Today significant health investment opportunities include health insurers, prescription drug fulfillment companies, hospital chains, specialized

treatment businesses, and a multitude of high-tech equipment and biotech opportunities. All of these non-pharma companies have franchise characteristics. At the same time, historically dominant franchises businesses—General Electric, IBM, General Motors—and whole industries such as newspapers have seen their values greatly diminished or even disappear. The ability to recognize franchises and to value growth accurately are essential and challenging skills required of modern Graham and Dodd investors.

For value investors as a group, the main historical reason for avoiding these “growth” companies is that they are rarely favorably priced. They are often the kinds of lottery-ticket, get-rich-quick stocks for which many investors repeatedly overpay. The evidence cited in Chapter 2 indicates that in almost all extended periods, in almost all countries, portfolios of “growth” stocks regularly underperform broad market indices. After years in which expectations drove prices beyond any reasonable estimate of underlying value, markets corrected. Some telling examples are Japanese stocks in the 1980s, technology and telecom stocks in the 1990s, and Chinese and other emerging market stocks more recently. By the time their prices declined and their valuations returned to earth, they had cost their enthusiastic shareholders substantial sums as well as vaporizing their unrealistic expectations. Value investors of any stripe need to be keenly aware of this pattern. The historical success of value investing has been based in large measure on the ability of value investors to avoid being caught up in these manic episodes. Growth stocks, chosen selectively, are not forbidden fruit for value investors, as Buffett and others have demonstrated, but they must be bought with great care and the skillful application of an appropriate valuation process.

The Traditional Graham and Dodd Approach to Growth Investing

Graham and Dodd recognized that the value of a franchise business is rooted in earnings power rather than assets. Being unwilling to pay extra for growth, they based stock purchase decisions on the margin of safety between the current earnings power value without growth and the market price of a company. Growth was treated as a qualitative factor that might justify a smaller margin of safety, but its calculated impact was never incorporated into quantitative valuations. In a world where investors have historically overpaid for growth stocks, this cautious approach has much to recommend it. Most forecasts of future growth

rates of company revenues, earnings, and thus value are systematically biased.

First, growth forecasts suffer from significant systematic biases. Short-term trend extrapolation is a staple of forecasting practice. Companies that have produced high recent growth rates are forecast to have high future growth rates; companies that have shown low or even negative growth rates are expected to continue on that path. But the realized growth rates for the formerly high flyers generally disappoint, while the slow growers exceed expectations. These forecasting biases have been widely reported in the academic and professional literature.¹ Yet most investment analysts persist in projecting the most recent past into the future. The traditional Graham and Dodd approach avoids these pitfalls by not relying on quantitative growth forecasts.

Second, growth stocks will be overpriced when estimated values depend on the growth of companies whose supposed competitive advantages are hard to identify or, if they exist at all, are likely to be short-lived. In these cases, the allure of growth overwhelms the fact that only franchise growth creates sustainable value. For example, many companies in some emerging markets show rapid earnings growth, but most of them do not benefit from sustainable competitive advantages. In emerging economies, consumer tastes evolve quickly, an unstable preference that undermines customer captivity. Also, especially in Asia, emerging market countries have high population density. Thus even in local service operations such as retail, new entrants can be viable at relatively low levels of market share. With fragile market share advantages and low customer captivity, competitive advantages, or moats, will be narrow and often insignificant. At the same time, there is usually a large roster of powerful companies in developed countries hungry for the growth opportunities offered in emerging markets. Emerging market companies that are export-oriented may also expand rapidly, but they are necessarily competing in large global markets that are highly competitive. The ability of companies in emerging markets to transform rapid growth into value creation is likely to be far less than is often forecast.

A similar reality affects markets for exciting new products and services, the emerging markets of product geography. The often touted “first mover advantage,” meaning entry into these markets in their early rapid growth phases, does not usually lead to sustainable

¹ The striking extent of these biases is documented, with deserved sarcasm, in Montier, James. (2010). *The Little Book of Behavioral Investing: How Not to Be Your Own Worst Enemy*, Hoboken: John Wiley and Sons.

long-term competitive advantage and high profitability. Rapidly changing technology is the enemy of customer captivity. Early adopters are constantly on the lookout for newer and better alternative products. Subsequent adopters who look at the range of products available are unattached and non-captive customers. During early rapid growth periods, most potential users are not yet customers. Their numbers are usually sufficiently large compared to existing users that in the race for economies of scale, later entrants are not materially disadvantaged. Finally, rapid technological change tends to undermine and hence minimize the advantages of proprietary technology. Firms that ultimately dominate these markets have generally been those that enter at a time when technology has stabilized and user numbers are a high enough proportion of the ultimate market to produce sustainable scale advantages, a development for which Google in the search engine market is a prime example. Judging whether this point has been reached and what ultimate value will look like is difficult even for expert industry observers. It is almost certainly beyond the capacity of most general investors.

A third problem that besets growth investing is that growth stocks are usually priced at very high multiples of current earnings or cash flows. For example, in late 1999, Microsoft traded at 70 times any reasonable estimate of current sustainable earnings. At that time, Microsoft paid no dividends. Any direct cash returns to investors, as opposed to potential capital gains, were being deferred into the far distant future. Microsoft's earnings were growing at more than 25% per year, an attractive rate but one that was clearly not going to last indefinitely. To establish a value for cash returns to investors, let us assume that Microsoft were to start distributing half its reported earnings to shareholders and could do so without impairing its growth rate. Assume further that the growth rate were to average 25% in the intermediate future, up to ten years out, and investors were to discount future cash returns at a cost of capital of 10% per year, a relatively low rate for a then high-risk tech stock as Microsoft. Under these assumptions, the value of cash received over the 10-year period from 2000 through 2009 would be worth roughly 10 times Microsoft's current earnings. This discounted dividend payout would represent about 15% of Microsoft then market price of 70 times earnings. The remaining 85% of the price paid for Microsoft in 1999 depended on payouts and performance in the years after 2009.

For a company operating in a rapidly changing economic and technological environment as Microsoft in 1999, accurately anticipating events in 2009 and beyond was difficult if not impossible. Such a growth

stock purchase in 1999 would have been, in Benjamin Graham's terms, a speculation, not an investment. Not surprisingly, it would not have worked out well. After the tech bubble burst in 2000, it took Microsoft stock 16 years to return to its price in 1999, although there were dividend distributions along the way. Similarly expensive growth stocks are usually poor investments, meaning they have returned less than the market overall. Given upwardly biased growth rate forecasts, the limited duration of many competitive advantages, and the high prices at which growth stocks sell when compared to current earnings or cash flow, it is hard to disagree with the traditional Graham and Dodd aversion to paying explicitly for growth.

Despite these difficulties, Warren Buffett, Charlie Munger, and several other modern value-oriented investors have earned consistently above market returns by investing in growing franchise businesses. They have demonstrated that a disciplined value approach to investing in growth opportunities can be successful. A traditional value investor will sell when a company's market price reaches or exceeds its value. If the value is static, the investor sells when the price has recovered from what is often a temporary sell-off or setback. This recovery approach defines a fixed horizon for earning a decent return. But if the value of the company increases continuously, it may be many years before a company's value and market price converge. A longer holding period means lower trading frequencies, lower transaction costs, and deferral of capital gains taxes for many years. As a consequence, while investing intelligently in growth stocks is difficult, it is an essential and worthwhile extension of the traditional Graham and Dodd approach to investing.

An investment based on a firm's prospects for growth must satisfy two requirements. First, the growth must create value, which is no sure thing. For most companies—more precisely for the shareholders of most companies—growth is at best a break-even proposition. As we noted briefly in Chapter 4, this sobering news applies equally to high-tech firms that expand by selling more superbly engineered tools or aesthetically elegant gadgets, to online retailers who multiply their offerings, and to restaurant franchisers that expand by opening more outlets. Growth creates value only when the investments required to support the growth earn more than the cost of capital. In the distinction we made in Chapter 3, in Figure 3.2, it is only Case C, where earnings power value significantly exceeds assets values, that growth will produce net distributable cash after investment to existing investors.

Second, not all growth—even growth that is worth something—can be appraised with enough precision to permit an accurate valuation.

Because value investors demand a margin of safety, they will buy growth only at a substantial discount from its estimated value, a discount large enough to make up for the greater uncertainty in valuation. The ideal price for growth is zero: pay in full for the current assets or earnings power and get the growth for free.

Despite these warnings, there are circumstances under which even disciplined value investors may pay something for growth and be rewarded provided they keep these two requirements squarely in view: (1) that the growth will in fact create value; (2) that there is a measurable margin of safety included in the price paid for it. Though the first rule is uncomplicated, it is not well understood by most investors. The second calls for a major readjustment in the Graham and Dodd approach to valuation. We will discuss the two requirements one at a time. In this chapter, we will identify the circumstances under which growth genuinely creates value, as distinguished from those in which either growth contributes nothing and can therefore be ignored in the investment decision or those in which growth actually destroys value and investment should be avoided entirely. In Chapter 8 we will develop an approach for calculating an appropriate margin of safety for growth stocks.

When Growth Creates Value

One of the advantages that stocks are thought to have in comparison with bonds—"fixed income securities"—is that they are a claim on a potentially growing stream of income. Within equity investing, the shares of a "growth company," one with appealing future prospects, typically command a higher price relative to current earning—a higher price-to-earnings (P/E) multiple—than those of a company that is not increasing its revenues or income. The reasoning is obvious. If the growth company can generate \$1 per share in earnings this year, \$1.20 next year, \$1.44 in the year after that, and continue at that rate for some time into the future, the shares of that company are more attractive than those in a company producing a level stream of \$1 per share in perpetuity.

But this view ignores a key element of growth: it comes at a price. Growth in virtually every instance requires additional investment, and that investment must be paid for. For the example above, where earnings grow at 20% per year, let us assume that that growth needs \$0.50 per share

Table 6.1 Growth with reinvestment

Year	1	2	3	4
Earnings	\$1.00	\$1.20	\$1.44	\$1.73
Re-invest	\$0.50	\$0.50	\$0.50	\$0.50
Distribute	\$0.50	\$0.70	\$0.96	\$1.23

each year for added plant capacity and working capital.² We will make a second assumption, that the additional investment is funded by retained earnings. That means that each year, the company can distribute \$0.50 less to shareholders than would be the case if it did not have to reinvest that money. So in the first year, of the \$1.00 in earnings, shareholders get a dividend of \$0.50; the other \$0.50 is retained and invested (see Table 6.1). The reinvestment continues each year, reducing the amount of earnings that can be distributed.

By contrast, a non-growing company that earns \$1.00 each year does not require new investment and can distribute the entire amount to shareholders. In this example, that \$1.00 per year is more than the investors in the growth company receive until year four.

The crucial question for determining the net value of growth for a company is the relationship between the amount of investment needed to pay for the growth compared with the additional amounts and timing of distributions that the growth may produce. It should be intuitively clear that if the growth is slow enough and the investment requirements high enough, the current costs will not be offset by future gains, and the net value of the growth will be zero or even negative. Identifying a break-even growth rate that is high enough to create value for investors when compared to the additional investment required seems at first glance to be a complicated question, calling for intricate calculations. However, if this question is posed correctly, the answer is immediately apparent. The growth rate is not the main issue. In examining this relationship, the cleanest approach is to concentrate on cash investments required to support growth. Focusing on attention-grabbing growth rates and the high future earnings they may generate demands a complicated analysis that most growth investors short change. The allure of riches is one of the reasons investors tend to overpay for growth.

² To keep the example simple, we will ignore the fact that as the company grows, the amount of reinvestment required will also grow.

A simple yet general example illustrates the point. Assume a company decides to spend \$100 million to support its plans for growth. The investment might consist of new plants, new products, distribution facilities, and sales offices, all acquired for the express purpose of generating additional revenues and increasing earnings. Or it might be spent on acquisitions of new businesses or competitors in existing markets. A third possibility is that the money goes into the additional working capital—accounts receivables and inventory minus accounts payables—necessary to support growing demand in existing product lines and markets. Whatever the use, some investment is required to fund growth: no new investment means no growth. Growth that requires no supporting investment is extremely rare.

The investors who provide the \$100 million need to be compensated for the use of their funds. This point is obvious when the money is raised from outside investors. If it comes in the form of debt, the creditors will receive interest and ultimately the repayment of principal. If it comes from the sale of additional stock, then the new equity investors will be rewarded with their share of the earnings of the overall enterprise. But even when the funds come entirely from retained earnings, they still represent an additional investment. The existing shareholders are in effect providing the capital by accepting management's decision to make the investment to fund the anticipated growth. A decent management will take this fact into account in evaluating the investment and require a return at least equal to what those shareholders would demand if they were making the investments voluntarily. No matter the source, the cost of capital is the same.³

In each case, the return that must be offered to attract the funds voluntarily is the cost of capital associated with the \$100 million investment in growth. That cost, in theory at least, depends on the level of risk inherent in the investment. For our simple example, we will assume that the required return is 10% and ignore for the moment the difference between debt and equity financing. Thus the cost of attracting \$100 million voluntarily from investors is \$10 million per year. This cost is the downside of growth, and if it is high enough, it can turn growth from a profitable into a losing proposition.

The upside is the future cash flows that the \$100 million will produce. Again, to keep things simple, we will assume that this cash flow

³ This discussion ignores tax-related distribution issues, but such considerations should not change this requirement significantly.

Table 6.2 Return on investment and value created

Investment capital	\$100M	\$100M	\$100M
Return on investment (%)	5%	10%	20%
Return on investment (\$)	\$5M	\$10M	\$20M
Cost of investment	\$10M	\$10M	\$10M
Net income created	(\$5M)	\$0	\$10M
Net value created	(\$50M)	\$0M	\$100M

occurs as a constant percentage throughout the future and unlimited life of the investment. We will also assume that this rate varies across companies; as with all investments, some will have higher returns than others. Our cost of capital is 10%, and the annual cost of the \$100 million investment is \$10 million (see Table 6.2).

For an investment that returns 5%, the \$5 million of additional earnings is only half of the \$10 million that must be paid to the investors who provided the \$100 million in capital. Growth at this rate of return generates an annual loss of \$5 million and destroys value for existing shareholders. If we value the annual \$5 million loss at 10%, that produces a total loss in value of \$50 million, or half the original \$100 million investment. The company will have growing sales, assets, and operating income. If the \$100 million comes predominantly from retained earnings, it will have increasing net income and earnings per share. Nevertheless, value has been unambiguously destroyed by the 5% return on investment.⁴

Next consider a company that earns 10% on the \$100 million and thus generates an additional \$10 million in distributable cash flow. All of this money goes to the investors who provided the \$100 million; there is no additional cash for the original owners of the firm. Despite the additional reported earnings, no value will be created, and none will be destroyed. Unlike the first example, where growth was destructive,

⁴ A common practice for companies that indulge in these kinds of value-destroying investments is to fund them with “low cost” debt financing. If the interest rate on the \$100 million debt is only 4%, then the investment appears to generate a net return of 1% (\$1 million) per year. This calculation is illusory. The cost of debt must always include the expected cost of future operational impairment from higher leverage in bad times. A company with a sound debt policy will take on debt up to the point where the overall debt cost just equals the cost of equity financing. At the margin, debt and equity financing should be equally costly. If a company has underused debt financing so that the total cost of debt is indeed below the cost of equity, the extra debt may be warranted. But the proceeds of that debt should be distributed to business owners, not squandered on value destroying “growth” projects.

growth that increases returns by precisely the cost of the new investment is neutral and therefore irrelevant. It represents the boundary between growth that destroys value and growth that creates it and helps us separate those investments that we do not want to consider from those we do.

Our third example is a company for which the \$100 million produces a still greater increase in sales and with higher profit margins. This project may return as much as 20% per year, which translates into \$20 million in extra earnings.⁵ Paying the \$10 million cost of the investment still leaves \$10 million left for the original owners of the company. The growth produced by this project will create substantial total value for the existing shareholders or other stakeholders. Even after the new investors have been paid \$10 million, there remains an additional \$10 million per year in cash flow. If that is valued at the 10% cost of capital, this project creates \$100 million in extra value on its \$100 million investment in growth, meaning \$100 million in added value. If \$100 million can be invested every year with the same results, then the value of growth is prodigious and long lasting, as the investment pays for itself and still leaves an additional \$10 million for every \$100 million invested. The key is the rate of return on the invested capital. High rates lead to value creation just as low or negative rates, such as 5%, lead to value destruction.

The answer to the question, “Is growth a good or bad thing for existing shareholders?” is that it all depends on whether the return on the growth investment is less than, just equal to, or more than the cost of capital required. Our three simple examples provide the entire set of possibilities. What does not matter is the size of the investment or the specific cost of capital. The same consideration applies to an investment of \$1,000 as to one of \$1 billion. It applies to a cost of capital of 4%, 10%, or 15%. It is the cost of that capital when compared to the return the investment generates that counts.⁶

Which Growth Is This?

How do we tell in advance whether the potential growth for this investment will equal, exceed, or even fall short of the cost of capital? Consider a firm that seeks to expand in markets or product lines where it has

⁵ We are using distributable cash flow and earnings interchangeably here. Properly measured, the two should be equal.

⁶ It is less obvious but can be demonstrated mathematically that the result applies to any pattern of returns, not just the constant perpetuity version we assumed in our examples.

no sustainable competitive advantages and therefore benefits from no barriers to entry. In its projections, the firm may anticipate a return on investment that will exceed its cost of capital—say a 15% return on capital that costs 10%. But other companies will be scouting for similar opportunities, and they will locate this one, especially if the first firm's investments turn out to be highly and visibly profitable. With no barriers to entry, these followers will join the party, driving down the initially high returns of the first firm. These returns will continue to fall until all the attractive profit opportunities have been eliminated. Sooner or later, and generally sooner, every competitor will earn only its cost of capital, at best. At that point, all the benefits of growth will vanish, and the brief phase of higher returns will have added precious little to the value of the company. More damaging still, this limited period will often be more than offset by a longer stretch in which overcapacity from all the new entrants will force returns down below the cost of capital. The lesson is clear: without competitive advantages and barriers to entry protecting the returns of the initial firm, growth will create little or no value.

Firms that seek to grow by entering a highly profitable market in which incumbents enjoy sustainable competitive advantages over potential entrants face even worse results. The high returns are a temptation; if the existing companies are flourishing, why not enter the market and enjoy a similar profitability? But the returns are high precisely because of the competitive advantages and barriers to entry that protect the incumbents from would-be entrants. A company trying to penetrate this market is on track to earn substandard returns or to fail in its attempt to establish a presence. A new, small, and aggressive entrant may be tolerated by larger and stronger incumbents, but its operations will be limited in scale. Since sustainable competitive advantages ultimately rest on economies of scale, it is difficult to see how the undersized firm can flourish. The likely result is that the entrant earns returns below its cost of capital. Any growth for a firm operating at a competitive disadvantage or one with low quality management is likely to destroy value.

The only growth that creates value by producing a return above the cost of capital is in markets where the firm enjoys an actual competitive advantage or has the potential to create one, typically markets it can dominate in the future. A sustainable competitive advantage produces current high returns, and the barriers to entry protect the firm from having them eroded by would-be entrants in the future. The three levels of returns from growth we described in Table 6.3—below, equal to, or above the cost of capital—correspond respectively to the three instances of firms operating with regard to competitive advantage: those who operate at a

Table 6.3 Returns and competitive advantage

	Returns relative to cost of capital	Competitive advantage	Earnings power value
Case A	below	others have it	below reproduction cost of assets
Case B	at	level playing field	at reproduction cost of assets
Case C	above	benefit from it	above reproduction cost of assets

competitive disadvantage; those competing on a level playing field; those who benefit from a competitive advantage. And these correspond to the three cases, A, B, and C, we identified in Chapter 3, in which earnings power falls below, equals, or exceeds the asset value of the enterprise.

Thus, the Graham and Dodd approach to valuation, when coupled with a thorough grasp of competitive advantage, allows the investor to extend the analysis from existing operations to growth in the future and provides a direct and comprehensive answer to the question, “When does growth create value?”

Understanding the competitive conditions that determine the value of growth provides insight into a host of growth-related issues:

1. The value of organic growth that is driven by increasing market demand as distinguished from growth that is actively pursued by the firm;
2. The consequences of other favorable developments, such as new technologies that reduce costs and lead to growth in earnings even if they do not directly increase demand;
3. The value of growth options, secondary opportunities that a company can pursue if warranted and avoid if unattractive;
4. The consequences of market shrinkage as opposed to expansion;
5. The differences between growth in core markets compared to growth in new markets.

We will examine each of these issues in turn.

Organic Growth

The most desirable kind of growth for any company is revenue growth that arises organically from market conditions as opposed to growth

in response to active investment initiatives. The investment involved is minimal. Additional capital consists chiefly of net working capital, largely additional inventory and accounts receivable offset by growth in accounts payable and accrued liabilities. Fixed investments in incremental production capacity tend to be small relative to the average cost of capacity. When Walmart's revenue increases because more customers, due to local population growth, are spending more per capital, thanks to rising income levels, the incremental returns are usually higher than those produced by new store openings. This is especially true for markets in which growing demand results in higher prices. Consultants and books on business strategy invariably stress the benefits of operating in high growth markets, rather than those that are stagnant or growing slowly.

However desirable and profitable, the benefits of organic growth and the value creation that ensues depend on the existence and sustainability of barriers to entry, meaning incumbent competitive advantages. Without these barriers, higher initial returns arising from organic growth will encourage existing firms to expand and draw new entrants into the market. The heightened competition will then drive down returns on invested capital for everyone. Entry and expansion will continue until returns no longer exceed the appropriate cost of capital for the market in question. They may persist even longer if there is an enthusiastic overreaction to the initial opportunities. The end result will not be value creation for the individual firm but rather a greater number of companies in the market and more capital employed. Since this capital earns its cost, there will be no value creation in the long run, despite the initial high rates of returns from organic market growth. Only for firms in markets protected by barriers to entry will organic growth lead to sustained value creation.

Favorable Economic Developments

In addition to high organic growth, there are other favorable economic developments that lead to higher returns. Two common examples are technologies that lower costs and innovations in marketing and distribution that increase operating efficiencies. Personal computers, tablets, and smart phones have all benefited from rapidly improving performance at declining costs. Again, the extent to which these developments create value for the firms that make them rather than their customers depends on the same factors that govern the value of organic growth: barriers to entry (sustainable competitive advantages). Without these barriers, the higher profitability that comes from lower costs, product improvements,

and more efficient distribution will merely attract new entrants and additional capacity, both of which will intensify competition and drive down profitability. As in the case of organic growth, entry will slow down only when returns fall to the cost of capital appropriate for the market in question. All of these improvements will not lead to sustainable value creation for the existing firms unless there are effective barriers to entry that limit the impact of new competition.

Growth Options

Much has been written in finance about the value to firms of entering markets with extensive new potential opportunities that are created by unanticipated developments. The basic thesis is that firms that spot these opportunities can enhance their returns by responding aggressively to them. Conversely, they can minimize the impact of unfavorable developments by responding defensively. Again, as we should expect by now, in markets not protected by barriers to entry, the value of these growth options is severely limited by the increased competition that is sure to follow a successful innovator. Take the case of a smart-phone manager who notices that certain combinations of features generate unusually high demand among particular consumers. For example, the ability to download and play music is strongly appealing to teenagers and young adults. The company adds that feature, raises the price of the phone to more than cover the additional cost, and sees sales rise rapidly among the targeted demographic groups with profits growing in line. If the company had not originally anticipated this kind of response, the serendipitous profits earned represent a classic growth option.

What happens next? We already know the answer. Without barriers to entry, competitors copy the innovating firm, add those features and perhaps a few additional ones, and market aggressively to the targeted groups. Competition as always drives down profitability, and new firms continue to arrive until returns have fallen below the cost of capital. This growth option then does not lead to sustainably higher earnings. If a new product feature fails, then the innovator eats the loss and competitors avoid the same mistake. In this case too, the growth option has no long-run value. Also, a market with attractive profit potential and extensive growth options, unless they are unrecognized, will draw more competitors into that market in the first place, not a positive setting for high returns on investment. Without barriers to entry, this negative effect should on average offset the benefits of the potential growth options.

Like other ostensibly favorable market factors, growth options generate significant economic value only in markets protected by barriers to entry.

Shrinking Markets

Markets shrink as well as grow. Creative destruction applies to markets as well as to firms. Tastes and technological changes induce consumers to spend their money on items that may not have existed five or ten years ago. Where are the great turntable manufacturers of the 1960s?⁷ The Betamax tape format fell victim to a supposedly inferior VHS standard; each is now a collector's item as are the smaller tape recorders that replaced both of them. So markets do shrink and even disappear. Consumers do not have limitless amounts of money to spend and will shift their purchases from one place to another. The digitalization of so many things has eliminated a number of physical products and many of the firms that produced them.

The consequences for firms operating in a shrinking market are in important ways the inverse of what occurs when markets are growing. It is the companies with competitive advantages in markets protected by barriers to entry that are hurt the most and experience the most value destruction. By contrast, firms in competitive markets with free entry suffer less from a decline in demand. Lower earnings in the future are offset by recoveries of capital, assuming firms are rational and willing to shrink or exit with dignity. Plants and equipment can be allowed to wear out or sold into other industries that might find them useful. Inventory can be reduced. If the capital recovered was previously earning around its costs, then the amount realized from the disposition and shrinkage of assets should roughly equal the value of the earnings that will no longer be generated by this investment. For example, if a company's earnings are reduced by \$2 million but it can recover \$20 million from the surplus assets, it has essentially realized the same 10% return it was making before.

Firms with competitive advantages are more hard hit by shrinking markets for two reasons. First, since they have been earning more than their cost of capital on investments, the capital recoveries from declines in demand are smaller relative to the income they had generated. The \$20 million recovered may have been earning \$4 million per year, for example; its recovery does not fully offset the value of that \$4 million in annual earnings. Second, sustainable competitive advantages are

⁷ There has been some revival of vinyl records and the equipment necessary to play them, but so far this is a niche market for audio-purists.

generally associated with economies of scale. Shrinkage leads to declining operating margins that are not compensated for by a reduction in capital employed. Fixed costs remain, and overall incremental declines are limited. When powerful monopolies with significant local economies of scale, such as local newspapers, see their markets shrink, the magnitude of the value destruction is far larger than it is for local restaurants and other service businesses that had operated on a level playing field and were not protected by barriers to entry. Also, there is a better market for the restaurant's equipment and leases than the newspaper's presses, which may have to be sold for scrap.

Expanding into New Markets

A company with a competitive advantage in its core market may be tempted to expand into markets that are not a part of that core. The geography may be different, the product itself may be novel, it may even be crossing industry boundaries. For some reason, the management thinks that it can bring with it the competitive advantages it enjoys in its current locational and product markets. This decision almost always turns out to be a mistake. It is in the new market that the firm must achieve a competitive advantage for value to be created. No matter how strong its competitive position in its core market may be, if the firm has no competitive advantages in the new markets into which it expands, the growth involved will not create value. IBM and Eastman Kodak both entered the copier market in the 1970s. Each had what they considered technology that would allow them to compete successfully with Xerox, the market leader. But these choices did not work out as anticipated. Xerox had a competitive advantage, at least at that time, and even these two technology giants could not break in.

When Walmart adds stores in its core markets, it does not need to increase advertising or recreate the distribution and management infrastructure already in place. It benefits from the barriers to entry erected by the regional economies of scale it already enjoys in these areas. Therefore, the stores, if they are properly located, should earn returns above the cost of the additional capital necessary to open and run them. The company can extend these advantages and profitability into adjacent markets that are within reach of its overhead infrastructure in advertising, distribution, and regional management. But the situation has been quite different when it has moved far afield and cannot rely on its existing infrastructure and captive customers. In these cases it is often an established local firm that enjoys the benefit of a competitive advantage. Growth in these

distant markets will almost surely destroy value, even before considering the difficulties of implementation and the need to overcome cultural differences. Walmart's actual experience of global expansion beyond adjacent markets in Mexico and Canada has confirmed this principle.

For growth to create value in new markets, the firm has to benefit from competitive advantages within these markets. The advantages it enjoys at home are irrelevant. They may be harmful in producing overconfidence and the idea that the capital it will need to invest is somehow cheap, having been provided by retained earnings in the core business where the return on investment is high. As we have shown, this calculation is an illusion.

Growth Opportunities for Value Creation

Most of the situations we have described about the profit potential from growth strategies are cautionary tales. Profitable growth is hard to pull off in part because it sends a powerful signal to other firms that returns above the cost of capital are being earned by someone, and they think, why not me? Competition erodes superior returns. So the most positive opportunity for growth related investments are those that are concentrated within core or adjacent markets where the firm's existing competitive advantages may allow it to prevent an onslaught of potential competitors and continue to earn returns above its cost of capital. In geographic terms, this involves filling in areas within core regions that a company already dominates and then expanding at the edges of those regions. This has been Walmart's strategy and that of other successful retail firms. It also describes the growth of Southwest Airlines, the most consistently profitable US carrier. Southwest began with a single hub at Love Field in Dallas, focusing initially on flights only within Texas. It then extended its reach to adjacent states and secondary airports that it could dominate.

Turning from geographic to product space, Microsoft serves as both a positive and a cautionary example. As it added essential office software—Excel, Word, PowerPoint—around its core operating system market, Microsoft was behaving much as Walmart did geographically. The result was a long period of consistent and large increases in profits and value. When Microsoft jumped into unrelated markets, such as game consoles, the results were far less successful. Walmart has done much less well in its ventures into Europe and Asia than it did extending from its home base in Arkansas.

The second most positive growth opportunities come from entering markets where there are no existing dominant competitors who have both captive customers and economies of scale advantages. An outside firm with a focused strategy that it executes well may succeed in dominating these markets, as Walmart came to dominate Arkansas and the mid-South in its early years or Microsoft controlled operating systems for personal computers, aided of course by its adoption by IBM. These became the companies with competitive advantages, protected by barriers to entry, and they earned returns above the cost of capital that did lead to value creation. Though it all seems so inevitable looking back, it is important to remember that there were other firms in those markets, each of which had, at least at the start, the same opportunities as Walmart and Microsoft. Firms seeking to dominate these virgin markets must be extremely disciplined—focus on one market at a time—and superb at execution.

Other than these two situations, the prospects for adding value through growth are at best limited and often negative. If the new market has no current barriers to entry and, given its size, structure, and other features, no likelihood that anyone will be able to create competitive advantages for itself, then the firm has at least the chance of doing well through superior management within a competitive environment. But if barriers to entry do exist and other firms are already taking advantage of them, under no circumstances should the firm try to enter. This warning seems so obvious that it should hardly need stating. Yet history has demonstrated repeatedly that the temptation to enter is strong and resisting it more difficult than it should be. The major firms in these markets are reporting high returns on invested capital, and many managers wonder why they too cannot rake in these profits and the rewards that accompany them. But these returns would not be sustainable without significant barriers to entry, so the newcomer is operating at a major disadvantage from the start. And should it somehow succeed in breaching the barriers and joining the inner circle of profitable firms, the implication is that those barriers are no longer intact and other firms will follow, leading to the competition that takes its toll on profit margins and returns on investment. In a sense, the fact of a successful entry should lead not to celebration but should raise the question, “If we can do it, why not others?” Markets with strong barriers to entry are like the classic Groucho Marx definition of a desirable club—one that would not admit him as a member.

A Last Cautionary Word on Paying for Growth

Any investor who “pays up” for growth by buying stocks selling for a high multiple of earnings or net assets in anticipation that future growth will create more than enough value to justify the price of the shares should remember what behavioral finance has learned about investing in these kinds of companies. First, they tend to be lottery-ticket stocks with get-rich-quick potential at impossibly long odds. Therefore, as a group, they are systematically overvalued. Second, when investors or security analysts forecast growth rates, they extrapolate recent experience into the future. For high growth rates, as we noted earlier, this is an invitation to disappointment. Third, the estimates of value created by growth are extremely sensitive to the growth rates themselves. The value is great when the growth rates are high but drops considerably as the growth rates decline. Even a small shortfall in rates has a major impact on value. Therefore, even after verifying that the anticipated growth will actually create value—that the market is protected by actual or potential barriers to entry, from which this firm will benefit—value investors must proceed with caution when they invest in these companies. Growth in value there may be, but its ultimate size is difficult to ascertain. As we indicated at the start of this chapter, growth investing in franchises businesses has become an inescapable part of a value approach, but it must be practiced with extreme care.

7

“Good” *Businesses*

Three rules summarize our discussion thus far of the implications of growth for valuation. First, for firms with asset values above earnings power value, generally firms with poor managements or those operating at a competitive disadvantage, growth is destructive. It erodes value and is yet another reason for replacing current management as rapidly as possible. Second, for firms with competent managements in competitive markets, that is, firms with asset values roughly equal to earnings power values, growth is at best irrelevant and at worst a distraction. It neither creates nor destroys value. For these firms, we need not consider growth at all. Third, only for firms protected from competition by sustainable competitive advantages, those that can earn above their cost of capital on growth investments, does growth create value. And even these “good” (from here on we will dispense with the quotation marks) businesses are likely to create value only when their growth is within or adjacent to protected markets. So the first question we must address in evaluating growth is whether a company meets these criteria: its earnings power value is above its asset value; it operates in a market in which it has sustainable competitive advantages; it currently earns above its cost of capital; and it is likely to make growth investments in current or adjacent markets to which its competitive advantages can be extended.

The term value investors use to describe such businesses, following Warren Buffett, is “franchises.” Because they will have current earnings power values well above the value of the assets, assets values will not be critical to determining what they are worth. The enhanced importance of earnings and the need to value growth means that we must apply a tailored valuation approach to franchise businesses. We can no longer

rely just on asset and earnings power values as a check on each other and as a test of whether we have a franchise business. This chapter will describe the process of identifying franchises. The recognition that franchise businesses are different from others and must be valued in their own way, not just using a one-size-fits-all DCF approach, is one of the critical insights of modern Graham and Dodd investing.

Investors use several indicators to identify good businesses. They look for high profit margins that return large amounts of cash to their owners even as they are growing rapidly. High margins may be high returns on sales—gross margins, operating margins, or net income margins—especially relative to competitors in the same industry. But ultimately, good businesses are ones that earn high returns on capital invested. There are alternative ways to measure returns on capital invested: (1) returns on equity (ROE, or net income divided by the book value of the equity); (2) return on capital (operating income divided by net debt plus equity invested); (3) after-tax returns on capital (net income plus net interest paid divided by net debt plus equity). Good businesses steadily earn returns of 15–20% or more by these measures, well above a typical 10% cost of capital. These returns should increase rather than erode over time. Good businesses are also ones that grow. Growing firms can have either high levels of investment per unit sales (asset heavy) or low levels of investment (asset light). The asset-heavy firms are desirable because they can deploy a lot of capital at high rates of return. The asset-light firms are capable of earning especially high rates of return because, requiring so little capital to support growth, they can distribute most of their profits. In both cases, the key to these sustainably high returns and the value of growth lies in the economic characteristics of franchise businesses, the qualitative factors that enable them to protect these returns from competitive erosion.

What Good Businesses Look Like

There is widespread agreement that good businesses in market economies are those that enjoy sustainable competitive advantages over other companies. Sustainable competitive advantages, if held by incumbent firms, create barriers to entry. Barriers to entry are what Warren Buffett and other value investors refer to as “moats.” Franchise businesses, in the language of value investing, are simply businesses whose operations are protected from competition by wide, long-lived moats. Thus sustainable incumbent competitive advantages, barriers

to entry, moats, and franchises are all terms that refer to the same underlying features of a business environment.

The only problematic term in these expressions that are more or less synonymous is the role of incumbency in qualifying relevant competitive advantages. In certain situations, incumbents operate at competitive disadvantages. For example, where technology is changing rapidly, a newcomer may be able to leapfrog an incumbent and set up operations with a later generation of equipment at lower cost. Situations such as these are inherently unstable and benefit no one, except consumers. Today’s entrant is tomorrow’s incumbent. If the technology continues to evolve, the entrant’s competitive advantage will itself be undermined by a newcomer whose kit is even more advanced. The result is that no company establishes sustainable barriers to entry that might protect it from relentless competition. There are no good businesses in a competitive environment with relentless change.

There may be widespread agreement that competitive advantages, barriers to entry, moats, and franchises are the essential economic factor for defining a good business. There is less agreement on what competitive advantages look like. One common view is that good businesses are synonymous with strong brand images. If customers perceive a brand as superior to competing products, it should be able to command a premium price. These premium prices should lead to the high profit margins characteristic of good business. A careful examination of the history of actual businesses suggests that things are not so simple. In Table 7.1, the left column contains businesses that have historically earned high profits. These are matched with high-profile brands in the right column that, for all the luster associated with their names, have not generally been nearly so profitable.

Individual consumers certainly do not expect to bask in the reflected glory of products like Colgate toothpaste, Tide detergent, Purina Dog

Table 7.1 Profits and brands.

Highly profitable businesses	Highly regarded brands
Coca-Cola	Rolls-Royce
Colgate Toothpaste	Mercedes-Benz
Purina Dog Chow	Sony, Hoover
Buffalo Evening News (historical)	NY Times, Wall Street Journal
Verizon	ATT
Federal National Title Insurance	Revlon
Walmart	Bloomingdales, Macy’s

Chow, the Buffalo Evening News, or a Walmart shirt. On the other hand, showing up in a Mercedes-Benz, a Rolls, or a Cadillac may turn heads and elevate the status of the owner, at least in some eyes. These are the iconic brands, widely used to connote quality and class. They have become metaphors: the “Rolls-Royce” of stoves, or the “Cadillac” of lawn-mowers. Nobody calls something the “Coca-Cola” of handbags, or even of vodkas. Sony is widely recognized for the quality of its products. The Hoover name was so widely recognized that in many parts of the world, it became synonymous with vacuum cleaners, the Kleenex of vacuums, we might say. Yet for all their iconic appeal, these brands and the companies that produce them have not been extraordinarily profitable, especially when compared with the less glamorous brands in the column on the left. Rolls-Royce and General Motors, the maker of Cadillac, have gone through bankruptcies. In the US cell phone market, it was not the former monopoly with the universally known brand, ATT, that won out, but its regional offspring and then competitors, Verizon and Southwestern Bell—which eventually bought ATT and took its name—that came out on top.¹ Local newspapers, like those owned by Gannett, have historically been more profitable than revered national names like *The New York Times* and *The Wall Street Journal*, even as the latter have dominated the competition for journalist awards and recognition. Whole industries, such as cosmetics, with high levels of advertising and brand awareness, are often far less profitable than no-name businesses like title insurance, in which FNF is a leading competitor. Brands simply do not tell anything like the whole story about what constitutes a sustainable competitive advantage and therefore a good business.

First mover advantage is a second frequently cited competitive advantage that looks even worse as an indicator of historical profitability. Table 7.2 arrays first movers with the companies that ultimately came to dominate their industries and achieve much higher profitability. In many cases, such as airlines (Pan Am and TWA), automobiles (Ford), and personal computers (Atari, Apple), pioneers have given way to competitive markets in which no one earns the returns we expect of a good business. In other instances, such as personal computer operating systems, ISPs (internet service providers), Internet search

¹ Before it bought AT&T, Southwestern Bell had already acquired Pacific Bell and a host of other telecoms, some of which were offshoots of the original AT&T, and changed its name to SBC.

Table 7.2 First Mover Advantages

	First movers	Dominant competitors
PC operating systems	CP/M	Microsoft
Search engines	Alta Vista, Yahoo, Inktomi	Google
Personal computers	Atari, Apple	none
ISPs	AOL, Earthlink	local cable companies
Automobiles	Ford	none
Airlines	Pan Am, TWA	none
Online auctions	eBay	eBay
Radio and TV networks	NBC	none

engines, and television networks, pioneers have been displaced by later, now dominant, competitors. In a handful of industries, notably online auctions, eBay, the first mover, has managed to remain the dominant competitor. First movers have been no better than companies with esteemed brands in achieving sustainable competitive advantages and the profitability that we expect from good businesses.

Other approaches, rooted more in the field of business strategy than investing, typically depend on many factors or long lists of potential competitive advantages to identify good businesses and appropriate business strategies. Probably the most notable example of this approach is Michael Porter’s 5 forces analysis, which looks not just at barriers to entry but also at intra-industry competition, supplier power, customer power, and substitute products.² The problem with these approaches is that they are unwieldy, difficult to apply in practice, and often lead to ambiguous conclusions that can be used to justify conflicting judgments of the strength of any particular competitive advantage. When the creator of this approach sought to identify good generic business strategies, the 5 forces tellingly played no role in his analysis.³ Since understanding the special characteristics of a franchise business is so central to modern Graham and Dodd valuation approaches, we would like a simple, transparent, and accurate approach to identify the competitive advantages that distinguish good franchise businesses from the others. We will begin with an example that seeks to classify the mechanisms of competition and thus the ultimate factors that protect company returns from its consequences.

² Porter, Michael. (1980). *Competitive Strategy*, Free Press.

³ In Chapter 2, the author describes generic good strategies as either niche or low cost; there is no reference here to the 5 forces enumerated in Chapter 1.

Toasted Earnings

Consider a hypothetical company—we will call it Top Toaster because toasters are what it makes—that has been particularly successful. For the last five years it has earned around \$20 million each year. To keep things simple, we will state that there is no substantial difference between reported net income and the amount of earnings it can distribute to shareholders. Investors in the company are willing to accept a 10% annual return on their investment. Applying the EPV equation we developed in Chapter 5, the EPV of the toaster company is $\text{Earnings} \times 1/\text{cost of capital}$ (\$20 million/10%), or \$200 million. Assume next that the company's asset value is \$80 million. This figure includes all the tangible assets—cash; accounts receivable; inventory; and plant, property, and equipment, each adjusted to reflect any differences between the figures on the company's books and the actual money a competitor would need to spend to reproduce them—and the intangibles as well, including consumer recognition and reputation, product design, production know-how, worker training, distribution channel development, whose reproduction also requires expense. There is a \$120 million discrepancy here between the asset value and the EPV. And if the stock market should value the company at \$300 million, then the gap above the asset value would be wider still.

Unfortunately for Top Toaster, the EPV of \$200 million and the market value of \$300 million will be a green flag to potential competitors. Some bright entrepreneur, perhaps someone already in the small appliance business or an executive at another toaster company with access to investment capital and experience with retail outlets, will recognize that they can produce this income stream worth \$200 million for an investment of only \$80 million. Or, to put it differently, they can earn \$20 million a year on an \$80 million investment, a 25% return that almost qualifies as exorbitant.

The entrepreneur enters the toaster business by opening a new plant, expanding the capacity of an existing one, or finding a contractor to produce the toasters. He or she finds or buys an acceptable design, develops attractive packaging, and hires experienced sales agents to hawk the stuff. Assume for the moment that toasters are a commodity-like product, meaning that they are more or less interchangeable with one another and are selected entirely on the basis of price. Increased competition means more toasters on the market, and more toasters means lower toaster prices. As toaster prices fall, the profits of all toaster companies fall in tandem with one another. As the earnings of Top Toaster start to

shrink, so does its earning power. After the first competitive wave hits, assume that earnings decline to \$16 million.

Unfortunately, the waves continue to break. Earnings of \$16 million, with a desired return of 10%, value the company at \$160 million. The assets still cost \$80 million to replicate. That leaves an \$80 million gap, sufficient inducement for a second entrepreneur to start stamping out toasters. In fact, only after the gap disappears and the EPV falls to \$80 million will new entrants—or incumbents expanding their production—stop their invasion of Top Toaster’s inviting financial territory. The process ends, in other words, when there are so many indistinguishable toasters on the market that prices and profits have fallen to the level at which none of the suppliers earns more than the cost of capital that investors demand on their money.

For many toaster companies, life will be even nastier, shorter, and more brutish. The final industry price, at which any incentive to enter will have been eliminated, will be one at which only the most efficient toaster producers will survive. Inefficient incumbent producers, those who cannot meet this price, will leave the business. It is the entry of the lowest cost, most efficient producers that drives prices, not the threat of future inefficient ones, who will simply stay away. In this competitive process, it is the strongest competitors that matter, not the weaker ones.

The problem of declining profits in the face of increased price competition has challenged thousands of companies that cannot distinguish their goods or services from those of other players. It is a truth universally acknowledged that all sensible people abhor commodity businesses, the archetype of “bad” businesses. The standard advice for avoiding this fate is to differentiate your product or service from all the others. For Top Toaster, that means spending money on advertising, adding product features, or changing the design. All of these may work to temporarily insulate Top Toaster from the immediate pressure to cut its price.

Unfortunately, these moves do not solve the fundamental problem. The competitors are still there, aiming for a share of Top Toaster’s attractive market. Nothing prevents them from introducing competing designs, adding their own features, and matching Top Toaster’s spending on advertising. Inevitably they take away some of Top Toaster’s business. Even if its prices do not decline, Top Toaster sells fewer units, but fixed costs such as product development, package design, and advertising do not decrease proportionately. In the face of active competition, they may actually rise. At the same time, Top Toaster is not going to be able to raise prices, and its variable costs are not declining. If anything, intensified competition for resources will drive variable costs

up. Top Toaster finds itself staggered by two blows. It is selling fewer toasters, and, because its fixed costs are spread over fewer units, its profit margin on each one has contracted. Overall earnings decline, as in a commodity market, until EPV and asset value are equal.⁴ Differentiating the toaster is at best a short-term fix.

New entrepreneurs will keep showing up, or old competitors continue expanding. Toaster production will keep increasing, and products will proliferate until the profit opportunity motivating it disappears, until, that is, the \$80 million of assets produces an income stream of \$8 million (provided investors still will take a 10% return on their capital). At that point, the EPV and the reproduction cost of the assets are equal to one another, and there is no easy money to be made by entering or expanding in the toaster business.

It doesn't matter if the toaster is a commodity and sells only on price or a differentiated product and sells on features, providing Top Toaster with discretionary control of prices. So long as Top Toaster cannot separate itself from the competition, new entrants will appear until the EPV becomes equal to the asset value. This equality is not an accident. It is a fundamental economic condition, and it results from the corrosive influence of competition on prices and profit margins. The process of competition again displays itself by the ability of other firms to enter or expand in quest of a profit opportunity. The proximate causes of profit erosion for a firm with differentiated products like Top Toaster are not identical to what would be the case with pure commodities. Rising unit costs and declining unit sales are the villains, not declining prices. But the end result is the same. Only the most efficient toaster producers survive, and in the long run returns on capital are driven down to the cost of capital. Moreover, in both commodity and differentiated product competition, periods of temporary prosperity will attract enthusiastic entry. This will often be followed by extended periods of overcapacity due to the long-lived nature of the capacity the new entrants have added. So even though average results over the long term should just equal the cost of capital, there may be significant fluctuations around this level. Although this process is rough on the Top Toasters of the world, and on their investors, it is a benefit for consumers, meaning all those who had been financing the difference between the EPV and the asset value with the price they paid for toasters.

⁴ In a true commodity market, such as that for wheat, only price matters. In Top Toaster's market, it is possible to differentiate products. But without barriers to entry, profitability still shrinks until EPV equals asset value.

Automobiles: Value in the Nameplate?

Top Toaster and its competitors are figments of our imagination. But, in the absence of barriers that impede entry, the process through which the value of the firm in the long run comes to equal the value of the assets is real enough. And this process works even in situations where companies and products benefit from strong brand images. An example from recent history confirms the theory.

In the entire world there are few brand names as widely recognized as Mercedes-Benz. It is universally associated with a superior product, high in quality and prestige. The cars themselves are immediately distinguishable from other automobiles by the Mercedes-Benz star, which serves as both hood ornament and icon for the corporate commitment to excellence. By all rules of product differentiation, Mercedes-Benz ought to enjoy a strongly protected market position and, as a consequence, high profitability. Yet for the three years 1995–1997, before the company acquired Chrysler, Daimler’s pre-tax return on the identifiable assets in its automotive business averaged 7.2%. If there is a franchise here, it does not show up in the returns being generated; they are at or below any reasonable cost of capital for the automobile industry.⁵

We can also analyze this situation from a value perspective. Suppose that the cost of capital for Daimler-Benz is 10%. In 1997, the automotive division earned 3.5 billion DM, pre-tax. Based on these figures, its EPV would be 35 billion DM, again pre-tax. The reproduction value of its assets is equal, at least, to these amounts. The book value of capital was approximately 30 billion DM at the end of 1997. This excludes the cost of reproducing the engineering knowledge, images, dealer network, and organizational experience of the corporation. At a minimum, that would amount to an additional 10 billion DM, representing 3 years of R&D. A new entrant would gain precious little above this asset value of 40 billion DM by entering Mercedes-Benz’s market.

Mercedes-Benz is not unique. The history of the automobile market in the United States illustrates the same characteristics. General

⁵ The investment bank Goldman Sachs, in its fairness opinion regarding the merger of Daimler with Chrysler, estimated the cost of capital for comparable companies at between 10 and 12%. See DaimlerChrysler AG, Initial Tender Offer Statement, Form SC 14D1, September 24, 1998. In the five years ending in 2018, Daimler had an average ROE of almost 15%. But given the amount it spent on brand development and other intangibles, its equity is understated. Also, it had an average asset-to-equity ratio of over 4. For the same period, Apple had an average ROE of over 40%, with less leverage (an asset-to-equity ratio of 2.6).

Motors was a master of product differentiation. Its range of brands from Chevrolet at the low end, through Pontiac, Oldsmobile, Buick, and Cadillac was intended to segment the market based on images tailored to particular demographics and income levels. In the 1960s, this strategy produced annual returns on capital that averaged 46%, implying an EPV well in excess of asset values. European car manufacturers looked at this return as a measure of the profit opportunities in the American car market. They judged that GM enjoyed neither superior technology nor fundamental advantages in reaching customers nor, in a market becoming increasingly global, advantages of scale. Over the course of the 1960s and 1970s, Volkswagen, Mercedes, BMW, Jaguar, Rover, Citroën, Renault, Peugeot, Volvo, and Saab all expanded aggressively in the United States. General Motor's average return on capital fell to 28% for the 1970s. But the invasion did not stop. Starting in the 1970s, the Japanese entered the US market in force. By the 1980s, Japanese manufacturers had the luxury car market in their sights. Honda started with its Acura line, followed by Toyota with Lexus, and finally Nissan with Infiniti. By the 1990s, GM's return on capital had fallen to 6 to 7%, and in 2009, during the great recession, the company went bankrupt. In a competitive market, it was no longer an efficient producer. Globalization of the car market, including even the most luxurious and highly differentiated brands of the luxury car market, created irresistible competitive pressure. In practice, as well as in theory, the absence of barriers to entry meant that product differentiation alone could not sustain a good business.

The Nature of a Franchise

The critical element in both the Top Toaster fable and the history of the luxury car market is the process of entry and the ability of entrants to compete with the established firms. So long as newcomers can develop and distribute new products on an equal footing with incumbents—they have, for example, an equal ability to differentiate—then all products are effectively commodities. Companies that have above average earnings for sustained periods of time benefit from the inability of competitors, both actual and potential, to do what they do, and they are generally awarded high valuations in the market. None of this means that brands have no value. A strong brand is an asset like any other; its value is equal to its reproduction cost. However, if the value of a brand is equal to what it costs to create, then branding by itself is not the source of value.

Value is only created when the incumbent has abilities that new entrants cannot match. When a potential entrant sees Top Toaster’s earning \$20 million per year on assets of \$80 million, it must recognize that it will not be able to do as well. Top Toaster must enjoy a competitive advantage over would-be rivals. The newcomer will stay out of the market if it sees that it cannot compete on equal terms. Top Toaster’s competitive advantage acts as a barrier to entry and puts a brake on the profit-eroding process that occurs when entrants are able to compete on potentially equal terms. Another way to say the same things is that the continued existence of Top Toaster’s profitable franchise depends on the existence of competitive advantages it enjoys; these act as barriers to entry and deter competitors. They are the major sources, in a modern market economy, of any value that exceeds the cost of reproducing a firm’s assets.

Competitive Advantages

Contrary to popular management discourse, there are only a few types of competitive advantages. Examples of sustained competitive advantages in the business world are the exception rather than the rule. The simplest form is the advantage created by the government when it grants a license to one or several firms to engage in some kind of business, leaving everyone else excluded. Cable franchises, broadcast television stations, telephone companies, and electric utilities all enjoyed exclusive local franchises. Potential competitors were deterred by law. As technology has changed, so have the regulatory and licensing regimes under which these firms operate, and some formerly protected businesses have had to learn to compete. It is unlikely that all forms of exclusive, governmentally generated franchises will disappear.

However, governments in modern constitutional democracies are not generally committed to making business owners rich. Exclusive government franchises tend to be restricted to the few industries—electric utilities, cable TV, and local telephone service—where the underlying cost structures mean that the most efficient configuration is to have a single provider and the duplication of basic infrastructure is inherently wasteful. Economists describe these industries as “natural monopolies.” In exchange for awarding exclusive franchises, governments usually get to control prices and profits to make sure that the selected companies do not earn excessive returns. To the extent that governments are successful in their oversight, the firms will earn just their cost of capital on

both future and past investments. As a result, these regulated monopolies should have asset values equal to EPVs and should create no long-term value through growth. From a value perspective, therefore, we are going to be much more interested in franchises that arise outside the protection of government regulation than those that are subject to it.

Cost and Revenue Advantages

The list of non-government competitive advantages follows from the basic profit equation of any business: Revenues less costs equals profits. One key term in the equation is costs. Potential competitors of Top Toaster or Mercedes-Benz might be deterred because they are unable to meet the (low) costs of these firms. The only way such cost advantages are sustainable is if the incumbents possess production techniques or products that the entrants cannot match. Patents, for example, whether on the products themselves or the process of producing them, create one kind of cost-based competitive advantage. Know-how, sometimes described as the downward sloping learning curve, is another important advantage. Even as it gains experience, the entrant will always trail the incumbent in the necessary expertise it takes to make things efficiently. The test here is whether the required technology, including the human-based skill, is accessible to the entrant on the same terms as it is to the incumbent. Neither Top Toaster nor Mercedes-Benz has access to technology or other knowledge that is not equally available to their potential competitors.

Another possible cost advantage is access to cheap resources like labor and capital. This advantage is hardly ever found in practice. Most resources are mobile and plentifully available on a global basis, and there is nothing that inhibits entrants from acquiring them on the same terms as incumbents. Some firms with unionized labor or other constraints may be forced to operate with high resource costs, but the incumbent has to worry about its most efficient competitor, not its least. Firms that operate in countries with low wage costs, such as China, are often thought of as having a competitive advantage based on their ready access to cheap labor. But this advantage is illusory. Their strong competitors are in China and other low wage countries, and they all have access to similar labor pools.

Some companies make the argument that their advantage is access to cheap capital. This argument takes two forms. Large companies often refer to advantages associated with “deep pockets,” implying access to

large amounts of capital at costs their competitors cannot match. The competitors they have in mind are much smaller, the mom-and-pop businesses that play such a large role in American economic folklore. But these small firms are their weakest competitors, not the relevant strongest competitors who are invariably large firms with pockets equally deep. Companies that talk about deep pockets as a competitive advantage end up emptying those pockets much more frequently than they fill them.

A second common version of the cheap capital argument arises when companies have sold securities in favorable capital market conditions and their competitors are faced with the need to raise capital on less attractive terms. Consider a firm that issued long-term debt at an interest rate of 2%. Suppose that the yield on comparable long term debt had risen to 7%, the price competitors would have to pay to raise funds today. The fortunate company may feel that with a cost of capital of 2%, it has a competitive advantage over the companies paying 7%. Again, this supposed advantage is an illusion. Devoting the proceeds of the 2% bond issue to projects yielding 2%, when these funds can reliably earn 7% elsewhere, buying the competitor's bonds, for example, is gross stupidity, and stupidity is never a competitive advantage. The true cost of these funds is today's 7% opportunity costs, not the original 2% contract cost.

Genuine cost advantages related to inputs are usually based on specialized resources, such as a mine with particularly high-grade ore, a store with a prime location, or a special talent, like Beyoncé, signed at a favorable price. But three points need to be made about even these resource advantages. First, the mine, the store location, or the talent is often controlled by someone outside the company—a government, a real estate company, or the talent herself. At some point the terms will be renegotiated and the competitive advantage will be priced away. Specialized resource advantages are often short-term ones. Second, such specialized resources are fungible. There is no particular reason why they reside in a particular company. They can, therefore, be regarded as special assets rather than competitive advantages. Third, and related, these special resources are not barriers to entry unless they are truly irreplaceable and fully occupy a market. Unless an efficient copper mine is capable of providing so much supply that there is no room for other, higher cost entrants, it will not represent a barrier to entry. The same applies to store locations. New stores will be built either near the special location or in competing areas. These stores will attract customers, and despite its favorable location, the existing company will see some of its customers shop elsewhere. Similarly, Beyoncé will face competition as

new musical talent appears all the time. It makes more sense to think of special resources as assets rather than competitive advantages. An important distinction is that these kinds of assets only rarely generate growth-related value.

Proprietary technology is a more common source of cost advantage. But it also tends to be fairly limited both in the range of industries in which it is significant and in the extent to which it contributes to barriers to entry. In industries subject to rapid change, both patent protection and proprietary experience-based advantages tend to be short-lived. Technological advantages are usually specific to particular technologies. They disappear when a technology is replaced by a better alternative. In high-tech industries, this sort of generational change occurs frequently, and current advantages must be constantly renewed to make them durable. A company's ability to maintain its lead through many generations, like Intel, is by no means guaranteed. As a consequence, technological advantages in the high-tech sector are generally of short duration.

At the other end of the spectrum, in low-tech service businesses such as retailing, the pace of technological change is glacial. After a time, most innovations that were once leading edge become embedded in industry-wide practices, spreading from the developers or early adopters to all the firms that operate efficiently. It is primarily in those industries with intermediate rates of technological change that proprietary technology represents a significant competitive advantage.

Customer Demand Advantages

The other key term in the profit equation is revenues, which depend on customer demand. If the incumbent is to have a competitive advantage here, it must have superior access to those customers that are the source of that demand. For this incumbent demand advantage to persist, customers must in some way be captive to those incumbents. In an open and competitive economy, there are only a limited number of ways in which customer behavior leads to captivity. Habit, usually associated with high purchase frequency, is probably the most powerful. For a soda company to compete with Coca-Cola, it must induce Coca-Cola drinkers to stop drinking their favorite beverage. This is no easy task. Consumer studies and historical experience suggest that Coke drinkers are fiercely attached to their Cokes. By comparison, the attachment to Budweiser, another

leading beverage brand, is weaker. When diners go to Chinese, Japanese, or Mexican restaurants, they have no reluctance to order a beer from that country, but the chances of them asking for a local cola are slim.

For items that are not on the weekly shopping list, there are other processes through which customers are made captive to specific products or services. If the cost of searching for an alternative to the existing supplier is high, then new entrants have a difficult time attracting customers who are not actually dissatisfied with their current arrangements. Take the case of the residential insurance market. There are many dimensions to a policy in addition to its cost: coverage, deductibles, levels of service, exceptions, credit worthiness of the carrier, and several others. Unless they are strongly motivated, few homeowners are going to take the trouble to search for a replacement. Their aversion to change is reinforced in this example by the painful results that may ensue from an inappropriate choice; it hurts to have the wrong insurance carrier or policy after the disaster strikes.

Given these difficulties and dangers, a new entrant into this market will find it difficult to induce customers who have had decent experiences with their existing carriers even to begin searching for a replacement. Practically the only way the entrant can make any inroads is by offering to write the insurance at premiums substantially lower than the incumbents are charging. As long as incumbent pricing policies are neither inflexibly unresponsive to competition nor overly aggressive, this is almost always a losing strategy. Entry in situations like this stop well before the high profit levels of the incumbents are completely eliminated. In the model we have proposed here, the high search cost limits the arrival of new entrants, and the gap between the asset values and the earnings power values of the incumbents does not disappear.

High switching costs are the third and probably the most common source of customer captivity. If it costs money, time, and effort for a customer to switch from one supplier to another, incumbents have an advantage over entrants. For example, when a company changes software systems for payroll, benefits management, internal communications, funds transfer, or other important functions, they have to spend not only on the software but also on extensive retraining of their staff. That is bad enough; even worse, the error rate on the new installation still goes up. It is no wonder that there is a powerful bias toward keeping the current system. And if this is true for functions such as payroll, it is 10 times truer for the aptly named "mission critical" or even "enterprise critical" systems that manage order entry, purchasing, production,

inventory, shipping, billing, and accounts receivable. The corporate graveyard is filled with firms that bet the business on introducing a new, improved, integrated, and full featured system, and lost.

Switching costs are a powerful source of competitive advantage for companies such as Microsoft and, in a previous computer era, IBM. An entrant going after this business will not be playing on a level field. Because most users are already familiar with Microsoft systems, any successful entrant must overcome the costs to those customers of switching to an unfamiliar alternative. Additionally, as computer users increasingly communicate with each other, any single user incurs additional costs of switching to a new software supplier unless all others do so at the same time. It doesn't pay to have the best communication program in the world if there is no one else connected to it. Less dramatically, there is a cost to switching any time the new supplier has to master the particulars of the customer, client, or patient. This applies to new lawyers, banks, and service companies such as mechanics who are familiar with the existing systems, health insurance plans, and drugs for a doctor who must learn the risks and potentials of the new medication. It also applies to Google competitors, who must learn to process search queries for users as effectively as Google, which has had many years of individual experience to adapt its search procedures to the meet the needs of each customer.

Neither the fictional Top Toaster nor the very material Mercedes-Benz and its luxury car rivals are likely to benefit from significant customer captivity. Purchases are infrequent and not driven to any significant degree by habit.⁶ Search costs are low for Top Toaster. They may even be negative for luxury automobiles, where search involves visiting attractive showrooms, being catered to by attentive sales staff, and taking test drives in luxurious new cars, hardly an onerous process. Finally, once a decision has been made to replace an existing toaster or automobile, the switching costs are generally the same no matter which alternative a customer selects. In the language of marketing, it is brand loyalty, not brand image, that is critical. Neither Top Toaster nor Mercedes-Benz are likely to be brands with high loyalty and repeat purchases.

⁶ Car companies actively try to raise the frequency of purchase decisions by encouraging customers to acquire cars through short-term lease arrangements. This tactic has so far done relatively little to create the kind of habitual behavior we see in repeated purchases of soft drinks.

We contended earlier that although the term “barriers to entry” was frequently used by investment analysts, generally to justify the high price of a particular security, in a world where businesses fiercely compete with one another, real barriers come in very few forms. The clearest cut are governmental privileges, such as licenses, patents, copyrights, or other protections that keep potential competitors at a safe distance. The other barriers we have described stem from either cost (supply) or customer (demand) advantages. These consist chiefly of proprietary technology or customer captivity. Given that they address the two halves of the profit equation, it is natural to think that they exhaust useful categories of competitive advantage. They do not. Especially where we emphasize the sustainability of competitive advantages, there is a third important category that has to be identified and discussed in detail.

Economies of Scale and Sustainable Competitive Advantages

The downside to the advantages of customer captivity and proprietary technology is that they have limited lives. As we discussed above, technology advantages are inherently transitory. They erode in the presence of both rapid and slow change. Customer captivity is generally longer lived, but like technologies, customers eventually die. Sometimes this is literally true; at other times they simply move on as tastes change and buying habits mature. Children are captive customers with strong habitual preferences, but they don’t stay children forever. A company marketing to youngsters has to capture each generation of children anew. For a company to benefit from barriers to entry over an extended period of time, it must enjoy competitive advantages in the process for acquiring new generations of technology and of customers. This is what advantages of scale provide.

The benefits of scale advantages are straightforward. Current and potential competitors cannot match an incumbent’s performance because they cannot match its scale of operation. The advantages of scale have to be found either in lower costs or greater access to customers. For the advantages to be durable, the company has to be able to defend its position. If competitors can easily steal away customers, then the scale advantage evaporates. Finally, the advantages conferred by greater scale must be material; smaller competitors must operate at a significant disadvantage. If the advantages of scale are largely achieved

with a mere 2–3% market share, then it will be difficult to deny them to any potential competitors.

The cost side of economies of scale is related to the proportion that fixed costs account for in the entire cost structure. When fixed costs are a major part, the more units a company makes, the lower its average costs, provided that variable costs do not increase with scale, because the fixed cost burden is shared over more units. In our age, the archetypical example is off-the-shelf software, programs that are sold to thousands and millions of users. It costs a fortune to organize the operation, design the program, write the code, debug it, and get it tested by hundreds of users. It costs virtually nothing to publish one more box of CD-ROMs, even less to distribute it via Internet downloads or to have it available in the electronic cloud for a subscription fee. Under these conditions, a competitor who can acquire a majority share of the market will enjoy lower unit costs than other firms in the business.

On the demand side, economies of scale take the form of network effects. This refers to situations in which the more customers who use a particular product, the more valuable it becomes to each individual customer. Microsoft has been the beneficiary of these network effects, both for the Windows operating system and its business suite of application programs—Word, Excel, PowerPoint, and Outlook. If these are the common tools of white-collar workers in businesses and other organizations, then schools are going to train their students on them so that graduates will be ready when they transition to work. And, to complete the circle, businesses that hire these students will also use Microsoft products because the cost of training is minimized, compared to what it might be with more esoteric competitive products, no matter how technically superior they might be. Similarly, if most online individual sellers of stuff they want to unload, or even sell commercially, operate through eBay using PayPal as a preferred payment option, more buyers will be attracted because of the greater range of choices than elsewhere and will need to master only PayPal as a payment mechanism. The buyer experience is simplified and improved for auction-like purchases from a widely dispersed universe of sellers. The same conditions apply on the seller side; more customers, and only one payment system to master. The network effect helps both buyers and sellers deal more efficiently with larger markets. These same features apply even more powerfully in aptly named “social networks.” Facebook is the most obvious example, and would be even without a movie credit. It has reinforced its barriers to entry by buying potential competitors such as Instagram and WhatsApp before they could mount serious challenges.

The critical feature of these scale advantages is that they apply equally to existing and potentially new technologies and customers. In the case of proprietary technology, consider the competition between Intel and Advanced Micro Devices (AMD) to produce the next generation of microprocessors for personal computers. If Intel has the dominant market share today and expects to continue to dominate the microprocessor market in the future, it will have resources 8 to 10 times the size of AMD. If the two companies spend the same proportion of revenue on R&D, meaning the same level per each unit sold, Intel will be able to spend 8 to 10 times as much in pursuit of the next generation of microprocessors as AMD. AMD may sometimes get lucky and produce a microprocessor that temporarily outperforms Intel's current product line, but given the disparity in spending on R&D, Intel will soon have a new generation of products that outperforms AMD. In the long run, the race for new technology, given this resource disparity, will always be won by Intel. Intel will also have advantages in attracting new customers, meaning new PC manufacturers.

Intel has the choice of spending less than AMD as a percentage of sales, say only 5 times in total more than AMD, as has historically been the case, and with lower unit costs, charge PC manufacturers lower prices. Also, Intel benefits from significant network effects. Because the Intel/Microsoft standard dominates the PC market, applications programmers tend to write programs that work especially well in that environment. The result is a greater range and quality of applications programs for PCs with Intel chips. This numerical and operational superiority induces manufacturers to prefer Intel's chip architecture and wait for its next generation of processors, rather than switch to an AMD chip with currently superior technology. Because of this advantage with customers, the Microsoft/Intel system has remained dominant through 20-plus generations of technology over 30 years. This is the sustainability of economies of scale in action, even in a sector with robust technological change.⁷

⁷ We originally did this analysis almost two decades ago for the first edition of this book. In 1998, Intel had sales more than 10 times those of AMD, it spent 4.6 times as much on R&D, but its R&D as a percentage of revenue was only 44% of AMD's. In 2018, Intel's revenue was 10 times that of AMD, it spent 9 times as much on R&D, and its R&D as a percentage of revenue had risen to 86% of AMD's. Much of that spending was on chips for mobile computing and other non-PC markets, where it faced competition from other makers who had much stronger positions competing with Intel than did AMD. See the detailed discussion of Intel in Case 4.

Economies of scale are also sustainable in low-technology markets. Consider a retailer such as Walmart in a local market, such as Arkansas where it has 60% of the market share. Local distribution, warehouses, and trucks to carry goods to individual stores, all benefit from local economies of scale. With higher sales, Walmart warehouses will have greater throughput rates and lower unit costs than those of smaller competitors. Its trucks will have shorter routes for the same volume and thus lower unit delivery costs. Local advertising is a fixed cost that Walmart can spread over many more dollars of sales than its Arkansas rivals. Experienced Walmart managers who supervise numbers of local stores will spend far less time traveling and thus have more time to manage than the executives of competitors with a less dense network of stores. In an industry such as retailing, with relatively high turnover of low-skilled labor, management effectiveness and involvement are critical to efficient operations. The overall result is that Walmart has lower costs and often a better store experience than its rivals. Walmart has used these scale advantages to support an “everyday low price” strategy that its local competitors cannot match. The “everyday low prices” in turn give Walmart a significant advantage in attracting new customers, sustaining and even adding to its dominant market share over time. Again, economies of scale advantages, in this case without any network effects, lead to long-run sustainable advantages through multiple generations of customers.

For manufacturers, development costs are usually fixed by product line, making market share within the product line what counts for economies of scale. General Electric is a giant company, but it only benefits from economies of scale where it dominates the market for a particular product line. Competitive advantages grounded on regional and product line economies of scale are the ones we are most likely to encounter in practice. Only rarely do firms achieve national or global advantages, and these hardly ever spread across a range of product lines. IBM, to cite the most obvious example, could not extend its dominance in mainframe computers into the desktop era, even though it established the standards for the microcomputer. That advantage went to Microsoft, which itself has not been successful in dominating markets beyond operating systems and standard office applications. Those, as we know, have been enormous.

A second important aspect of economies of scale is implicit in both these examples. The scale involved is not overall size but the size relative to the particular markets in which companies, in our examples Intel and Walmart, operate. Intel has scale advantages in the market for PC microprocessors. It does not have comparable scale advantage in making signal

processing chips for smart phones, graphic chips for game consoles, or memory chips for personal computers. The scope of the relevant market is defined by the fixed costs involved. If Intel were to attempt to enter any of these markets, it would incur a whole new set of R&D costs that would have to be spread over its signal processing, graphics, or memory chip sales. With minimal current share in those markets, Intel would have no scale advantages. Network effects are similarly market specific. Applications developers for smart phones or game consoles are not writing programs for an Intel/Microsoft environment. Other firms dominate in these areas, and Intel would be operating at a competitive disadvantage. Intel's scale advantages are global in geographic reach thanks to the global nature of the PC market, but they are decidedly local in product space.

In the Walmart example, distribution, advertising, and regional management are specific to particular geographic areas. It is not Walmart's overall size that matters but its density—its local share—in each regional market, as its advertising blankets a region and its distribution and management systems serve the region efficiently.⁸ Expanding to countries where it has neither adjacent facilities, as it does in Mexico and Canada, nor dominant local share has been consistently unprofitable for Walmart. An identical situation applies to health maintenance organizations (HMOs) and other service businesses. An HMO that has enrolled 60% of New York Metropolitan Area patients will spread its local fixed costs of marketing and administration over a much larger customer base than its competitors. It will also benefit from powerful local network effects. With 60% of patients, virtually all local doctors and hospitals will find it necessary to join the HMO's network. With such a comprehensive population of affiliated doctors and hospitals, the HMO will offer a better range of treatment choices than its competitors, reinforcing its hold on patients. None of these scale advantages are affected by the HMO's market positions in Chicago, Miami, Dallas, or even Boston. For service businesses, it is local geographic scale—density—not global size that matters.

A third feature specific to economies of scale is that they cannot stand by themselves. To benefit from scale advantages, a company must be able to defend that advantage from competition. It is not just that

⁸ Many analysts claim that Walmart's absolute size provides economies of scale in its purchasing power. The evidence argues strongly against this claim. As Walmart has grown bigger but less dense, its profit margins have fallen, not risen. Walmart's international expansion outside North America has been notably unsuccessful. And a careful analysis of the available accounting numbers indicates that Walmart generally pays suppliers the same prices as its large competitors.

Intel has a higher market share today that allows it to outspend AMD or other potential entrants on R&D. It can afford to do that because it expects to have the same dominant scale in the future. Suppose that AMD expected to capture the whole market with a “better” chip. In that case, AMD would be happy chasing the same future scale as Intel and could, in theory, afford to spend the same amount as Intel on R&D. Similarly, Walmart earns advantages from the greater use of its advertising, distribution, and management infrastructure only if it can keep its customers in the face of competition. If a new entrant matching Walmart on prices could divide the customers equally with Walmart, then Walmart’s scale advantages in using its infrastructure of warehouses, delivery trucks, and experienced management, would disappear and the barriers to entry with them. Scale advantages have to be acquired; they also have to be defended.

The defense depends on some degree of customer captivity or at least customer inertia. The demand advantages can be small, but they must rule out the possibility of a new competitor gaining an equal share of the market. Suppose an incumbent company whose cost structure benefits from economies of scale has won a disproportionate share of the market. If it can retain its customers at a price, product quality, and marketing budget that just match those of potential competitors, it will be able to defend its dominant market share. And, thanks to the lower unit costs it achieves through its scale advantages, it will have higher margins and profitability than its rivals, which should provide the basis for continued market dominance. A strategy of matching whatever innovations entrants employ will always be a successful one. For network effects, there is a natural element of customer captivity since customers would have to move to a competitor in a coordinated way to undermine the incumbent. Nevertheless, where economies of scale are involved, consistent vigilance in protecting these advantages is always called for. The first priority of management is to protect market share, matching or anticipating rivals’ product features, advertising campaigns, or pricing initiatives. All this is essential because once market share starts to erode, underlying cost and network advantages shrink with it.

In our Intel and Walmart examples, Intel is protected by customer captivity. Its customers are leading PC manufacturers who have learned to depend on Intel to deliver leading-edge, quality chips in high volumes. Switching to any other untried suppliers, even one with the resources of an IBM, is a risky undertaking. The manufacturers have a natural preference for Intel that Intel consistently seeks to reinforce by its relentless pursuit of leading-edge technology and high quality, coupled with its

“Intel Inside” and subsequent campaigns to lock in end-use customers. Walmart benefits from a degree of habitual behavior that characterizes most shoppers, but it too seeks to reinforce this behavior with its “everyday low price” policy even at the expense of current profits.

The “Good Business” in Sum

We can now summarize what a “good” business enjoying a sustainable competitive advantage looks like. It must enjoy economies of scale advantages buttressed by customer captivity and, in some cases, also proprietary technology. Economies of scale are defined on a market by market basis. It is not size but size in a specific market that matters. One final example should help make this clear. Coca-Cola is a classic franchise business. It has earned extraordinary returns on capital for more than a century. It is widely regarded as a quintessential global brand, but in fact it makes essentially all its profits in a relatively small number of national markets that it dominates. Local bottling and distribution involve significant fixed costs that are specific to each local market served. Advertising and marketing to retail outlets are also locally bounded expenses. In the markets where Coca-Cola is dominant, it spreads these fixed costs over far more units than its competitors. As a result, its overall costs are relatively much lower than those of its competitors. It can charge less for its products and spend more on promotion than its competitors and still enjoy high profit margins. Given the deep attachment of Coca-Cola drinkers to the brand, even if Coke were to raise its prices, it would lose very little in market share. In these high profit national/local markets, Coca-Cola has powerful competitive advantages that create strong barriers to entry independently of any “secret formula.”

We can quantify the width of Coca-Cola’s moat. Start with the required market share an entrant would have to attain in order to earn anything like its cost of capital. This will depend on the size of Coca-Cola’s economies of scale advantages in each local market. We can get a rough estimate of that by looking at the profitability of beverage companies around the world as a function of their local market share. This varies with the locality and the nature of the beverage. For example, high-priced craft beers or products distributed jointly with brands like Coca-Cola may require less, but in general mass-market beverages like Coca-Cola must have at least a 20% local market share to be economically viable. The second element of the moat is customer captivity, which determines how hard it is to acquire that target market

share. Again, there are many variables involved. If an incumbent vigorously defends its position, then a market share goal will be harder to reach than if an incumbent firm is more accommodative and more concerned with its own current profit. We can look at historical market share stabilities for caffeinated soft drinks over a range of local markets and time periods. On average no more than 0.2% of market share changes hands in any given year. This is not surprising given the deep brand loyalty of Coca-Cola drinkers.

With market share changing 0.2% annually, it will take 100 years for a competitor to reach a 20% share. This is a wide moat indeed, and it is enhanced by Coke's generally aggressive defense of its territories. This defense takes two forms. First, Coke can increase advertising and improve distribution in ways that increase fixed costs for itself and, more importantly, for its competitors. Higher fixed costs in a stable market raise the target share that an entrant must reach to be viable. Second, it can implement pricing policies and loyalty programs that reinforce customer captivity and reduce the rate at which entrants can acquire market share.

The global market for luxury cars is at the other end of the spectrum in moat size. As part of a large global market, car companies that originally sold almost exclusively within their national markets can now be economically viable with as little as 2% global market share. Share stability is also much lower in this market than it is for colas. We have seen changes of 0.5 to 1% in a single year, so getting to 2% might only take two years. The moat in the luxury car markets, then, is narrow and bridgeable.

These calculations are based on historical evidence about competitive advantages and franchises businesses. They reveal that franchises are overwhelmingly niche businesses. Large global markets are difficult if not impossible to dominate. If, as in the global automobile market, entrants can be viable at a market share of 3% or less, it is going to be difficult to keep anyone out. As the history of the personal computer industry demonstrates, it has been the niche businesses that have prospered, not the highly visible, broadly based PC makers. IBM, Dell, Sony, Hewlett-Packard/Compaq, and even Apple with its distinctive operating system have ultimately seen limited profits or have exited the business altogether. In contrast, Microsoft with its operating system and adjacent office suite, Oracle with its database expanding to adjacent applied business software, Google with its search engine, Adobe in fonts and graphics, and Intel in microprocessors after it abandoned memory chips have all enjoyed high profitability and long-lived dominant positions in their niche markets. Highly profitable service businesses such as Walmart,

Verizon in cell phones, and United Healthcare in health insurance have all dominated particular geographic regions and been careful to expand at the edges of those regions.

Sustainable competitive advantages are grounded in economies of scale, a proportion of fixed costs, and/or network effects. In each local market, it will be the largest or dominant firm that possesses these advantages. This in fact is the primary characteristic of a franchise business—that it dominates its local market. The scale advantages will be protected and supplemented by customer captivity, proprietary technology, and sometimes government licenses or other restrictions on entry.⁹ These firms will produce high returns on investment over many years, during which they are able to sustain their dominant market positions. There will be high share stability within their markets and a history of failed attempts, or in some cases no attempts at all, at entry. The analysis here must be done on a market-by-market basis. Markets with no dominant competitor and/or a large and fluid population of companies are rarely markets with sustainable competitive advantages. For franchise businesses, value-creating growth is growth that occurs in markets in which the company's existing competitive advantage apply, usually within or adjacent to its existing dominant market position.

In assessing competitive advantages, it is usually clearest to begin by looking at the more qualitative dimensions of a franchise business—market dominance, failed entry, economies of scale, customer captivity, and proprietary technology. Only after these qualities have been identified is it worthwhile to examine the more quantitative measures of a franchise business—returns on investment, market share stability, and years of moat duration. Clarity rarely begins with complex numerical calculations.

Two final points are worth noting. First, as with Coke and Pepsi in the United States or Airbus and Boeing in the global airframe market, a dominant market position may be shared by more than one competitor. Under these conditions, the competitive dynamics between those firms is a critical determinant of profitability and the value of growth. The best management strategy is cooperation. Shared competitive advantages lie between situations of individual dominance on the one hand and competitive markets with no barriers to entry on the other. Cooperative managements, provided they steer clear of anti-trust law violations, will produce a level of total industry profitability resembling

⁹ For financial firms, informational advantages, which also tend to depend on local scale, are an important additional source of barriers to entry.

that of a single dominant company. Growth will also be value creating, provided each of them restrains the natural desire to grab it all. Failed cooperation and testosterone-driven competition will reproduce a competitive market environment that will undermine the value of any competitive advantages and lead to growth that does not create value.¹⁰

Second, as the previous point illustrates, management quality matters to valuation. Good management is certainly an advantage, but there is nothing built into competitive situations to assure that a company's superior talent will endure. As Warren Buffet has pointed out, when an industry with a bad reputation meets a management with a good reputation, it is generally the reputation of the industry that survives. Taking that wisdom to heart, we have focused on the industry's or, more precisely, on the market's reputation. We will address issues of management in a later chapter.

The symbols by which securities are identified in the public markets do not include labels that allow investors to differentiate companies with genuine franchises from those without. No visitor to the corporate headquarters needs to climb over barriers to entry, should they exist, in order to meet with company executives. And of the many advantages a firm proclaims for its flagship products, durable competitive advantages are generally not among them. Spotting franchises is a difficult skill, one that takes time and work to master. Also, it is not easily extensible. Value investors like to operate within their own circle of competence, where the knowledge they have accumulated about newspapers, insurance companies, cable stocks, bankrupt bonds, or other types of investment can be applied in a new but still familiar situation. When Benjamin Graham went scouring financial statements looking for his net-nets, it did not concern him that he may have known little about the industry in which he found his targets. All he was concerned with were asset values and a margin of safety sufficiently large to protect him from too frequent losses. But in a world where market prices already exceed asset values, and the margin of safety by that measure is negative, a contemporary value investor had better be able to identify and understand the sources of a company's franchise and the nature of its competitive advantages. Otherwise he or she is just another punter, taking a flier rather than making an investment.

¹⁰ The points made in this chapter about the crucial importance of understanding the competitive conditions that apply in the relevant market and a company's position vis-à-vis barriers to entry are discussed in greater detail in Greenwald, Bruce, and Kahn, Judd. (2005). *Competition Demystified: A Radically Simplified Approach to Business Strategy*, New York: Portfolio Books.

8

The Valuation of Franchise Stocks

Franchise businesses are the only ones for which growth creates significant long-term value. We have described the characteristics of these companies and explained why they alone benefit from growth. Identifying these companies is only half the task of making sound investment decisions; the other part is determining how much to pay for growth. The historical record confirms that investors as a group have substantially overpaid. Returns on portfolios of high growth, “glamour” stocks trading at rich valuations have generally been lower than the returns on a market portfolio.¹ Graham and Dodd investors purchasing growth stocks must be able reliably to distinguish those that are overpriced from those that offer genuine value. This chapter describes a value approach to growth stock investing that extends the Graham and Dodd philosophy and methodology without violating its basic principles.

Growth investors face a fundamental problem. The intrinsic values of growth companies cannot ordinarily be calculated accurately enough to be useful. The standard formula for determining the value of a constantly growing cash flow stream with a constant cost of capital applies a multiple to the initial level of that stream. In an elementary example, assume the initial cash flow level is \$100 million. The multiple is equal to one

¹ See Chapter 2.

divided by the difference between the constant cost of capital, say 10% per year, and the constant growth rate, say 5% per year. The difference between 10% and 5% is 5%. One divided by 5% is 20. That makes our constantly growing perpetual cash flow stream worth \$2 billion, 20 times \$100 million.

Though simple and clear, this formula is precarious. Even minimal changes in the underlying variables subject the calculation to wide swings. Suppose that instead of growing at 5% per year, our cash flows stream were to grow at only 4% per year. And instead of the cost of capital averaging 10% in the future, it turns out to be 11%. Then our multiple, one divided by the 7% difference between 11% and 4%, is reduced to 14, which leaves the value of our cash flow stream at \$1.4 billion. Alternatively, if the cost of capital were 9% and the growth rate 6%, then the valuation multiple would be 33 (one over 3%), and the value of our cash flow stream becomes \$3.3 billion. These slight changes produce an error range of \$1.9 billion (\$3.3 billion – \$1.4 billion) on an average value of \$2 billion, or roughly 100%.

In practice, it is difficult to avoid errors of at least 1% in estimating both long-term future growth rates and the long-term cost of capital. Value estimates for growth stocks based on this approach are routinely subject to margins of error of 100% or more. Identifying a comfortable margin of safety under these circumstances is a challenging exercise. This problem does not depend on either the particular numbers we have chosen or the constant growth and cost of capital assumptions. It is a consequence of the effects of compounding. We are valuing this company as a perpetuity, and it is the great number of years that gives compounding time to work. On the growth rate side, small differences in growth trajectories lead ultimately, via compounding, to large differences in cash flows far in the distant future. On the time value of money side, small differences in the cost of capital used to generate discounted future values lead to large differences in the values of far future cash flows. The interaction of these two factors for growth stocks, whose value depends significantly on those far distant future cash flows, is what leads estimates of current values to vary so markedly with small changes in growth rates and costs of capital.

To avoid this problem, we need another method for making investment decisions that does not depend on error prone estimates of intrinsic value. Instead, we will compare the estimated *returns* attainable from buying a growth security at its current market price to reasonable alternative *returns* available to us. This approach has two advantages. First, future returns can generally be estimated more accurately than future

values. Second, the connection between the underlying determinants of future returns—growth rates, discount factors, franchise erosion rates—and the returns themselves are more straightforward and transparent than their relatively complicated and non-linear relationship to intrinsic value. A major benefit of a Graham and Dodd approach to investing is the use of value measures that are clearly and directly related to underlying assumptions.

Despite that disclaimer, intrinsic values do affect future returns as we calculate them here. Intrinsic value, after all, is the Graham and Dodd term for the economic value of the business enterprise in contrast to the price the market is willing to pay for that enterprise at any moment. The ratio of intrinsic value to market price affects the value creation/growth return independently of future growth rates in earnings. Even though we cannot precisely compute an intrinsic value, we can start with a simplifying assumption that intrinsic value and market value are equal. This calculable benchmark return will understate actual return if intrinsic value is greater than market value—the traditional Graham and Dodd target. If intrinsic value is lower than market value—we have overpaid for the stock—the benchmark return will overstate the actual return. And we can tell which situation holds by comparing the benchmark return to the cost of capital. When the benchmark return exceeds the costs of capital, intrinsic value is greater than market value and the return created by growth is greater than the growth rate. If the benchmark return is below the cost of capital, the actual return is below the benchmark return.

Nevertheless, focusing on returns for growth stocks has its costs. By concentrating on returns, we surrender the ability to compare asset values with earnings power values, which practice, as we have seen, is an essential check on earnings-based valuations. But because growth stocks are by definition franchise stocks (see Chapter 7), the existence of moats protecting earnings means that asset values are only a secondary determinant of value. For companies such as Coca-Cola, calculating the asset value contributes relatively little to estimating overall intrinsic value.

A second problem involves sell decisions. Buy decisions are based on current market prices. Sell decisions require selecting a price at which to sell. When growth does not significantly affect value, as is the case for non-franchise businesses, intrinsic value can be estimated with reasonable accuracy. A sell decision based on a price that approaches or exceeds intrinsic value will be well-defined. For growth, meaning franchise, stocks, there is no solid estimate of intrinsic value to serve as grounds for a decision to sell. We could decide to sell at the price at which the calculated future return on that stock falls below some

predefined level, such as an average market return. However, estimated returns are often insensitive to changes in market prices. Thus, a wide range of market prices will yield calculated future returns that are broadly acceptable. A return-based approach, though it improves the buy decision for franchise/growth stocks, does not solve the sell decision problem. It is not surprising, therefore, that successful Graham and Dodd-oriented growth investors uniformly regard the sell decision—when and at what price—as their greatest challenge.

Estimating Growth Stock Returns

Though it may be impossible to estimate precisely the intrinsic values of growth stocks, it is possible to calculate with reasonable accuracy the *return* that can be earned by buying a stock at any given market price. A productive method is to break down the calculation into component parts, each of which, in the Graham and Dodd tradition, can be clearly related to important underlying assumptions about the future of the business under analysis. The three components we will highlight are (1) the cash return from current earnings power; (2) “organic” growth in earnings, and thus value, that results from market developments external to the company; and (3) “active” growth in value that arises from management’s allocation of retained capital above the amount required to support organic growth.

Dividing a company’s earning power by the market price of its stock² provides a sustainable earnings return. Multiplying that earnings return by an estimate of the fraction of earnings the company is expected to distribute in the future to shareholders through dividends and stock buy-backs yields an estimated cash return. The future growth rate of that cash return can be estimated from likely future organic growth in the company’s core markets and the likely future returns on the fraction of earnings reinvested. The overall growth rate in cash distributions and earnings, assuming a constant pay-out policy, should translate into a rate of growth in value and ultimately the market price of the stock.

² More precisely, if there is substantial debt and/or cash and other non-operating assets, we should divide net operating profits after tax (NOPAT) by the enterprise value of the firm (the market value of the equity plus the debt, broadly defined, less non-operating assets, including cash). See the discussion in Chapter 5 of enterprise versus equity valuations. Because growth applies largely to a company’s core enterprise, an enterprise value approach is generally better for evaluating growth stocks. However, for expositional simplicity, we will focus here on equity value.

The estimated total return from buying the stock will then be the sum of the estimated cash return plus the estimated growth rate in value. This figure can then be compared to likely average returns from investing in a market portfolio of stocks, which should be a relevant alternative for equity investors. A sensible investment decision can then be based on whether the margin of safety in returns between an estimated growth stock return and the overall market return is sufficient to compensate for the risks of a growth stock investment, especially the risk of future franchise erosion due to changing economic conditions, which we will refer to as franchise “fade.”³

1. Cash Return on Current Earnings Power

A simple example will illustrate the process involved. Consider a company we will call Consolidated General (CG) that dominates a particular product market. It has current sustainable revenues of \$3 billion and a sustainable net profit margin of 8% on sales. These figures produce a sustainable earnings power of \$240 million. It is entirely equity financed, holds no excess cash or other assets, and has a current market price of \$4 billion. The earnings return from buying all of CG at the current price is 6% (\$240 million divided by \$4 billion). Management determines the fraction of these earnings distributed to shareholders. Likely future distributions can be estimated from an explicitly articulated policy or from historical distribution practices. In the hypothetical case of CG, we will assume management has a declared objective of distributing 60% of earnings to shareholders, a target consistent with historical practice. This policy makes the estimated future cash return \$144 million per year, 60% of \$240 million, which amounts to a return of 3.6% (60% of 6%) on the \$4 billion invested. This 3.6% is the cash return from the current earnings power. The remaining \$96 million of earnings (2.4% of the return) will be retained and reinvested in company operations.

2. Organic Growth

Growth that creates value, for CG as well as any other firm, is growth that generates returns above the cost of capital on the funds invested to support that growth. Typically, this will mean only growth within the

³ This term was coined by Matthew McLennan of First Eagle Investment Management.

core markets in which CG enjoys sustainable competitive advantages, that is, markets within CG's moats. What we are calling organic growth has two sources. The first is growth in the company's core markets. This includes population growth (more customers), rising household income (more demand per customer), and sometimes the continuous exercise of pricing power. The second is cost reductions from improved productivity. These will raise profit margins in those markets where barriers to entry prevent any erosion in prices from intensified competition. Since these higher margins will produce an earnings increase above the growth in revenue, they will add to the rate of organic growth in value.

Future organic revenue growth in CG's core market can be estimated by looking at the historical trend of direct measures of organic growth. These will be measures of demand growth within its existing geographical footprint and product profile. They do not include the effects of expansion beyond these areas, as that is usually the result of an active investment decision. Same store sales growth for mature stores with stable product lines is one such measure for retail firms. Revenue growth for established product lines in existing markets is a more general measure. If the historical growth rates of these measures have been stable, then we can reasonably assume they will roughly continue. If growth has been accelerating, estimated organic growth may exceed the historical rate. If growth has been slowing, then estimated future organic will be less than the historical rate. In any case, it is the long-term rates that we focus on, not recent changes to them. For CG, we will assume that historical measures of organic growth have been stable for many years at roughly 3% per year.

An alternative approach is to estimate future organic growth in revenue based on the likely impacts of underlying trends in population, income, and customer behavior. The natural place to start is with overall growth in economic activity as measured by GDP. This macroeconomic growth rate should then be adjusted to account for the specific geographic, demographic, and product markets in which a company operates. For example, the Southeast and Southwest regions of the United States have been growing more rapidly than the Northeast and the Midwest, a trend that is likely to continue. In both the United States and globally, the incomes of affluent households have for many years been increasing 2–3% more rapidly than average incomes. Incomes of less well-off households have been growing around 2% less than the overall economies. In most developed economies, spending on services such as housing, medical care, education, travel, and entertainment has been increasing faster than spending on goods. For example, in 1970, an

average US household spent 23% of its income on food, 18% went for groceries, and 5% on restaurant meals. In 2010, spending on restaurant meals remained at 5% but only 9% went for groceries, half the figure for 1970.

A third alternative is to look at historical growth rates in demand for markets or products that are similar to those of our target company. For example, if we are attempting to forecast organic growth for a budget airline with a dominant market share on European routes, we can look at historical growth rates for similar operations in the US or Asia, adjusted for differences in regional GDP growth.

We will assume that CG sells an average mix of services and goods to an average mix of all households. In this case, organic growth should correspond roughly to expected future GDP growth of about 3% (in nominal terms). We will also assume that this 3% figure tracks well with CG's history of a stable 3% constant footprint revenue growth. In this case, both the firm's own history and projections driven by economic trends lead to an estimate annual future organic growth in CG's revenue of 3%.⁴

Because sustainable barriers to entry are ultimately rooted in economies of scale, organic revenue growth within core markets should lead to increased operating leverage and thus larger profit margins. Improved technology should also enhance margins. Organic growth in earnings, then, should exceed organic growth in revenues. This effect is usually small but should be detectable by examining historical margin growth within CG's core market. We will assume that core market earnings growth for CG has on average been about 0.5% above historical revenue growth. Adding this margin benefit to the projected revenue growth of 3% produces an estimate of annual organic growth in earnings, and thus in value, of about 3.5%.⁵ This organic growth in value is the second component, above the cash returns of 3.6% from current earnings, in the overall return to buying CG's equity at \$4 billion.

Organic growth does require some investment, usually in net working capital. An organic growth rate of 3% in CG's revenue of \$3 billion

⁴ If recent same facilities revenue growth is substantially above what the underlying economic trends suggest, then it is probably better to rely on the underlying economic trend figure. If same facility growth is significantly below what the trends suggest, then there may be a management issue that requires investigation. Poor management can be improved; bucking fundamental economic trends is more difficult.

⁵ If there is no trend to increased margins in the historical data, we should ignore the theoretical impact of higher operating leverage and technology and assume that future earnings growth will equal revenue growth.

produces additional sales of \$90 million in year one. We will assume that each additional dollar of revenue requires an extra \$.15 in accounts receivable and inventory and no significant fixed investment; this figure is partially offset by \$.10 in higher accounts payable and accrued liabilities. That leaves required net investment at \$.05 for each extra dollar of sales. For an increase in sales of \$90 million, required net investment is \$4.5 million.⁶ This money has to come out of the \$96 million in earnings that CG retains on an average sustainable basis at current levels. The remaining \$91.5 million is available for active investments that should increase total earnings growth above the 3.5% organic growth rate.

3. Active Investment Growth

This additional increase in growth, and in value, will depend on how effectively CG's management deploys the \$91.5 million available for active investments. If it is wasted in attempts to penetrate highly profitable markets dominated by others and protected by wide moats, attempts that have a low probability of succeeding, then management will earn far less than the cost of capital if it earns anything at all. If the company earns 5% on these investments when its cost of capital is 10%, then each dollar reinvested generates only \$.50 worth of value. In these circumstances, the \$91.5 million reinvestment rate will add only \$45.75 million per year to a firm's intrinsic value. On the other hand, management could use the reinvested \$91.5 million on major improvements in efficiency in its franchise markets or to add facilities and/or extend its product line in ways that benefit from existing economies of scale in advertising, R&D, distribution, and management. In this alternative scenario, returns on the new investment may substantially exceed the company's cost of capital since they are protected from competitive erosion by its current moat. If the company earns 20% on incremental investment when its cost of capital is 10%, then each dollar reinvested is worth \$2 and the active investment driven growth

⁶ The portion of organic growth that arises from increased margins may in theory require some investment. However, the portion of the margin increase related to increasing returns to scale requires no such investment. And returns on investments implementing continuous technology-based process improvements are in practice very high, on the order of 50% or more, so that these investments tend to be negligible (see Sloan Foundation studies).

in value is \$183 million (\$91.5 million times 2). For convenience, we will refer to the value created per dollar of active reinvestment as the value creation factor associated with management's capital allocation initiatives. A value creation factor of less than one (e.g., 0.5 in our initial example) indicates that management is destroying value; a factor above one means management is creating value.

The value created by active reinvestment is simply this value creation factor multiplied by the amount of active reinvestment. This dollar figure can be converted into a reinvestment return by dividing the value created by the \$4 billion market price paid for access to that value. With poor capital allocation and a value creation factor in this example of 0.5, the \$45.75 million in value creation represents an active investment return of only 1.14% ($\$45.75/\$4,000$). With good capital allocation, the \$183 million in added value raises this return to 4.58% ($\$183/\$4,000$).

The quality of capital allocation can have a major impact on the future returns likely to be realized from investing in growth stocks. Company managements that devote funds, time, and attention relentlessly to improving operational efficiency, which is generally the greatest source of investment-driven value creation, will add much more to returns than growth initiatives that are far outside core markets and do not benefit from existing competitive advantages. Calculating the value creation factor by a thorough assessment of capital allocation strategies is a critical and difficult element in successful growth stock investing. The analysis entails a degree of complexity well beyond that associated with estimation of current cash flow returns and organic growth rates. We will discuss it in more detail in a separate section later in the chapter.

For the moment, we will assume that CG management's allocation of the \$91.5 million of active investment funds is good but not great, that every dollar of active investment generates \$1.60 of value. The active investment, then, produces \$146.4 million of value (a value creation factor of 1.6 times \$91.5 million), which is a 3.66 % return on the \$4 billion paid for CG. As with any investment, this rate of return depends on the amount paid for CG; the less paid, the higher the return.

We can now calculate our total return for CG. The cash return is 3.6%. Organic growth in earnings adds 3.5%. Active investment of retained earnings contributes another 3.66%, or 3.7% to avoid specious precision. The total annual return to an investment in CG is 10.8%, or between 10.5 and 11%. These calculations are summarized in Table 8.1.

Table 8.1 Basic return calculations for Consolidated General.

Consolidated General	Amount (millions)	% of Market Value or as noted	Assumptions
Sustainable revenue	\$ 3,000		basic assumption
Sustainable earnings	\$ 240		sustainable margin 6%
Cash return total	\$ 144		distribution 60 %
Cash return percent		3.6%	market value \$4,000
Organic growth revenue	\$ 90	3.0%	basic assumption \$3,000 Revenue
Organic growth margins		0.5%	basic assumption \$3,000 Revenue
Organic growth return		3.5%	revenue plus margin growth
Reinvested earnings	\$ 96		\$240 minus \$144
Organic growth investment	\$ 4.5		5 % net working capital * \$90 of revenue growth
Active Investment	\$ 91.5		\$96 – \$4.5 for organic growth
Value creation factor	1.6		basic assumption
Active reinvested value creation	\$ 146.4		1.6 * \$91.5
Active reinvestment return		3.7%	market value \$4,000
Total return		10.8%	sum of cash (3.6%) organic growth (3.5%) active reinvestment (3.7%)

Growth Rates and Growth Returns

The total CG return of 10.8% consists of a cash return of 3.6% and a growth return of 7.2%. The growth return is the sum of organic growth of 3.5% and the results of active reinvestment of 3.7%. In this and similar examples, we have implicitly assumed that the returns generated by growth are equal to the related growth rates, that a 3.5% organic growth rate produces a 3.5% organic growth return. But this assumption is not generally accurate, and understanding why is critical to understanding a return-based approach to evaluating franchises businesses.

To highlight the difference between growth rates and growth returns, we will consider first the effect of organic growth. For the constant-growth-rate with constant-cost-of-capital world that we have assumed, a 3.5% organic growth rate in Consolidated General's earnings leads to 3.5% increase in intrinsic value, which for this franchise business is based on earnings power value. Thus, if CG has an intrinsic value of \$3 billion, a 3.5% increase in earnings adds \$105 million (3.5% times \$3 billion) to shareholder value. But the resulting percentage return to shareholders depends on the price paid for CG's business. If CG trades

in the stock market at a price of \$4 billion, then 3.5% organic growth produces a shareholder return of just 2.63% (\$105 million divided by \$4 billion). If CG's market price were \$2 billion, then 3.5% organic growth in earnings would yield a 5.25% organic growth return (\$105 million divided by \$2 billion).⁷ Only when CG's intrinsic value and market price are equal will a 3.5% organic growth rate in earnings equal 3.5% organic growth return (\$105 million divided by \$3 billion).

A simple but important formula is at work here. For any business, let us designate the rate of organic growth in earnings by g , the percentage organic growth return by r , the intrinsic value by V , and the market price by M . The dollar addition to shareholder value will be gV (g times V). This represents a return percentage of gV/M . Thus,

$$r \equiv \text{organic growth return} = gV/M = g(V/M). \quad (1)$$

In words, the organic growth return is equal to the growth rate in earnings multiplied by the ratio of intrinsic value to market price. If CG stock is a bargain, available in the market for less than the intrinsic value, then the percentage return from organic growth will exceed the organic growth rate in earnings. If CG stock is overpriced, selling for more than the intrinsic value, then the percentage return to organic growth will be below the rate of organic growth. When intrinsic value and market price are equal, then the default assumption that the return from growth equals the growth rate will apply. These relationships are depicted in the diagram below.

In Case 1, undervaluation of the stocks amplifies the effect of the organic growth rate on returns. In Case 2, accurate valuation leaves the effect of organic growth on returns unchanged. In Case 3, overvaluation attenuates the impact on returns of the organic growth rate.

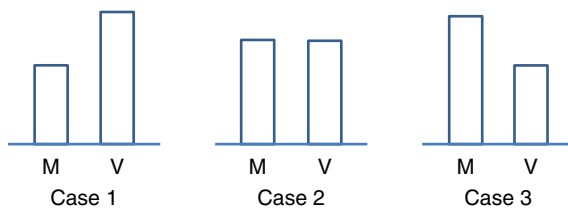


Figure 8.1 Market Price and Intrinsic Value Compared

⁷ The difference in market prices should not significantly affect CG's underlying business operations.

Conventional return calculations, such as those of a dividend discount model, automatically, but rarely explicitly, assume that market and intrinsic values are always equal, so that growth rates and growth related returns are always equivalent. In reality, and especially in Graham and Dodd reality, the divergence of market price from intrinsic value is at the heart of the investment process. As a consequence, we must consider explicitly the impact of the ratio of intrinsic to market values on future returns. Our difficulty is that we are concentrating on returns precisely because for growth stocks, we cannot reliably measure intrinsic values. As a result, we cannot calculate the ratio of intrinsic to market value to use as an adjustment in translating growth rates into growth related returns. Therefore, we will for the moment base our return calculations on the simplifying assumption that intrinsic and market value are equal and that the ratio of V to M is 1. Later we will address in detail the effects of relaxing this assumption.

An additional consequence of equation (1) applies to our calculation of active investment returns. We calculated this part of the growth return directly, dividing the value created from active investment (amount invested times the value creation factor) by the market price of CG. The growth rate in earnings corresponding to this return must be calculated from equation (1) since the growth return, r , is the growth rate, g , times the ratio of intrinsic value to market price. Conversely, the growth rate, g , is the growth return, r , divided by the ratio of intrinsic value to market price. If CG stock is undervalued, then the earnings growth rate corresponding to our 3.7% active investment return will be less than 3.7% (the ratio V/M is greater than one). If CG stock is overvalued, then the growth rate in earnings from active investment will be greater than 3.7%. Again, the growth return of 3.7% and the growth rate of earnings will only be equal when CG's market price and intrinsic value are equal. Since we cannot measure intrinsic value accurately, we will for the moment assume this equality holds.

Under this assumption, our active investment return of 3.7% should correspond to a 3.7% active investment-related growth in earnings, or a 7.2% earnings growth overall. This figure can be compared to average historical earnings growth rates for CG as an important check on our assumptions about organic growth and active investment returns. But we must remember that this simple comparison only properly applies when a company's market price and intrinsic value are equal. For the moment, let us assume that growth rates in earnings for CG have averaged around 7% over the last 10 to 20 years. With the caveat cited above, this would

indicate that our estimated growth returns are broadly consistent with past history.⁸

A final check on the validity of our growth estimate of 7.2% is that it is not so high as to have unreasonable implications concerning the future role of CG in the overall economy. If we were to project a long-term growth rate above 15% for the firm, compared to nominal GDP growth of 3 to 4%, then CG would eventually become a conspicuous part of the overall economy, which is not a realistic prospect for most companies. In addition to the cash return of 3.6%, the 7.2% projected growth in value yields an overall return of 10.8 %, or more realistically 10–11%, on our \$4 billion investment.⁹

4. Comparison Returns

A return of 10–11% speaks directly to the attractiveness of a CG investment. As we write, interest rates run from below zero on much short-term government debt to between 2% and 3% on long-term government bonds to 4% on relatively safe long-term corporate debt. By these standards, 10–11% looks to be a highly desirable return. But CG's return should be judged against returns on comparably risky assets and equity markets, whose risks more closely resemble those of a CG investment, appear likely to return 6–7%. This alternative implies a significantly smaller return advantage for CG. It is clearly important in making investment decisions to calculate appropriate alternative returns as accurately as possible.

This step is a virtual requirement for Graham and Dodd investors, for whom a margin of safety between value and price is at the heart of the investing process. For investments with measurable intrinsic values, the margin of safety is the difference between value received and price paid; the price paid represents the value of foregone alternative opportunities. For a decision based on returns, the calculated future return captures the value received while the appropriate alternative return represents

⁸ If CG's stock is currently undervalued, then (1) the return generated by the 3.5% organic growth will be greater than 3.5% and (2) the growth rate in earnings corresponding to the 3.7% active-investment return will be below 3.7%. Thus, if future growth were to track a historical 7.2% rate, returns to CG investors would exceed 7.2%. For undervalued stocks, historical growth rate comparisons to estimated growth returns are conservative, a desirable quality from a traditional Graham and Dodd perspective.

⁹ Estimates of intrinsic values and returns always involve some uncertainty. One should be wary of extremely precise calculations, whether one's own or others.

the opportunities foregone. Both parts of this comparison must be estimated as accurately as possible so that the difference between them can provide a sound basis for making investment decisions. The nature of a franchise business makes it particularly important to have as precise as possible a measure of the margin of safety in returns. No franchise lasts forever. When it is impaired, the damage to its values is far worse than for competitive businesses. (See Chapter 7.) In declining competitive markets, capital recoveries should largely offset lost profits. For franchise businesses, decline brings substantial value destruction. A margin of safety based on return estimates must be large enough to anticipate the inevitable but not predictable possibility of franchise impairment, what we have called “fade.”

There are two ways to estimate a benchmark return against which the 10–11% expected return from CG can be compared, although a 10–11% return from an unlevered franchise business will be attractive under most circumstances. The first is to estimate a cost of capital for CG, as we discussed in Chapter 5, “Earnings Power Value.” Since we are assuming that CG is all-equity financed, its cost of capital will be its cost of equity. We will assume that at the time of purchase, the range of possible equity returns runs from a venture capital return of 12% (high risk equity) to a return on a contemporary B– rated bond of 5% plus 1%, or 6% total (low risk equity with near bond returns). The midpoint of this range, which is a rough estimate of average equity risk, is 9%. (See Chapter 5.) We will also assume that given its lack of leverage and stable franchise position, CG’s risk falls just within the lower half of this range, or roughly 7.5%, midway between 9 and 6%. This figure is safely below our estimated 10–11% return from purchasing CG at a market price of \$4 billion.

A second alternative return can be calculated by looking at current potential returns from the stock market as a whole. Suppose that the average cash return on stocks is 2.5% per year. We find this figure by taking all the dividends and net share buybacks during the past year from a broad market basket of stocks, such as the S&P 500, and dividing by the cost of buying that market basket today.¹⁰ Average future growth in value for this basket will be driven by the average future growth in earnings. Earnings growth, in turn, will be based on growth in aggregate US or global output and incomes and by any trend in the share of profits as a

¹⁰ If the past year, these figures were exceptional, then we would have to use our estimate of “sustainable” pay-outs here.

fraction of overall income. Assume nominal GDP growth is expected to be about 3.5% per year, a level roughly equal to actual average growth in the early 21st century. This figure is the combination of around 1.5% annual inflation, 1% annual growth in the labor force, and about 1% annual growth in productivity. Between the late 1980s and the early 21st century, profits as a share of overall national income in the US increased from around 18.5–13.5%, an annual rate of 1.5–2.0%. Combining this increased share of GDP with estimated GDP growth indicates corporate profit growth of 5.0–5.5%. Adding this rate to the cash return of 2.5% produces an overall future US equity market return of 7.5–8.0%. Since CG is a stock with a below average level of risk, it should have a cost of capital of about 7%, which is close to our direct cost of capital estimate of 7.5%.

An alternative approach to estimating the benchmark equity return is to base it on an aggregate market earnings return. Suppose that the S&P 500 index trades at 17 times earnings for the stocks in the index. This price-to-earnings multiple inverts into an earnings return of roughly 6%. On average, reinvested earnings should earn something very close to the cost of capital of the firms making these investments. There will be some net value creation from the active investments of franchise firms, protected by barriers to entry from the leveling effect of competition. These should be offset in part by poor capital allocation that occurs because these same firms' managements are protected from the full force of competition. The same equalizing process occurs in non-franchise firms: good management makes good investments, poor management makes poor investments. At the same time the organic real growth of some franchise businesses will be offset by the fading franchises of others. The only organic growth we need worry about is that relative to low levels of inflation (high levels of inflation are associated with economic disruptions that have historically had significant negative impacts on corporate profit performance). The early 21st century environment has been one of a relatively benign inflation rate of 1.5%. Adding this growth in nominal earnings to the 6% earnings returns produces a likely future market return of 7.5%. This figure closely aligns with our other estimate of 7–8%.

Because we are assuming that CG is a low risk stock, the rate of return against which it should be compared is slightly lower than the market return, around 7%. That is the figure we will use for computing a return-based margin of safety. Our expected return of 10–11% on the \$4 billion investment in CG leaves a margin of safety of between 3 and 4%.

Table 8.2 Comparison returns for Consolidated General.

Method	Calculation	Cost of equity	CG cost of capital
Weighted average cost of capital (Cost of equity for all-equity firm)	$R = (1) \times \text{cost of equity}$	range: 6% to 12% CG low risk (no leverage) 7%	7%
Return relative to market			
Dividends plus growth	market dividend return + growth	market = 2.5% growth = 5.0% total = 7.5% CG below market risk	7%
Earnings plus low inflation	market = $1/PE$ + low inflation	market = 6% (1/17) + 1.5% = 7.5% CG below market risk	7%

These calculations are summarized in Table 8.2.

Franchise Fade

This 3–4% margin of safety must, as always for a Graham and Dodd investor, provide protection against miscalculations and unexpected future developments. There is an additional concern when deciding to purchase a franchise business. No franchise is eternal. Some, such as Coca-Cola, which has been around for well over a century, may last for a very long time, but others, like Xerox, local newspapers, or the daily racing sheet, a favorite early example of Warren Buffett’s, had established positions but were subject to significant erosion over relatively short periods of time. Our margin of safety in returns for franchise businesses must also provide protection against such possibilities.

We have referred to these possibilities as franchise “fade,” and we have included this “fade” in our margin of safety discussion because “fade” rates are difficult to calculate precisely and are better considered as “qualitative” factors affecting the desired margin of safety rather than “quantitative” factors that are included in the calculated rate of return. Clearly, a company such as Coca-Cola, with a durable franchise, requires a smaller margin of safety than even an established technology company such as Intel or one in a rapidly changing industry, such as Tesla, if indeed it is a franchise business at all. Nevertheless, it will be useful to consider how important “fade” rates are in quantitative terms.

The simplest although perhaps not the most realistic way to think of franchise fade is to assume that the business and its values dies at an exponential rate of say 2% per year. For this case, “fade” is a subtraction from growth. Net growth would then be the sum of organic growth plus active investment growth minus fade, taking care to distinguish between the growth rate of a business and the return from the growth obtainable by purchasing the business at its current market value. For this simple case, we can compare the margin of safety for CG directly to its fade rate, with the difference between the two being a “net” margin of safety. Also, for this case, when we look at overall historical growth rates in a business, the fade rate will already be incorporated as part of the total.

A more realistic way to look at franchise fade is in terms of survival probabilities for the franchise. Franchises, such as Xerox and newspapers, are generally undermined by social or technological developments that arrive within discrete, relatively short periods of time. Assume that the likely arrival rate for such fatal developments is roughly constant at a rate of 2% in any given year. In this case, the probability that the firm is alive at the end of this year is 98%; at the end of two years, the survival probability is 98% times 98%. If the value of CG’s business, provided it survives, grows by 7% per year, then, incorporating the fade rate, the expected value of the business at the end of the year is 98% times 107% of the value at the start of the year, or approximately 105% of its original value.¹¹ This more realistic fade process affects the growth in CG’s value—its expected value—in exactly the same way as the less realistic exponential decay we discussed above. In general, these episodic “fade” rates can be treated as subtractions from basic growth rate estimates, just as we did for constant erosion.¹²

¹¹ For simplicity, we assume here that the value of the business after the fade event is zero, although it is easy enough to incorporate non-zero values into the analysis. The expected end-of-period value of the business is then $(1 - f)V$, where f is the fade rate, V is the pre-fade event value of the business, and $1 - f$ is the probability that the business survives the period (i.e., the fade event does not occur). Suppose instead the post-fade event value of the business is mV rather than zero, where m is positive but less than one. Then the end-of-period expected value of the business is $(1 - f)V$ (the fade event does not occur) plus $f \times mV$ (the fade does occur). This can be rewritten as $V(1 - f + fm)$, or $V(1 - (1 - m)f)$. The term $(1 - m)f$ equals f when $m = 0$ (no residual value). Otherwise, the fade related rate of depreciation in expected value is $(1 - m)f$. A non-zero residual value reduces the effective fade rate from f to $(1 - m)f$.

¹² Even if a post-“death” business declines slowly after a “death” event, the market value of the business is likely to drop immediately as the further deterioration is anticipated and rapidly incorporated in the market value.

There is an important empirical implication of this more realistic view of “fade.” The historical growth rate of the business is in this case a pre-fade rate of growth since the “death” event has usually not yet materialized. The historical growth must be adjusted by the fade rate to provide an estimate that will incorporate fade possibilities going forward.

It is difficult to estimate precisely this episodic fade rate for a franchise business. One way to start is by thinking of the half-life of the business, the number of years in the future at which its survival probability declines to 50%. For a durable franchise such as Coca-Cola, this might be 80 years or more. For a more recent tech dependent franchise such as Apple or Intel, it might be as brief as 15 or 20 years. The half-life can be translated into annual fade probabilities using the rule of 72. An 80-year half-life corresponds to an annual fade rate of roughly 0.9% (72 divided by 80) which will usually be immaterial (i.e., equivalent to an annual downward adjustment of 1% in the organic growth rate of value). But an 18-year half-life corresponds to a 4% fade rate (72/18), which is far more serious. In general, half-lives of 25 years or less, meaning at least 3% a year, need to be taken seriously.

These annual fade rates (or extinction rates) are negative annual returns and must be covered by the margin of safety calculated without fade. It should be clear that they are substantial and must be taken into account in making franchise investment decisions. If we believe CG operates in a highly stable economic environment, with a franchise half-life of 50 years, then the fade rate is 1.5%, which is covered by our 3 to 4% margin of safety. On the other hand, if its likely half-life is only 20 years, then the fade rate is 3.6%, which eliminates the entire margin of safety. From a Graham and Dodd perspective, a particularly attractive feature of this approach is that the fade rate, calculated from its half-life, is linearly and clearly related to the margin of safety required of the returns. Episodic fade rates should be subtracted from the estimated return to calculate the overall expected return because it is not part of the historical growth experience. This overall calculation is important since our benchmark return (including fade) is conservative only if it is greater than the cost of capital. Successful growth investing is by definition investing in franchise businesses. Ignoring the fade rate overstates the margin of safety and increases the risk that the decisions will be based on overoptimistic return forecasts, which is the main vulnerability of all growth stock investments.

The Return Formula

The calculations that we have just worked through for the case of Consolidated General are based on the familiar dividend discount model for estimating stock returns.¹³ In that model, the overall return per share for an equity investor consists of a dividend and a rate of capital appreciation. Formally,

$$r = \text{return per share} = (d/p) + g, \quad (2)$$

where d is the current dividend per share, p is the market price of a share of stock, and g is the anticipated growth rate in the share price. In practice, this estimate of the return assumes that the anticipated growth rate in the stock prices is driven by the future growth rate in dividends, which is itself driven by the growth rate in earnings and ultimately the growth rate of the revenue and profitability of the enterprise.

The CG calculations differ from those of the traditional dividend discount model in several ways. First, for CG, we looked at the return to owning the company as a whole rather than the return to a single share. In the case of an all-equity financed CG, this distinction is inconsequential. If there are no share repurchases, total dividends divided by market value is identical to dividend per share divided by price per share. Also if the number of shares is constant, then the percentage growth rate in total dividends will be exactly the same as the growth rate in dividends per share. In this case, the company level calculation and the traditional per share calculation are exactly equivalent.

The complicating factor here is net share buybacks. A repurchase of 2% of the shares outstanding will, other things being equal, increase dividends per share by 2%, since the number of shares has declined by 2%. From a company level perspective, repurchase of 2% of the shares outstanding at the current market price will add 2% of the market value (price times 2% of the number of shares outstanding) to the total cash return. From both perspectives, the individual share and the company's total equity, the cash return will increase by two percent. The change in total return in both cases will be this 2% less the impact of the loss of the distributed cash on future growth, which is also the same in both cases.

We prefer the company level analysis because it is better adapted to analyzing returns to owning a proportionate share of a core enterprise

¹³ See Gordon, Myron. (1959). "Dividends, Earnings and Stock Prices." *Review of Economics and Statistics*, 41 (2): 99–105, for an early description of this model.

than the single share approach.¹⁴ For a franchise business, an enterprise valuation approach, as opposed to an equity only approach, is particularly important. The essential value of a franchise business, especially the value created by growth, is produced by its core operations that are protected from competition by barriers to entry. Adding a growth component to the returns from non-core assets—such as net cash, securities, real estate, and business segments that operate in competitive markets—is a mistake. We should evaluate a franchise business by calculating the return it produces on the net cost of the core business itself: the cost of the company, equity plus debt, minus the value of non-franchise assets. In this framework, distributions to investors includes dividends, net stock buybacks, net interest payments, and net debt repurchases, which must be made out of the net operating after tax profits of the core business.¹⁵ None of these calculations are easy to adapt to the traditional dividend discount model.¹⁶

The Impact of Assuming Market and Intrinsic Values Are Equal

As we noted above, when we calculated growth rates from the underlying characteristics of a business, the value impact of these growth rates applies to the intrinsic value of the business, not its market price. The true total return formula is

$$r \equiv \text{return} = D/M + g(V/M), \quad (3)$$

¹⁴ Another advantage of company level analysis is that it highlights the differences in returns from buybacks between selling and non-selling shareholders. The traditional per share return calculation tends to obscure these differences. In general, if the company buys back stock at a price above intrinsic value, then selling shareholders benefit at the expense of non-selling shareholders. If the company buys back stock at a price below intrinsic value, the benefits are reversed and those who hold on profit at the expense of sellers. However, any shareholder can protect against being on the wrong side of the trade by selling their holdings in the same proportion as the overall buyback.

¹⁵ If there are non-core, non-franchise businesses, the cash distributions from them must not be included as part of the net after tax operating profits of the core franchise operations. We already account for these earnings when we subtract the value of these businesses from the price paid for the company to arrive at the price paid for the core franchise business.

¹⁶ There are certain instances, notably financial service companies, where equity values are most pertinent. Since the distinction between operating and non-operating debt (e.g., accounts payable versus formal borrowings) is not clear-cut, we have to look at the core enterprises from an equity perspective. But even then we should perform a company level and not a single share-based return analysis.

where D represents the total cash return, g is the growth rate in earnings, V is the firm's intrinsic value, and M is the price at which it trades in the market. This same formula applies equally to the dividend discount model, which should properly be

$$r = d/p + g(v/p), \quad (4)$$

where d is dividend per share, g is the growth rate in earnings, v is the intrinsic value per share, and p is the market price per share. The v/p term, which has been ignored in the standard version of the dividend discount model, corrects for one of the obvious problems of that model. Without it, the growth part of the return is the same regardless of the price paid for the firm's stock. Even an infinitely high price would not reduce the growth return, and hence the overall return, to zero. But unless we can measure a firm's intrinsic value, which we have acknowledged we cannot, the return formula of equation (3) is unusable. That is why, in calculating returns, we start with the assumption that price and value are equal.

But we do not stop there. We will refer to the return calculated on this basis as the "benchmark" return and, for ease of reference, designate it as r_b .

By definition,

$$r_b = D/M + g. \quad (5)$$

Consider next the effect of basing an investment decision on the benchmark return, which can be calculated, just as we did in the case of CG. For a growing franchise business, one in which g is greater than zero, the difference between the actual return, r , and the benchmark return, r_b , is

$$r - r_b = D/M + g(V/M) - (D/M + g) = g(V/M - 1). \quad (6)$$

The first equality in equation (6) simply substitutes the definitions of r and r_b (equations (3) and (5)) into the expression $r - r_b$. The second equality follows from cancelling out the D/M terms. If the growth g is positive and V/M is greater than one, then $g(V/M - 1)$ will be positive. This means that the actual albeit unobservable return (r) will be greater than the benchmark return r_b . For growing franchise businesses,¹⁷ the actual return will exceed the benchmark return if we purchase the company at a market value (M) below its intrinsic value (V).

¹⁷ For shrinking franchises, far distant cash flows diminish rapidly in value, and conventional DCF calculations will usually be accurate.

In the appendix to this chapter, we show that whenever a benchmark return exceeds a business's cost of capital, the intrinsic value of the business is greater than its current market price. This means that the actual return exceeds the benchmark return, and using the benchmark return minus the cost of capital as the margin of safety understates the true margin of safety. If the benchmark return and cost of capital are equal, then the market and intrinsic values are equal, which means in turn that the actual returns and the benchmark returns are equal. If the benchmark return is below the cost of capital, then a business's market value will exceed intrinsic value, and the actual return will be below the benchmark return. As a consequence, if we purchase a company where there is a margin of safety between the benchmark return and the cost of capital, the margin of safety between the actual return and the cost of capital will be still larger. Using a benchmark return and insisting on a benchmark margin of safety is therefore a conservative investment strategy.

In taking advantage of this relationship, any franchise "fade" rate should be included as part of the overall growth rate, g . The actual future growth rate of expected earnings, including the impact of future franchise erosion, must be used to calculate the benchmark return. If we choose to exclude the fade rate in our benchmark growth calculation, then the necessary standard for conservatively judging an investment is that this pre-fade return should exceed our cost of capital by more than our estimated fade rate. This is the criterion we used in evaluating the CG investment. Properly applied, our approach to evaluating growth stocks on the basis of the benchmark return remains a conservative one.¹⁸

Active Reinvestment Returns Revisited

As we pointed out earlier, calculating returns to active reinvestment of retained earnings involves both practical and theoretical complexities

¹⁸ We can refine our margin of safety estimate by using an iterative process. Start by estimating the benchmark return r_b . In the appendix to this chapter, we show that the ratio of intrinsic-to-market value for a company is equal to the ratio of its expected future return to its cost of capital. We can approximate this ratio by using the ratio of the benchmark return, including the impact of franchise fade, to the cost of capital. We can use this ratio instead of $V/M = 1$ as a growth multiplier to recalculate a second, more accurate, benchmark return, say r_{b2} , where $r_{b2} = (D/M) + g(r_b/r^*)$ with r^* denoting the cost of capital. And we would iterate again using the ratio of r_{b2} to the cost of capital, and so on. In practice, there appear to be diminishing returns to this procedure after the first iteration.

that require separate, more detailed discussion. Estimate of the current cash return is straightforward. For the benchmark return, calculating the organic growth return, which in the benchmark case equals the organic growth rate in earnings, is also relatively uncomplicated. We described the process in the Consolidated General example. For the active reinvestment component of returns, the value created is the amount invested times the value creation factor associated with each dollar reinvested. The associated return contribution is the value created divided by the price paid for the company in question. The math is uncomplicated. The difficulty lies first in defining and identifying the amount that a company actively reinvests and then in defining and identifying the value creation factor associated with that investment. We will deal with each of these issues separately.

The complication in calculating the amount reinvested arises because some active investment does not get identified as such in standard financial accounts. Expenditures on research and development, new customer acquisition, and improved operational efficiency typically do not get treated as investments for accounting purposes. To the extent that this spending is to sustain the quality of a firm's product portfolio, customer base, and operating structure, leaving them at the end of the year just as capable as they were at the start of the year, no net investment is involved. The new investment merely offsets the depreciation of unaccounted for intangible assets as products become obsolete, customers leave, and operating discipline deteriorates. However, in cases where these "hidden" investments generate net business improvements, they are in effect net investments in intangible assets. If the spending were capitalized rather than expensed, operating margins and net income would be higher. In that framework, this spending represents active reinvestment.

Fortunately, where the level of hidden "net" investment relative to the size of a business is stable, we need not account for its effect explicitly. The associated returns will show up as steady earnings growth above the growth attributable to organic market factors and explicitly identified active investments. The earnings growth at issue will often be included as part of organic growth in earnings. If we believe we can identify significant levels of hidden "net" investment in a business, we should be able to account for it as part of a historically elevated R&D or other operating expenses, which accompany higher than average growth rates. Assuming net "hidden" investments continue at past levels, we can simply incorporate the higher historical growth rate in our estimate

of future growth in value.¹⁹ If the historical growth does not exceed our conventionally calculated organic plus active reinvestment growth rate, then any “hidden” investment appears unlikely to be making a significant value contribution, in which case, its impact can be safely ignored.²⁰

The Value Creation Factor

Estimate of a valuation creation factor is a complex task representing both practical and theoretical issues. It will be useful, therefore, to keep certain broadly applicable principles in mind. First, value creation factors—the ratio of the return to the cost of capital—greater than 1 apply only to investments protected by sustainable existing competitive advantages, or moats. New business initiatives, whether by acquisition or direct investment, that do not occur within a franchise will at most have value creation factors of 1. Historically, more often than not, they have been significantly less than 1, representing major value destruction. Financial investments in competitive markets will also have value creation factors of 1 or, if they are subject to double taxation, less than 1.

Only well-focused investments in operational efficiencies and those that extend existing competitive advantages, geographically or in product space, will have value creation factors above 1.

Second, precise value creation factor estimates will not be necessary for companies that distribute more than three-quarters of their earnings. Suppose a franchise trades at an earnings return (sustainable earnings divided by market price) of 6.5%, which is a sustainable multiple of 15, a potentially attractive rate for a growing franchise business. If it distributes 75% of its earnings, *retained* earnings will equal roughly 1.6% of the price of the firm (6.5% times 25%). Some small amount will be required to support organic growth. At most 1.5% of the market price will be actively reinvested. Average capital allocation, a value creation factor of 1, will add 1.5% to the overall benchmark return. Poor capital allocation, say with a value creation factor of 0.5, will add 0.75% to the benchmark return. Good capital allocation, a value creation factor of 1.5,

¹⁹ We are calculating benchmark returns here, which assume that market price and intrinsic value are equal. In this case, growth rates and growth-related value creation percentages are also equal.

²⁰ Temporary increases in “hidden” investment, such as the vast expenditure on Geico’s ad campaigns, are more difficult to account for. But firms deviating from normal behavior generally fall into the too-tough-to-call investment pile.

would add 2.25%. Variations of 0.25 in the value creation factor around these levels will affect the active investment return by less than 0.4%, an amount that should not be critical to a sound investment decision. For companies that distribute three quarters or more of their earnings, broad judgments of the quality of capital allocation should be adequate.

Third, if a company retains one-third or more of earnings and the growth in operating earnings can be accounted for largely by organic growth (i.e., the overall historical earnings growth rate is 1% or less above that implied by organic growth alone), then the value creation factor has historically been close to zero. Without clear evidence of improvement in capital allocation practices, a value creation factor of zero or close should be used.

Finally, retained earnings are often held in cash or used to retire debt that presents no major threat of impairing business operations. The returns to these actions depend on what is ultimately done with the cash or the debt capacity. In the best case, the available funds will be distributed to shareholders. In the worst case, they will be expended on value-destroying investments, for which a value creation factor of less than one-half is appropriate. But even in the best case, the tax consequences of holding cash—the interest is taxed—or paying down debt—reducing the tax shield of paying interest—means that the value creation factor is less than 1. In practice, these kinds of investments warrant a value creation factor of around 0.8 (sparing the reader the detailed calculations underlying this figure).

We have to know how management allocates active investment among initiatives that (1) improve efficiency, (2) extend either the geographic footprint with new stores or production facilities or product lines, or (3) acquire new business or invest in other new revenue acquisition projects such as expensive marketing campaigns. For each of these distinct types of investments, we need to estimate a range of returns relative to the associated cost of capital. In some cases, these returns can be gleaned from broad industry data. For example, efficiency-improving investments have historically had payback periods of between six months and two years, a pre-tax annual rate of return from 50 to 200%. A value creation factor of five or more is not excessive for these investments, which usually constitute only a small fraction of overall capital budgets.²¹ In some instances, industry experts have

²¹ See Sloan Foundation Industry Studies. One must be careful not to double count these returns. Efficiency improvements are often included as part of organic cost reduction trends. In these situations, to include them as part of active investment returns would be to count them twice.

produced detailed return calculations that are publicly available. There is an extensive literature on the returns to Walmart from new superstore openings and the conversion of existing stores to the superstore format. The returns to acquisitions can often be estimated from post-acquisition financial reports. (See the Intel case below.) A weighted average of these individual category returns will then yield an overall value creation factor for the company under consideration.

A theoretical problem with estimating value creation factors arises because the ultimate value of any active investment depends on what happens to the returns generated by that investment. The default assumption is that they are fully distributed to shareholders. In this case, each dollar of return is worth exactly 1 dollar, and we need not worry further about what happens to active reinvestment returns. But if all or part of the initial returns is actively reinvested in other projects, the value per dollar of the initial returns depends on the value creation factors associated with these secondary investments. But the value of these secondary investments depends in turn on what is done with their returns and so on in a theoretically infinite sequence. We cannot calculate a value creation factor unless we can calculate the returns of the proceeds of an active investment that depend on an as-yet unknown value creation factor for those proceeds. One exit from this difficulty is to simply make the default assumption that net active investment proceeds are fully distributed to investors. But it would still be useful to know how this assumption is likely to affect our calculated value creation factors when a substantial part of the active reinvestment returns are themselves reinvested.

We address this problem in the appendix to this chapter. Here we will simply state the results of that investigation for two polar cases. In the first case, secondary reinvestments neither create nor destroy value. The assumption is that all of the value-creating investment opportunities are fully funded from the primary retained earnings, essentially that the level of retained earnings is set to meet this objective. This situation is equivalent to the default assumption that active investments returns are fully distributed to investors since in both cases the value creation factor is 1. With a value creation factor of 1, it makes no difference whether earnings are distributed or fully reinvested. Thus, the incremental return per dollar of active reinvestment—the value creation factor—is equal to incremental earnings plus the organic growth rate in the incremental earnings.

In the second polar case, the value creation factors on the original active reinvestment and on the reinvestment of the proceeds of the original investment are equal. This assumes that the original active

reinvestment level does not exhaust or even diminish value-creating opportunities. In other words, the returns from the first round of capital allocation are compounded by the second round (and subsequent rounds) with an equivalent value creation factor. If the initial capital allocation creates value, a value creation factor greater than 1, under the assumption of Case 1 above, then the actual value creation factor is even greater than the Case 1 value creation factor. Thus, use of the value creation factor of Case 1 is conservative. By contrast, if the initial capital allocation under the assumptions of Case 1 destroys value, then in a Case 2 reality, things are even worse. Management with consistently bad capital allocation of both initial and secondary returns should generally be avoided unless current cash returns are likely to be sustainable and by themselves exceed a business cost of capital.²²

Recap

We began this chapter by identifying the need for an approach to investing in franchise/growth businesses that did not rely on estimates of current intrinsic value. The values of such businesses are extremely sensitive to variations in growth rates and costs of capital. As a result, they cannot be estimated with the degree of precision required to provide a sound basis for investments decisions. An alternative approach is to calculate an expected future return from buying the business at its current market price and to compare that return to an appropriate cost of capital. The process developed in this chapter for implementing this approach should mitigate the problem of extreme sensitivity to growth rates and costs of capital, leading to better franchise/growth investing, especially for value investors who have shied away from investing in growth stocks.

The return approach has several other advantages. First and most important is the direct, linear relationship among the critical factors that determine the margin of safety in returns. The estimated return is the simple sum of a growth rate and a current cash return. A 1% increase in the forecast growth in earnings raises the expected return by 1%.²³ This relationship holds for all current levels of earnings, all costs of capital,

²² In the appendix to this chapter, we show that if V_1 is the Case 1 value creation factor and V_2 is the Case 2 value creation factor for a firm, $V_2 = b V_1 / (1 - (1 - b) V_1)$, where b is the fraction of earnings distributed to investors.

²³ More precisely, it is 1% minus the usually minor impact of the associated increase in the required investment to support organic growth.

and all fade rates. A 1% increase in organic growth increases the overall growth rate by 1%, regardless of the level of active investment. A 1% change in the cost of capital changes the margin of safety by 1%. Analysis of the sensitivity of investment decisions to changes in these underlying factors is simple and straightforward. This simplicity is in marked contrast to the complicated, interdependent relationships between these factors and any measure of intrinsic value. For intrinsic value, the impact of a 1% change in growth rates depends on the level of the cost of capital, the level of cash returns, and, if it is even recognized, the franchise fade rate.

Changes in earnings power estimates do have more complicated implications, but even in this case the sensitivity analysis is far more direct than for intrinsic value. A 1% change in earnings power leads, other things being equal, to a 1% change in cash distributions. If the cash distribution return is 3%, the 1% change adds 3 basis points to the expected return. Added complexity arises because retained earnings and active reinvestment also increase. But if we have a well-formed idea of the value creation associated with active reinvestment, then we merely multiply the increase in retained earnings by the value creation factor and divide by the market price of the company to arrive at the change in return.

A second benefit of the return approach is that it isolates the individual factors that determine the attractiveness of a franchise company investment so that research efforts can be focused efficiently on those that are most important. Current earnings power, distribution and reinvestment policy, organic growth, active investment effectiveness, franchise fade, and the appropriate cost of capital can each be investigated independently. Many of these factors are often overlooked in a conventional intrinsic value analysis. The distinction between organic growth and the results of the capital allocation decisions that constitute active reinvestment often goes unrecognized. The effect of distribution policy on organic growth is minimal, but the effect on the amount available for reinvestment is significant. The difference between current franchise strength, which affects margins and earnings power, and franchise stability, which may be imperiled by the fade rate, is also often ignored. Using a return framework, these differences are clearly recognized, and the impacts of the individual factors are relatively easy to trace. The return approach highlights the particular importance of franchise fade, which is rarely analyzed effectively, if at all, by conventional intrinsic value methods.

Finally, a return approach by its nature focuses on the very long-run future. When we use a constant growth assumption, the growth rates

should be an average for the whole of the future that plays a significant role in determining today's value. That may be 40 years or more. Fade rate assumptions do the same thing. A 40-year half-life corresponds to an annual fade rate of 1.8%. A very long-run focus is essential for investing in franchise businesses and taking advantage of the lower taxes and transactions costs that come with long holding periods.

Decisions based on intrinsic value calculations, especially DCF-based decisions, typically consider only events occurring within a 7- to 10-year horizon. At that point, a terminal multiple is applied to the contemporary earnings. This multiple is usually chosen to be conservative, but without considering fade rates and long-run declines in earnings growth, it is not clear what a truly conservative multiple looks like. Take railroads as an example. For the last two decades, railroad earnings have done extremely well, thanks to the combination of deregulation, operating efficiencies, and rising fuel costs that have enhanced the competitive position of trains over trucks. Taken together, these events have led to steadily higher earnings margins. But looking forward, there are long-term trends that are far less favorable. The economic transition of consumption from goods manufactured in concentrated industrial sites to services delivered locally will lower the demand for railroad services. Developments such as 3-D printing will decentralize manufacturing and adversely affect transportation as a whole. Smart trucks that can travel in convoys without drivers will capture many of the current benefits of railroads while taking advantage of the shared rights of way that are the highway system and the ability to target more precisely the final local destination. None of these developments are likely to have a major impact in the next 7–10 years, but in the longer term they will be important. A constant growth return model will take these factors into account. An intrinsic value model limited to the next 7 or 10 years will not.²⁴

The return-based valuation also has limitations. First, given a market price, returns can be calculated going forward, and informed buy decisions can be made. Sell decisions must be made at prices chosen by other investors. Typically, in the Graham and Dodd tradition, this would be done by comparing intrinsic value to market price. But for franchise-based growth businesses, we do not have a usable estimate of intrinsic value. We could choose a level of calculated return, knowing that the expected return declines as the market price of the stock

²⁴ For a firm with an 8% cost of capital growing at 4% per year, only about one-third of the value arises from cash flows in the first 10 years.

risers. Unfortunately, this approach provides a very inexact discipline. Our calculated benchmark returns, as in the CG case, are usually very insensitive to changes in market prices. If most of the return arises from growth and predominantly organic growth, then doubling the market price might reduce the calculated benchmark returns by as little as 10%.

The return approach improves buy decisions for franchise businesses. The sell decision for those stocks remains a thorny and intractable problem, given the fundamental difficulty of calculating intrinsic values for franchises businesses.²⁵ It is not surprising, therefore, that value-oriented growth investors resort to rule-based and somewhat arbitrary sell protocols. Warren Buffett, for example, typically plans never to sell. Other value investors often use arbitrary limits on the multiple of sustainable earnings (e.g., 25 times) above which they will no longer hold franchises businesses.

Second, the approach will not work for all growing franchises. Forecasting future returns, including capital gains, requires that the future be reasonably stable and predictable. Otherwise, estimates of growth rates cannot be made with any useful degree of accuracy. Franchise businesses growing at 15% or more per year cannot sustain those growth rates forever. Sooner or later growth rates will decline to no more than a few percentage points above the long-term rate of the overall economy. Otherwise, as we have noted, one franchise business would take over the greater part of the whole economy. It is impossible to predict accurately when the inevitable transition to sustainable rates of growth will occur. It is also difficult to forecast the return impacts of shorter term developments, such as better management, economic fluctuations, potential major acquisitions, or innovative competitor behavior. In these fluid circumstances, at least some competitive advantages may shift in unpredictable ways. As a result, attempting to estimate returns for these companies will be, in the language of Graham and Dodd, a speculation rather than an investment judgment. These opportunities will usually land in the too-tough-to-call pile. In franchise investing, as in value investing more generally, it is critical to identify what is accurately knowable and, in the Graham and Dodd tradition, focus on those factors in making investment decisions.

²⁵ At this point, it is worth asking whether the return approach should also be applied to non-franchise businesses. Doing so would sacrifice the ability to compare earnings power and asset values. In franchise businesses, the independence of earnings from assets makes this a small sacrifice. For non-franchise businesses, it means giving up a major advantage of the Graham and Dodd approach.

Appendix to Chapter 8: Return Calculations for Franchise Businesses

The procedures described in Chapter 8 for calculating returns on investments in franchise businesses depend on a number of assumptions about those businesses. It is important to understand these assumptions, how much they may differ from reality, and how they compare to the assumptions that are employed by alternative approaches, such as ratio-based or DCF valuations. This appendix identifies the assumptions explicitly, highlights their implications, and formally links them together.

At the top of the list is the assumption that the future unfolds in a constant fashion. Future organic growth rates of revenue and profits are assumed to be constant, as is the fraction of earnings distributed and the amount of capital required to support each dollar of organic growth. The value created per dollar of active investment is assumed to be constant, which implies consistency or offsetting variations in both the quality of investment opportunities and the efficiency with which management addresses them. On the other side of the equation, we will assume that the cost of capital to which we compare our calculated return is constant. Taken together, these assumptions mean that the growth rate in value is constant. Finally, franchise fade probabilities, although not explicitly used in the return calculation, will implicitly be assumed to be constant for the

purpose of assessing whether the margin of safety between a calculated return and the cost of capital is adequate.

In reality, history does not proceed this steadily, and these variables will fluctuate in unpredictable ways. The assumptions of a constant future will not be precisely accurate. If we interpret our forecasts as expected values around which future fluctuations take place, then so long as they properly capture average future trends, they will not significantly distort valuations. Ratio valuations and DCF calculations of terminal values embody the same assumptions of constancy. Interim variation assumptions before terminal values generally have only minor effects on DCF valuations. Thus, constant growth and operating ratio assumptions are effectively common to all approaches to valuing franchise businesses.¹

The dividend discount model, which is the basis of our return estimates, usually calculates returns on a per share basis. Our version of the model looks at the enterprise as a whole. For us

$$R = D/M + G, \quad (1)$$

where R is the return from purchasing all of a business, D is the total current cash distribution to investors including both the proceeds of net share buybacks and disbursements to debt holders, M is the market price paid for the firm including both equity and debt, and G is the growth rate in firm value, which, given our constant growth rate assumptions, is equal to the growth rate of earnings. As we argued in Chapter 8, this company level formulation is fully equivalent to the traditional per share approach for an all-equity firm.

For a business with debt, equation (1) can accommodate net debt distributions, including sales and redemptions, in the current cash flow term D , and the outstanding value of net debt in the market purchase price M . The earnings whose growth is measured by G will then be the operating profits after tax of the core business. For franchise businesses, an enterprise perspective is essential because it is the core franchise

¹ There is one qualification to this point. For rapidly growing franchise businesses, some analysts attempt to identify periods of accelerating and then tapering growth (the common S-curve of new product introductions). In practice, these break points are almost impossible to predict, as the poor track record of this approach demonstrates. The situations of these businesses are often ones in which their franchise positions are not yet fully established. Still, they generally trade at elevated prices. Moreover, even if they do successfully erect substantial barriers to entry, the growth rates of the nascent markets in question have not yet stabilized and therefore are usually difficult to forecast. These investments almost invariably fall into the too-tough-to-call category.

operations that are responsible for creating both current value and its growth over time.

The Assumption That Market Price and Intrinsic Value Are Equal

As we discussed in Chapter 8, the proper form of equation (1) is

$$R = D/M + G(V/M), \quad (2)$$

since the growth in value that arises from earnings growth applies to the intrinsic value of the enterprise, V , not to its market price. Equation (1), like the traditional dividend discount model, embodies the assumption that V and M are equal, so that V/M equals 1. In our process for estimating returns, we began with this assumption. But Graham and Dodd investing is about exploiting differences between V and M , making it essential to understand how returns change if V and M are not equal.

Consider a firm in our constant growth world, with current after-tax operating earnings, E . A constant fraction, C , of earnings is distributed to investors; the remainder is reinvested in the business to contribute to the growth rate, G . The cost of capital appropriate to the enterprise will be denoted R^* . The intrinsic value of the firm will equal the present value of all future earnings distributions, which in the constant growth case is

$$V = CE/(R^* - G), \quad (3)$$

which, since C , R^* , and G are assumed to be constant, grows at the rate of earnings growth, G . Thus G times V represents the growth in investor value over the course of a year, and CE represents the current cash return. The total return, R_T , to owning the firm is the sum of the 2 returns, namely,

$$R_T = CE + GV. \quad (4)$$

Dividing R_T by the market price M yields the percentage return equation (2).

From the valuation equation (3), multiplication of both sides by $(R^* - G)/V$ yields

$$(R^* - G) = CE/V. \quad (5)$$

We can rewrite the return equation (2) as

$$R = (V/M)(CE/V + G), \quad (6)$$

where we have substituted CE for D and factored out the (V/M) term. Substitution from equation (5) for CE/V into equation (6) yields

$$R = (V/M) (R^* - G + G) = (V/M) R^*. \quad (7)$$

Division of both sides of (7) by R^* yields the useful result that

$$R/R^* = V/M. \quad (8)$$

The ratio of the intrinsic value of a business to its market price is equal to the ratio of the return to buying the business at the market price to the business's cost of capital. Returns above the cost of capital imply an intrinsic value above market price. Returns below the cost of capital mean the intrinsic value is below the price being paid.

Because V cannot be accurately estimated, we cannot use either equation (2) or equation (5) to calculate R . Instead, we start our return estimation process using a benchmark return, R_B . The benchmark return is based on the assumption that M and V are equal. Thus, the benchmark return is

$$R_B = (CE/M) + G. \quad (9)$$

The error in using R_B as an estimate of the true return, R , is

$$R - R_B = (CE/M + G(V/M)) - (CE/M + G) = G((V/M) - 1), \quad (10)$$

where the second equality comes from cancelling out the (CE/M) terms. We have established in equation (8) that V/M equal R/R^* , so that equation (10) becomes

$$R - R_B = G(R/R^* - 1) = (G/R^*)(R - R^*). \quad (11)$$

Simply subtracting R^* from each term on the left-hand side of (11) yields

$$(R - R^*) - (R_B - R^*) = (G/R^*)(R - R^*), \quad (12)$$

or, rearranging terms,

$$(R - R^*) (1 - G/R^*) = R_B - R^*. \quad (13)$$

The term $(1 - G/R^*)$ is equal to $(R^* - G)/R^*$. Division of both sides of (13) by this term produces the result that

$$(R - R^*) = (R^*/(R^* - G))/(R_B - R^*), \quad (14)$$

which defines the relationship between the true margin of safety in returns and the margin of safety using the benchmark return of

equation (9). The term $R^*/(R^* - G)$ is always positive since R^* must be greater than G . Otherwise, our franchise business would have infinite value. For growing franchises, G is greater than zero. Thus R^* will be greater than $(R^* - G)$, and $R^*/(R^* - G)$ will be greater than 1. Given these relationships, for growing franchise businesses, the true margin of safety exceeds the benchmark margin of safety as long as the benchmark margin of safety is positive. Insisting on a benchmark margin of safety as a condition for investment in growing franchise businesses is therefore a conservative strategy.

One final observation. It might be tempting to use the formula of equation (14) to calculate a “true” return for a franchise business. However, the term $1/(R^* - G)$, which creates so much imprecision in DCF terminal values, also enters the right-hand side of equation (14) for the “true” return. The qualitative implications of equation (14) are useful. Its quantitative implications are not.

For shrinking franchises, for which G is less than zero, the estimated return will be above the true return. But shrinking franchises do not present the same valuation challenges as growing ones. Consider a franchise business whose earnings, assumed to be fully distributed, are shrinking at 5% per year with a 10% cost of capital. The earnings power value multiple will be 1 divided by 15% (10% minus negative 5%, or 10% plus 5%), or 6.7 times. If we are off by 1% in our estimates of the cost of capital and the growth rate, this multiple will range from 7.7 times ($1/(9\% + 4\%)$) to 6 times ($1/(11\% + 6\%)$), an error of only about 25%. With a shrinking franchise, cash flows in the far distances carry little weight, and a conventional Graham and Dodd valuation, adjusted for shrinkage in earnings power, or even a DCF, will be a reasonably accurate basis for investment decisions.

Determination of the Value Creation Factor for Active Investment

For many kinds of active reinvestment, especially stand-alone investments outside a firm's core franchise, there are straightforward ways of estimating the value created per dollar of actively investment retained earnings. Acquisitions of whole businesses can be evaluated like another equity/enterprise investment. Non-franchise business acquisitions can be evaluated by the methods we developed in Chapters 3 through 5. An asset value/earnings power approach can be used to estimate the intrinsic value of the acquired business. Division of the intrinsic value

estimate by the acquisition cost gives an estimate of value acquired per dollar invested, which is the value creation factor for this investment. Stand-alone projects such as new store openings can be evaluated in the same way they are for capital budgeting purposes. The estimated value produced can then be divided by the investment cost involved to get a value per dollar invested, which again is a value creation factor. If the estimated value creation factor is greater than 1, the return produced exceeds the project's cost of capital.

For stand-alone projects, there are only 2 ways in which active investments can produce a value creation factor greater than 1. In the first case, exceptional management—that is, management with a long history of operating more efficiently than its competitors in a market not protected by barriers to entry—invests to expand in competitive markets. Provided it maintains its focus and continues to outperform, its investments will earn more than the cost of capital. In the second case, a company with an established competitive advantage expands within its core franchise or into adjacent markets, adjacent either geographically or in product space, and manages to extend its competitive advantage into this new territory and remains protected by barriers to entry. If, in the first case, management is merely competent rather than exceptional, it will not achieve a value creation factor much greater than 1, no matter what detailed estimates show. If, in the second case, the firm moves into markets where it cannot extend its franchise, or, even worse, into markets where existing competitors have competitive advantages, the value creation factor is going to be less than 1, again no matter what detailed estimates indicate.

When active reinvestment initiatives produce incremental earnings that are additions to those of the core franchise, calculating a value creation factor is more complicated. There is a self-referential element in the estimated returns that requires detailed analysis. Like any investment in a growing franchise business, active investments cannot be evaluated on a DCF or other estimate of intrinsic value. The weight of uncertain far distant cash flows means that these value estimates will be too imprecise to use. We must calculate value creation factors for these investments by estimating returns and dividing them by the business's cost of capital. Estimated returns will be the sum of 3 elements:

1. A cash return equal to incremental earnings per dollar invested times the portion of earnings distributed to investors;
2. The value created by organic growth of these incremental earnings;
3. The value produced by “secondary” active investments of the part of the incremental earnings that are retained and actively reinvested.

The last of these elements presents potential difficulties. If both the original and “secondary” reinvestments are made under the same conditions, then they ought to produce the same returns. But if this is the case, we cannot compute the return on the secondary investment without knowing the return on the initial reinvestment to which it is equal. And we cannot compute the return on the initial active reinvestment without knowing the secondary reinvestment return, which is potentially an important part of that initial return.

This is not merely an academic issue. Active investments in this category are the ones whose earnings get distributed in the same proportion and grow organically at the same rate as the core franchise earnings of a business. To the extent that they are retained and actively reinvested, they produce the same amount of value per dollar as core franchise earnings. These investments include cost reduction and process improvement initiatives focused on core franchise operations and basic product and geographic extensions of a franchise business. In practice they will usually account for a significant share of actively reinvested retained earnings.

To address the circular nature of the return calculation for these kinds of active investments, we will assume that the project produces incremental earnings of e dollars per dollar invested. Because these incremental earnings are indistinguishable from core franchise earnings, they are assumed to grow organically at the same rate as core franchise earnings. The distribution fraction C for core franchise earnings will apply to e and a fraction $(1 - C)$ of those earnings will be retained. The return to those retained earnings will be the same as the return to all other franchise earnings actively reinvested. As a result, the growth rate in e will be equal to the growth rate of core franchise earnings, G . Because they are simple additions to core franchise earnings, the risks associated with the incremental earnings should be the same as the risks associated with core franchise earnings. Thus, the cost of capital associated with e will be equal to R^* , the cost of capital for the firm’s core franchise business. The value created per dollar of incremental investment will then be

$$\nu = Ce / (R^* - G), \quad (15)$$

where C , G , and R^* are respectively the distribution fraction, the growth rate, and the cost of capital of core franchise earnings (i.e., using the same notations as the previous section of this appendix). Given the constant growth rate assumptions that apply to the incremental earnings, e , the value created per dollar reinvested, ν , grows at the same rate, G , as earnings. Thus, the return per dollar invested generated by this incremental investment is

$$r = Ce + G\nu, \quad (16)$$

where Ce is the cash return from e , and Gv is the incremental value produced.

Equations (15) and (16) look exactly like equations (3) and (2) of the previous section except that the cost of acquiring e is by definition \$1, so that we do not have to divide by a market value, M . We can, therefore, simplify equations (15) and (16) exactly as we did for equations (3) and (2) to obtain the result that

$$v = r/R^*. \quad (17)$$

Unfortunately, we cannot use equation (15) to estimate v with a useful degree of precision because the term $1/(R^* - G)$ is extremely sensitive to small changes in our estimates of R^* and G . Without an estimate of v , we cannot estimate the value creation factor r from equation (16). So, as before, we will define a benchmark return, r_b , based on an initial assumption that v equals 1, equivalent to the assumption V/M equals 1 that we made previously. Thus

$$r_b \equiv Ce + G. \quad (18)$$

Again, equations (15), (16), and (18) look exactly like equations (2), (3), and (9) except that V/M has been replaced by 1. Thus, simplification of (15), (16), and (18) in the same way as we simplified equations (2), (3), and (9) leads to the result that

$$(r - R^*) = (R^*/(R^* - G)) (r_b - R^*), \quad (19)$$

which defines the relationship between the true return, r , produced by the incremental earnings e , and the benchmark return r_b .

The interpretation of equation (19) is quite different from the interpretation of equation (14). The return $(r - R^*)$ on the right-hand side of equation (19) is a measure of the quality of this secondary capital allocation. If $(r - R^*)$ is greater than zero, then r is greater than R^* , and r/R^* is greater than 1. From equation (17), r/R^* is the value creation factor for a core franchise investment producing e in incremental earnings per dollar invested. When r is greater than R^* , v is greater than 1, and the investment of retained earnings is value creating. If r equals R^* (i.e., $r - R^*$ equals zero), then v equals 1, and the reinvestment neither creates nor destroys value. If r is less than R^* , then v is less than 1, and management is destroying value when it reinvests a portion $(1 - C)$ of earnings. What equation (19) says is that our measurable benchmark return, r_b , is a good qualitative surrogate for the unmeasurable true return, r . Since $(R^*/(R^* - G))$ is positive, $(r_b - R^*)$ has the same

sign as $(r - R^*)$. If $(r_b - R^*)$ is greater than zero, then $r - R^*$ is greater than zero, and ν is greater than 1. For growing franchises (G greater than zero), $(R^*/(R^* - G))$ is greater than 1. Thus in this case, r is greater than r_b , and the estimated value creation factor using r_b (i.e., r_b/R^*) is a conservative estimate of the true value creation factor.

On the other hand, if the benchmark return is below the cost of capital, the true return is below the benchmark return. In this case, the benchmark return underestimates the true degree of value destruction. Managements who produce an e that is so low that r_b is significantly below R^* are destroying value. The conservative strategy in this case is simply to assume that the value creation factor is between 0.5 (50% value destruction) and zero if active investment seems qualitatively to be extremely misguided. To summarize the implications of equation (19), in slightly different terms, we can define a benchmark value creation factor, $\nu_b = r_b/R^*$, then if R^* is greater than r_b ,

$$1 > \nu_b > \nu. \quad (20a)$$

If r_b is greater than R^* ,

$$\nu > \nu_b > 1. \quad (20b)$$

What we have considered so far is active reinvestment of the earnings of a franchise business directly into that core franchise business. A second possibility is that active reinvestment produces franchise-type earnings that are different from core franchise earnings. This scenario is less likely than that of reinvestment in the core franchise business, but it is not completely improbable. This investment might be a product extension or geographic expansion to an adjacent market to which some of the core market competitive advantages carry over but with different organic growth characteristics and reinvestment opportunities than those of the core market. In this situation, the relevant cost of capital, r^* , need not equal R^* ; the organic growth rate need not equal the core market organic growth rate, and the fraction of incremental earnings paid out, c , need not equal C . Thus the overall growth rate of incremental earnings, g , may differ from G .

In this case, active reinvestment that produces e in incremental earnings per dollar invested will have a value creation factor of

$$\nu = ce/(r^* - g). \quad (21)^2$$

² Equation (21) assumes that the investment necessary to support organic growth of incremental earnings is zero. If it is significantly positive it can be treated as simply a reduction in incremental earnings reinvested.

The return per dollar to this investment will be

$$r = cd + g \times v, \quad (22)$$

and, as before, we can show that

$$v = r/r^*. \quad (23)$$

The self-referential problem that V depends on v will again apply. Thus, we must make assumptions about secondary investments returns and will use two polar cases to help with the analysis.

Case 1 is to assume that secondary reinvestment neither creates nor destroys value. In this case, it does not matter for valuation purposes whether incremental earnings are distributed or reinvested. Thus, the return equation becomes

$$r = ce + (1 - c)e + g = e + gv, \quad (24)$$

where g is the rate of organic growth of incremental earnings. Next, as before, since r and v are not measurable with useful precision, we define a benchmark return

$$r_b = e + g, \quad (25)$$

which assumes v is 1.

The structure of equations (21) to (25) is exactly the same as equations (15) to (18) with the additional assumption of neutral secondary reinvestment imposed in equation (24). Thus,

$$(r - r^*) = (r^*/(r^* - g))(r_b - r^*) \quad (26)$$

defines the relationship between the measurable, r_b , and the relevant but unmeasurable, r .

In this case, the benchmark return is simply the sum of incremental earnings plus the rate of organic growth in those earnings. The difference between the benchmark return and the cost of capital has the same sign as the difference between the actual return and the cost of capital. Thus, if capital allocation looks good in the sense that the benchmark return exceeds the cost of capital, actual capital allocation is better because the actual return exceeds the benchmark return. For organically growing franchises (i.e., $g > 0$), the factor $r^*/(r^* - g)$ is greater than 1. Thus the benchmark value creation factor (r_b/r^*) is a conservative estimate of the actual value creation factor (r/r^*). On the

other hand, poor capital allocation destroys value at a rate greater than the benchmark value creation factor indicates. In this case, $(r - r^*)$ is negative and greater in magnitude than $(r_b - r^*)$.

A second polar assumption about secondary reinvestment for active franchise investments outside a company's core business is to assume that the qualities of secondary and primary reinvestment of retained earnings are the same. In this case, the value of active secondary reinvestment per dollar is v rather than 1. The amount of secondary reinvestment per dollar of primary investment is $(1-c)e$, where e is the incremental earnings per dollar from the primary active investment and c is the fraction of those earnings distributed to investors. The value created by this reinvestment per dollar of primary investment is $(1-c)ev$. Thus the primary active investment return is

$$r = ce + gv + (1 - c)ev, \quad (27)$$

where ce is the cash return, gv is the organic growth return and $(1 - c)ev$ is the secondary active investment return. This can be rewritten as

$$r = ce + (g + (1 - c)e)v, \quad (28)$$

where $g + (1 - c)e$ represents the overall growth rate in earnings from both organic growth in e and secondary reinvestment of a fraction $(1 - c)$ of e . We will designate this overall growth rate as h (equals $g + (1 - c)e$). Then equation (28) takes the familiar form

$$r = ce + h \times v, \quad (29)$$

and we can define a measurable benchmark return, assuming v equals 1,

$$r_b = ce + h = ce + (1 - c)e + g = e + g, \quad (30)$$

where the second equality in (30) substitutes the definition of h (i.e., $(1-c)e+g$), and the final equality adds ce and $(1-c)e$ together. The striking aspect of this is that the benchmark return in this case is the same as the benchmark return in the neutral reinvestment case. The reason lies in the definition of benchmark returns. They are calculated on the assumption that v equals 1, which assumes that primary reinvestment neither creates nor destroys value. For the case assuming primary and secondary returns are equal, this means the secondary return is also neutral, which is the assumption made in the neutral secondary case. So under the benchmark assumptions the two cases are the same and the benchmark returns are equal.

The relationship between the benchmark and actual returns in the second case is again

$$(r - r^*) = (r^*/(r^* - h)) (r_b - r^*). \quad (31)$$

Here because h is greater than g , the factor $(r^*/(r^* - g))$ is greater in the case of common secondary reinvestment performance than in the case of neutral secondary reinvestment. Thus, the gap between the benchmark and actual returns is greater both positively and negatively in this common case than in the neutral case. If benchmark capital allocation is good, the assumption that secondary reinvestment is also good will add more to the actual return than the assumption that secondary market reinvestment is neutral. If benchmark capital allocation is poor, the assumption that secondary reinvestment is equally poor will amplify the true level of value destruction.

The approximate magnitude of the effect can be captured in a relatively simple formula. The value creation factors for the neutral and common secondary reinvestment cases will be designated by ν_1 and ν_2 respectively. From the definition of ν_2 multiplied by $(r^* - g)$,

$$ce \equiv \nu_2 (r^* - h) = \nu_2 (r^* - g) - \nu_2 (1 - c)e. \quad (32)$$

From the definition of ν_1 multiplied by $(r^* - g)$,

$$e = \nu_1 (r^* - g). \quad (33)$$

Substitution from (33) into (32) yields

$$c\nu_1 (r^* - g) = \nu_2 (r^* - g) - \nu_2 (1 - c)\nu_1 (r^* - g). \quad (34)$$

Dividing out $(r^* - g)$ yields

$$c\nu_1 = \nu_2 - \nu_2 (1 - c)\nu_1 = \nu_2 (1 - (1 - c)\nu_1). \quad (35)$$

Dividing both sides of (35) by $(1 - (1 - c)\nu_1)$ then yields

$$c\nu_1 / (1 - (1 - c)\nu_1) = \nu_2, \quad (36)$$

which expresses ν_2 , the value creation factor in the common reinvestment case, directly as a function of ν_1 , the value creation factor for the neutral reinvestment case.

With minor rearrangement this becomes

$$\begin{aligned} (\nu_2 - 1) &= [c\nu_1 / (1 - (1 - c)\nu_1)] - 1 \\ &= (c\nu_1 - 1 + (1 - c)\nu_1) / (1 - (1 - c)\nu_1) \end{aligned} \quad (37)$$

or

$$(v_2 - 1) = (v_1 - 1)/(1 - (1 - c)v_1). \quad (38)$$

The gap between v_2 and 1 represents the gain per dollar from reinvestment under the common secondary return assumption. The gap between v_1 and 1 represents the net gain per dollar from reinvestment under the neutral secondary return assumption. In order for the reinvestment values to be finite it must be the case that $(1 - c)v_1$ is less than 1 and hence that $1 - (1 - c)v_1$ must be greater than zero. Therefore, if neutral reinvestment creates value (i.e., $v_1 > 1$), it must also be the case that common reinvestment also creates value (i.e., $v_2 > 1$). If neutral reinvestment destroys value ($v_1 < 1$), then common reinvestment also destroys value ($v_2 < 1$). The factor $(1 - (1 - c)v_1)$ must be less than one. Both $(1 - c)$ and v_1 must be greater than zero (assuming $c > 0$), and thus their product must also be greater than zero. Therefore, $[1/(1 - (1 - c)v_1)]$ must be greater than one, and the effect of assuming common as opposed to neutral reinvestment returns is to magnify the difference $(v_1 - 1)$, whether for good (i.e., $v_1 > 1$) or ill (i.e., $v_1 < 1$).

To give a simple illustration of the magnitudes involved we can consider the case where half of incremental earnings are further reinvested. Then the multiplier is $1/(1 - \frac{1}{2}v_1)$. If v_1 were to be 1.5, then $1 - \frac{1}{2}v_1 = 1 - \frac{3}{4} = \frac{1}{4}$. Therefore, $(v_2 - 1)$ equals 4 times $(v_1 - 1) = 4 \times (1.5 - 1) = 2$. The value of v_2 is, therefore, 3, which is double the assumed neutral creation factor of 1.5. If v_1 equals $\frac{1}{2}$ (i.e., 50% value destruction), then $1 - \frac{1}{2}v_1 = 1 - \frac{1}{4} = \frac{3}{4}$. This means $(v_2 - 1)$ is $\frac{4}{3}$ times $(v_1 - 1)$, or $\frac{4}{3} \times -\frac{1}{2}$, which equals $-\frac{2}{3}$. Therefore v_2 equals $\frac{1}{3}$ (i.e., 67% value destruction). The differences are substantial.

Example Three: *WD-40*

One way of tracking the evolution of our approach to value investing between the first edition of this book, written largely in 1999–2000, and the current edition is to re-examine some of the investment opportunities that we analyzed then. For Hudson General, a classic Case A opportunity with asset value well in excess of earnings power value, there has been little change. But for franchise businesses like WD-40, our understanding of the valuation process has evolved considerably. We will re-examine WD-40 at separate dates to highlight developments both in the company and in our analytical process. First, we will look at WD-40 at the end of September 1998, when the annual report for FY 1998, ending on August 31, became available. This snapshot revisits the principal valuation analysis of the first edition. Second, we will look at WD-40 in September 2000, a second date we considered in the first edition. The differences in valuations at these two times provide a useful measure of the additional details in our current approach. Finally we will look at WD-40 in September 2018, when financial reports for the most recent fiscal year became available, in order to demonstrate our new approach to contemporary valuations.

WD-40 had been a one-product company for over 40 years until it purchased the 3-in-One brand from Reckitt and Colman in December 1995. The product WD-40 is a lubricant that unfreezes sticky screws, removes rust, dissolves adhesives, stops squeaks, and comes to the aid of the home handy person or the machine shop professional whenever anything small needs to be lubricated or protected from rust. The name WD-40 means “water displacement, 40th attempt,” suggesting the trials and effort that went into development. The familiar blue can is practically ubiquitous in homes, factories, and repair shops in the United States, United Kingdom, and many other parts of the world. According to the company, four out of five American households have a can, and more people use WD-40 each week than use dental floss. While this may be good news for dentists, it leaves WD-40 with few untapped targets for new business in its domestic market.

Through 2000, WD-40 had been an extraordinarily profitable company: consistent and high operating margins (25–30%), net income margins (15–18%), and returns on equity (35–42%), even without adjusting for excess cash. But growth in the mid-1990s had been more difficult, dropping from around 10% a year to slightly more than 5%. Net income, always a more variable figure, held up until after 1995, when it also tailed off.

WD-40 took three steps starting in 1995 to invigorate its growth. It bought 3-in-One; it internally developed T.A.L.5, a more powerful lubricant that it sought to market to commercial users, without success; and it purchased the Lava brand of hand soap from Block Drug in April 1999. It was too early, from our 2000 perspective, to say whether the last new effort would pay off, by which we mean if it would spur growth in sales and income as well as maintaining the high level of profitability. Instead, we will focus initially on the financial numbers for the fiscal year ending in August 1998, before Lava entered the picture, and write from that vantage point.

WD-40 contracted out the manufacture, packaging, and shipment of all its lubricants. In fiscal year 1998, it needed only 167 employees to support \$144 million in revenues. Primarily a sales and marketing organization, it spent around 10% of its revenue on advertising and promotion. It had no secret formula for any of the lubricants and other products, nothing that Dow, DuPont, Exxon, 3M, P&G, Unilever, or any other firm with a modicum of competence in the chemical business could not duplicate. No patents protected these products. Yet WD-40 had for many years earned an exceptionally high return however measured, as return on sales, return on assets, or return on equity. The

company referred to WD-40 as a “fortress,” and one of its goals was to extend the fortress beyond the US and UK markets. When the company, in order to comply with environmental concerns, switched its aerosol propellant to CO₂ (yes, that CO₂, better than the ozone-depleting chemical it was previously using,) it was able to pass on the increased cost without blinking. WD-40 had all the characteristics of a powerful franchise business, including, it seems, significant unexploited pricing power. Unlike Hudson General and Magna, we have to look at WD-40 as an earnings-generating machine for which growth creates value, and assets play a lesser role in valuation. In other words, WD-40 is likely to be a Case C company for which current earnings power value significantly exceeds asset value. We need, therefore, to focus primarily on earnings power, growth potential, and the sustainability of current levels of profitability. We have to understand its strategic position. Where were the competitors who should have been entering into markets and stealing away customers or forcing WD-40 to lower its prices? If there were nothing magic in its lubricants elixir, what kept the challengers at bay?

WD-40 as the Reward of a Well-Conceived Search Strategy

In looking at a franchise business, we have to concentrate on the core enterprise that is the key locus of value creation, meaning that we start with an enterprise valuation rather than valuing only the equity. This begins with the total market value of WD-40's equity. At the end of September 1998, WD-40's common stock was trading at \$26.50. The fully diluted number of shares outstanding as of August 31, 1998, was 15.7 million for a total market value of \$416 million.¹ From this figure we must subtract the value represented by WD-40's net cash holdings of \$13 million. That leaves the value of the ongoing enterprise, what we would have had to pay for the company at the current market price minus the value of the net cash acquired, which is cash minus debt, at \$403 million. To this we should next add the cost of legacy liabilities, such as unfunded pension liabilities, that we would, in principle, have to pay off, along with the debt, in order to acquire an unencumbered claim on the underlying business. We should subtract the offsetting value

¹ In principle, we should adjust this figure for the potential proceeds from the exercise of outstanding options, a subtraction, and the option value of unexercised options, an addition. But, for WD-40, that net figure is negligible, so we will ignore it to simplify the discussion.

of non-core assets, such as investments in unrelated enterprises like low income housing that could, again in principle, be sold to pay off debt or legacy liabilities. For WD-40, the net total is near zero, so we will ignore these adjustments. At the end of September 1998, we would have had to pay \$403 million to acquire the core enterprise of WD-40 at then-prevailing market prices.

For our \$403 million, we would acquire a business that generated roughly \$34 million of operating earnings before interest and taxes (or EBIT) in FY 1998. With an average tax rate of roughly 36%, this falls to around \$22 million in potential after-tax earnings, a 5½% return on our \$403 million investment. Using a more traditional metric, the WD-40 enterprise was trading at 12 times EBIT (\$403 million divided by \$34 million) or 18 times after-tax earnings (\$403 million divided by \$22 million). By both measures, WD-40 was a relative bargain compared to other contemporary investments. In September 1998, the overall stock market, including non-franchise businesses, was trading at 22 to 23 times earnings. However, by longer-term historical standards, where the average P/E ratio was 17–18 to 1, WD-40 was less attractive. In the year prior to September 30, 1998, WD-40's stock price was flat to slightly down in the midst of a generally rising stock market. Overall, therefore, WD-40 was slightly disappointing, slightly cheap for a growing franchise business, and with a small market cap of \$416 million not closely followed by financial analysts. On the other hand, its product was widely known and admired. WD-40 might well have been worth looking at in an inflated market environment with few obvious bargains, but we should not be surprised if it turned out to be fairly valued. It was not an obvious bargain, unlike Magna International in March 2009.

The WD-40 Franchise

The first step in evaluating any franchise business is to assess the nature, magnitude, and sustainability of the business's competitive advantages /barriers to entry. Of the three fundamental sources of competitive advantage, we have already addressed the question of proprietary technology. WD-40 had no significant patents, learning-curve advantages, or proprietary process technologies.

What WD-40 did enjoy was a significant degree of customer captivity. Customer loyalty, built up over many years of advertising and customer satisfaction, ensured that a competitor would have a difficult time persuading users to try something new. Here the absence of

technological change benefited WD-40. Even WD-40's own "new" lubricant T.A.L.5 did not represent a sufficiently large improvement over the established product to attract customers. Relative to the price of the product, search costs were strikingly high. A can of WD-40 typically sold for under \$5 and lasted for many years. Trying an alternative product at a potential savings of \$1 per can would have minimal appeal since a single can lasted so long. The benefits of repeat purchases of a superior alternative were limited by the low replacement frequency. If an alternative proved unsatisfactory, then the consumer's loss would be the full \$4 purchase price. Even with a success probability of 75%, this was a losing proposition: an expected gain of \$.75 (75% times \$1) compared to an expected loss of \$1 (25% times \$4). Beyond this marginal gain at best, the time and trouble of finding an alternative would have had to be considered.

The economics for retailers were likely to be even worse. The 20% discount minimally necessary to attract WD-40 customers inevitably meant a lower retail margin or an even greater wholesale price reduction than 20% by the entrant brand. Given the unlikelihood that a competitive entrant would attract large numbers of WD-40 customers within a reasonable period of time, any resulting investment in promotion and shelf space at the retail level was almost certain to be unprofitable. The absence of retail support for alternative brands reinforced WD-40's customer captivity.

Finally, WD-40's market saturation meant that only a relatively few potential customers were not already WD-40 users. Their number, even if they were freely available targets for new entrants, were likely to be too small to support the fixed costs the newcomer would have to cover.

These hurdles for a potential competitor led naturally to the most important source of competitive advantage, economies of scale. In order to enter each separate geographic market for WD-40, an entrant would have had to establish a local sales and advertising presence sufficient to compete with WD-40. Advertising and promotion expenses for WD-40, with basically a 100% market share, were 10% of sales. These costs were fixed for each geographic market regardless of sales. Suppose the market in southern Florida was \$30 million per year. Then the 10% fixed marketing and sales costs would be \$3 million. If an entrant succeeded in capturing a 25% market share at a price 20% below WD-40's, its sales would be \$6 million (one quarter of a market of \$24 million in revenue at the 20% lower price). The \$3 million fixed overhead would then consume 50% of sales revenue. WD-40's cost of goods sold, its variable unit production and distribution costs for local

contractors, was at least 40% of sales. If the entrant had those same costs with a 20% lower price, then its cost of goods sold would be 50% of the its revenue (80% of WD-40's unit price divided by 40% of WD-40's unit price). Together, the fixed local sales overhead plus cost of goods sold would consume 100% of the entrant's revenue with nothing left for product management, administrative overhead, or local management supervision. This "opportunity" was not an economically attractive proposition for any entrant with a calculator or a spreadsheet.

Looked at another way, suppose an entrant could be economically viable at 30% market share and that the entrant would capture 0.5% market share per year, an optimistic figure for a product like WD-40 with a low price, low purchase frequency, and high search cost. Reaching a 30% market share would take 60 years. A 60-year path to economic viability would likely deter any rational potential entrant and bankrupt any irrational one. The strength of WD-40's economic franchise depended paradoxically on the small size of its market. Remember the best defended monopoly is a single store in a town too small to allow for a second one. WD-40's existing market looked in 1998 very much like a one-store town.

The financial history provides strong support for this argument. By 1998, which is our vantage point for this analysis, WD-40 had enjoyed immunity from entrants for over 40 years while earning outsized returns on invested capital. On August 31, 1998, WD-40 had a book value of equity of \$55 million, including \$13 million in net cash and an additional \$3 million invested in low-income housing, which at the time provided tax advantages. The book value of equity investment in the operating business was only about \$40 million. Net operating income after taxes of \$22 million represented a 55% return on equity. In theory, this high a return should have attracted swarms of competitors. The fact that no successful entry took place over 40 years indicates that powerful barriers to entry were in place.

Valuing WD-40 in 1998

Earnings power for WD-40

For a business like WD-40 with a sustainable franchise, the reproduction costs of the assets play only a secondary role in determining intrinsic value. But it is also true that earnings power values by themselves do not determine the attractiveness of a franchise investment. In contrast to

competitive businesses, growth in a franchise may add significant value. Nevertheless, estimating the earnings power is the proper starting point for evaluating franchise businesses (see Chapter 8). Table Ex3.1 presents a history of that relevant financial data for WD-40 from 1990 through 2002.

Throughout this period, revenues rarely declined. Even in the recession year of 1991, revenues fell by only about 1% and then quickly recovered in 1992. Given the absence of cyclical variation, current year revenues should be sustainable revenues. Margins were also highly stable and do not appear to vary with the business cycle. In 1991, operating margin actually rose. There were year-to-year margin fluctuations that should be averaged out to calculate sustainable margins. From 1990 to 1998, operating margins averaged 27.0%.

There was a slight downward trend in margins during this period, with margins in 1998 particularly low. We should be careful not to use the fact that margins in 1999 and 2000 continued to drop since that information was not available in 1998. However, it is notable that the decline seems to have coincided with a rising concern regarding stagnant sales in the early 1990s that led management to develop T.A.L.5 and acquire 3-in-One. Increased marketing expense to boost sales growth was reflected in rising levels of overhead expense relative to sales, a large part of the reason for the declining margins. If we believed that this increase in marketing spend was permanent, then sustainable margins from 1998 going forward would be well below the historical average of about 27%. If we thought the higher costs were temporary, then the historical average should apply. Since we were uncertain, we split the difference and used an estimated sustainable earnings margin of 25%, halfway between 27% and the more recent levels below 24%. Thus, our 1998 estimated sustainable operating earnings before accounting adjustments was \$36 million (25% of sustainable revenues of \$144 million).

Few accounting adjustments to this number were required. WD-40 took only a single exceptional charge of \$12.6 million between 1990 and 1998, the result of a litigation loss in 1994. This averaged out to \$1.4 million per year over the 9 years. Counterbalancing this loss, depreciation charges between 1990 and 1998 regularly equaled or exceeded total capital expense. Depreciation seems to have exceeded any reasonable estimate of replacement capital expense by at least \$1 million per year. Since this second adjustment is largely protected from income tax by the depreciation tax shield, it almost exactly offsets the post-tax cost of the average write-down of \$0.9 million (100% less the 37% corporate tax rate times a pre-tax cost of \$1.4 million). Our sustainable operating

Table Ex3.1 WD-40 Financials Performance, 1990–2002

	(\$ millions)												
	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002
Revenue	\$ 91	\$ 90	\$ 100	\$ 109	\$ 112	\$ 117	\$ 131	\$ 138	\$ 144	\$ 146	\$ 147	\$ 164	\$ 217
Gross Income	\$ 51	\$ 50	\$ 58	\$ 65	\$ 66	\$ 67	\$ 73	\$ 79	\$ 81	\$ 82	\$ 77	\$ 84	\$ 109
Gross Margin	56%	56%	58%	59%	59%	57%	56%	57%	57%	56%	52%	51%	50%
Operating Income	\$ 25	\$ 25	\$ 29	\$ 32	\$ 32	\$ 33	\$ 33	\$ 33	\$ 34	\$ 34	\$ 34	\$ 28	\$ 42
Operating Margin	27.7%	27.9%	29.5%	29.1%	28.5%	27.9%	25.5%	24.2%	23.8%	23.4%	23.3%	17.1%	19.4%
Tax Rate	38.5%	39.0%	38.7%	39.1%	38.0%	37.3%	36.2%	35.9%	36.7%	35.4%	34.0%	34.0%	31.0%
Net Income	\$ 15	\$ 15	\$ 18	\$ 19	\$ 13****	\$ 20	\$ 21	\$ 21	\$ 22	\$ 22	\$ 21	\$ 16	\$ 25
Net Margin	17.0%	17.0%	18.1%	17.7%	11.6%	17.5%	16.3%	15.5%	15.2%	15.0%	14.3%	9.8%	11.3%
Net Debt *	\$ (22)	\$ (25)	\$ (19)	\$ (22)	\$ (23)	\$ (25)	\$ (8)	\$ (12)	\$ (16)	\$ 3	\$ 8	\$ 75	\$ 84
Equity	\$ 39	\$ 41	\$ 45	\$ 46	\$ 42	\$ 45	\$ 47	\$ 51	\$ 55	\$ 56	\$ 53	\$ 55	\$ 83
Return on Equity	40%	38%	40%	42%	31%	46%	45%	42%	40%	39%	40%	29%	30%
Depre/Amort	\$ 0	\$ 0	\$ 1	\$ 1	\$ 1	\$ 1	\$ 1	\$ 2	\$ 2	\$ 2	\$ 3	\$ 5	\$ 2
Capex	\$ 0	\$ 1	\$ 1	\$ 1	\$ 1	\$ 1	\$ 1	\$ 1	\$ 1	\$ 1	\$ 2	\$ 1	\$ 1
Dividends	\$ 15	\$ 13	\$ 16	\$ 18	\$ 18	\$ 19	\$ 19	\$ 19	\$ 20	\$ 20	\$ 20	\$ 18	\$ 16
BuyBacks**	\$ (0)	\$ (0)	\$ (2)	\$ (1)	\$ (1)	\$ (0)	\$ (1)	\$ (2)	\$ (1)	\$ 1	\$ 4	\$ -	\$ (7)
Total Paid Out	\$ 15	\$ 13	\$ 14	\$ 17	\$ 17	\$ 18	\$ 19	\$ 18	\$ 19	\$ 21	\$ 24	\$ 18	\$ 9
Payout Ratio	98%	85%	79%	88%	132%	89%	87%	82%	86%	95%	114%	113%	37%
Shares outstanding (millions)***	15.1	15.1	15.3	15.3	15.4	15.4	15.4	15.6	15.6	15.6	15.5	15.5	16.1
EPS (\$/share)	\$ 1.03	\$ 1.01	\$ 1.18	\$ 1.26	\$ 0.84	\$ 1.33	\$ 1.38	\$ 1.37	\$ 1.40	\$ 1.41	\$ 1.35	\$ 1.02	\$ 1.54
DPS (\$/share)	\$ 1.01	\$ 0.86	\$ 1.07	\$ 1.15	\$ 1.15	\$ 1.21	\$ 1.24	\$ 1.25	\$ 1.28	\$ 1.29	\$ 1.29	\$ 1.16	\$ 0.99

*Includes LT Investments in Cash. Negative Figures indicate Net Cash

**Negative numbers = net shares sold.

***Fully diluted

****After \$12 million pre-tax, (\$8 million after tax) Write Down of due to a law suit in 1994.

Source: Data from WD-40 Annual report

earnings of \$36 million pre-accounting adjustments were slightly above post-adjusted figure. The average rate of corporate income tax incurred in the five years from 1994 to 1998 was roughly 37%. If we assumed that this was a sustainable future tax rate,² then the after-tax earnings power for WD-40's core franchise business in 1998 was \$23 million (63% of \$36 million).³ This represented a 5.7% earnings return on the \$403 million cost of buying the core business.

Earnings power value (2)

At this point in the first edition of the book, we proceeded to calculate earnings power value, just as we have done in this edition for Magna International and Hudson General. We simply divided the \$23 million earnings power by the cost of capital appropriate to an investment in the core WD-40 business. The cost of capital is the cost of voluntarily attracting the funds necessary to support the business, a weighted average of the cost of attracting both cheap debt financing and expensive equity financing. The weights in question should reflect the mix of actual financing for a firm's operations going forward, with the notable exception of companies that are overly reliant on debt. In these cases, the cost of the debt should properly include the potential cost of debt-related impairment of company operations, including the potential cost of bankruptcy. This cost is difficult to calculate and often overlooked. As a result, the calculated cost of capital for firms with a heavy reliance on low-cost debt will be inappropriately low. The standard solution to this problem is to reduce the amount of debt from actual historical levels to a level at which it would not be expected to significantly impair firm operations. In many finance contexts, this is assumed to be an optimal level of debt (e.g., 20% as opposed to an actual debt level of 80%) with an associated optimal level of equity (e.g., 80% as opposed to an actual level of 20%).

Some finance practitioners recommend applying a similar "ideal" capital structure in calculating cost of capital for firms with too little debt. But in this case, the "true" cost of debt is not being ignored and there is no such hidden cost to too much equity. WD-40 had generally

² Long-term averages for tax rates are inappropriate since it is only recent tax rates and trends that are likely to apply to the future.

³ Remember that we are looking at the earnings power of the core franchise business so that interest payments do not apply. We effectively include net interest paid by adding net debt (subtracting net cash) to the equity cost of buying that core franchise business.

been all-equity financed, often with negative net debt (positive net cash). Moreover, there was little indication that this would change in the future, short of significant activist intervention. Thus, for WD-40, the weighted average cost of capital was the cost of equity financing.

The cost of equity financing in 1998 lay between the contemporary cost of slightly risky (say BAA) debt and the cost of venture capital funding, typically the most expensive source of equity. In the fall of 1998, this range ran from 7 to 15%. Given its highly stable economic position and all-equity financial structure, WD-40's cost of equity should have fallen near the bottom of this range. Thus, our estimate of the cost of capital for WD-40 would have been roughly 8%, producing an earnings power value of \$288 million for the underlying operation (\$23 million divided by 8%). To this we must add the value of net cash and other extraneous investments of \$16 million for a final earnings power value of \$304 million for WD-40's equity. This figure corresponds to a value per share of \$19.36 (\$304 million divided by 15.7 million shares), which was below the market price of \$26.50. An orthodox Graham and Dodd investor would not have invested in WD-40. But for businesses such as WD-40, given the powerful nature of its franchise, growth does create incremental value. This added value might make an investment attractive for a value investor willing to include value-creating growth in the investment decision.

The Value of Growth for WD-40

In the first edition of this book, we devoted three short paragraphs to the possibility of profitable growth, for which we used a highly abbreviated DCF model. First, we noted that WD-40 paid \$1.28 per share out of its earnings of \$1.40. We assumed this amount captured the likely future sustainable cash distributions out of sustainable earnings (roughly also \$1.40 per share), since net share buybacks were negligible. Next we estimated without detailed analysis that recent earnings growth rates of about 3% would be sustainable in the future and that, with a constant payout ratio, dividends would grow with earnings. Finally, with our 8% cost of capital and 3% growth rate, we capitalized the \$1.40 per share payout at 20 times ($1/(8\% - 3\%)$). This yielded a per-share value of \$25.60, still slightly below the market price of \$26.50.

The shortcomings of this calculation, then and now, are its painful uncertainty. If the growth rate in earnings were 4.5%, a mere 1.5% off our 3% estimate, a figure closer to the long-term earnings growth rate,

then the capitalization factor would have been 28.5 (1 divided by 3.5%) for a per-share value of \$36.48, a nearly 30% margin of safety given the contemporary price of \$26.50 per share. On the other hand, an earnings growth rate of 1.5%, equal to the most recent few years, would lead to a multiple of 15 (1 divided by 6.5%), for a price of under \$20 per share, leaving WD-40 a truly unattractive investment. As we pointed out at the beginning of Chapter 8, this indeterminacy is fundamental to growth stock evaluation. In order to make useful growth stock investment decisions, we should look at returns, not values.

WD-40 Returns

Analysis of sustainable future returns begins with the future cash return. This is the quantity of earnings actually distributed to stockholders out of the 5.7% sustainable earnings return that we calculated above. This figure itself depends on the distribution policy of WD-40's management. In six years through the end of August 1998, WD-40 distributed a net \$18 million per year on average to shareholders in dividends minus, in this case, net equity sales (see Table Ex3.1). Net income averaged \$21 million per year after making appropriate adjustments for over-depreciation and the one-time charge in 1994. Net cash remained roughly constant; the resulting 86% payout ratio was financed from earnings and therefore likely sustainable. A payout of 86% of the 5.7% sustainable earnings return represented a roughly 5% cash return going forward to investors in WD-40 stock purchased on September 30, 1998, at \$26.50 per share. To this we must add the returns from growth, which would have arisen from two basic sources: (1) organic growth in sales and earnings of WD-40's existing product lines, distribution arrangements, and productivity improvements; and (2) returns to active reinvestment initiatives such as the purchase for \$15 million of 3-In-One products in December 1995. We will consider each of these sources of the value of growth in turn.

Organic Growth Returns

Table Ex3.2 shows historical nominal growth rates in WD-40 revenues and earnings relative to overall US inflation rates.⁴ For the period prior to

⁴ These US rates closely tracked inflation rates in WD-40's major overseas markets.

Table Ex3.2 WD-40 annual growth rates, 1980–95.

Period	1980–85	1986–90	1990–95	1995–98	1980–95
Sales	10.2%	9.9%	5.2%	7.3%	8.4%
US inflation	4.9%	4.2%	2.8%	2.2%	3.9%
Real sales	5.3%	5.7%	3.4%	5.1%	4.5%
Net income	1.1%	11.7%	9.8%	2.2%	7.4%

Source: Data from WD-40 Annual report

the acquisition of 3-In-One, 1980–95, nominal revenues grew by 8.4% per year, real sales grew by 4.5% per year, and net income grew by 7.4% per year. Growth in sales was entirely internally generated by a combination of rising organic demand, expansions to new geographic markets financed out of current income, and, for profits, from daily improvements in operating efficiency. Capital investment was minimal and the one active investment initiative, the development of T.A.L.5, was a flop. We consider this growth organic in the sense that it arose naturally in the ordinary course of doing business. Profit growth was slightly below revenue growth, but that was largely attributable to developments prior to 1985. If the extraordinary spending on T.A.L.5 were to be eliminated, the historical growth of net income in tandem with revenue should resume.

The question then is what would revenue growth going forward look like. To answer that, we need to focus on the underlying determinants of WD-40's revenue growth. Throughout the late 1990s, WD-40 exports from the US, UK, and Australia to emerging economies rose sharply. Demand from those areas was in no sense saturated, meaning that WD-40's growth would not be limited by increases in the developed economies of Europe and North America. WD-40 appealed to households across the income spectrum and would not be greatly affected by secular changes toward greater income inequality. On the other hand, WD-40 was a product rather than a service and would suffer from the global trend in consumption away from goods and toward services. Taken together, these forces suggested that demand for WD-40 going forward would grow slightly more slowly than global GDP, perhaps 4% in real terms compared to an expected real GDP growth of 5% from 1998 going forward. This GDP figure would correspond to the population growth of 1½% per year and global productivity growth of 3½%. On top of this real growth, global inflation was likely to run at about 1.5%, continuing a deceleration that began in the early 1980s. Nominal GDP growth of

6 to 6.5% per year should translate into growth in demand for WD-40 of at most 4.5%. That would closely resemble the pre-1995 historical growth rate of 4.5%. If we allow for the further possibility that a fraction of the recent decline in margins, prior to 1998, would be permanent, an organic growth rate in WD-40 net income of 4% a year would be a reasonable estimate.

The investment required to support this organic growth would consist almost entirely of working capital. WD-40 outsourced all its manufacturing and distribution. Historical accounts receivable and inventory had amounted to roughly 20% of sales, partially offset by accounts payable and accrued liabilities equal to 6% of sales. Using these ratios, net working capital of \$.14 would be required for each extra dollar of sales. The 1998 base level of \$144 million in sales growing at 4.5% would mean an extra \$6.5 million in sales. The additional net working capital required would be about \$1 million (14% of \$6.5 million), or about 4.5% of sustainable 1998 earnings of \$21 million. If 86% of earnings continued to be distributed to shareholders and 4.5% was needed to support organic growth that would leave roughly 10% of earnings for active reinvestment. To sum up: on top of the cash return of 5%, organic growth should contribute a further 4% annual increase in value, a total of 9%, while leaving 10% of earnings to generate further value through active reinvestment.

Active reinvestment

After distributing \$19 million to shareholders, 86% of \$22 million, and using \$1 million to support organic growth, management would be left with \$2 million sustainable earnings per year for active reinvestment. In the case of WD-40, this money had recently gone to the acquisition of new product lines. Prior to 1998, the sole acquisition had been 3-In-One in December 1995. Including acquired debt, WD-40 paid \$16 million for brand value and other intangibles plus a small amount of inventory. Assuming that WD-40 could employ these assets with the same effectiveness as had Reckitt and Colman, the prior owner, incremental 3-In-One revenues and earnings for WD-40 should be the same as they were before the sale. For 1994 and 1995, Reckitt and Colman reported average 3-In-One sales of roughly \$12.5 million and average operating income after amortization of roughly \$6 million. The reported \$6.5 million in cost included cost of goods sold and direct marketing cost equal to 15% of sales with no provision for other overhead. For a

product very similar to WD-40, the implied operating margins of 48% seems improbably high compared to WD-40's pre-1995 operating margins of 27%. With 3-In-One operating margins adjusted to that level, the average operating income acquired by WD-40 would have been \$3.4 million (27% of \$12.5 million). At a tax rate of 37%, after-tax incremental profit would have been \$2.1 million, a roughly 13% annual return on WD-40's \$16 million investment before attributing any value to growth in 3-In-One revenues and profits.

If 3-In-One were indeed similar to WD-40, then organic growth should add considerably to the returns from the acquisition. However, the limited history published before the deal revealed flat to slightly declining sales and margins. It seems realistic to ignore growth in evaluating the 3-In-One acquisition. The 13% return on investment, even with no growth, was well above a likely cost of capital of 8 to 9%, similar to WD-40's. This return represents a value creation factor of 1.5 (13% divided by 8.5%) on the acquisition. If this transaction reflected WD-40's skills in actively investing retained earnings, then each dollar reinvested should create \$1.5 of value for WD-40. Thus, the \$2 million in sustainable reinvested annual earnings would add \$3 million in annual value, a further return of about 0.75% on the \$403 million purchase cost of WD-40.

For this reinvestment scenario, the total return on an equity investment in WD-40 at \$26.50 per share would be almost 10%: a 5% cash return, 4% from organic growth, and 0.75% from active reinvestment. The low level of this active reinvestment return is due to the low level of actively reinvested earnings. Compared to WD-40's cost of capital of 8%, this leaves roughly 25% as the margin of safety on returns.

In the late 1990s, average equity returns in the US equity market were around 6%. The market multiple of sustainable earnings was roughly 25 times, a real earnings return of 4%. Inflation added another 1.5% to reach nominal estimated returns of 5.5%. Alternatively, we could have estimated the market return from cash payouts plus overall growth. Average payouts on US equities were 2%; with nominal growth adding perhaps another 4.5%, the total prospective nominal US returns were 6.5%. Averaging the two gives an estimated market return of 6%. In comparison, a return close to 10% on low risk equity such as WD-40 would have seemed attractive. But value investors should be wary of these kinds of return comparisons in euphoric markets like that of the late 1990s. Short-term contemporary return comparisons are relative value comparisons. True value investor should compare the 10% return to the long-term cost of capital estimate embodied in our 8% figure. Even with

that caveat, a careful accounting for growth in terms of returns reveals that WD-40 in late 1998 was a much more attractive opportunity than indicated by the abbreviated DCF analysis in our first edition.

This return approach turns out to have been remarkably accurate. Table Ex3.3 presents realized returns to a WD-40 equity purchase on September 30, 1998, over the following 5-, 10-, and 15-year periods. It also includes estimated returns for the period, which we computed before looking at the actuals (no peeking). The only substantial difference is in the 10-year period that terminated on September 30, 2008, in the middle of the financial crisis. Over the entire 15 years, the S&P 500 index returned slightly less than but close to our 6% estimate, which by historical equity return averages is not what Graham and Dodd would have regarded as a satisfactory return. However, these figures overstate the true margin of safety of the WD-40 investment since the SP returns are unusually low. They highlight the danger of relative return decision-making, particularly when investing in richly valued markets.

A second piece of these return comparisons also overstates the attractiveness of a WD-40 investment in 1998. Looking forward, there was a low but not zero probability that changing economic conditions would undermine WD-40's franchise position. As we discussed above, barriers to entry into WD-40's markets appeared formidable. The retail infrastructure through which WD-40 sold its products looked highly stable. Internet-based retail, whose consequences we will discuss below, did not yet appear as a threat. Technology did not seem likely to undermine

Table Ex3.3 Estimated and realized returns for WD-40 (percent per annum).

		5 years	10 Years	15 Years
	Estimated	Actual		
Purchase September 30, 1998				
WD-40	9.7%	9.2%	4.9%	10.8%
SP 500	6.0%	1.4%	3.4%	5.3%
Excess return	3.7%	7.8%	1.5%	5.5%
Purchase September 30, 2000				
WD-40	9.0%	8.9%	9.0%	13.7%
SP 500	5.0%	-2.0%	-0.8%	3.8%
Excess return	4.0%	10.9%	9.8%	9.9%

Source: Data from WD-40 Annual report

the utility of WD-40's products, which by 1998 had enjoyed an unchallenged market position for roughly 50 years. If we use as a rule of thumb that future franchise life should equal existing franchise life, we would have estimated a franchise fade rate for WD-40 of 1.4% (72 divided by 50 years to get a franchise half-life). Our 2% margin of safety slightly exceeded this potential rate of loss, but taking this fade rate into account would have made WD-40 a significantly less attractive investment.

As the future unfolded between 1998 and 2013, the WD-40 franchise remained intact. The temptation, therefore, is to treat the fade probability as zero and to accept the realized returns in Table Ex3.3 as an accurate measure of the likely return on the original WD-40 investment. The ex-ante odds certainly favored this outcome. With a half-life of 50 years, WD-40's franchise was unlikely to be undermined by events within 15 years. There was, however, an undeniable element of luck in this outcome. Today, as retail through the Internet threatens to become the dominant marketing channel for WD-40, the future of WD-40's franchise looks far more uncertain than it did in 1998. This evolution might easily have arrived 5 or 10 years earlier. The fact that it did not may represent good fortune for WD-40, but it was hardly inevitable. Thus, the returns in Table Ex3.3 do not fully reflect the cost of franchise fade and to that extent overstate the return to the 1998 WD-40 investment.

WD-40 on September 30, 2000

From \$26.50 on September 30, 1998, the price of a WD-40 share fell to \$21.88 on September 30, 2000. At a time when the US stock market as a whole was doing extremely well, the annual return to a WD-40 investment over these two years was negative 9%. A stock price decline of 9% per year was partially offset by dividends. In the near term, the 1998 investment in WD-40 would have been a disappointment. One of the key assumptions underlying our 1998 valuation turned out not to be accurate in the short run. We assumed that management would not persist in the value destroying pursuit of growth, whether through acquisitions like 3-In-One or internal developments like T.A.L.5. We were wrong.

In April 1999, WD-40 acquired LAVA brand soap, a heavy duty hand cleaner. A year later, it bought Solvol hand cleaner. In April 2001, management paid \$72.9 million for three household products—2000 Flushes, X-14, a toilet bowl and hard surface cleaner, and Carpet Fresh. These acquisitions, the results of a policy of unfocused growth that became apparent between 1998 and 2000, were both more expensive

and less closely related to WD-40's existing products than the 3-In-One acquisition. The consequences are evident in the operating performance data in Table Ex3.1. From FY 1998 to FY 2000, both gross and operating margins declined noticeably while revenues increased only slightly. The decline in tax rates moderated the drop in net income, which still fell from \$22 million to \$21 million. Payout ratios continued to be high, at least until the April 2000 acquisition. But net cash fell by \$24 million from \$16 million of net cash to \$8 million of net debt, so that high payouts were sure to be subject to balance sheet pressures in the future. Management continued to stress the pursuit of growth.

There were indications, even prior to September 1998, that this might happen. As we noted above, operating margins had declined from roughly 28% in FY 1994 and FY 1995 to 24% in 1998, as marketing and overhead expenses increased relative to sales in pursuit of more rapid revenue growth. These efforts had been disappointing. After an earlier unsuccessful growth policy from 1980–5, WD-40's management had refocused on operational efficiency, and margins had recovered. In our valuation, we assumed something similar would happen going forward from September 1998. Our error in making this judgment naturally raises the question of whether, at the reduced price of \$21.88 in September 2000, WD-40 remained a sound investment.

If management persisted in pursuing ill-focused growth, earnings power was likely to fall from \$22 million to \$20 million. The cost of acquiring WD-40 for its operations was now \$340 million (15.6 million shares times \$21.88 per share plus \$8 million of net debt). The decline in the capital necessary to buy WD-40 made the sustainable earnings return 5.75%, very close to the 5.7% earnings return in September 1998. The pay-out ratio in 2000 was actually more than 100% of earnings, which did not appear sustainable. Let us assume that pay-outs decline from 86% of earnings power to 76% of earnings power. This yields a 4.5% cash return. Organic growth should continue to be 4% per year with an investment cost equal to roughly 4% of earnings. This would leave 20% of earnings for active reinvestment (100% minus 76% payout minus 4% organic growth investment). The quality of the active investment had undoubtedly dropped by between 1995 and 2000. It seems unlikely that acquisitions in the areas of toilet, hand, and rug cleaner would be as accretive in value as the 3-In-One acquisition. Exactly how much value would be destroyed per dollar invested would have required a detailed acquisition-by-acquisition analysis, similar to what we did for 3-In-One. But let us simply assume that each dollar invested produces only about

\$0.50 of value.⁵ Then the \$4 million of actively invested earnings (20% of \$20 million) produces \$2 million of additional value (50% of \$4 million). That, in turn, represents a further 0.6% return on the \$340 million market purchase price (\$2 million divided by \$340 million).

A rough estimate of the total return from an investment in WD-40 in 2000 is thus 9% per year (4.5% cash return, 4% organic growth and 0.6% active investment return). The decline in management performance was largely balanced by the decline in WD-40's market price. Meanwhile, the overall level of US stock prices had risen, lowering a reasonable estimate of the S&P 500 return from 6% to about 5%. In relative terms, WD-40 had by September 2000 become slightly more attractive (see Table Ex3.3). But, in critical absolute return terms, with its likely fade rate taken into account, WD-40 now represented a return just equal to its cost of capital, leaving no margin of safety.

As things worked out, WD-40 in the long run outperformed these reduced expectations. At the end of 5 and 10 years, actual returns were almost exactly 9%; the 15 year return was over 13%. In the long run, WD-40 regained its touch by returning to what it did best and abandoning its unfocused growth efforts. From 2012 to 2015, even as WD-40's revenues grew by only 3.3% per year, close to the organic growth rate which had slowed slightly with the decline in global GDP growth, its operating margins rose from 15.1% to 17.3%. By 2018, the operating margins were 19.2% and pay-outs were averaging over 100%; these were financed from a rising level of net debt (see Table Ex3.4). This change illustrates an essential point about franchise investing. The long run matters, and in the long run, unless there is strong evidence to the contrary, stability is more likely than change.

WD-40 in 2018

On September 28, 2018, the last trading day in September, WD-40 closed at a price of \$172.10 per share. Operating margins appear to have stabilized at 19%, and revenues were \$409 million. As a result of a dramatic recent reduction in corporate tax rates, corporate taxes in 2018 were only slightly above 12% of net income. In the long run, corporate tax rates appear unlikely to continue at such a low level, if only because US federal tax rates reductions will be partially offset by rising

⁵ This is a general measure of the costs of unrelated acquisitions.

Table Ex3.4 WD-40 financial performance, 2003–18.

	(\$ millions)																
FY Aug 31	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	
Revenue	\$ 238	\$ 242	\$ 263	\$ 287	\$ 308	\$ 317	\$ 292	\$ 322	\$ 336	\$ 343	\$ 369	\$ 383	\$ 378	\$ 381	\$ 381	\$ 409	
Gross income	\$ 122	\$ 126	\$ 129	\$ 138	\$ 149	\$ 148	\$ 145	\$ 165	\$ 168	\$ 169	\$ 189	\$ 199	\$ 200	\$ 214	\$ 214	\$ 225	
Gross margin	51%	52%	49%	48%	48%	47%	50%	51%	50%	49%	51%	52%	53%	56%	56%	55%	
Operating income	\$ 50	\$ 45	\$ 47	\$ 46	\$ 49	\$ 43	\$ 40	\$ 55	\$ 54	\$ 52	\$ 57	\$ 64	\$ 65	\$ 71	\$ 76	\$ 79	
Operating margin	20.9%	18.6%	17.9%	16.0%	15.9%	13.6%	13.7%	17.1%	16.1%	15.2%	15.4%	16.7%	17.2%	18.6%	19.9%	19.3%	
Tax rate	34.0%	34.0%	35.2%	34.3%	33.1%	34.3%	31.3%	32.6%	32.0%	30.3%	30.1%	30.5%	29.0%	27.7%	29.1%	13.3%	
Net income	\$ 29	\$ 26	\$ 28	\$ 28	\$ 32	\$ 28	\$ 26	\$ 36	\$ 36	\$ 36	\$ 40	\$ 44	\$ 45	\$ 53	\$ 53	\$ 65	
Net margin	12.0%	10.6%	10.6%	9.8%	10.4%	8.8%	8.9%	11.2%	10.7%	10.5%	10.8%	11.5%	11.9%	13.9%	13.9%	15.9%	
Net debt*	\$ 53	\$ 56	\$ 27	\$ 30	\$ (8)	\$ 1	\$ (14)	\$ (55)	\$ (47)	\$ (26)	\$ (18)	\$ 9	\$ 6	\$ 15	\$ 37	\$ 47	
Equity	\$ 105	\$ 113	\$ 130	\$ 156	\$ 168	\$ 164	\$ 173	\$ 197	\$ 201	\$ 186	\$ 180	\$ 169	\$ 158	\$ 140	\$ 139	\$ 156	
Return on equity	27%	23%	22%	18%	19%	17%	15%	18%	18%	19%	22%	26%	28%	38%	38%	42%	
Dep./Amort.	\$ 2	\$ 2	\$ 3	\$ 4	\$ 4	\$ 4	\$ 4	\$ 4	\$ 4	\$ 5	\$ 5	\$ 6	\$ 7	\$ 7	\$ 7	\$ 8	
CapEx	\$ 2	\$ 2	\$ 3	\$ 3	\$ 2	\$ 6	\$ 3	\$ 2	\$ 3	\$ 3	\$ 3	\$ 4	\$ 6	\$ 4	\$ 20	\$ 12	
Dividends	\$ 13	\$ 14	\$ 14	\$ 15	\$ 17	\$ 17	\$ 16	\$ 17	\$ 18	\$ 18	\$ 19	\$ 20	\$ 22	\$ 24	\$ 27	\$ 30	
Buy backs**	\$ (6)	\$ (8)	\$ (3)	\$ (7)	\$ 8	\$ 13	\$ (1)	\$ (4)	\$ 21	\$ 33	\$ 27	\$ 42	\$ 28	\$ 31	\$ 30	\$ 22	
Total paid out	\$ 8	\$ 6	\$ 11	\$ 8	\$ 25	\$ 30	\$ 15	\$ 13	\$ 39	\$ 51	\$ 46	\$ 62	\$ 50	\$ 55	\$ 57	\$ 52	
Payout ratio	27%	22%	39%	29%	78%	107%	58%	36%	108%	142%	115%	141%	111%	104%	108%	80%	
Shares outstanding (millions)***	16.6	16.9	16.8	16.9	17.3	16.8	16.7	16.7	17.0	16.1	15.6	15.2	14.7	14.4	14.1	14.0	
EPS (\$/share)	\$ 1.71	\$ 1.50	\$ 1.65	\$ 1.66	\$ 1.83	\$ 1.64	\$ 1.58	\$ 2.15	\$ 2.14	\$ 2.20	\$ 2.54	\$ 2.87	\$ 3.04	\$ 3.64	\$ 3.72	\$ 4.64	
DPS (\$/share)	\$ 0.80	\$ 0.80	\$ 0.84	\$ 0.88	\$ 0.97	\$ 1.00	\$ 1.00	\$ 1.00	\$ 1.08	\$ 1.14	\$ 1.22	\$ 1.33	\$ 1.48	\$ 1.64	\$ 1.89	\$ 2.11	

*Includes LT investments in cash. Negative figures indicate net cash.

**Negative numbers = net shares sold.

***Fully diluted

Source: Data from WD-40 Annual report



state corporate income tax rates. Because the trend toward lower corporate income tax rates has been long-lived and global in nature, we will assume a sustainable corporate tax rate of 20%. These figures produce an earnings power estimate for WD-40 in 2018 of \$62 million (\$409 million times 19% times 80% after-tax). Given management's return to its long-standing policy of distributing a large fraction of earnings and to focusing retained earnings on improving core business operations, we will assume that all earnings, aside from those required to support organic growth, are either distributed to stockholders or, if actively reinvested, earn a return equal to WD-40's cost of capital. In terms of value delivered to shareholders, these alternatives amount to the same thing. Organic growth, which given the low level of active reinvestment should have tracked overall growth in the past few years, appears to have declined from 4% to roughly 3% per year, mirroring slowing global economic growth in both developed and developing countries. At 3% organic growth, WD-40's 2018 sales growth should be \$12.3 million per year (3% of \$409 million). Since each dollar of extra revenue continues to require about \$.15 of net investment, \$2 million of retained earnings must be devoted to supporting organic growth. That leaves \$60 million of sustainable earnings for distribution to investors.

The cost of buying WD-40's core business at the end of September, 2018, consists of \$2,392 million in equity (\$172.10 times 13.9 million shares outstanding) plus \$37 million in net debt, for a total enterprise value of \$2,429 million. The \$60 million sustainable annual distribution to shareholders then represents roughly a cash return of about 2.5% (\$60 million divided by \$2,429 million). Adding 3% for organic growth yields a total return of 5.5%, since we are assuming that active reinvestment will be negligible. This figure is clearly below our long run cost of capital of 8%, meaning that intrinsic value is below market value and the 3% organic growth return exaggerates the true organic growth return, further reducing our "true" return.⁶

In addition, the shift of retail to the Internet has almost certainly narrowed WD-40's moat. Rival products no longer need to pursue the difficult and costly process of in-store placements and distribution. They can now appeal directly to masses of consumers. At the same time, online

⁶ Remember organic growth rates multiply intrinsic value (V), not market value (M), and must therefore be multiplied by V divided by M to calculate the "true" organic growth contribution to returns. If our calculated return is below the cost of capital, then V will be less than M . Thus, we multiply the organic growth rate by a factor (V divided by M) which is less than one in order to calculate the true contribution of organic growth to value.

user reviews should provide some measure of assurance about the quality of each new rival product. Finally, Internet retailing greatly facilitates price comparisons and increases price competition. As these changes become embedded in markets and consumer behavior, the potential for WD-40 franchise erosion should rise markedly. In our terminology, “fade” rates could well increase from 1–1.5% per year to as much as 2.5–3% (from a franchise half-life of 50 years to one of 25–30 years). With no margin of safety left in the current WD-40 stock price, there is no protection against franchise “fade” of this magnitude. At the end of 2018, WD-40 is clearly overvalued. The technological innovations that can threaten even a well-entrenched franchise like WD-40 may emerge not from changes in product improvement or process technology but from the infrastructure of sales and distribution.

Example Four: Intel

By any measure, Intel is one of the great success stories in business history. We first wrote that sentence in the first edition of this book in January 2000, when Intel, which had been founded in 1969, had a market capitalization of \$275 billion. It stood at the pinnacle of the semiconductor industry, which was to the world economy in the last quarter of the 20th century what the railroads had been to the 19th century and electricity and automobiles to the early 20th century: the great engine of growth and productivity. In 1999, Intel sales were \$29 billion, having grown from \$1.4 billion in 1985, a compound annual rate of roughly 24%. Profit margins on sales had averaged more than 20% and returns on equity, properly measured, more than 40% during those 15 years. Buying Intel stock at the IPO in 1971 would have been a stroke of genius. So would an investment 10 years later in 1981, and 10 years after that in 1991. Twenty-seven years later, at the end of 2018, Intel remained an outstanding if not *the* outstanding company in the semiconductor industry. Sales in 2018 exceeded \$70 billion. Margins on sales and on equity, again properly measured, were both over 30%. Yet Intel's market capitalization in December 2018 was only \$218 billion, well below its January 2000 level. An investment in Intel stock over these 19 years would have produced average annual return of just 2%. Notwithstanding the company's extraordinarily strong long-term performance, there were many times during these years when an investment in Intel would have been a disappointment. The critical question

is whether the value-investing process and the valuation tools we have developed in this book can enable a value investor to navigate successfully the turbulent waters of a stock such as Intel. This chapter attempts to provide an answer by applying the full range of those tools to a decision to invest in Intel at several critical points in its history. For every decision point except the most recent, we have the advantage of being able to compare our estimated returns with those actually realized by an investor, a potentially chastening experience for everyone who makes predictions.

Intel History: The Very Short Version

The series of events that gave birth to Intel read like a chapter from Genesis. In the beginning was Bell Labs. There, in the late 1940s, John Bardeen, Walter Brattain, and William Shockley invented the transistor, a solid state replacement for the vacuum tube that had been a key component of the ENIAC generation of computers. The three earned the Nobel Prize for their efforts in 1956. In that year, Bell Labs begat Shockley Laboratories, as Shockley left to set up his own firm. One year later, eight of Shockley's best engineers departed to start a new firm with the financial backing of Sherman Fairchild. And so Shockley Laboratories begat Fairchild Semiconductor. At Fairchild, Robert Noyce managed to combine multiple transistors on a single piece of silicon, inventing what became known as the integrated circuit. Fairchild, by contrast, began to disintegrate as talented employees left to strike out on their own. In 1967, Fairchild begat National Semiconductor when Charles Sporck founded this new firm. One year later, Noyce and Gordon Moore, head of research and development, decided it was time for them to leave, frustrated that their most promising innovations seldom came to fruition. They made a telephone call to Arthur Rock, a venture capitalist before the term was invented, who raised enough money in two days for them to start on their own. Thus, Fairchild Semiconductor begat Intel.

Intel's initial business was designing and manufacturing computer memory chips. Although it developed its first microprocessor, the 4004, in 1971, the company's bread and butter was integrated circuits that replaced magnetic cores as the memory components of mainframe computers. The chips were smaller, faster, and cheaper than magnetic cores. These three virtues, which became the dynamic behind continual innovation and growth in the computer and related industries, made

the memory business very profitable for Intel. The founders had not discovered the particular technology that made that early generation of memory chips successful, nor was Intel the only company in the neighborhood—soon to be known as Silicon Valley—able to produce them. But it successfully combined product design, process engineering, and customer service so that it quickly emerged as the largest player in the memory chip game. Fairchild Semiconductor, from which both the engineers and the inventions had sprung, became less significant.

In 1971 Intel raised \$7 million by selling 300,000 shares in an IPO. It lost money on its operations in its first year, but then began a succession of profitable years that lasted until 1985. The company flourished by turning silicon, an inexpensive raw material, into valuable and essential finished products: memory chips and later microprocessors. It accomplished this alchemy through the organized application of human capital in the form of scientific and engineering knowledge.

We will telescope the first three decades of Intel's history into a few sentences. It made a lot of money in the 1970s on memory chips, even though it was often not the market leader or the most efficient producer. Around 1980 one of its microprocessors, the 8088, was chosen by IBM to be the central processing unit of the soon to be launched IBM Personal Computer. At the same time that the PC revolution took off, Intel started to lose money on its memory chip business. Large Japanese conglomerates like Hitachi and Fujitsu beat Intel in every aspect of the dynamic random access memory game; they produced better chips (fewer duds) more cheaply and then sold them for less money to manufacturers with whom they had worked more closely.

After considerable soul searching—Intel's history was interwoven from birth with that of the memory chip—Gordon Moore and Robert Noyce decided in 1985 to abandon the memory business. Since then, Intel has concentrated on microprocessors, both the original PC chip and its much more powerful successors, and on a range of other processors for network, industrial, and computer uses. The company has been continually profitable since it left the memory chip period in the mid-1980s. It has continued to grow, although not at the rate as in its early years when the silicon world was young. Throughout its history, Intel has invested heavily in developing its intellectual capital. Its R&D expenses averaged over 11% of sales for the period 1971–1998. And Intel defended its intellectual capital. It went to court repeatedly to contain rivals and former employees when it thought they were infringing on its patent-protected domains.

Table Ex4.1 Intel sales and income growth, 1975–1998 (millions)

Year	Net sales	Net income
1975	\$ 137	\$ 16
1980	\$ 855	\$ 97
Change per year	44%	43%
1985	\$ 1,365	\$ 2
Change per year	10%	–56%
1990	\$ 3,921	\$ 650
Change per year	23%	234%
1995	\$ 16,202	\$ 3,566
Change per year	33%	41%
1998	26,273	6,068
Change per year	13%	14%
1975 through 1998	24%	28%

Intel’s giddy success in growing its business and keeping it profitable, with minor exceptions, is apparent from a few numbers and charts. Table Ex4.1 and Figure Ex4.1 display these results. In the figure, the scale is logarithmic; each horizontal bar represents a 10-fold increase from the line below.

Several aspects of this history are common to many rapidly growing franchise businesses like Intel. They make Intel a particularly useful example for illustrating how to value such opportunities.

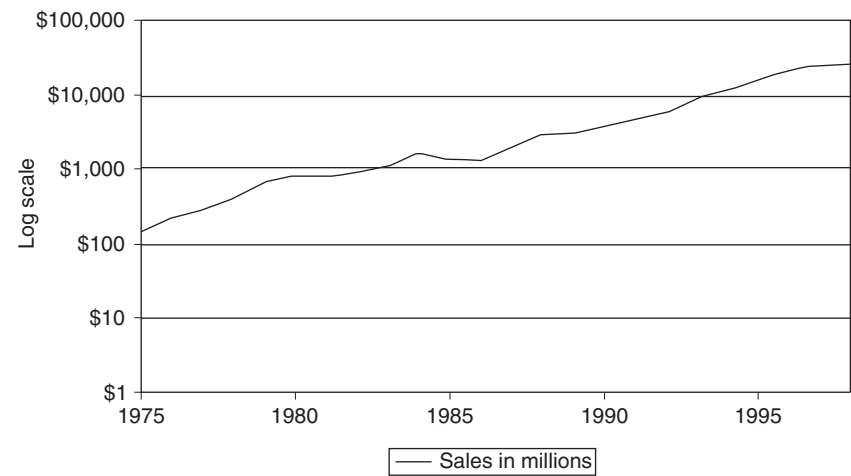


Figure Ex4.1 Intel revenue growth, 1975–1998.

Source: Data from Intel annual report

First, new industries, especially those in high technology, are far from stable. Intel has never been WD-40. It began life as a maker of memory chips, but this episode was cut short by the entry of powerful global competitors with equivalent technology. At inception, the memory chip market was small and required highly specialized expertise. Under these circumstances, early entrants such as Intel enjoyed temporary advantages. They had a head start in moving down the learning curve to produce cheaper and more reliable chips. In the confines of the small initial market, they possessed significant scale advantages in R&D, marketing, and distribution. Finally, they were well positioned with customers who depended critically on a reliable supply of high-quality memory chips. For these customers turning to untried alternative suppliers was highly risky. As a result, early entrants were protected from competition by strong customer captivity.

However, as the market grew, the technology became more widely available and customers more sophisticated. Early competitive advantages dissipated. A larger market created more room for entrants to achieve viable scale at relatively low market share. Technology diffusion eliminated any proprietary, learning-based supply advantages. Increasing customer sophistication reduced customer captivity. The consequence, as Intel experienced, was intense global competition from a number of highly efficient firms. The easy profitability of Intel's early years could not be sustained. By 1985 and 1986, Intel was suffering serious operating losses. (See Table Ex4.2 for Intel's operating history during these years.)

Intel survived this change, unlike many memory chip makers, due to a number of fortunate circumstances. At the time that memory

Table Ex4.2 Intel's operating history, 1984–1990 (millions).

Year (Dec 31)	1984	1985	1986	1987	1988	1989	1990
Revenue	\$ 1,629	\$ 1,365	\$ 1,265	\$ 1,907	\$ 2,875	\$ 3,127	\$ 3,921
Gross margin	46%	31%	32%	45%	48%	45%	51%
Operating income	\$ 250	\$ (60)	\$ (195)	\$ 246	\$ 594	\$ 557	\$ 858
Operating margin	15.3%	NM	NM	15.1%	36.5%	34.2%	52.7%
Net income	\$ 198	\$ 2	\$ (203)	\$ 248	\$ 453	\$ 391	\$ 630
Net income Margin	12.2%	0.1%	NM	13.0%	15.8%	12.5%	16.1%
Equity	\$ 1,360	\$ 1,421	\$ 1,245	\$ 1,276	\$ 2,080	\$ 2,549	\$ 3,592
Return on equity	14.6%	0.1%	NM	19.4%	21.8%	15.3%	17.5%

Source: Data from Intel annual report

chip profitability disappeared, Intel had a significant side business making microprocessors for the IBM PC and clone machines. By the mid-1980s it had become, along with Microsoft's MS-DOS operating system, the personal computer industry standard. Second, Intel had sufficient resources to support a transition from memory to microprocessor chips. Third, Intel had a brave and visionary management that abandoned not only its now non-viable memory chip business but also a promising initiative in parallel computing microprocessors in order to focus intensely on PC microprocessor chips. Investments in less favored memory chip firms would have not earned Intel-like returns. As a first step, investing successfully in nascent rapidly growing businesses such as Intel requires both highly specialized knowledge of the relevant industry economics and management capabilities, or extremely good luck.

Even after making the transition to PC microprocessors, Intel's success, although it looks inevitable in retrospect, was hardly guaranteed. Achieving a sustainably dominant position in the microprocessor market depended both on the nature of that business and the skill with which Intel's management addressed opportunities. If the microprocessor market were not to follow the path of memory chips, it would have to possess several distinct characteristics. First, as semiconductor technology advanced, the complexity of memory chips did not increase proportionately. Manufacturing leading-edge memory chips did not require steadily increasing R&D expense. In contrast, the demand for microprocessor functionality—what microprocessors were asked to do—rose alongside of semiconductor advances. PCs required microprocessors able to perform more complex operations, more rapidly, in small packages, and using less energy in order to keep up with user demands. As a result, the scale of required microprocessor R&D tended to increase over time relative to the size of the microprocessor market. This meant that the space available for viable competitors was decreasing rather than increasing, unlike the situation for memory chips.

Second, effective memory chips could be designed without reference to particular software programs and operating system. Microprocessor efficiency depends importantly on the sequences in which and the frequency with which particular operations are performed. Component and subsystem placements within a microprocessor must be optimized to facilitate the most frequent operations and sequences of operations. These in turn depend on the software environment in which a microprocessor operates. Thus, Intel's experience interacting with Microsoft programs, which constituted the dominant PC software environment,

gave Intel important advantages in microprocessor design that it had not enjoyed in memory chips.

Third, to nurture these potential competitive advantages to establish dominant scale-protected positions in the PC and server microprocessor markets would require a management focused almost exclusively on these products and able to cooperate effectively with the major software providers. Given the often messianic ambitions of tech managements, Intel's ability to maintain the necessary narrow focus and cooperative attitude was by no means certain, although by the mid-1980s Intel did have a good track record in this regard. Again as with the memory chip companies, successful investors in microprocessor companies such as Intel would have needed a detailed understanding of both the industry economics and the capabilities of managements.

Finally, as it gradually became apparent in the late 1980s and early 1990s that Intel would achieve a dominant market position in PC and later server microprocessors, its stock price inevitably rose. A successful investor in Intel stock would have had to be able to determine whether, at such prices and in the face of divergent future possibilities, Intel was an attractive investment. This is a much more complicated undertaking than analyzing WD-40. At each stage in Intel's history, alternative future trajectories based on different industry evolutions and management behaviors would have to be considered. At any given time, Intel might be a Case A investment (asset value above EPV due to poor management and a vanished franchise, as was the case of several other memory chip competitors), Case B (capable management in a competitive industry), or Case C (growing franchise) company. In evaluating Intel we must apply all three valuation techniques, asset value, earnings power value, and estimated returns, that we have developed. Only a liquidation analysis seems unnecessary.

Valuation I Reproducing Intel's Assets

Once Intel's competitive advantages in memory chip production had evaporated, its value would have been rooted in the value of its assets. Without barriers to entry, earnings levels above those necessary to provide an adequate return on these assets would be rapidly eliminated by the entry of efficient competitors. Since these competitors would themselves need to earn an adequate return on their own investments in memory chip production, they would not enter the market if industry earnings were insufficient to provide this return. Thus, if the investments

required by efficient entrants were comparable to those of Intel, meaning that Intel itself was an efficient operator, then the value of earnings would have to cover the cost of reproducing Intel's own assets. The value of Intel's future earnings would therefore be largely determined by that cost, which is the natural starting point for valuing Intel.

One simple way of estimating that reproduction value is to take Intel's accountants at their word and use Intel's accounting book value. A decision on whether to purchase Intel stock could then be made by comparing the book value of Intel's equity to its market value, whether for the company as a whole or on a book value per share compared to price per share basis. The shortcoming of this approach is that without careful analysis and adjustment, book values may be a very poor measure of the reproduction value of a company's assets. Still as we noted earlier (see Chapter 2) buying stocks at substantial discounts to book value has been a successful investment strategy. It is also easy to do. It is therefore an appropriate strategy for investors who don't want to do a lot of work. Unfortunately few successful businesses are available for sale at book value and even fewer at a discount sufficient to provide an adequate margin of safety.

Just to be certain, Table Ex4.3 presents the book value and market value figures for Intel over 5-year intervals starting in 1975.

A graph of the market-to-book ratio over the entire period makes the relationship more apparent (see Figure Ex4.2). For the years between 1980 and 1995, Intel traded at between two and four times the book value of its equity. These are all end-of-year numbers; during each year, Intel stock price hit highs and lows that differed, sometimes significantly, from the price at the year-end. But only in the general market run-up in the first half of 1987 did the company's market value exceed more than four times the value of its equity. This relationship lasted until the end of 1995, when the investors decided that they could pay more than four times the book value for a company as successful as Intel and still make money.

Table Ex4.3 Intel book and market values, 1975–1998 (millions).

	Dec-75	Dec-80	Dec-85	Dec-90	Dec-95	Dec-98
Book value	\$ 74	\$ 433	\$ 1,421	\$ 3,592	\$ 12,865	\$ 23,578
Market value	\$ 503	\$ 1,760	\$ 3,447	\$ 7,812	\$ 50,167	\$ 197,761
M/B ratio	6.8	4.1	2.4	2.2	3.9	8.4

Source: Data from Intel annual report

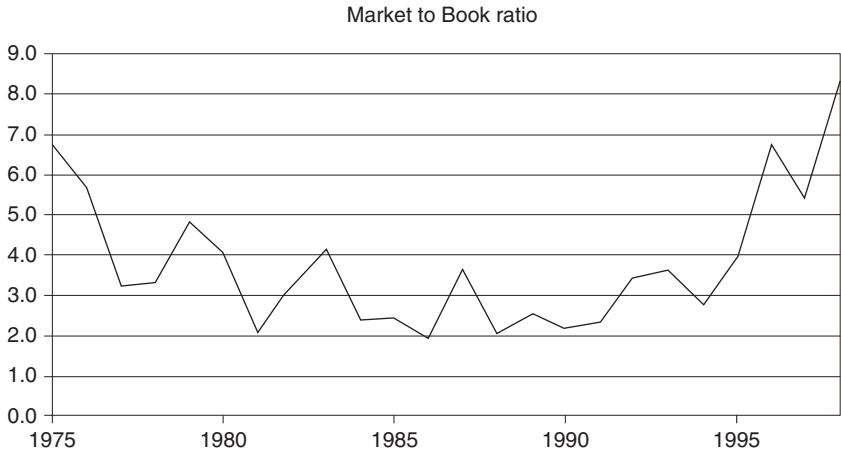


Figure Ex4.2 Intel Market-to-book ratio, 1975–1998.

Source: Data from Intel annual report

The market-to-book analysis is quick and sometimes a bit muddy, if not downright dirty. To get a more precise take on what a competitor would need to spend in order to get into the business, we have to look at the assets line by line and calculate—sometimes estimate—what their replacement costs would be. We went through this exercise in Chapters 4 and 6, and the guidelines for adjustment remain the same. Table Ex4.4 presents Intel's assets as listed in its financial statements for 1975.

We have made virtually no adjustments to the assets as reported on the balance sheet. Intel did not use last in, first out (LIFO) accounting to value its inventory, and it had not purchased another company in a transaction that would have landed Goodwill on its books. That leaves property, plant, and equipment (PPE) as the account that needs scrutiny.

Intel owned chip factories (*fabs*, in industry parlance) in the Santa Clara region, and it had stocked them with the sophisticated equipment and clean rooms essential to turn out integrated circuits. It was a young company in 1975, so none of the plants or equipment would have been very old, although it had bought its first building, used, from Union Carbide. On the other hand, the industry was changing rapidly, and the rate of obsolescence in semiconductor capital equipment may have been faster than the depreciation Intel was charging itself. A competitor might have been able to duplicate its facilities for less than the book value of its assets.

We can test how realistic is the stated value of PPE by comparing it with the actual capital expenditures Intel made (see Figure Ex4.3). For almost every year in the entire public history of the firm, Intel's

Table Ex4.4 Intel's assets, 1975 (values in millions).

Assets	Book value	Adjustments to arrive at reproduction costs	Adjustment amount	Reproduction cost
Current assets				
Cash and	\$ 19.3	none		\$ 19.3
accounts receivable, net	\$ 29.9	add bad debt allowances; adjust for collections	1	\$ 30.9
Inventories	\$ 20.1	add LIFO reserve, if any; adjust for turnover	0	\$ 20.1
Prepaid expenses	\$ -	none	0	\$ -
Deferred taxes	\$ -	discount to present value	0	\$ -
Other current assets	\$ 4.8		0	\$ 4.8
Total current assets	\$ 74.1			\$ 75.1
Property, plant, and equipment, net	\$ 28.5	original cost plus adjustment	0	\$ 28.5
Goodwill	\$ -	relate to product portfolio and Research and Development	0	\$ -
Total assets	\$ 102.7		1	\$ 103.7

Source: Data from Intel annual report

Table Ex4.5 Intel adjusted book value, 1975–1998 (values in millions).

	Dec-75	Dec-80	Dec-85	Dec-90	Dec-95	Dec-98
Book value	\$ 74	\$ 433	\$ 1,421	\$ 3,592	\$ 12,865	\$ 23,578
R&D adjustment	\$ 27	\$ 190	\$ 500	\$ 1,149	\$ 3,202	\$ 6,377
Marketing adjustment	\$ 39	\$ 260	\$ 698	\$ 1,616	\$ 4,822	\$ 7,822
Book value adjusted	\$ 141	\$ 883	\$ 2,620	\$ 6,357	\$ 20,889	\$ 37,777
Market Value	\$ 503	\$ 1,760	\$ 3,447	\$ 7,812	\$ 50,167	\$ 197,761
Market to adjusted book ratio	3.6	1.9	1.3	1.2	2.4	5.3

Source: Data from Intel annual report

net PPE had been more than the sum of its last four years of capital outlays and less than the sum of the last five years. Only if one thinks that a competitor could replicate Intel's entire production and research facilities with substantially less than four years' worth of its expenditures is the net PPE figure overstated. Conversely, if Intel's net PPE number understated the real (market) value of its fixed assets, it would not have needed to spend the equivalent sum every five years. The net PPE figure stands up as a reasonable figure when measured against capital outlays.

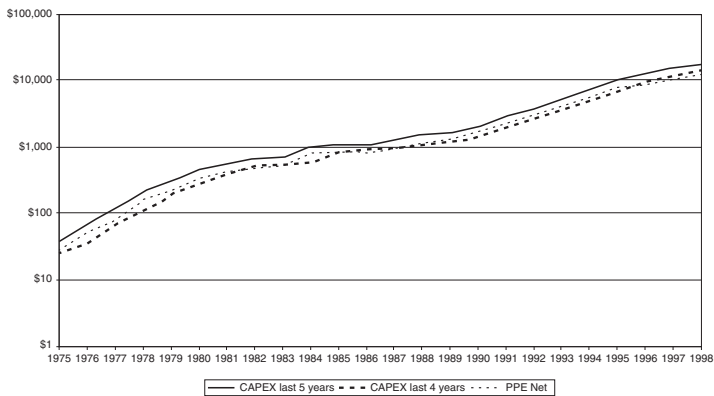


Figure Ex4.3 Intel capital expenditure and PPE, 1975–1998.

Source: Data from Intel annual report

If there are other assets that a competitor would have needed to produce to compete with Intel, they are not found on the balance sheet for 1975. That does not mean they didn't exist. Intel, we should not forget, was a New Economy stock long before the New Economy had a name, at least in its post-1920s incarnation. As an early and major manufacturer of memory chips and then microprocessors, Intel invested in knowledge-based resources—the science and engineering skills needed to design and fabricate semiconductors—and supplied knowledge-enhancing products, which were the memory and brains of computers and industrial equipment, to its customers. But Intel's investments don't appear on the balance sheet because under accounting rules, R&D is generally treated as an annual expense rather than a capital expenditure. Unlike utility bills, computer paper, or real estate taxes, money spent wisely on R&D should continue to earn profits for the company long after the checks have been cut. The expertise essential to design and produce these chips, everything from Andrew Grove's 1967 book *The Physics and Technology of Semiconductor Devices* to the months of trial and error involved in increasing the yield on some stubborn manufacturing process, does not come cheaply; once purchased, it has lasting value. And not all of that knowledge is reflected in the R&D expense account; some of it no doubt disappears into the cost of goods sold as a manufacturing expense.

The investment in knowledge that shows up in Intel's income statement has been considerable. We wrote earlier that R&D averaged 11% of sales for the period 1975–1998 (see Figure Ex4.4). Any company trying to compete with Intel would need to spend considerably to build up an equivalent expertise.

How much would be enough? Some analysts have suggested treating R&D as a capital investment and depreciating it on a straight line basis over five years. If we simplify and say that this past year's outlays should be fully valued as an asset, last year's at 80% and so on, we can calculate the value of an off-balance sheet intangible asset that estimates what a competitor would need to spend just to get into the business. For Intel in 1975, that amount would have been \$27 million. This would have increased the reproduction costs of the assets by 40%, and the book value of equity from \$74 to \$101 million, a gain of 37%.

There are other ways to gauge the costs of reproducing the knowledge base. If we used the sum of the last three years spent on R&D, the figure would be somewhat higher than our depreciated total; using the sum of the last two years would give us a slightly lower number. All of them might be understating the reproduction cost. Perhaps a

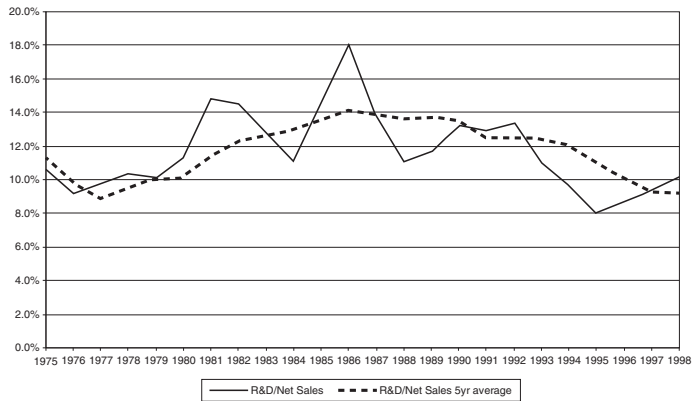


Figure Ex4.4 Intel R&D/net sales, 1975–1998.

Source: Data from Intel annual report

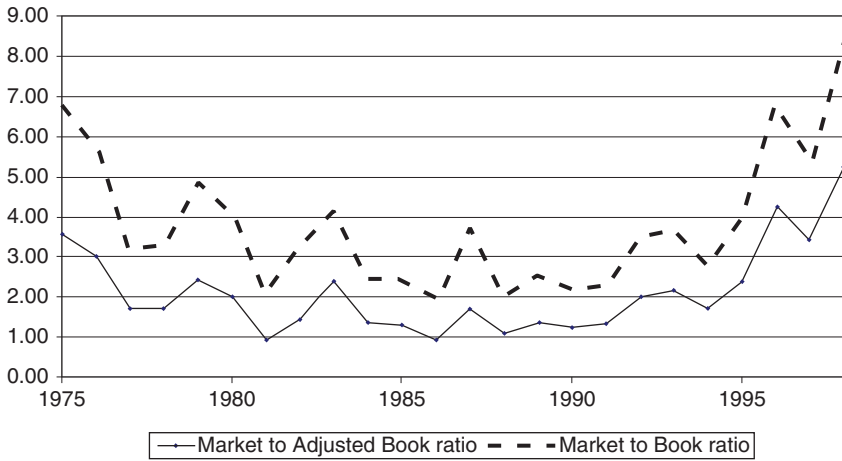


Figure Ex4.5 Intel market-to-book and adjusted market-to-book ratios, 1974–1998.

Source: Data from Intel annual report

competitor would need to spend five years' worth of R&D to become viable, or hire away some key Intel employs at a premium and then pay the court costs when Intel sued. We will use the five year depreciated figure as a conservative estimate of the reproduction costs of Intel's storehouse of knowledge.

There is another non-tangible, off-the-balance-sheet asset a competitor would have to create in order to compete with Intel. Almost every company needs to spend money marketing its wares. In Intel's case, though, it took several decades before they started to spend serious money telling consumers about the advantages of having "Intel Inside" their computers. From the start they sold highly technical products to a large number of sophisticated purchasers. This sales effort involved more than posting a list of specifications and prices for the semiconductors they offered. Sales executives had to work with customers to understand their needs and to win contracts. Any new competitor would have had to develop the rapport with customers and their specific requirements to go head to head against Intel. All this effort costs money; it takes time to build the relationships with the engineers in the customer firms.

It is not possible to produce a hard figure for the magnitude of Intel's spending on marketing, which would include primarily the salaries and commissions of the sales staff and money spent on advertising and other forms of promotions. The amount for advertising is broken out, but until 1990 it never amounts to even 8% of the marketing, general, and

administrative (MGA) expenses. To arrive at a reasonable number for what a competitor would have to spend to draw even with Intel requires making some estimates:

1. To even out the annual variations, we take the average of MGA as a percent of sales for the most recent five years and apply that to the current sales figure.
2. We assume that it would take three years of marketing expenses to get up to speed with Intel.
3. We set the share of MGA spent on running the business at half the total, leaving the other half for marketing.

Each of these assumptions can be challenged and refined; our aim is to arrive at a reasonable number to add to Intel's assets as an indication of the reproduction costs a new entrant would face. The amount is significant, slightly larger in most years than the R&D adjustment. (See Table Ex4.4 and Figure Ex4.4.)

When we include an adjustment for both R&D and marketing, we lower the market-to-book ratio for Intel by a considerable amount. There were several years in the period after 1975 when Intel could have been purchased at adjusted book value or slightly less. (Again, we are looking at year-end equity values and year-end prices. In some years, the interim prices were below the previous year-end.)

Starting in 1982, Intel began to invest in other companies, generally with aim of advancing Intel's overall strategy, which was to encourage broad demand for microprocessors. Intel marked to the market the stock of publicly traded companies in this strategic portfolio; for privately owned companies, it valued the shares at its cost. We see no reason to challenge the company's practices here. Only if the privately owned shares had declined substantially since Intel's investments would it be necessary to adjust shareholder equity downward.

We have made no modifications to the liabilities on Intel's books. Over the years, the company did build up a sizeable deferred taxes account; if that figure were discounted to the present, the liabilities would decline and the equity would therefore rise. But the adjustment is minor and only becomes visible in the years after 1995, when the market value of Intel greatly outstripped the book value.

Measured against book or adjusted book value, the three best times in the history of this period to have purchased Intel shares would have been

1. In January 1982, when the market value of the whole company fell briefly to around \$925 million, compared with an adjusted book at year-end 1981 of just over \$1 billion.
2. In August 1986, when the market value was about \$2 billion, versus an adjusted book value of \$2.5 billion.
3. At the end of 1988, when the adjusted book value of \$3.9 billion was about even with the market value (see Figure Ex4.5).

Had an investor been disciplined or fortunate enough to load up on Intel every time its market value came within hailing distance of its adjusted book, the results would have been excellent. At no time subsequent to the purchase would the value of the shares been lower by any meaningful amount, and the gain over the next five years would have been substantial. Now, all of these dates occur within the extended bull market that began in August 1982 and persisted into the year 2000. The investor in Intel would have been sailing with the wind. Still, purchasing shares at the start of 1982 and holding them into the one very rough period in Intel's history would have provided a decent return, though slightly below the returns on the S&P 500 index. (See Table Ex4.6.)

Nevertheless, it is important not to be lulled by the comfort of hindsight into thinking that these decisions were easy to make. In early 1982, the coming of competition to the memory chips began to seem inescapable, and the progress toward dominance of Intel's microprocessor position was hardly certain. The IBM PC had only entered the personal computer market in the late 1981, and there was always a very real possibility that IBM might produce its own microprocessors. If Intel could not acquire sustainable competitive advantages either in memory chips or in microprocessors, then as a competitive player, asset

Table Ex4.6 Intel share price appreciation (selected periods and adjusted for splits).

Dates	Price at start	Price at end	Compounded annual return	S&P 500 comparison
1/82-1/87	\$.47*	\$.895	13.8%	14.6%
8/86-8/91	\$.82	\$ 3.14	30.7%	6.5%
12/88-12/93	\$ 1.48	\$ 5.29	39.5%	17.5%

*Prices reflect subsequent share splits.

Source: Data from Intel annual report

reproduction value would capture the full value of future earnings that Intel could expect to enjoy. In 1986, Intel was incurring large losses, and in late 1988 Intel's earnings began to decline once again. Having the confidence to invest in Intel under these circumstances would have required, as we pointed out above, a highly developed understanding of Intel's market position and management.

On the other hand, the fact that Intel was trading at or below net asset reproduction value in a viable industry provided significant downside protection. Assuming that Intel's management was capable of making efficient use of these assets, taking a flier on Intel's future would not have entailed serious losses even if that future did not pan out. The key here would be Intel management's ability to make efficient use of its assets, which would be reflected in the long-run earnings power that Intel was able to generate. Thus, the next important step is to look at Intel's earnings power value.

Valuation 2: Earnings Power Value

Here we will follow the approach we took in valuing WD-40 based on earnings power. The underlying assumptions are that the company will not grow, that its current earnings are sustainable for a long period of time, and that a shareholder, as an owner of the company, will receive as his or her return a proportionate share of the company's distributable earnings. The no-growth assumption works easily with WD-40, but it certainly does not fit Intel, a dynamic company in a rapidly changing and expanding industry. Therefore it requires a considerable dose of strategic analysis to produce a reasonable estimate of Intel's constant earnings power in any particular year. Since the question we want to answer is whether and when Intel shares presented an opportunity to investors adhering to the valuation criteria we have established, we can look only at information that would have been available at the time.

As we have seen, an estimate of the company's current intrinsic value on the basis of its earnings power requires two steps: first, adjustments to the reported earnings to arrive at a figure that represents the cash the investors can extract from the firm and still leave it functioning as before; second, selection of a discount rate that reflects both interest rates and the riskiness of the firm relative to other investment alternatives. Dividing the discount rate into the adjusted earnings gives us our earnings power value (EPV).

Adjusted Earnings: Special Charges, Business Cycles, R&D, and D&A

We will concentrate on the years from 1987 on, after Intel abandoned the memory chip business and just as the personal computer revolution began to hit its stride. As with WD-40, we begin with operating earnings and make our adjustments from there. The first adjustment concerns special charges. These are the write-offs companies make when they revalue assets, such as inventory, equipment, or other investments they have made, or make provisions for lay-offs, plant closings, and the like. The justification for keeping these charges separate from operating earnings is that they are singular events that do not affect the company's permanent earnings capacity. But in fact each of them represents the accumulation of real expenses the company has incurred in the course of its business. If these charges persist from year to year, it means that the company is understating its true operating costs. To smooth out the erratic nature of these special charges, we take an average of the charges for the current and four prior years and deduct that from operating income.

Second, unlike WD-40, Intel's sales and earnings were not immune from cyclical swings. When Intel was involved in the memory chip business in the 1970s, its operating earnings margins were as high as 30% and as low as 20%. But in the microprocessor business, where there was less competition and the market was growing, it seems unlikely that the margins vary as much. In the years 1987 through 1991, as sales grew rapidly, operating margins stabilized at somewhat more than 20% and even increased during the recession of 1990. So we think it is conservative to use 20% reported EBIT as a base for this period. (See Table Ex4.7.)

Third, for Intel, R&D was a major expense each year, averaging more than 12% of sales from 1987 through 1991, and hardly falling off after that. These were growth years for Intel, and we have to assume that some of that R&D was spent to support that growth. But in the fast-changing world of integrated circuits, a company will need a large R&D budget just to run in place. So to get a more accurate picture of the earnings power with zero growth, we ought to add some of the R&D expense back to operating earnings. But how much? We can take the approach we used in calculating the replacement cost of the assets, which was to treat R&D as a capital investment and depreciate that over five years. Each year's expense would be the depreciation charge, which is somewhat less than the actual R&D cost. A second method is to estimate a maintenance R&D on the same basis we estimated maintenance CapEx, by capitalizing R&D into an asset, finding a sales-to-asset ratio,

Table Ex4.7 Intel adjusted after-tax operating earnings, 1987–1991 (millions).

	Dec-87	Dec-88	Dec-89	Dec-90	Dec-91
Sales (net)	\$1,907	\$2,875	\$3,127	\$3,921	\$4,779
EBIT as reported	\$ 246	\$ 594	\$ 601	\$ 858	\$1,080
EBIT at 20% of sales	\$ 381	\$ 575	\$ 625	\$ 784	\$ 956
Special items average adjustment	(\$14)	(\$10)	(\$30)	(\$21)	(\$11)
Add back 25% of R&D	\$ 65	\$ 80	\$ 91	\$ 129	\$ 155
Add back 25% of SGA	\$ 154	\$ 194	\$ 212	\$ 283	\$ 346
EBIT adjusted	\$ 587	\$ 838	\$ 899	\$1,176	\$1,445
EBIT adjusted margin	31%	29%	29%	30%	30%
After tax of 38%	\$ 364	\$ 519	\$ 557	\$ 729	\$ 896
Add back 25% of D&A	\$ 43	\$ 53	\$ 59	\$ 73	\$ 105
Income as adjusted	\$ 407	\$ 572	\$ 616	\$ 802	\$1,000
Income as adjusted margin	21%	20%	20%	20%	21%
Income as reported	\$ 248	\$ 453	\$ 391	\$ 650	\$ 819

Note: Figures are in millions except percentages.

Source: Data from Intel annual report

and using that ratio times the dollars of additional sales to arrive at the growth portion of R&D. Maintenance R&D is simply the other portion. A third way is to look at the R&D spent by the closest competitor, in this case Advanced Micro Devices, and use that number as the necessary Intel expense. Finally, we can take a guess and say that at least a certain percent—we will use 25%—of Intel's R&D can be attributed to growth and therefore should be added back to EBIT. Having done all of these calculations, we are going to use the last one here. It is the most conservative in that it produces a smaller increase in current earnings than any of the alternative methods, and it is the simplest to implement.

Fourth, the adjustment for R&D can also be applied to selling, general, and administration (SG&A), a substantial part of which is spent on winning new business. We will add back 25% of the total, a rough estimate but again a justifiable and conservative one.

Fifth, we can't escape taxes. For all these years we will assume a tax rate of 38%. This is higher than the accounting tax rate reported in Intel's audited financial statements, and it gives Intel no credit for whatever clever management of its tax liabilities it achieved. It is a valid and conservative estimate of taxes on operating income.

Sixth, we need to adjust for depreciation, amortization, and maintenance capital expenditures. Here some knowledge of the industry helps. The cost of semiconductor capital equipment, the big and expensive

machines that Intel used to make its microprocessors, had declined over the years when improved capacity is taken into account. As a result, Intel's depreciation expense, which is based on the historic cost of its machinery, overstates the amount Intel would have to pay to keep its production capacity level. Rather than adding back all of depreciation and amortization (D&A) to EBIT after tax and then subtracting the maintenance portion of capital expenditure, we will simplify the calculations and add back 25% of D&A, assuming that the other 75% will be more than enough to cover maintenance CapEx. There is nothing magic about the 25% figure in all these calculations; we use it because it seems both reasonable and conservative.

Making all these adjustments gives us the following adjusted and distributable operating earnings for Intel in the years 1987–1991, as shown in Table Ex4.7. After all that hard work, the differences between reported net income and adjusted and taxed operating income is not enormous, but that is only a rough check on the legitimacy of the estimations.

From Adjusted Earnings to Earnings Power Value

We have expressed skepticism about the ability of analysts to derive with sufficient precision the rate at which the earnings of a company should be discounted to calculate a useful earnings power value. In our current case, we are looking at Intel's earnings over a period in which long-term interest rates fell from around 12% to under 7%, the risk-free rate from more than 13% to under 6%. Because the company rarely had any net debt on its books (debt left after deducting cash and short-term investments), its weighted average cost of capital would not have benefited from the fact that it costs less to borrow than to raise equity, especially after the deductibility of interest payments for tax purposes is taken into account. Thus, as in the case of WD-40, the appropriate cost of capital would be the cost of equity. In the late 1980s and early 1990s, the returns that venture capitalists expected to earn and felt they had to earn to attract funds were between 16 and 18%. Long-term BAA debt paid an average of about 9%. The cost of equity for Intel at that time would have fallen within this range. With an established but threatened core business, Intel equity represented a substantially riskier investment than WD-40. But with low leverage and a long-term history of success built around that core business, the risk would have been well below that associated with venture capital investments. At the time, cash returns on equity market investment as a whole were about 3%. If we add 4% for

real growth and another 3–4% for inflation, we would have an estimated overall stock market return of 10–11%. Assuming an investment in Intel was a slightly higher than average risk level, the cost to Intel of attracting capital voluntarily would have been around 12%. We will use this figure to discount Intel's earnings.

With our earnings power (adjusted after tax EBIT) and discount rate in hand, we are almost ready to calculate an earnings power value for Intel. Before we compare that figure with the asset and market values, we will need to make one more set of adjustments. Both the asset value we calculated and the market value we are citing refer to the equity portion of Intel's capital and exclude the debt. We specifically subtracted this debt from the reproduction value of Intel's assets to arrive at the asset value. To be consistent, we need to do the same thing here and reduce earnings power value by the amount of the debt outstanding. On the other hand, Intel had a lot of cash on its books in these years, substantially more than it needed to run the operations. When we did our asset valuation, we included this cash hoard. Since we did not build this extra cash into the earnings power—operating earnings omit the interest on cash balances—we should add surplus cash to the earnings power value. This cash is definitely incorporated into the market value of the equity. Anyone buying the whole company would own this money along with all the other assets. Our cash and debt adjustments, which make all three values comparable, will be to subtract the book value of the interest-bearing debt and to add back all cash¹. With these last modifications, we arrive at these figures for Intel during the period 1987–1991 shown in Table Ex4.8. Figure Ex4.6 displays these values and extends them back and forward a few years to illustrate the transformation of Intel in this period.

We have already seen that the mid-1980s were a time of instability and transition for Intel. This is reflected in the greater fluctuation in earnings power value compared to asset value. But the critical point here is that for the years 1987 and 1988, Intel was trading at and slightly below net asset reproduction value, and the average earnings power value at \$3.35 billion was essentially equal to the average asset reproduction value of \$3.31 billion. Intel's management did indeed seem to be

¹ Since Intel had little debt in its capital structure, we have concentrated on the equity values of the assets and the earnings power. When a firm has significant financial leverage, the appropriate starting point is the enterprise value, which includes both debt and equity and then subtracts cash. Equity values are less stable because of the financial leverage. In Intel's case, the debt is insignificant.

Table Ex4.8 Intel EPV, adjusted book value, and market value, 1987–1991 (amounts in millions).

	Dec-87	Dec-88	Dec-89	Dec-90	Dec-91
EPV at 12%	\$ 1,891	\$ 4,768	\$ 5,137	\$ 6,684	\$ 8,337
Less interest-bearing debt	\$ (750)	\$ (696)	\$ (569)	\$ (623)	\$ (536)
Plus cash	\$ 619	\$ 871	\$ 1,089	\$ 1,785	\$ 2,277
Total EPV	\$ 1,760	\$ 4,843	\$ 5,657	\$ 7,846	\$ 10,078
Adjusted book value	\$ 2,755	\$ 3,893	\$ 4,781	\$ 6,357	\$ 7,671
Market value	\$ 4,779	\$ 4,285	\$ 6,513	\$ 7,812	\$ 10,240

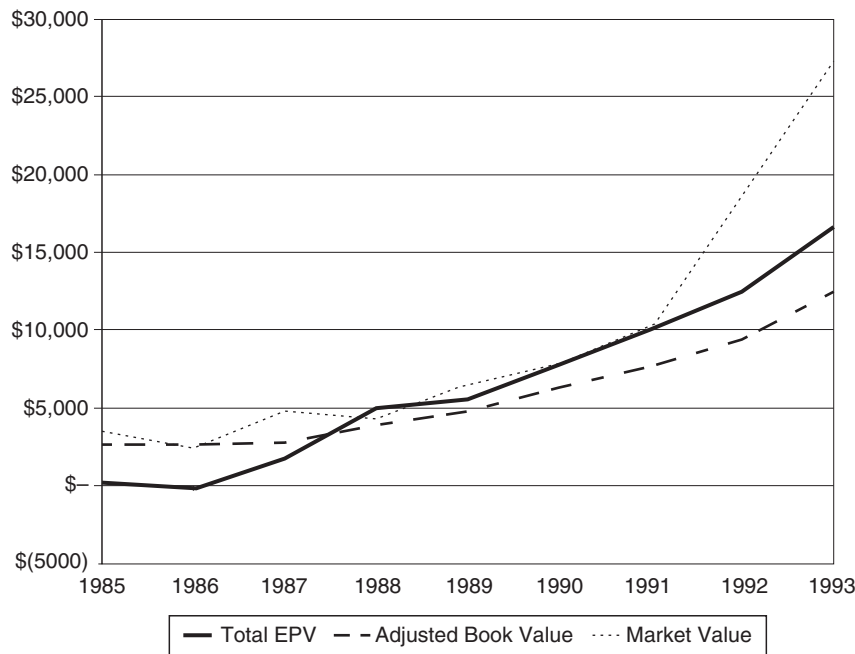


Figure Ex4.6 Intel 1985–1993 EPV, adjusted book value, and market value (in millions).

Source: Data from Intel annual reports.

making efficient use of Intel’s assets, a conclusion generally confirmed by our qualitative review of Intel’s management behavior. Any downside to Intel’s value during this period was largely protected by the value of the company’s assets. Under these circumstances, an investment in Intel based on future developments whose value would have been impossible to measure precisely but whose average expected value would have been

significant, would have been fully justified. In the language of Graham and Dodd, it would have been an investment and not a speculation.

Intel as a Growth Stock

As Intel's earnings grew with its dominance of the microprocessor industry in the 1990s, its stock price also rose rapidly. By the late 1990s, Intel's market value was typically between four and six times any reasonable estimate of the reproduction value of its assets. Making an investment in Intel during this period would not have been justified without earnings well in excess of those supported by Intel's assets and significant value attributed to the continuing growth of those earnings. In other words, Intel from the late 1990s on would have had to be evaluated as a franchise business with growth potential or avoided altogether by more traditional value investors.

The Intel Franchise 1998–2017

The first step in looking at a situation like this is to understand the nature and durability of the competitive advantages / barriers to entry that protected Intel's earnings. By the late 1990s there was little doubt that Intel, as the world's dominant microprocessor producer, was in a powerfully protected competitive position. Intel enjoyed the benefits of the most durable competitive advantage—significant economies of scale secured from erosion by strong customer captivity.

Intel's customer captivity arose from the risks facing any PC manufacturer who considered offering a product built with a non-Intel processor. Intel's chips were the standard around which Microsoft designed its virtually ubiquitous Windows operating systems. Alternative chips were unlikely to work as well in this software environment. Inferior microprocessor performance would degrade PC performance, which no manufacturer could risk in the highly competitive personal computer market. A new supplier could also not be counted on to deliver the large number of high-quality chips that Intel had historically always provided in a timely manner. Any disruption of high-quality chips supplied would mean costly delays in PC production. Any impairment in chip quality might permanently damage a PC manufacturers' reputation

for quality products. Finally, the “Intel Inside” advertising campaign had raised awareness of and comfort with the value of Intel chips in the minds of PC buyers. Personal computers without the “Intel Inside” label risked the loss of customers even if an Intel competitor were able reliably to supply large numbers of high-quality, high-performance chips.²

What this meant for competitors such as AMD and potentially IBM and the Japanese firms was that even if they were to produce a superior chip, they would in the short term capture only a small fraction of the microprocessor market. At the high-end of the market where Intel was dominant, new generations of microprocessors arrived roughly every 18 months. By the time a competitor’s chips achieved widespread acceptance, their products would usually be obsolete. The short windows of opportunity that characterized the PC market greatly increased the difficulty of cutting into Intel’s dominant market share. In deciding whether or not to invest in a new chip technology, Intel competitors could not expect to achieve sales levels that would justify the substantial R&D budgets necessary to compete with Intel’s technology. And, unless these companies could offer significantly cheaper or higher quality chips than Intel, Intel chips would always be the safe, preferred choice of PC manufacturers.

This is where Intel’s scale advantages came into play. Design of microprocessors, especially as they get faster, denser, and more energy efficient, is a large fixed cost, independent of how many units of each microprocessor get sold. The more units sold, the less the design costs per unit. Table Ex4.9 shows the levels of sales, R&D spending, and operating margins for Intel and its major competitor AMD throughout the 1990s. By the late 1990s, Intel’s R&D budget was four to five times as large as AMD’s even as Intel’s R&D spending per dollar of sales, at about \$0.10, was less than half of AMD’s over \$0.20. Furthermore, the accounting figures for R&D represented only part of the fixed cost of bringing a new chip design successfully to market. Additional R&D-type expenses were included in cost of goods sold every time a new chip fabricator line was put into operation and adjusted until chip yields (the fraction of chips suitable for sale) and chip quality met company standards. Marketing expenses were incurred to introduce the new chip and its capabilities to PC manufacturers. It is not surprising that by the late 1990s, Intel’s operating margins were four times those of AMD.

² In 2006, around the middle of this period, Apple switched its microprocessor source to Intel, having used Motorola and then IBM since the inception of the Macintosh in 1984.

Table Ex4.9 Intel and AMD: Sales, R&D, and EBIT margins as adjusted (in billions).

	Intel sales	R&D	EBIT margin	Sales	AMD R&D	EBIT margin
1990	\$ 3.92	\$0.52	31%	\$1.06	\$0.20	9%
1991	\$ 4.78	\$0.62	32%	\$1.23	\$0.21	22%
1992	\$ 5.84	\$0.78	36%	\$1.51	\$0.23	28%
1993	\$ 8.79	\$0.97	47%	\$1.65	\$0.26	30%
1994	\$11.52	\$1.11	40%	\$2.13	\$0.28	33%
1995	\$16.20	\$1.30	38%	\$2.43	\$0.40	26%
1996	\$20.85	\$1.81	42%	\$1.90	\$0.40	3%
1997	\$25.07	\$2.35	46%	\$2.30	\$0.47	11%
1998	\$26.27	\$2.67	40%	\$2.50	\$0.57	10%

Note: Figures for both Intel and AMD are firm-wide. Intel had a broader line of products and spent on R&D for products in which it did not compete with AMD, so this is not a perfect comparison. Amounts are in billions of dollars, except for percentages.

Source: Data from Intel and AMD annual reports.

Higher costs were not AMD's or any other competitors only disadvantage. Even at the higher R&D cost per chip apparent in Table Ex4.9, AMD was spending only one-fifth as much as Intel on R&D. With new generations of microprocessor chips appearing every 18 months, a large part of RD was devoted to the development of these next generation chips. In the race for the next generation, AMD was being outspent by a factor of five. Under these circumstances, Intel was the inevitable winner of this repeated competition. When on occasion AMD did manage to produce a superior microprocessor, Intel quickly superseded that chip with its next-generation product. In the meantime, most PC manufacturers chose to wait for Intel—customer captivity at work—and even AMD's "successful" chips were never able to capture more than 30% of the market. At its existing level of sales, if AMD were to match Intel's level of R&D spending, it would typically have to spend between 50 and 100% of its revenues on R&D, a painfully uneconomic prospect. It is not surprising, therefore, that AMD was never able to make significant inroads into Intel's dominant market position. Economies of scale protected by customer captivity converted temporary competitive advantages in each generation into a permanent advantage over many generations. From the 1990s on, Intel's franchise was a powerful and durable one.

The historical evidence confirms this assertion. Tables Ex4.10 and Ex4.13 present operating histories for Intel from 1991 to 2002 and from

2002 to 2017. Returns on equity, even without adjusting for excess cash, understated earnings due to expensed investments in growth, consistently average more than 20%. Properly measured, they have exceeded 30% for many years. Given these high returns, well-financed competitors including IBM, Japanese technology giants and more recently Samsung have tried to enter the market. They have invariably failed, except at the very low end of microprocessor performance. Intel's market share, averaging 90% or better, has if anything grown over time. This is what a powerful sustainable franchise looks like.

The Return on an Intel Investment in Early 1999

On March 1, 1999, when Intel's results for FY 1998 were becoming available, Intel shares traded at a price of \$29.70.³ With 6,940 million shares outstanding, the market value of Intel was \$206 billion. If we subtract Intel's net non-core liquid assets—cash plus investments minus debt—of \$12.1 billion, the net cost of acquiring Intel's core operating business was roughly \$194 billion. Table Ex4.10 presents Intel's operating history between 1991, when Intel's microprocessor franchise became clearly established, and 1998, with the years 1999–2002 added. Revenues in 1998 of \$26.3 billion had grown at a rate of 5% over the previous year and 24½% over the prior five years. Operating margins had risen steadily from 22.6% in 1991 to 39.4% in 1997, before declining to 31.9% in 1998. Over the five years from 1994, a year of slowing sales growth and declining margins, to 1998, Intel's average operating margin was 33.9%. Intel's average tax rate over the five years through 1998 was 36%, declining from 37% in 1994 to 34% in 1998 as corporate tax policy became slightly more favorable to tech companies such as Intel.

Given this history, sustainable revenues in 1998 were roughly \$27 billion. This accounts on the positive side for the growth in revenues over the course of the year. No downward adjustment for the possibility of future recession years seems necessary since 1998, the year of the Asia crisis, was already a slightly difficult year for Intel, and revenues still increased. The sustainable operating margin appears to be about 33%, above Intel's actual 1998 margin but slightly below the average of the previous five years. Because of the generally increasing trend, a reasonable estimate of sustainable margins might have been as high as 35%,

³ All stock prices and shares outstanding have been adjusted for splits to convert to 2018 levels.

above the five year average. In both cases, the margins appear sustainable in the face of potential competitors. Assuming fixed costs represented 20% of Intel's revenue and Intel's sales were $2\frac{1}{2}$ times the size of its nearest competitor, both conservative assumptions given the strength of Intel's franchise in 1998, Intel would have had a fixed cost advantage of 30% over that competitor; fixed cost of 50% of revenue for the competitor ($2\frac{1}{2} \times 20\%$) compared to 20% for Intel. That would certainly support an operating margin for Intel in the 33–35% range.

Sustainable revenues of \$27 billion times a sustainable operating margin of 33% yields sustainable operating earnings for Intel of \$9 billion in 1998. A reasonable sustainable tax rate is around 35%, midway between the 1998 rate and the five year average rate, since favorable business climates for taxes tend not to last forever. Thus, sustainable net operating profit after tax for Intel in 1998 would have been roughly \$5.9 billion (65% of \$9 billion).⁴ This represents an earnings return of 3% on the cost of acquiring Intel's operating franchise at its early March 1999 market price.

Cash Return (1)

The fraction of this return that Intel would likely distribute as cash was relatively difficult to pin down. Management had not articulated a long-term distribution policy. Historical net cash distributions to shareholders were variable. Until 1994, net equity sales by Intel exceeded its relatively paltry dividend distributions (see Table Ex4.10). More recently, net distributions as a percentage of net income had grown from 20% to 40% in 1997 before jumping to 80% in 1998. An 80% distribution level left barely enough retained earnings to support Intel's organic growth for the slow growth year of 1998. At higher organic growth rates, Intel would be unable to distribute this level out of current earnings. For this reason, a 60% distribution seems appropriate and also corresponds to the average distribution fraction over 1997–98. With 60% of net income

⁴ We have not adjusted these earnings figures for any fixed asset over- or under-depreciation or for the portion of R&D and SG in spending that supported growth. In the former case it is because the discrepancy between accounting depreciation and maintenance CapEx was small. From 1990 to 1998, total accounting depreciation was \$11 billion; total capital expense was \$21.7 billion. Revenue growth was \$25.2 billion. Net fixed assets investment per dollar of revenue averaged about \$0.45. Therefore the \$25.2 billion in additional sales would have required \$11.3 billion in growth capital expense. Subtracting that amount from total capital expense leaves \$10.4 billion of replacement capital expense, very close to but slightly less than the \$11 billion of accounting depreciation.

Table Ex4.10 Intel financial history, 1991–2002.

ANNUAL INCOME STATEMENT (\$millions)												
Year ending December 31	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002
Revenues	\$ 4,779	\$ 5,844	\$ 8,782	\$ 11,521	\$ 16,202	\$ 20,847	\$ 25,070	\$ 26,273	\$ 29,389	\$ 33,726	\$ 26,539	\$ 26,764
Gross margin	52%	56%	63%	52%	52%	56%	60%	54%	60%	63%	49%	50%
Operating income \$	\$ 1,080	\$ 1,490	\$ 3,392	\$ 3,387	\$ 5,252	\$ 7,553	\$ 9,887	\$ 8,379	\$ 9,767	\$ 10,395	\$ 2,256	\$ 4,382
Operating income margin %	22.6%	25.5%	38.6%	29.4%	32.4%	36.2%	39.4%	31.9%	33.2%	30.8%	8.5%	16.4%
Tax rate	32%	32%	35%	37%	37%	35%	36%	34%	35%	30%	41%	26%
Net income	\$ 819	\$ 1,067	\$ 2,295	\$ 2,288	\$ 3,566	\$ 5,517	\$ 6,945	\$ 6,068	\$ 7,314	\$ 10,535	\$ 1,291	\$ 3,117
Net income margin	17%	18%	26%	20%	22%	26%	28%	23%	25%	31%	5%	12%
EPS	\$ 0.13	\$ 0.16	\$ 0.34	\$ 0.34	\$ 0.54	\$ 0.78	\$ 1.06	\$ 0.91	\$ 1.10	\$ 1.57	\$ 0.19	\$ 0.47
Div. per share	\$ -		\$ 0.01	\$ 0.01	\$ 0.02	\$ 0.02	\$ 0.03	\$ 0.04	\$ 0.06	\$ 0.07	\$ 0.08	\$ 0.08
Dep. amort.	\$ 418	\$ 518	\$ 717	\$ 1,028	\$ 1,371	\$ 1,888	\$ 2,192	\$ 2,863	\$ 3,597	\$ 4,835	\$ 6,460	\$ 5,334
Cap expense	\$ 948	\$ 1,228	\$ 1,933	\$ 2,441	\$ 3,550	\$ 3,024	\$ 4,501	\$ 4,032	\$ 3,403	\$ 6,674	\$ 7,309	\$ 4,703
Dividends	\$ -	\$ 21	\$ 84	\$ 92	\$ 116	\$ 148	\$ 180	\$ 217	\$ 336	\$ 470	\$ 538	\$ 533
Net buybacks	\$ (112)	\$ (180)	\$ (91)	\$ 432	\$ 757	\$ 985	\$ 2,727	\$ 4,658	\$ 4,049	\$ 3,210	\$ 3,246	\$ 3,333
distribution				23%	24%	21%	42%	80%	60%	35%	293%	124%
Net cash		\$ 2,271	\$ 3,629	\$ 3,658	\$ 3,365	\$ 8,230	\$ 10,276	\$ 12,130	\$ 18,514	\$ 16,450	\$ 11,545	\$ 12,466
Equity	\$ 4,418	\$ 5,445	\$ 7,500	\$ 9,267	\$ 12,410	\$ 16,872	\$ 19,295	\$ 23,377	\$ 32,535	\$ 37,322	\$ 35,830	\$ 35,468
ROE	19%	20%	31%	25%	29%	33%	36%	26%	22%	28%	4%	9%
Avg. shares (MLN)	6,688	6,872	7,056	6,992	7,072	7,101	7,179	7,035	6,940	6,986	6,879	6,759
Employees	25	26	30	33	42	49	64	65	70	86	83	79
(THOUS)												

Source: Data from Intel annual report

distributed to shareholders, the cash return on an Intel investment in early 1999 would have been \$3.5 billion (60% of \$5.9 billion) or just 1.8% (60% of 3%) on the investment of \$194 billion.

Organic Growth Returns (2)

For Intel, with its successive generations of new chips, organic growth is hard even to define, let alone measure, especially when compared to a stable product company like WD-40. Technically, each new generation design is a new product with its own initial market of new high-performance PCs, workstations, and servers. But for practical purposes, growth within the microprocessor industry is organic growth. The computer manufacturers, their main customers, and even users of computing power in non-traditional applications such as automobile controls, historically have constituted a reasonably stable population. The design and production challenges have also had a high degree of continuity, so that the supply side of the market also was served by a relatively stable group of firms, primarily Intel. Organic growth in this arena of all microprocessor and microprocessor applications has not been entirely driven by external economic factors such as economic growth and technology. For example, Intel has always sought out new applications for an extension of its chip technology. But like organic growth, this process was largely a continuous one that was not the subject of large capital allocation decisions. The vast majority of the investment involved—R&D and marketing—was expensed for accounting purposes and independent of explicit allocation decisions related to retained earnings.

By this definition, Intel's growth from the late 1980s through the end of 1998 was almost entirely organic growth. During that time Intel made no significant acquisitions. Flash memories, its major product extension outside of microprocessor and related chipsets, had by the end of 1998 produced very limited sales and no profits. Yet even with this broader definition of organic growth, it was still difficult to forecast.

Actual growth in revenue and operating earnings from 1991 to 1998 averaged 28% and 34% respectively. Management's announced growth targets, generally in the range of 20% or more, reflected this experience. For the very long term, these growth rates were not going to be sustainable, despite Intel's observable performance. For analysts, one way out of this difficulty is to define a high-growth period of fixed duration and then a subsequent future of much lower stable long-term growth. For some companies, such a strategy might work well. As an example, Google is a company whose technology replaces the operation

of a well-defined existing market segment, information advertising. A target level for Google's mature sales can be estimated by adding current information advertising revenues spent in various media on a global basis excluding China, where Google is not dominant. These consist of Yellow Pages, newspaper classified ads, business directories, focused consumer publications, some direct-mail, and so on. They have historically grown about 1% faster than global GDP. The next steps are to estimate how much of this revenue will move to the web—at least 80%—and about what market share Google might capture, say 50%. Those calculations yield a “mature” revenue level for Google. The length of Google's high growth period could then be calculated as the time it would take to reach this “mature” level of revenue at a high growth rate extrapolated from Google's historical experience. Beyond that point, the growth of Google's revenues would be predicted to decline to 1% above global GDP. Unfortunately, there is no such well-defined revenue that Intel will displace. Information technology in all its forms is in the process of remaking the whole of industrial society.

For Intel the only realistic alternative is to assume a long-term organic growth rate that (a) generates a finite value for Intel future earnings, and (b) embodies a level of growth above that of global GDP. How far above that figure is difficult to say, but in the history of the modern industrial economy, no industry has grown faster than 3% of GDP for more than 50 years. Thus, a reasonably long-term organic growth rate for Intel would be 3% above the growth rate of global GDP.

In 1999 global GDP was forecast to grow at 5 to 6% in nominal terms.⁵ If we add 3% to this figure, we get an organic revenue growth rate for Intel going forward of 8.5%. Intel's operating profit margin was essentially stable from 1993 to 1998. So operating earnings and net operating profit after tax were likely to grow at the same 8.5% rate as revenues. Thus, organic growth should, as a first guess, contribute 8.5% to investor returns.

Active Investment Returns (3)

The amount of investment necessary to support organic growth naturally depends upon the rate of growth. The 20% growth (roughly Intel's 1990s average) would have added \$5.4 billion in the first year to Intel's revenue from our estimated sustainable revenue level of \$27 billion. The amount

⁵ In 1999 China and India were much smaller economies than they were in 2018 so their contribution to global growth rate was smaller.

of fixed capital required per dollar of revenue averaged 45% from 1991 to 1998, with a downward trend in the early years and an upward trend more recently. Use of the average 45% figure yields an investment level of \$2.43 billion to support 20% organic growth.⁶ A 40% earnings retention level in 1998 would provide \$2.34 billion out of \$5.05 billion sustainable earnings, leaving nothing for active investment. At our lower long-term organic growth rate of 8.5%, the investment required to support organic growth would have been \$1 billion (45% of \$3.2 billion in incremental revenue). Subtracting the lower-level organic growth investment from retained earnings would have left about \$1.3 billion for active investment.

During the 1990s, Intel was highly disciplined in its focus on its core microprocessor business. In the five years from 1993 through 1997, Intel made no acquisitions. In 1998, Intel paid \$321 million for Chips and Technologies, a maker of graphic products that Intel hoped would enhance the graphics capability of its chipsets. It paid another \$585 million for the semiconductor operations of Digital Equipment, including \$475 million in capital assets and some mobile microprocessor technology. On December 31, 1998, Intel had no significant goodwill on its balance sheet. For investments classified as current expense, flash memory was the only non-core product developed. This activity formed part of Intel's Computer Enhancement Group (CEG), which consisted mostly of chipsets and products to install microprocessors in non-computer environments. In 1998, CEG, of which Chips and Technologies was a part, reported operating earnings of \$358 million on sales of \$4.0 billion compared to earnings of \$9.1 billion on sales of \$21.5 billion in the core microprocessor business. Essentially all of CEG's profits were due to natural extensions of the core microprocessor business that we treated as organic growth. To the limited extent that Intel actively invested for growth outside its organic franchise, the returns appear to have been negligible.

Any expensed investments, therefore, were either oriented toward organic growth, in which its return was included, or they were devoted to new initiatives for which returns appear to have been zero. From 1996 to 1998, despite CEG revenues that increased from \$3.6 billion to \$4.0 billion, CEG operating earnings fell from \$940 million in 1996 to \$529 million in 1997 and \$358 million in 1998. Under these circumstances,

⁶ Intel required essentially zero net working capital in 1998. Working asset accounts receivable and inventory of \$5.1 billion on December 31, 1998, were almost exactly offset by working liabilities, accounts payable, and accrued liabilities of \$5.0 billion. We will assume that investment in working capital to support organic growth was zero.

we do not need to add back our former 25% of R&D and SGA to operating earnings and then trace its impact through active reinvestment. The impact was nil.

Generally, retained earnings by Intel did not go to new initiatives. They went to cash and outside investments. From year-end 1994 to 1998, Intel's net cash plus investment position increased from \$3.7 billion to \$12.1 billion. The value of these investments depends on how long the cash stays in the company and where it goes when it gets spent. While it is in the company, its value to the investor is reduced by 35% of the amount earned on the cash due to the double taxation of returns, once at the corporate level when they are earned and again at the individual level when they get distributed. Without compelling investment opportunities, the best final outcome is that the funds get distributed to investors, the sooner the better. If they are distributed within five years, the loss in value on the amount earned may be as little as 10%. If they are ultimately invested in unrelated initiatives as Intel's core business deteriorates, a common practice of managements trying to maintain growth, the value destruction may be total. Toward the end of 1998, Intel announced two relatively large acquisitions in non-core communications technologies, as well as new priorities in this direction. Thus, by early 1999, the second path for cash use had become a far more likely outcome. In the long run, as organic growth slowed, this tendency could only get worse. Taking that probability into effect, we will reduce the value of additions to cash by 50%. Our \$1.3 billion of notional active reinvestment should therefore add \$650 million in active investment returns to our overall return, which represents about an additional return of 0.3%.

Total Return, Intel's Cost of Capital, and Intel as a Value Investment

These calculations are summarized in Table Ex4.11. The total return estimated for an investment in Intel in March, 1999, was 10.6%, consisting of 1.8% cash return, 8.5% from organic growth, and 0.3% from active investment. Intel's cost of capital, which we estimated at 12% for the late 1980s, should also be the likely cost of capital in 1998. An investment in Intel stock, even after its dominant position in microprocessors had been established, remained a highly risky one. The great majority of the return embodied in the 8.5% organic growth lay in a distant uncertain future. Moreover, unlike the risk associated with WD-40, Intel's risk should be largely undiversifiable since future

Table Ex4.11 Estimated returns for Intel, 1999–2018.

	Mar 1999	Mar 2003	Mar 2018
Cash return	1.8%	3.0%	4.2%
Organic growth			
Initial	8.5%	4.5%	4.0%
Adjusted	5.8%	2.4%	2.6%
Active investment	0.3%	0.7%	0.0%
Total return			
Initial	10.6%	8.2%	8.2%
Adjusted	7.9%	6.1%	6.8%
Cost of capital	12.0%	11.0%	10.0%
Fade rate	2.4%	2.4%	1.8%
Post-fade returns			
Initial	8.2%	5.8%	6.4%
Adjusted	5.5%	3.7%	5.0%

Source: Data from Intel annual report.

growth in Intel's earnings would critically depend on future growth in the global economy as a whole. Overall, therefore, Intel was an investment likely to earn a below-cost-of-capital return. It certainly provided no margin of safety to protect against franchise fade.

By early 1999, Intel's franchise position in microprocessors had a duration of 10 to 12 years, much shorter than that exhibited by WD-40. Its industry was experiencing rapid change, and there were already indications that parts of the Intel franchise was subject to serious competition. In the portion of the microprocessor market for lower power devices, such as smart phones (originally personal digital assistants with the phone feature added later), notebook computers, and laptops, R&D costs were much less. Entrants could therefore be viable at much lower market shares. Also low-end devices tended to use older, more seasoned technology, reducing customer captivity. By the end of 1998, companies such as AMD had established sustainable competitive positions in the low end of the market. If this portion came to occupy more of the overall microprocessor market as the value of incremental technological performance became subject to the law of diminishing returns, Intel's competitive position could erode rapidly. Under these circumstances, the assumption that Intel's franchise had no more than 20 to 30 years of remaining life would not have been extreme. Lives of this length correspond to the franchise fade rates of $2\frac{1}{2}$ – $3\frac{1}{2}$ % per year. If we were to subtract these fade rates from our estimated return

of $10\frac{1}{2}\%$, the net return offered by the Intel investment would fall to between 7 and 8%, well below Intel's cost of capital. Franchise fade rates were a serious negative factor for an Intel investment in early 1999.

Matters were even worse than this. The organic growth rate applies to the intrinsic value of Intel, not to its inflated market value. To calculate a true return to organic growth, we must multiply the contribution of organic growth rate by the ratio of intrinsic value to market value. As we derived in Chapter 8 on growth valuation, the ratio of intrinsic value to market value should approximately equal the ratio of the estimated future return to the cost of capital. If we assume that all-in estimated return includes an average fade rate of 3%, then the estimated return for Intel will be 7.5%. This is just 62.5% of our 12% cost of capital. If we multiply the organic growth rate of 8.5% by 62.5%, we get a "true"/adjusted organic growth return of about 5.5%. This further 3% point reduction in our estimated return gives us an adjusted Intel return estimate of just 4.5%, a very poor return indeed compared to Intel's 12% cost of capital.⁷ Intel in March 1999 was not a good value investment however attractive its near to intermediate term prospects looked.

As it turned out, Intel's franchise did not wholly evaporate over the next 15 years, but its growth rate fell more than we projected. Nevertheless, our 4.5% estimated return would have significantly overestimated subsequent performance. Table Ex4.12 presents 5-, 10-, and 15-year returns for Intel stock going forward from March 1, 1999. The annualized returns were respectively 0.6%, -7.3% (in 2009 at the trough of the financial crisis), and 0.7% for the 15 years starting in March 2003 and ending in February 2018. Even our pessimistic estimate of Intel returns proved insufficiently negative. Investing in bubble stocks under bubble conditions is a practice value investors, and everyone else, are well advised to avoid.

Intel in March 2003

From a price of \$29.70 on March 1, 1999, Intel stock price rose to a peak of \$74.88 in late August 2000 before bottoming out at \$13.42 in early October of that year. By March 3, 2003, the price per share had recovered to \$16.66. At that point Intel might have been a reasonable candidate for a value investment. The company's stock market performance had been spectacularly disappointing. Its price-to-earnings ratio

⁷ Further iterations of this process would reduce this estimated return still further.

Table Ex4.12 Realized Intel returns.

	5 years	10 years	15 years
Investment			
March 1999			
Intel	0.6%	-7.3%	0.7%
S&P 500	-0.1%	-3.4%	4.7%
Investment			
March 2003			
Intel	5.6%	5.2%	10.2%
S&P 500	11.2%	6.2%	10.3%

was still high at 35 times, and its market-to-book ratio of slightly over 3 provided little asset protection. However, a case could be made that 2002 earnings were cyclically depressed so that sustainable earnings of \$1 per share were a more reasonable basis for estimating earnings power. This would have produced a price-to-earnings ratio of 16 or 17 to 1 for a company with a powerful franchise in an industry with good long-term growth prospects. Through the end of 2002, the last year for which data would have been available in March 2003, no significant impairment of Intel's barriers to entry/competitive advantages had taken place. Its long-term growth prospects looking forward from 2000 had deteriorated only slightly. (See Table Ex4.10 for Intel's financial history from 1990 through 2002.)

From 1998 to 2002, Intel's business performance mirrored the roller coaster ride of its stock price. Sales rose from \$26.3 billion in 1998 to \$33.7 billion in 2000 before falling back to \$26.8 billion in 2002. Operating margins went from 31.9% in 1998 to a high of 33.2% in 1999, then to just 8.2% in 2001 before recovering to 15.7% in 2002. The critical question concerning an Intel investment was whether these lower levels of earnings, revenue, and revenue growth were permanent or temporary.

On the positive side, Intel continued to dominate the microprocessor market with a share of over 90%. Its reputation for quality and its technical preeminence were unimpaired. Microprocessors, despite the bursting of the tech/telecom bubble in early 2000, remained at the heart developments in computing power that were generating innovations in automation, graphics interfaces, data handling, Internet communications, remote computing, and technical analysis. These innovations were still the engine of growth in the early 21st-century economy. Reasonable estimates of both sustainable earnings and sustainable long-run growth

would have been below those of March 1999 but not massively so. Adjustments for the artificial break in demand in 2000 and the cyclical developments of 2001 and 2002 suggest a sustainable revenue of \$30 billion. This estimate embodies an annual revenue growth rate of only 2.5% between 1998 and 2002. Since potential microprocessor sales were almost certainly growing at least as rapidly as GDP as a whole, perhaps 4–5% per year in nominal terms in the early 2000s, this figure would have allowed for some increase in our earlier sustainable revenue estimate of \$27 billion. Operating margins did appear to be in a long slow decline since the mid-1990s, as technological demands raised costs. But Intel's dominant position and pricing power should have limited this decline going forward. An estimated sustainable margin of 25% for 2002 would have represented both a significant decline from our 1998 estimate of 33%, in line with more recent experience, but at a level significantly above the depressed results of margins in 2001 and 2002. Sustainable operating earnings would then have been estimated at \$7.5 billion. Intel's average tax rate from 1998 through 2002 was 33%, which appears consistent with likely future developments in global, US federal, and state and local corporate income tax policy at the time. Applying this tax rate would have meant an estimated level in 2002 of \$5 billion in net operating profit after taxes.⁸

On a year by year basis, distribution to shareholders from 1999 through 2002 averaged 128% per year (see Table Ex4.10). This figure is inflated by the maintenance of a preset policy of dividends and buybacks despite declining earnings, which included a serious trough in 2001. If instead of taking the average percentages of annual distributions, we divide total distribution for 1999 through 2002 by total net income for those years, we get a distribution percentage of 70%, which is far more reasonable than the average annual percentage. Still, Intel's net cash position fell from \$18.5 billion at the end of 1999 to \$12.5 billion at the end of 2002. A continued distribution level of nearly all earnings was unlikely to have been sustainable in the long term. At the same time, as we will discuss in more detail below, Intel had begun to

⁸ Again there are no necessary adjustments for either excess (or under-) depreciation or expensed investments. In the case of depreciation, we have still further evidence that accounting depreciation quite accurately measures economic depreciation. For 2001 and 2002, revenues were essentially flat. During those two years, accounting depreciation and amortization was \$11.8 billion. Capital expense, presumably entirely maintenance CapEx, since growth was zero, was \$12.0 billion. The difference is negligible. As far as expensed investment was concerned, the arguments of 1999 continue to apply. The impact of these reinvested earnings would largely be captured by organic growth returns.

diversify its business activities, a move that would require significant capital both for acquisitions and to finance internal developments. Thus, we will assume that distribution policy from 2002 going forward would have been captured by our earlier estimate of 60%. This yields a sustainable cash distribution of \$3 billion per year. At a price of \$16.66 per share, Intel's market value would have been about \$112 billion, down substantially from \$206 million in March 1999. From this figure we must subtract Intel's cash and investment holdings, net of debt, of slightly over \$12 billion at the end of December 2002, to obtain the \$100 billion market cost of acquiring Intel's core franchise business on March 1, 2003. At this cost, the \$3 billion of sustainable annual cash distribution represented a 3% annual cash return, well above the 1.8% figure of March 1999. These estimates are summarized in Table Ex4.11. It is worth noting that calculation of the impact of varying revenue, operating margins, tax rates, and distribution policy assumptions in this context is simple and straightforward.

Organic Growth in March 2003

In the five years 1992 through 1996, growth averaged 34%. In the six years 1997 through 2002, growth dropped to just over 4% per year. From 1999 through 2002, it was reduced to 0.5% per year. Clearly by 2002, the years of extraordinarily rapid growth in Intel's core market were over. On the other hand, given the critical role of microprocessors in advanced technology sectors, extrapolation of the recent experience of 2002, with organic growth rates below the rate of global GDP growth, would have been overly pessimistic. Global GDP growth looking forward from 2002 appeared to be about 3–3.5% in real terms; with 1–1.5% inflation, that came to 4–5% per year in nominal terms. This is slightly below the 5–6% global growth rate we forecast in 1999. The rate of growth of productivity among the world's developed economies, and with that the overall rate of economic growth, had clearly slowed by the early 2000s. This decline was partially offset by the greater weight of rapidly growing economies such as China and India on total global economic activity. A reasonable assumption would have been that Intel's core market would grow at or near this rate. We will estimate Intel's future organic growth from March 2003 at 4–5%. We assumed that the long-term decline in Intel's margins would stabilize at 25%. Earnings growth, and therefore value growth, should occur at the same organic growth rate as revenues, 4–5%.

Active Investment Returns

At this growth rate, Intel's sustainable revenues from 2000 to 2003 would have grown by \$1.35 billion per year. The investment necessary to support each dollar of this growth appears not to have changed significantly since 1998. Net fixed capital in 2002 was roughly the same 45% of revenues as in 1998. Net working capital remained near zero. At this level of investment, the projected \$1.35 billion in organic revenue growth would have required \$0.6 billion (45% of \$1.35 billion) of reinvested earnings. With our sustainable net operating profit after tax of \$5 billion, distributing \$3 billion to shareholders leaves \$2 billion of retained earnings. After spending \$0.6 billion in support of organic growth, \$1.4 billion would have been available for active investment.

Looking at the value created per dollar of this active reinvestment, the results between 1998 and 2002 were hardly encouraging. In 1999, Intel spent more than \$5 billion, largely on communications-related acquisitions. In 2000, it spent \$2.3 billion, followed by \$0.9 billion in 2001. Fortunately there were no further significant acquisitions in 2002. But Intel continued to prospect in areas outside its microprocessor core business.

The returns to these efforts were dismal. From 2000 through 2002, Intel's non-core operations, its Wireless Communications and Computing Group and its Communications Group, had revenues of \$15.3 billion and a net operating loss of \$0.7 billion. Investment on acquisitions alone had been more than \$8 billion; the value creation appears to have been zero. At best, if we discount the 2001 and 2002 years as cyclically depressed and unrepresentative, the after-tax net operating profits in 2000 were \$0.6 billion. This represents a 7.5% return on the \$8 billion in acquisition-related expenses. With the cost of capital on these new, relatively unfamiliar initiatives likely to be at least 12%, the value created per dollar was about \$.60 (7.5% divided by 12%), making the value creation factor on actively reinvested earnings 60% at best. We will use a factor of 50%. At this level of return, the \$1.4 billion in active reinvestment would have created \$0.7 billion in value, which was 0.7% of the \$100 billion cost of Intel's core enterprise.⁹ This figure can be found in Table Ex4.11.

⁹ It should be clear from this discussion that given the relatively low level of Intel's active reinvestment, such investment was unlikely ever to be a significant source of value creation relative to the company's size.

Aggregate Returns, the Cost of Capital, and Fade Rate

The total estimated return to an Intel investment in early March 2003 would have been 8.2%, consisting of a cash return of 3%, organic growth of 4.5%, and an active reinvestment return of 0.7%. Intel's cost of capital in March 2003 would have been lower than its cost in March 1999, since microprocessor market conditions had stabilized, and less of the overall return was embedded in growth and thus in the far distant future. On the other hand, events since 1999 had demonstrated that future growth rates were highly uncertain and that Intel's results were more cyclically sensitive than anticipated. There was also significant uncertainty about the future direction of margins, which would depend on the development of the microprocessor industry and in turn, on the growth of the overall global economy. Under these circumstances, a cost of capital of 11% does not seem unreasonable.

Between 1999 and 2003, Intel's franchise had clearly not disappeared, but the probability of future fade remained. Intel itself was concerned that the incremental benefits of greater microprocessor power would diminish over time as basic computers became more and more capable. If this were to occur, then developments at the low performance end of the microprocessor market, where Intel already experienced substantial competition, would spread to the market as a whole. The consequences of these changes were beginning to be apparent in the mobile computer market. For mobile devices, communications bandwidth, not local computing power, was the limiting factor. Thus the demand for high performance microprocessors was significantly lower in the mobile market than in the basic IT computing market. ARM, the leading microprocessor design firm in the mobile market, had only a fraction of Intel's profitability and market value. The fade rate of 2.4% (a 30-year half-life) that we used in 1999 should continue to be applicable in 2003.

For Intel in 2003 to be a genuine value investment, the estimated return of 8.2% would have had to exceed the cost of capital, at 11%, by a margin of safety at least equal to the fade rate of 2.4%. Instead Intel offered a return below its cost of capital. Changing growth prospects, management behavior, and the original overvaluation in 1999 meant that, even after a price decline of roughly 45%, Intel in 2003 remained an unattractive investment. These figures are summarized in Table Ex4.11.

Moreover, our initial return estimate in 2003, like the estimate in 1999, was overly optimistic. The return formula applied the organic growth rate to the market value of Intel, not, as it should, to the intrinsic value. In March 2003 the post-fade return to an Intel investment was

5.8%, roughly half the cost of capital of 11%. The ratio of estimated return to the cost of capital should approximate the ratio of intrinsic to market value. The return ratio is 53% (5.8/11). If intrinsic value were therefore just 53% of Intel's market value, then the "true" value of Intel's organic growth was just 53% of the organic growth rate of 4.5%, or 2.4%. The reduction in future returns is 2.1% (4.5% minus 2.4%), which yields an estimated future return looking forward from March 1, 2003, of just 6.1%. This made Intel, at that time, an unattractive investment.

In this instance our estimated return tracked actual future Intel returns relatively well. Intel's franchise did not evaporate over the next 15 years, so that the appropriate comparison is to our estimated pre-fade rate of return of 6.1%. Actual average annual returns for the five years from March 2003 to March 2008 were 5.6% (see Table Ex4.12). Despite higher than market risk, Intel underperformed the S&P index by 5.6% per year. In the 10 years from March 2003 to March 2013, average realized annual returns were 5.1%, again underperforming the S&P. Both returns are reasonably close to our 6.1% estimate. However, from 2015 to 2018, Intel stock performed exceptionally well. In the 15 years post March 2003, the return was 10.2%, underperforming the S&P 500 by only 0.1% per year.

The next and final question we address is whether the stock market returns that underlay this 15-year performance related to actual business developments that we should have anticipated or to the kind of chronic overvaluation that, since the 1990s, boosted Intel's stock.

Intel in March 2018

Between 2002 and 2017, Intel's business continued to evolve.¹⁰ The low end of the microprocessor market for tablets and smart phones became unprofitable for the company. For these relatively low performance devices, Intel's advanced technology provided no advantage. However,

¹⁰ When we look at Intel as an investment in early March 2018, the official information available from the company runs through year-end 2017. The year-end annual report typically becomes available in mid to late March, but the report of fourth-quarter operations, which includes unaudited full-year financials, typically appears in mid to late February. As a compromise we will assume that we are considering buying Intel stock on the first business day of March 2018, and that the full 2017 annual report is available. This is the same practice we have followed in evaluating an Intel investment in earlier years. Intel's financial information for 2017 was in fact available on February 16, 2018. We use the beginning of March to allow for some variability in dates.

Intel managed to dominate the increasingly important market for server microprocessors where high performance was crucial. It also continued to enjoy a strong franchise position in the market for high-performance desktop PC microprocessors. Other initiatives, in vision systems, programmable microprocessors, computer security, and communications have been less successful. By 2017, Intel had distinct operations in two major business segments for PCs (the Client Computing Group) and servers (the Data Center Group) and three smaller segments for processors embedded in a wide range of smart equipment such as driverless cars, (the Internet of Things Group), for programmable processors (the Programmable Solutions Group), and for non-volatile memories.

The final results of this evolution are presented in Table Ex4.13. From 2002 to 2017, Intel revenues had grown by nearly 6% per year. Operating margins had risen from 16% in 2002 to a peak of nearly 36% in 2010, before declining to just under 29% in 2017. The price of an Intel share rose from \$16.66 in early March 2002, to \$47.84 in early March 2017. As a result, the enterprise value of Intel's core business grew from \$100 billion to \$232 billion. The question for investors was whether the improved performance of Intel's core business justified the substantial increase in market value.

Intel's Cash Return

By the end of 2017, Intel's revenues were running at a rate of roughly \$64 billion per year. That year was a relatively good one for Intel but not a cyclical peak. A sustainable revenue estimate of \$64 billion in March 2018 seems appropriate, although perhaps slightly generous.

Intel's average operating margins over the five years up to and including 2017 were 25%. Over the 10 years to 2017, operating margins averaged 26%, as they had for the 15 year period. They were remarkably stable, although with a small but persistent downward trend. We will assume that sustainable operating margins for Intel at the end of 2017 were 25%. This figure yields an estimate of sustainable operating earnings of \$16 billion (25% of \$64 billion).¹¹

¹¹ The error range for this estimate should run from \$14 billion (roughly 24% of \$60 billion) to \$17 billion (26% of \$65 billion). Average earnings outside this range would be incompatible with Intel's history. Unless there were to be a dramatic change in Intel's basic business, it is the range of realistic possibility. For mature franchises like Intel, dramatic improvements are rare. The possibility of dramatic deterioration is what the fade rate estimate should capture.

Table Ex4.13 Intel financial history, 2002–2017.

ANNUAL INCOME STATEMENT (\$millions)

Year ending December 31	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Revenues \$	\$ 26,764	\$ 30,141	\$ 34,209	\$ 38,826	\$ 35,382	\$ 38,334	\$ 37,586	\$ 35,127	\$ 43,623	\$ 53,999	\$ 53,341	\$ 52,708	\$ 55,870	\$ 55,355	\$ 59,387	\$ 62,761
Gross margin %	50%	57%	58%	59%	52%	52%	56%	56%	65%	63%	62%	60%	64%	63%	61%	62%
Operating income \$	\$ 4,382	\$ 7,533	\$ 10,130	\$ 12,090	\$ 5,652	\$ 8,216	\$ 8,954	\$ 5,711	\$ 15,588	\$ 17,477	\$ 14,638	\$ 12,291	\$ 15,347	\$ 14,002	\$ 12,874	\$ 17,936
Operating income margin %	16.4%	25.0%	29.6%	31.1%	16.0%	21.4%	23.8%	16.3%	35.7%	32.4%	27.4%	23.3%	27.5%	25.3%	21.7%	28.6%
Tax rate %	26%	24%	28%	31%	29%	24%	31%	23%	29%	27%	26%	24%	26%	20%	20%	53%
Net income \$	\$ 3,117	\$ 5,694	\$ 7,516	\$ 8,664	\$ 5,014	\$ 6,976	\$ 5,292	\$ 4,369	\$ 11,646	\$ 12,942	\$ 11,005	\$ 9,620	\$ 11,704	\$ 11,420	\$ 10,316	\$ 9,601
Net income margin %	12%	19%	22%	22%	14%	18%	14%	12%	27%	24%	21%	18%	21%	21%	17%	15%
EPS \$	\$ 0.47	\$ 0.86	\$ 1.17	\$ 1.42	\$ 0.87	\$ 1.20	\$ 0.93	\$ 0.79	\$ 2.06	\$ 2.46	\$ 2.20	\$ 1.94	\$ 2.39	\$ 2.47	\$ 2.18	\$ 2.04
Div. PS \$	\$ 0.08	\$ 0.08	\$ 0.16	\$ 0.32	\$ 0.40	\$ 0.45	\$ 0.55	\$ 0.56	\$ 0.63	\$ 0.78	\$ 0.87	\$ 0.90	\$ 0.90	\$ 0.96	\$ 1.04	\$ 1.08
Dep. Amort. \$	\$ 5,334	\$ 5,070	\$ 4,889	\$ 4,595	\$ 4,912	\$ 4,798	\$ 4,616	\$ 5,052	\$ 4,638	\$ 6,064	\$ 7,522	\$ 8,032	\$ 8,549	\$ 7,910	\$ 7,790	\$ 8,129
Cap expense \$	\$ 4,703	\$ 3,656	\$ 3,843	\$ 5,871	\$ 5,860	\$ 5,000	\$ 5,197	\$ 4,515	\$ 5,207	\$ 10,764	\$ 11,027	\$ 10,771	\$ 10,105	\$ 7,326	\$ 9,625	\$ 11,778
Dividends \$	\$ 533	\$ 524	\$ 1,022	\$ 1,958	\$ 2,320	\$ 2,618	\$ 3,100	\$ 3,108	\$ 3,503	\$ 4,127	\$ 4,350	\$ 4,479	\$ 4,409	\$ 4,556	\$ 4,925	\$ 5,072
Net buybacks \$	\$ 3,333	\$ 3,045	\$ 6,622	\$ 9,435	\$ 3,547	\$ (274)	\$ 6,090	\$ 1,362	\$ 1,149	\$ 12,088	\$ 2,654	\$ 559	\$ 9,132	\$ 2,135	\$ 1,479	\$ 1,845
Distribution %	124%	63%	102%	131%	117%	34%	174%	102%	40%	125%	64%	52%	116%	39%	62%	72%
Net cash \$	\$ 12,466	\$ 17,384	\$ 19,487	\$ 15,025	\$ 12,395	\$ 17,616	\$ 13,832	\$ 16,651	\$ 23,804	\$ 8,957	\$ 9,697	\$ 14,414	\$ 9,519	\$ 10,494	\$ 2,712	\$ (3,907)
Equity \$	\$ 35,468	\$ 37,846	\$ 38,579	\$ 36,640	\$ 37,210	\$ 43,120	\$ 39,546	\$ 41,704	\$ 49,430	\$ 45,911	\$ 21,503	\$ 58,256	\$ 55,865	\$ 61,085	\$ 66,226	\$ 69,019
ROE %	9%	15%	19%	24%	13%	16%	13%	10%	24%	28%	51%	17%	21%	19%	16%	14%
Avg. shares (MLN)	6,759	6,621	6,494	6,178	5,880	5,936	5,748	5,645	5,696	5,411	5,160	5,097	5,056	4,894	4,875	4,836
Employees (THOUS)	79	80	85	100	94	86	84	80	83	100	105	108	107	107	107	103

Source: Data from Intel annual report

Corporate income tax rates for Intel in the five years from 2012 to 2016 averaged 23%. In 2017, Intel's reported tax rate was an unusually high 53%, due to the one year effect of the 2017 US Tax Reform Act and the repatriation of accumulated earnings held off-shore. Ultimately, the 2017 legislation would lower, not raise, taxes for Intel. But as less business-friendly governments come to power, faced with demographic needs for higher retirement and medical expenses, these lower rates might not survive. After adjusting for the temporary effect of the tax reform and the one-time sale of the Intel Security Group, Intel itself estimated that its 2017 tax rate would have been 23%. We will use that figure as our estimate of Intel's long-term tax rate. The estimated sustainable net operating profit after tax is then \$12.3 billion (77% of \$16 billion).

Intel's historical cash distribution levels out of net income were high and relatively stable over five year periods. From 2003 through 2007, Intel distributed 91% of its reported net income in dividends and net share buybacks. From 2008 through 2012, the distribution fraction was 92%. From 2013 through 2017, it fell to 73% (see data on Table Ex4.13). There appear to have been two reasons for this decline. First, Intel's core business became more capital intensive over time. In the early 1990s, the balance sheet value of property, plant, and equipment was about 45% of annual revenue. By 2013 this ratio had increased to 60%. In 2017 it was 65%. The rise in capital intensity increased the amount of investment necessary to support organic growth, leaving fewer funds available for distribution to shareholders. This squeeze was exacerbated by extensive investments in areas beyond Intel's core business as management sought to diversify its operations.

Second, by 2017, Intel's habitual positive net cash balance (cash and investments minus debt) had evaporated. The net cash of \$14.4 billion at the end of 2013 had turned into a negative \$3.9 billion at the end of 2017. Continuing to distribute 90% or more of net income would require further borrowing and debt accumulation, a financial strategy that did not seem to appeal to Intel's management. For these reasons, we will assume a cash distribution level of 80% of net income; above the 2013 through 2017 average but below the long-term historical norms. This level yields a sustainable annual cash return of \$9.9 billion (80% of \$12.3 billion). If anything, this is a generous estimate; future distribution levels below the 73% level of 2017 appear to be far more likely than a

return to the historical levels of 90%.¹² The \$9.9 billion of sustainable cash distribution represents a 4.3% return on Intel's March 2018 enterprise value of \$232 billion, higher than the March 2003 cash return of 3.0% and well above the 1.8% estimated cash return in March 1999.

Organic Growth Return

For the six years between 2012 and 2017 (peak to peak), Intel's revenues grew by 2.5% per year. From 2005 to 2017 (again peak to peak) revenues grew at 4.1% per year. Recent average historical revenue growth appears to have been about 4% per year, at or slightly below global GDP growth. Going forward from 2017, the slowdown in productivity and labor force growth in China, coupled with continued slow growth in the developed economies, meant that future real global GDP growth would be about 3% per year. With roughly 1.5% annual inflation, this puts nominal global GDP growth rate at 4.5%, slightly below the nominal global growth rate for the years during which Intel's revenue growth averaged almost 4%. Therefore, organic revenue growth for Intel going forward from 2017 should be about 4%. Earnings growth, given the slow historical margin deterioration, should be slightly below this figure. To be optimistic, we will estimate organic growth in value for Intel at 4% per year, slightly below our March 2003 estimate of 4.5%. (See Table Ex4.11).

Active Investment Returns

With 4% organic growth in revenue, Intel's 2018 revenue should increase by \$2.6 billion (4% of \$64 billion). The investment necessary to support this growth at recent fixed capital levels of 62.5% of revenue would be about \$1.6 billion. Required working capital investment should be essentially zero, which has been the historical experience. Retained earnings at 20% (100% less the 80% payout ratio) of \$12.3 billion in sustainable earnings would be \$2.5 billion, leaving \$0.9 million for active investment after the \$1.6 billion necessary to fund organic growth.

In fact, Intel had been investing far more than this amount in new initiatives.

¹² We will, therefore, consider the potential impact of lower cash distributions and higher reinvestment levels out of retained earnings.

Spending on acquisitions averaged \$6 billion per year from 2011 to 2017. The largest purchases were Altera, a maker of programmable semiconductor chips, for \$14.5 billion in 2015 and Mobileye, a computer vision, data analysis, machine learning, and self-driving car company, for another \$14.5 billion in 2017.

The value created by these investments was highly questionable. In 2011, Intel acquired McAfee, a computer security software company, for \$7.7 billion. In 2017, Intel's sold a majority stake in its Intel Security Group at a valuation of \$4.2 billion. In the interim, the after-tax earnings of the group appear to have averaged no more than \$150 million per year. Assuming that the cost of capital associated with the McAfee acquisition was 10%, the value Intel received for its \$7.7 billion investment amounted to just \$0.50 cents per dollar invested.

The Altera acquisition looks equally disappointing. In the five years from 2010 through 2014, the last year for which Altera published independent annual financials, Altera reported operating income that declined from \$868 million to \$553 million. Revenues over that five year period were essentially flat, falling from \$1.95 billion in 2010 to \$1.93 billion in 2014. Intel's Programmable Solutions Group appears for the first time in 2016 annual report. Most of Altera's operations would have fallen within this segment. The group's reported revenues were \$1.67 billion in 2016 and \$1.9 billion in 2017. Operating income was negative \$104 million in 2016 and positive \$458 million in 2017. If we were to assume, generously, that Intel's incremental operating income from the Altera acquisition was level at \$600 million per year, then the return on its Altera investment would have been 4.1% per year (\$0.6 billion divided by \$14.5 billion). Assuming a cost of capital of 10%, this represents roughly \$.40 of value created for each dollar invested.

The Mobileye acquisition also appears to have been a mistake. In its last year as an independent company, Mobileye had revenues of \$358 million, 660 employees, and an operating income of \$120 million. The company may have had valuable technology, but it was operating in fields where technology was changing rapidly, and neither it nor any of its many competitors had established a dominant position. Under these circumstances, current earnings power/asset value was what Mobileye was likely to be worth. Since asset value was difficult to assess, earnings power value is our best measure of what Intel got for its \$14.5 billion. Assuming a 20% tax rate and 10% cost of capital, Mobileye's earnings power value would have been about \$1 billion (\$120 million times 80% divided by 10%), a mere 7% of the price Intel paid. This looks

like another example of value destruction brought about through Intel's active investment practice. Given this history, a value creation factor of 40% (\$.40 per dollar) for Intel's active investments would be optimistic. We will estimate the value creation factor at one-third. If Intel were simply investing the \$0.9 billion out of its retained earnings after financing organic growth, the net value creation would be \$0.3 billion (\$0.9 billion times one-third); an additional return of roughly 0.1% (\$0.3 billion divided by \$232 billion) to the company's results. But if Intel were to devote even \$2 billion per year to this kind of active investment, far less than the \$6 billion per year recent average, then ultimately \$1.1 billion would have to come from reduced shareholder payouts (\$2 billion less \$0.9 billion). In this case the net return would be negative \$0.7 billion (two thirds value destruction per dollar times the \$1.1 billion reduction in cash distributions). This figure would represent negative 0.3% of Intel's \$232 billion enterprise value. Rather than assume that such behavior would persist, given Intel's overall history of good management, we will simply assign a zero value to Intel's active investment return. (See Table Ex4.11.)

Total Return, Cost of Capital, and Fade Rate

Our total return estimate for Intel is 8.2%, composed of the cash return of 4.2% plus organic growth of 4.0%. The cost of capital against which this return should be measured would have been less than the cost of capital in 2003 for three reasons. First, interest rates in early 2018 were lower than those of 2003. The March 2018 yield on a 20-year US treasury bond was 3.1% compared to 4.1% in 2003. Second, given the increase in the cash return between 2003 and 2018, a smaller fraction of the return from 2018 would have depended on future growth and returns in the distant future. Third, the greater historical stability and longer track record for Intel in 2018 compared with 2003 would have reduced the uncertainty of an Intel investment. On the other hand, Intel remained a relatively risky investment that should have warranted an above-market average return. With required venture capital returns in 2018 at around 13%, and the overall stock market yielding around 7%, a 10% cost of capital, midway between these two figures, seems appropriate. This is 1% below our 2003 cost of capital estimates (see Table Ex4.11).

But even at a lower 9% cost of capital, Intel's roughly 8% return would have provided no margin of safety against franchise fade. And in 2018, fade risk remained a significant concern for Intel investors.

Regarding the fade rate, some good news and some bad news had emerged in the years since 2003. The good news was that a large part of Intel's franchise remained intact. This was especially the case in the relatively rapidly growing server market, driven by cloud computing, that Intel had succeeded in dominating. Also Intel continued to lead the microprocessor market for desktop PCs. By early 2018, Intel had maintained its leading position in the greater part of the microprocessor market for 25 to 30 years.

The bad news was the evidence from the mobile segment of the markets, especially smart phones and tablets, that demonstrated just how rapidly Intel's franchise position might disappear. By 2012, Intel's mobility segment, making microprocessors for smart phones and tablets, was down to \$1.8 billion in revenue with operating losses of \$1.8 billion. Two years later, revenues had dropped to \$0.2 billion while operating losses had risen to \$4.2 billion. In 2017 Intel's Client Computing Group, which made microprocessors for PCs including the old mobility products, had revenues of \$34 billion, stable since 2014, but with income of only \$12.9 billion, down from \$14.6 in 2014.

If the desktop market represented by the Client Computing Group followed the mobility market, then Intel as a whole would be in genuine trouble.

Balancing these two factors, the optimistic approach would be to assume that the expected future half-life of Intel's franchise had grown slightly, say from 30 to 40 years. This would mean a reduction in fade rate from 2.4% to 1.8% per year. The post-fade return to Intel's stock in March 2018 would then have been 6.4% (8.2% minus 1.8%). Compared to a cost of capital of roughly 10%, this would correspond to an intrinsic value to market value ratio of roughly two-thirds. Adjustment of the organic growth return to reflect this factor would lead to a further reduction in our estimated return of 1.3% (one-third of 4%). Thus our adjusted pre-fade return estimate would be 6.9% (8.2% minus 1.3%); including the fade rate, the return would be just 5.1%.

Intel stock in March 2018 was not a bargain and was almost certainly significantly overvalued from a long-term perspective. This ought not to surprise us. From March 2014 to March 2018, Intel stock price almost doubled; its average return over the period was roughly 22% per year. At the end of such a run, it was not the kind of disappointing, cheap, ugly stock that is representative of most value opportunities. Even assuming a sustainable franchise and decent prospects for growth, the initial starting price, relative to intrinsic value, posed a major challenge to a successful investment in Intel.

Conclusion

Intel stock has experienced dramatic changes in price over the last 30 years. There have been dramatic declines, such as that between 1998 and 2002, and rapid rises in response to improved operating performance, such as the period between 2014 and 2018. But price movements by themselves have not determined whether Intel is a worthwhile value investment. Value for a franchise business such as Intel depends not just on current earnings but also on future organic growth prospects, the quality of capital allocation, and the likelihood of franchise erosion. Balancing the impact of these complex forces on valuation can be effectively done only in the context of a return-based approach of the kind we have used here. DCF valuations capturing all these factors would be unavoidably opaque and imprecise.

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Research Strategy

A third step in the investment process, after search and valuation, is the active and intensive collection of information. When Benjamin Graham and David Dodd did their seminal work, many investors did not even perform detailed analyses of company financial statements. Indeed, until the Securities Act of 1933, even companies with publicly issued securities were not required to publish systematic financial information on a regular basis. Active research at that time consisted largely of ferreting out the information necessary to perform a basic valuation. For example, one of Graham's key insights into railroad valuation was that detailed financial statements and operating information, although not widely published, had to be filed with the Interstate Commerce Commission, the federal body responsible for regulating railroads, where they could be inspected by the public. Today basic financial and operating information for all companies is instantly available to all investors on the Internet. The SEC itself publishes corporate filings, including quarterly 10-Qs and annual 10-Ks, on its Edgar website. Services such as Bloomberg and Capital IQ facilitate use of this information by organizing data from many sources and compiling them into standard electronic formats for either quick viewing or further analysis. Modern investors look well beyond the basic data that underlies valuation. Still, even with this instantaneous availability, much important information is overlooked, and a good deal of effort goes into collecting information of only peripheral value. A systematic research strategy is an important part of any well-conceived investment process, especially since so much information is readily available to investors on the other side of the trade.

By research strategy, we refer to the effort spent actively investigating investment opportunities after a basic valuation has been done. A good research strategy will be both complete and efficient. By complete, we mean that it should cover all the important information, both specific to the company and more general, that bears on the likely success of the investment. In particular, it should address all of the collateral evidence that is readily available and often overlooked. Careful analysts should always know who owns or has recently sold a security in which they are interested, whether management is buying or selling the company's stock, how prevailing market opinion views the company and its prospects, and how well or badly the analyst him- or herself has done with past investments in similar opportunities and what was learned in the process. By efficient, we mean that an analyst's efforts should be focused on the information that is likely to have the greatest impact on valuation. This means identifying the key assumptions that underlie a valuation and concentrating research time on them. This last point may seem obvious, but many investment firms have research practices and information collection formats that must be followed regardless of their relevance to a particular investment decision.

Indirect Information

Collection of indirect information begins by looking at who owns the stock of a company. This information is regularly published both in the United States and in other major markets. One source is the company itself which usually publishes the identities of major shareholders. A second source is information on the holdings of major financial investors. In the United States investors managing more than \$100 million in US equities must reveal their equity holdings on a quarterly basis to the SEC. In addition, when a particular investor acquires more than a threshold percentage of a company's stock, that fact must often be disclosed in a separate regulatory filing both in the United States and abroad.¹ This information is generally available from all the major financial data services as well as the SEC's Edgar. For the holdings of particular investors, there are websites that are devoted to publicizing them, especially the subgroup of investors deemed especially worth following—usually investors

¹ In the US, the SEC requires buyers to file a report when they pass the 5% ownership threshold. More frequent and timely filings must be made by owners of at least 10% of the shares.

who have a history of outstanding performance. But a note of caution is in order. The information may not be up to date, and the prestigious investor may be mistaken. The fact that some notable investor owns the security is something to be noted, not blindly followed.

One obvious way to use this information is as part of a search strategy. Seeing what successful investors, such as Warren Buffett, currently hold and have recently bought and then following their actions is not a bad way to invest. If the stocks in question have declined since the most recent disclosure, so that an analyst can get a better price than an outstanding investor, then buying later may even enhance returns. In pursuing this strategy, it helps to identify those areas in which noted investors have been most successful. By virtually everyone's judgment, Warren Buffett is an outstanding investor. But he has done especially well investing in banks and insurance companies. He is a long-term student of both industries and has owned and run companies in them. His other selections, while successful, have not done as well. As another example of specialization, Mario Gabelli has been a successful student of the media and telecommunications for decades. If you are going to look to noted investors for ideas, concentrate on those areas in which they are most expert and/or have been most successful.

The benefits of identifying expertise apply to geography as well as industry. Suppose you decide to look at potential investments in Turkey because it has experienced a major market decline and is the source of a raft of bad economic and political news. The holdings and recent purchases of investors who have long-term experience and a record of successful investments in Turkey will be a good place to start. In looking at industries or regions that are not within your own circle of competence, developing a list of well-informed successful investors and examining their holdings is potentially useful.

A second benefit of looking at who owns a particular security arises after you have done an initial valuation. If the company in question is widely held by disciplined, value-oriented investors with high levels of relevant expertise, that is obviously a positive sign.² If they have been increasing their holdings in the face of a declining stock price, that is an even stronger one. It indicates that they were comfortable with their initial positive assessment and willing to take advantage of the decline in price to increase the number of shares they own, if not always the

² There are examples, such as Valeant in 2015, in which many reputable value investors made the same mistake, assuming that a business strategy was more durable, and perhaps even more legitimate, than proved to be the case.

dollar size of the position. On the other hand, if those same investors are reducing their holdings in the face of declining prices, you ought to be cautious and review carefully your valuation assumptions. If the shareholders of your target security have been historically less disciplined, less well-informed about the area, and less successful investors, then again you want to reconsider but not necessarily abandon any originally favorable view.

One set of investors who are particularly well-informed about a company's future prospects are the company's senior managers and directors, the classic insiders. Whenever they buy or sell company securities, they are required to disclose their actions, including the nature, timing, and size of any transaction. This information is then almost instantly disseminated over the Internet by the SEC and financial investment information services. Insider behavior has historically been a powerful predictor of the likely success of an investment. If they are buying, the chances are good that they believe the stock is undervalued and will appreciate in price. Sometimes insiders make arrangements to sell when a stock price reaches certain pre-specified levels. These levels are sometimes disclosed and sometimes can be inferred from repeated transactions. Higher pre-specified selling prices clearly represent a more positive view of the company's future.

Just as the holdings of some external investors in a particular stock may be more informative than the holdings of others, some insider actions are more revealing than others. Insider buy decisions seem to be more reliable indicators than sell decisions. Also, academic research suggests that the transactions of board members who are long-serving and live close to a company's headquarters are a better predictor of future company performance than those of other less well-connected directors. Similarly, the actions of very senior managers (e.g., CEOs and CFOS) may sometimes be less revealing than those of lower level executives. Given their greater visibility, higher managements are more exposed to criticism for their transactions and, as a result seem to be more circumspect in their behavior. It is important in using insider behavior for investment purposes to think in detail about which transactions will be most informative for any particular company. A careful review of the historical relationship between insider transactions and subsequent company performance is necessary to make the most of insider trading information.³

³ H. Nejat Seyhum. (1998). *Investment Intelligence from Insider Trading*, Cambridge, MA: MIT Press.

Another signal of a management's view of future prospects is a decision to issue more equity either through a secondary market offering, an acquisition for stock rather than cash, or by enriching stock-based compensation plans. Academic studies over many years strongly indicate that issuing more equity is a negative sign for future returns. First, shareholder-oriented managers will not issue equity today if they expect their business to outperform current expectations in the future. In that case, they would raise funds through debt for near-term needs and sell equity at a higher price later or simply defer external financing until the stock price improves. Second, rational managers should never sell equity when they think their stock is undervalued. These sales are a gift to outside investors at the expense of current shareholders. An equity issue usually signals the stock price is not undervalued, and raises the probability it is overvalued, a fact that the market fully understands.⁴ Third, strong companies are better able to handle debt than weak ones, and there are usually tax advantages to debt financing. Therefore, the decision to raise capital by issuing equity suggests that a company is in a relatively weak financial position (or that in management's view its stock is overvalued). The specific choice of financing from among the full spectrum of options reveals additional information; long-term covenant-lite subordinated debt issues, which may carry higher interest rates, indicate greater financial weakness than shorter-term, more restrictive, senior debt, which usually carries a lower interest rate.⁵ Like insider stock sales, the information conveyed by equity issues also depends on the context. For example, when overall stock prices are depressed, issuing equity is likely to be especially costly to existing shareholders. Under these conditions, usually going into or during a recession, equity issues have historically been rare. Companies selling equity at these times are companies with backs firmly against the wall, when the negative signal of an equity issue is especially potent. Investors should take this into account.

A third and perhaps the most valuable source of information is the development of a network of able, like-minded investors whose areas of expertise complement your own. For example, a critical issue for those investing in small- and medium-size Japanese companies is the quality of the corporate governance in its treatment of shareholders.

⁴ Equity issue announcements have historically triggered negative stock price movements that extend for a number of years.

⁵ An exception to this last point is companies *in extremis* that have no choice but highly restrictive senior debt. But these cases are readily identified.

Japanese companies are often driven by internal operational imperatives for growth, employee welfare, and industry position, goals that may be laudable in themselves but leave little room for shareholder interest. In many cases, the shareholders of successful companies see little of the benefit of that success in the form of cash distributions. This behavior is not immutable. Anticipating favorable changes in shareholder treatment ahead of the stock market as a whole is a key part of profitably investing in small- and medium-sized Japanese companies. But it is a difficult feat to pull off, because, quite apart from rules against revelation of insider information, many Japanese managements are largely inaccessible to outside investors, especially investors without a significant presence in Japan. As a result, among the most successful investors in small- and medium-sized Japanese Companies are a group of foreign investors who have lived and invested in Japan for 30 years or more. These people have tended to gravitate to different Japanese regions within which they have developed close relationships with many local Japanese companies. Thus, even without access to inappropriate inside information, they are able to understand current changes in company cultures and anticipate the consequences of these changes for shareholders. A crucial aspect of their success is that they meet regularly to share information so they are not limited to investments within a single region. They are also connected to similar investors in Europe and the United States who have valuable perspectives on small- and medium-sized companies in their own territories. The network as a whole enables its members to perform better than any individual member could do individually. Connection to these kinds of networks is an important part of an efficient and complete research process.

Consensus public perceptions of a company's situation are useful in a different way. They do not provide active investment guidance, unlike outstanding investors, corporate insiders or networks of able, focused, and informed investors. Investing with the herd will produce market returns at best. However, public perceptions are not always misguided. In many situations average crowd consensus is a better judge than any individual. If you are going to invest with a view of the future that is contrary to those perceptions, a posture that generally characterizes value investors, it is still important to become familiar with the consensus point of view. Sometimes that perspective will include things that you have overlooked. These overlooked factors may often have no bearing on your original analysis, in which case no harm is done. But at times they will, and those cases justify the exercise. Value investors often declare that they consciously avoid "Wall Street" market research, which

is a common characterization of broad market perceptions. Their fear seems to be that they will be negatively influenced by the siren song of the consensus. But genuine value investors should be able to look dispassionately at consensus views and reject them when warranted. For a disciplined value investor, there ought to be no downside to identifying the “variant perceptions” of the market “crowd” that differs from your own analysis.

Finally, the investor should look carefully at his or her own prior behavior. A large fraction of value opportunities are created by the irrational behavior of Benjamin Graham’s Mr. Market. The field of modern behavioral finance has identified a number of persistent common irrationalities, the exploitation of which are an important element of the search strategies we defined in Chapter 2. But irrationality is not confined to investors outside the value investing clan. Value investors are human beings with all the same underlying instincts as non-value investors. There will inevitably be particular tendencies that affect all but a few inhumanly disciplined value investors that can lead to bad investment decision-making. The important thing is not to allow these tendencies to spawn recurring mistakes of either commission or omission.

Mastering your own cognitive biases involves first identifying them and then developing strategies to deal with them. Identification requires careful monitoring. Every value investor should keep a record of decisions made to buy and to pass, to sell and to hold, with a note of the associated rationale. That record should be reviewed at regular intervals, at least yearly, to see if there are persistent patterns of behavior that undermine the processes and standards that the investor has established. Making a mistake once is excusable. Making the same mistake repeatedly smacks of carelessness. And this applies both to actions taken and not taken. Strategies for overcoming identified irrational behavior are many, but the simplest and most effective is to avoid completely those investment situations where you made poor decisions. The irrationality of others creates opportunities; your own irrationality creates losses.

Direct Information

The efficient pursuit of direct information is rooted in the process of valuation. For fixed income investments with clear payoffs—usually senior debt in distressed situations—the critical issue is timing. You know what you are going to get; the only question is when. For example, if the payoff is face value plus accumulated interest at a known coupon rate of

5% with unpaid prior interest of two years, the immediate payoff would be 1.1 times the face value. At one year the payoff will be 1.15 times, at two years just over 1.2 times. If the bonds in question are bought at 90% of face value, the immediate return would be 22% (1.1 divided by 0.9), and the rate of return would be very high indeed. At one year, the payoff is 28% (1.15 divided by 0.9) which is also the annual rate of return. Two years out, the payoff is 33%, but the annual rate of return falls to 15.5%. As the payoff recedes into the future, the return and the present value of the payoff, assuming the appropriate cost of capital is about 5%, decline steadily. For very long delays, the payoff may be unattractive. Unless there are issues of seniority or the ability of the company to cover the senior debt, timing is the most important determinant of the return and thus the proper focus of active research. If there is a question of legal seniority, that also requires attention, provided the answers have a material impact on the potential payoffs. If the ability of the company to cover the senior debt is in question, then the circumstances under which this might be of consequence are proper subjects of active research. In both these instances, the research will be relatively narrowly focused. Complete enterprise valuations across a range of scenarios, especially optimistic ones, will usually be a waste of time.

For businesses in competitive markets or regulated companies whose rates of return on investment are controlled at or near their cost of capital, growth is unlikely to create value. In these competitive situations, only companies with sustainable cultures of exceptional operating efficiency will make growth worthwhile to investors, and these companies rarely trade at bargain prices. For these companies, complex models of earnings, investments, and cash flow into the extended future will generally contribute little to identifying future returns. To the extent that these models show substantial benefits from higher growth rates, they are probably misleading and potentially harmful to sound investment decisions. At best they demonstrate a misallocation of research time and effort. Efficient research efforts for non-franchise companies should focus on estimates of current asset and earnings power values. Yet these areas, especially asset reproduction values, are often overlooked, as analysts focus on short-term earnings movements and future growth.

The Magna International case, which focuses on the period following the financial crisis of 2008-9, is instructive here. Magna, along with other automotive industry stocks, traded on the basis of earnings scenarios that were carefully researched but ultimately of little value. The North American automobile industry, including independent parts suppliers such as Magna, was going to be economically viable for a considerable

time. The industry was also going to be highly competitive as it had been for many years. This meant that future long-run profitability would have to support the reproduction value of the necessary assets. Some companies, such as General Motors, ultimately faced bankruptcy because the net reproduction value of their assets—net operating assets minus formal debt minus amount owed to suppliers, dealers, and workers (i.e., the book value of the equity)—was well below zero. The companies were not going to fail immediately because the creditors—workers, dealers, and suppliers—had a strong interest in their continued existence. But ultimately they were doomed since those creditors could not afford to write off the debts that were growing steadily over time, and a negative net asset reproduction value will not support profits in the face of competitive entry. Yet very few analysts bothered to compute asset reproduction values, including the estimated value of intangibles, for these companies. Other companies with positive net asset value also went bankrupt because high levels of leverage meant that any sustained downturn would drive the value of these below zero.⁶ Again, very few analysts concentrated substantial research time on these downside balance sheet scenarios. Instead, they were focused on complex forward-looking earnings projections that were unmoored from the economic reality of the automobile industry in the 2000s. For non-franchise businesses, a well-conceived research process should devote roughly equal time to asset and earnings power values since they contribute equally to value determination.

For franchise businesses, growth does matter, and asset values are relatively less important. Given this difference, the first research task is to determine the extent to which a company's operations are protected by barriers to entry. In principle, identifying the presence of barriers to entry is relatively straightforward, as we describe in Chapters 6 and 7. It is determining the extent, durability, and consequences of the specific competitive advantages that requires significant active research attention. Most important is a thorough understanding of the businesses involved, including the principal submarkets in which a company operates either in physical geography or product space. The research needs to extend beyond the company in question to market variables and, most important, the financial performance of other firms in the industry. A lengthy

⁶ The sliver of equity with which these firms started would be eroded by a decline in business and ultimately fall below zero. The earnings power associated with the reproduction value of these assets would not be enough to make payments on the liabilities. Sooner or later, these situations, without some kind of intervention, end in bankruptcy.

industry history is essential, ideally extending over more than one industry cycle, to identify long-term industry trends and any recent deviations from these trends. It is critical to determine whether current deviations are temporary or permanent. Conducting research at this level is a major undertaking, but it does have the advantage of being applicable to other companies in the industry that might later become potentially attractive value investments. The efficiencies that flow from this accumulated knowledge are a major argument for specialization.

Once the nature of a business's franchise is well understood, the relevant quantities that should be actively studied again arise directly from the valuation process. The key value-determining factors are (1) current earnings power, (2) distribution-retention policy for earnings, (3) organic growth rates in revenues and margins, (4) active value creation, (5) franchise fade rates, and (6) the cost of capital.

Measuring items such as current earnings power and the cost of capital may be fairly straightforward, but others, such as active value creation, franchise fade rates, and even future distribution policy, will require a thorough understanding of the capabilities, strategic understanding, and priorities of management. Active research time and attention should be directed accordingly. The most difficult issue is the value created by active investment, although it is less significant for companies that regularly distribute 85% or more of their earnings, leaving little to invest.

Management, Activism, and Catalysts

For both franchise and non-franchise businesses, management performance is central to both current earnings power and future value creation. Active research regarding management should be comprehensive. Good shareholder-friendly management performance involves more than just a high level of distributed earnings, which is the central demand of much "show me the money" activism. There are four major areas of management performance. They are (1) achievement of operational efficiency both in controlling costs and in areas such as marketing and product development, (2) the strategy for growth, especially capital allocation, which should avoid growth for its own sake and focus on those opportunities in which growth creates value, (3) financial structure and the distribution of cash flow to investors, and (4) human resource management. A complete management assessment looks at all of these areas.

By far the most important area is the pursuit of operational efficiency. A wide variety of studies, notably those financed by the Sloan Foundation, but including many others,⁷ have demonstrated that the most efficient firms in an industry typically have cost structures that are between one-half and one-third of the industry average. This represents a 100–200 difference in efficiency between leading and average firms. These differences exist even though leading firms typically have observable workforce characteristics—education, age, gender, and others—similar to those of other industry firms. Their levels of capital investment are also unexceptional. And leading firms tend to use seasoned rather than leading-edge technologies. The gap between the best and average firm efficiencies depends, then, not on resources but on management's commitment to and skill in running an efficient organization. Although more difficult to quantify than costs, other studies tend to show comparable cross-company differences in marketing and R&D efficiency. Finally, leading firms seem to improve performance at roughly the same rate as less efficient ones. The average performers catch up to the leaders very slowly, if ever.

The implication is that at any time almost all firms could harvest large potential returns from a concentrated management focus on operational improvements. Managements that succeed will create considerable value for shareholders. For investors, the ability to identify such managements early on, before their qualities are fully appreciated by the market, should produce significantly better than average returns. Two identifiable characteristics stand out. First, achievement of operational excellence is a marathon, not a sprint. It requires constant attention to incremental improvement. This in turn calls for a management organization focused on driving such improvements, including a CEO who makes it a high priority. Sudden short-term interventions from outside agents, like the notorious "Chainsaw Al" Dunlap, rarely lead to long-term value creation. Second, in any company, the time and attention of senior management is a scarce resource. Those managers who embrace aggressive acquisitions or international expansion or promote their own public visibility rarely achieve operational excellence. Managements that focus effectively on operational performance will generally have simple, narrowly directed overall corporate strategies.

The second important area of management performance is the growth strategy. As we have reiterated incessantly, only growth

⁷ See, for example, van Biema, Michael, and Greenwald, Bruce. (1997). "Managing Our Way to Higher Service-Sector Productivity." *Harvard Business Review* (July–August).

investments that produce returns above their cost will create value. This growth must benefit either from sustainable competitive advantages or long-term operational excellence. Since operational excellence only rarely generates significant growth, value-creating growth will generally be found in areas where the company currently enjoys sustainable competitive advantages. For starters, managers must understand where their competitive advantages exist, if they exist at all. The first place to direct growth is within those markets in which their existing franchise lies. When Walmart expands into groceries or Coca-Cola adds non-cola beverages to its product line, each is leveraging dominant existing local distribution infrastructures. When Microsoft adds features to its Windows and Office Suite programs, it is leveraging both existing products and customer captivity. While widening moats through improved offerings, the judicious exercise of pricing power also represents value-creating growth within existing franchises.

The next move for a firm is to expand into adjacent markets to which their core competitive advantages carry over. Historically, Walmart spread sequentially into adjacent geographic markets in which it could use its existing distribution, advertising, and management infrastructure. Oracle has followed a similar strategy in product space, adding industry focused functional software to its central core of database management systems.

Value creating growth is more difficult to achieve beyond these two areas. Pursuing previously underpenetrated specific markets in a disciplined way may make sense if there is the ultimate possibility of dominating these markets and economies of scale and customer captivity are sufficient to convert that dominant position into sustainable barriers to entry. Trying to accomplish this feat in many markets simultaneously is a recipe for failure. Finally, attempting to compete in markets that have dominant incumbents in the hope of achieving the return enjoyed by them is a plan for value destruction. The reason that those incumbents earn sustainably higher returns is because their returns are protected from competitive erosion by insurmountable barriers to entry.

A good management will pursue growth opportunities of the first two types. Pursuing growth without competitive advantages is a sign of poor management. Managers that direct their companies into global expansion without any clear rationale except that globalization is the wave of the future fall into this category. It also includes management who unthinkingly enter rapidly growing markets in emerging economies, such as India and China, or in emerging technologies because that is where the growth is. Without identifiable

market-specific competitive advantages such growth will not create value. Managers who regularly and enthusiastically announce initiatives of this sort are managers to avoid.

A third important area of management performance to examine is financial resource management. First, if there is no obvious way for a company to employ financial resources at equal to or above cost of capital returns, then these resources should be distributed to investors through either dividends or share buybacks. The double taxation of company financial returns at the corporate level and later when distributed at the individual level is one reason for this policy. Potentially more damaging is the historical record that reveals, sooner or later, managers with spare financial resources will be tempted to put them to do something with them. If there are no good investments available, then they will make bad ones. These include undisciplined expansions that destroy value or, even worse, throwing good money after bad in futile attempts to rescue failing initiatives or struggling divisions within a company. Good management will remove this temptation by distributing those enticing piles of cash, and most activist investor interventions are designed to push them or their replacements in this direction.

Another sign of good financial management is the acquisition of financial capital at the lowest possible costs. If debt is cheaper than equity either for tax reasons—because while interest payments are tax deductible for companies, dividends are not—or market reasons, such as government intervention that produces distortedly low interest rates, then well-managed companies will prefer debt to equity finance. They still need to be prudent. Much of the cost of debt arises from the mere possibility of financial distress in which the imperatives of debt repayment materially impair the non-financial operations of the firm. This threat may undermine hiring, customer relationships, suppliers' cooperation, and public perceptions of the company well before formal financial distress occurs. But there are sensible levels of debt at which the possibility of distress is remote or the consequences of distress are limited, such as tobacco companies with strong brands. Good managements in these circumstances will take on debt up to the limit where there is no potential for material interference with business operations. This policy, by lowering the cost of capital, will create long-term value for shareholders.

Finally, investors should look for good or improving human resource management policies. A history of steady, non-cyclical hiring is a positive sign. Companies that add staff energetically in good times, when labor markets are tight, and then resort to expensive lay-offs in bad times, when

laid-off workers have particular difficulty finding other jobs, are serving neither their employees nor their shareholders well. Ultimately, it will make it harder to attract and retain a high quality workforce, since among other things good workers are easiest to hire in bad labor markets. The result will be poorer business performance. A second responsibility of human resource management is succession planning. Good management should have an orderly process for replacing itself. Without one, any management quality will be relatively short-lived, and its impact on value will be reduced.

In assessing the impact of good management, it is important to remember that the past and current quality of management is already reflected in a company's financial performance. It is changes in management quality that have to be examined. Some change comes from gradual improvements in management performance and company culture, often as a result of a near-death or other adverse experience. This change will usually be visible in improving financials. The difficult question is one of duration: how deeply are the changes, for good or ill, becoming embedded in a company's culture and how persistent are they likely to be. Other changes, also for good or ill, may be the result of new top management. It should first be apparent in shifts in management focus. However, unless the new management has a clear mandate for change and the resources to implement it, like 3G in the Heinz takeover, the results will emerge relatively slowly. Thus, there is usually time to assess the performance of new management before making a major investment.

Changes of the 3G/Heinz sort are generally the result of activist intervention. Investments made in anticipation of activist intervention, whether on the basis of an announcement of intent, in response to a company's concessions to an intervening activist, or in the face of an imminent activist takeover, are common among value investors and obviously worth considering. However, not all activist interventions are created equal. The success of activist investing for either the activists or those who follow their lead depends on understanding the likely outcome of the intervention. That rests on two elements. First is the likely response of the target company, whether it accepts or rejects the changes activists propose. The second is the quality and scope of the intervention, which will determine the size of any benefit if the target company responds either voluntarily or, after a change of control, involuntarily.

The response is an issue of company culture, which varies from country to country. Asian companies are far less receptive to activist interventions than those in Europe. European companies, especially

where there is a history of family control, are more difficult to budge than North American ones. However, even if companies manage to resist a full change of control of the 3G/ Heinz variety, there is a range of corporate responses. Some companies are rigidly opposed to intervention. They refuse to engage with activists and change little or nothing in their current practices. Systematic academic research indicates that for these companies, there is no significant shareholder benefit to activist intervention unless an activist is able to seize control of the target company. However, many companies do respond to activists and establish a constructive interaction with them. For these companies, there is wide variation in the level of the company's responses, but significant beneficial change can occur without a complete change of control.⁸

The nature of the change and the size of the benefit, which is our second element, depends critically on the motives of the activists involved. Activists who are narrowly focused on changing corporate distribution policy, pushing for increased dividends and buy-backs, and attempting to intervene across a wide range of industries produce on average the smallest benefit. Industry-specialized activists, those who have taken the time to become well-versed in the economics of their companies' industries and who are concerned with improving management practices across the board, are more likely to intervene successfully and provide the greatest benefit when they succeed. Paul Hilal is a model of this activist type. Investing in response to interventions by these activists will produce the best returns. Understanding who they are and where their expertise lies is an important element in successful activism-related investing.

Activist intervention is one way that the underlying business value of a poorly managed company can be realized. Events that bring market prices into line with underlying values, such as takeovers, recapitalizations, and activist interventions, are referred to as "catalysts," and a great deal of investment research among value investors is devoted to identifying catalysts. Some value investors will not invest in undervalued companies at all unless they can identify the catalyst that will cause the underlying value to be realized in the near future. Before devoting research time to identifying and analyzing potential catalysts, it is important to understand the role that catalysts play in different kinds of value investing.

⁸ See Wei Jiang's research on hedge fund activism. For example, Bebchuk, Lucian A., Brav, Alon, and Jiang, Wei. "The Long Term Effects of Hedge Fund Activism." NBER Working Paper No. 21227, June 2015.

Benjamin Graham and David Dodd paid little if any attention to catalysts. Their importance was first made prominent by Mario Gabelli and other investors who rely on deal-based estimates of private market values to anchor valuations. Graham and Dodd had faith that although short-run prices are determined by the stock market acting as a “voting machine,” in the long run the market was a “weighing machine,” and intrinsic value would ultimately surface. In the meanwhile, an equity bought at a discount would continuously provide an above-market earnings return. Suppose an all equity-financed non-franchise business has an earnings power of \$50 million per year and an equity cost of capital of 10%. Its earnings power value, which should approximate its asset value, would be \$500 million, providing an earnings return of 10%. If it were purchased at a price of \$250 million, a margin of safety of 50%, then the earnings return would be 20%. Suppose, to keep things simple, all the earnings are distributed as dividends. Then the annual cash return on the purchase price of the stock is 20%. It would be nice if the price were immediately to jump to the intrinsic value of \$500 million, but even without this move the investment is an entirely satisfactory one. Moreover, if we were to sell it for \$500 million but could not find comparable bargains in which to invest the proceeds, \$500 million invested at the 10% cost of capital would produce the same annual earnings return as our stock without the increase in price. Hence the relaxed Graham and Dodd attitude toward accelerating the convergence of intrinsic value and market price.

Suppose that instead of distributing all earnings as dividends our company's management retains and reinvests half of earnings. If it does so in a way that neither creates nor destroys value, the \$25 million of retained earnings would generate \$25 million of value each year, which would increase future earnings. The dividend return on our purchase price of \$250 million would be 10% (\$25 million divided by \$250 million), but we would also receive another 10% worth of a valuable business whose future earnings and dividends we would get with no further investment. While waiting for the true value to surface, we would still be earning a 20% annual return. A problem does arise if management is a poor capital allocator and destroys half the value of the reinvested returned earnings. In that case, our returns will be well below 20%, and replacing management will be a more urgent task. A catalyst that can make that happen will now be of considerable interest.

The importance of a catalyst to remove poor management will be even greater if an investment was based on asset value without a comparable earnings power value. Those assets in the hands of poor

management are not producing an adequate current return, quite apart from any loss in the value of retained earnings through poor capital allocation. If management destroys half the value of the original assets, then our earnings return will be \$25 million. If half of the earnings are retained, creating only \$.50 of value per dollar, the value of those earnings will be just \$18.75 million (\$12.5 million + \$6.25 million), a 7.5% annual return compared to a 10% cost of capital. Delay in realizing the value of those assets may wear away any original margin of safety. A catalyst will then be an essential part of a successful investment.

This example illustrates the fundamental nature of catalysts. They are protection against bad management. If management makes good use of its assets and reinvests effectively, a catalyst is welcome but not essential to a successful value investment. With bad managements, which in practice are often capable of destroying value at a breathtaking pace, a catalyst is essential, and the identification of catalysts should be an important part of any research process.⁹

A final word ought to be said here about “value traps” and catalysts. There is a persistent belief among some value investors that unless there is an imminent catalyst that will drive market price to intrinsic value, then the value constituting the margin of safety will be “trapped” and never be realized. These are investors who never invest without catalysts. For them identifying and investigating catalysts is a central part of their research process. Given the Graham and Dodd analysis above, this outlook is misguided. For investments in companies that are well-managed, value does not get “trapped” even if the movement of price to full value takes a long time. True “value traps” are inextricably linked to poor managements. In that case, given the ability of bad managers to destroy value, it is a mistake to invest unless there is a clear, near-term prospect of management change. Otherwise, focusing heavily on catalysts is a distraction from other more important research areas.

Other Research Process Issues

There are two final points that need to be made about the research process. The first has to do with compulsive behavior. A compulsive attention to detail may be a useful asset for an investment analyst; maybe it is the last rock that hides the secrets of either hidden value or hidden

⁹ The same line of reasoning applies to franchise businesses, where there is more value to squander.

disaster. But in the real world of limited time, compulsive behavior can be detrimental. Before the outstanding investor Edward Lampert acquired a large position in AutoZone, he is said to have visited every one of its hundreds of stores. If more than an urban legend, this level of scrutiny would have been time-consuming and unnecessary. Visiting a carefully selected sample of 10% of the stores would have conveyed almost all the necessary information in a fraction of the time involved. Given Lampert's subsequent difficulties operating his Sears and Kmart stores, that time might have been better spent learning about the economic imperatives of retailing in general. Compulsive, formula-driven active research will often lead to the collection of much information of little incremental utility at the expense of the pursuit of more relevant data.

This criticism applies especially to investment companies that obsessively tabulate the number of their annual visits to company managements. Management interactions can be a useful source of information, but in a world of fair disclosure rules, the value of intermittent management visits has been greatly reduced. Public statements by managements about their strategies, their views of the economics of their markets, and their broad near-term prospects are widely and frequently disseminated. A careful reading of this record will usually provide 90% or more of what management has to offer. The reliability of these statements can and should be evaluated against the widely available public record of the management's past actions and performance. Meetings with management, which often require time-consuming travel, should occur only after the public record has been fully analyzed and any major points of ambiguity or uncertainty that remain have been identified. To calibrate the reliability of management responses to direct questions of this sort, it is useful to ask a number of questions to which you already know the correct answers. Questions that focus on competitive advantage and longer-term strategic thinking can provide important information about the future of the company. The topic of capital allocation is almost certainly a key part of this discussion. In practice, the most useful management interactions appear to arise from continuous, high-frequency contacts between knowledgeable industry analysts and a low-profile management.

The second point about research practice concerns the kind of information sought. The Graham and Dodd tradition focuses on continuity and long-run value. Market opportunities often arise from adverse events to which investors overreact by treating temporary adverse changes as if they were permanent. The Graham and Dodd approach is to look at these situations and identify those where there is underlying stability. If they do forecast change, it is on the basis of well-established long-term

trends for which there is a record of years rather than months. A much more common practice is for analysts to attempt to anticipate change ahead of the market. Enormous effort is devoted to trying to forecast near-term earnings fluctuations, business and/or industry cycles, the arrival of disruptive technology or the impact of the latest marketing innovations. Predicting the timing, duration, and magnitude of these changes is far more difficult than identifying stability. Serious recessions for either economies or industries are rare—probably less than 1 year in 7. Cyclical stability, therefore, characterizes the other 6 years. In addition, most recessions are temporary. The downward phase of a typical recession is 6 to 9 months long.

The history of the global crisis that began in 2008 should reassure investors that apocalyptic projections are generally wrong. If a firm selling at 15 times earnings loses two entire years of earnings, that is a loss in value of only about 15%, much of which has historically been offset by temporarily high post recovery earnings. It is best to enter an active research race with the fewest competitors. If most investment analysts are looking to forecast change, then concentrating on stability will provide a greater chance of success. Having a differentiated view about the next three to five years is more likely to lead to outperformance than intense focus on the current quarter's sales growth rate. For all these reasons, a Graham and Dodd approach that incorporates the same contrarian stance toward research as it does toward finding undervalued stocks is likely to produce superior results.

10

Risk Management and Building Portfolios

In the late 1970s, a major American corporation decided to evaluate its pension fund. The study began by reviewing the performance of the company's equity managers, of whom there were well over 100. The examination first looked at trading costs. To start, the analysts simply calculated how much the sales of one manager offset the buys of another during any given week. To the company's surprise, in a typical week approximately 90% of orders netted out. With hundreds of managers, positive and negative sentiment on any particular stock tended to be evenly distributed; selling by pessimists largely matched buying by optimists. Indeed, if the pension fund had not been steadily receiving more contributions to cover current employees than it was paying out to retirees, the percentage of offsetting trades would have been even higher. There was a similar issue with the net buys. The actions of hundreds of independent investment managers as a group assembled a portfolio that closely tracked the stock market. Managers who were attracted to a particular industry or stock, say a small oil company, were counterbalanced by other managers who felt just the opposite. As a result, more than 90% of net purchases simply mirrored the stock market as a whole.¹ Overall,

¹ This period may not have been entirely representative since at the time institutional portfolios were heavily concentrated in a relatively small number of large capitalization stocks, but comparable results have been found for other periods and other institutions.

more than 99% of all trading activity involved buying the equivalent of passive index funds while paying for active management.

The lesson here is not just that too many cooks spoil the broth. More carefully selected managers with designedly different perspectives might have reduced the extent of self-canceling investments. The real import is that for some purposes, decentralizing investment decisions cannot be done successfully. In the case of this large pension fund, requiring that orders be submitted to a central in-house clearing facility and then only executing net orders achieved significant savings. Buying index funds and acting only on deviations from the market portfolio would have saved even more without ignoring the “net” decisions of the individual managers.²

An equivalent problem arises when risk management decisions for a large portfolio are made by a number of independent investment managers. Some managers will be embracing risks that other managers are avoiding. The portfolio as a whole may own costly hedges put in place by one manager while another manager has intentionally chosen exposure to those same risks. Alternatively, if one manager is long a stock while another manager is short a similar stock, then the portfolio may be hedged without incurring the cost of explicit risk mitigation transactions. Wealth owners who delegate investing to multiple individual managers must know what their aggregate portfolios look like before they can effectively assess and protect against their vulnerabilities. Efficient risk management requires that wealth owners hedge their aggregate vulnerabilities and not allow hedging to take place at the level of each individual manager. The first fundamental principle of risk management is that the ultimate wealth owners or their designees must manage risk centrally.

Risk management must be customized as well as centralized. Investment management firms that attempt to handle risk with a single model for all their clients will do this job badly for several reasons. First, each wealth owner is likely to have allocated funds to a different set of investment managers. The relationship between one investment manager's portfolio and the portfolio of each of the client's other managers will vary from client to client. What may represent valuable risk management for one client may represent unnecessary diversification for another. This problem is exacerbated because many wealth owners, both individuals

² As commercial products, index funds, which have become so central to equity investing, were just getting started in the mid-1970s, but the company could easily have created an in-house market portfolio for their own use.

and institutions, have large and essentially illiquid holdings that constitute a major portion of their overall assets. Families with enough assets to require this level of professional help have likely earned their wealth from focused entrepreneurial ventures that continue to represent a major part of their holdings even after they have surrendered control. Other families have been successful in service businesses or real estate in particular geographies and continue to have major illiquid holdings in these places. Institutions such as universities or hospitals have core activities whose economic health often depends critically on local, national, or industry economic conditions. Individual investors have jobs that generate a major fraction of their wealth, and these jobs generally have exposure to particular industry and geographic risks. Proper risk management must take these commitments into account in building liquid investment portfolios. Investment managers usually don't have the knowledge, the expertise, or the time to do this for each individual client. An effective risk manager must be able to adjust to the needs of different wealth owners.

A second critical aspect of centralized risk management is that it is better done separately from investment management. Wealth owners are the parties most familiar with their own overall holdings, commitments, and attitudes toward risk. They should, therefore, be the ones directing risk management activity. Given that most wealth owners are not qualified to handle this task well, they need professional risk managers to monitor the risks of the subordinate investment managers and manage overall risk for each client separately. This imperative is increasingly well recognized in the wealth management business. Individual or shared family offices often apportion assets among investment managers but manage overall risks for each client. Large institutions have portfolio management offices that perform the same functions. Smaller institutions are beginning to get into the equivalent of multi-family offices. However, individual investment managers have not fully adapted to this reality. Too many of them deal with risk as if they were managing all of each client's wealth. As a result, they are often insufficiently specialized and hold portfolios that are far too broadly diversified when part of a large pool of assets. In what follows, we will talk about risk management while clearly delineating the distinct roles of wealth and investment managers.

Defining Appropriate Measures of Risk

The first step in describing an effective risk management process is to define properly the risk that must be managed. Currently the

most commonly used risk metric is portfolio volatility measured by the variance of the value of the wealth owner's total holdings.³ In theory, this overall portfolio should include the illiquid assets of the sort we described above. In practice, variances are usually calculated and managed only for the security portfolios of wealth holders because assessing variances for the values of illiquid holdings is hard to do. Beyond this exclusion, there are other serious problems with using security portfolio variances as measures of wealth-holder risk.

First, the range of possible wealth levels becomes wider as we move further into the future. Likely changes in the value of holdings between today and the end of next week should be small. Likely changes between now and the end of next year will be larger. Likely changes between today and 10 years from today will be much larger still. Thus, when we look at wealth variances—and by this we mean future wealth variances—we must specify the length of time over which we are considering future changes, which is our investment horizon.

Common practice identifies only a single variance without specifying the period to which it applies. The reason for this omission is an important assumption that underlies conventional variance measures, namely, that unanticipated and hence uncertain future changes in value are unrelated between one period and the next. If tomorrow's value happens to increase, one assumes that this indicates nothing about what happens the day after tomorrow. In this case, the two-day variance is just the sum of two independent one-day variances. If we further assume that the daily variances are all equal, then the two-day variance is two times the daily variance. For a month of 21 trading days, the monthly variance will be thus 21 times the daily variance. The annual variance will be 12 times the monthly variance or 252 (12×21) times the daily variance. Under this assumption, once a "standard" time period has been defined, usually a year for the purposes of calculating a "standard" variance, the variance of any other period can be calculated by simply multiplying the "standard" variance by the ratio of the length of the other period to a year. With this in mind, we can sensibly talk about the "standard" variance as "the" variance of a portfolio.

Unfortunately, financial market reality does not reflect the independence assumption outlined above. In the short run, generally up to

³ The variance of an uncertain quantity is the average squared distance of each possible value from the average future value, weighted by the probability of occurrence for each of these possible values. The standard deviation, another commonly used measure, is the square root of the variance.

six months, daily changes in value tend to be positively correlated. If stock prices are up today, they are more likely to be up tomorrow. As a result, the two-day variance is more than twice the size of the single day variance. For periods of up to six months, longer period variances are generally proportionally greater than short period variances. Thus, the period to which a variance applies has to be specified and calculated separately. For periods longer than a year, changes in asset values tend to be negatively correlated. If this year was up, next year is more likely to be down. Thus, the two-year variance, because of this offsetting change, will be smaller than the single year variance. There is regression to the mean in the longer run. Again, variance cannot be specified without reference to the period over which it applies. If variance is a measure of risk, then an investor has to decide whether to be concerned with relatively short-run risk—say one year—or long-run risk. Graham and Dodd intuitively recognized this fact and explicitly focused on the long run, an approach that remains dominant among value investors.

The second issue with variance as a measure of risk is that it fails to distinguish between upside and downside deviations from the average expected return. The assumption is that distribution of returns is symmetrical; with upside and downside variations equal, there is need for only a single variance measure. In reality, return distributions are not symmetrical. They have much greater upside than downside deviations. Positive returns are potentially unlimited while negative returns are capped at minus 100%. Variance measures of risk are dominated therefore by upside uncertainty. Yet when most people think of risk, they usually see it as the possibility of losses, not outsized gains. Graham and Dodd were also well aware of this shortcoming. In assessing the risk of an investment, they recommended focusing solely on possibilities of loss. Together with their concentration on long-term returns, their preferred measure of risk was the likelihood and size of potential permanent loss. In their words, risk was the possibility of a “permanent impairment of capital.”

Unfortunately permanent impairment of capital is not as easily quantifiable as variance. However, it can still serve as a sound basis for formulating risk management strategies and ultimately quantifying the overall risk exposure of the complete holdings of any particular wealth owner. Because of the temporal differences in variance and its deviation from what most people understand as risk, we will use “permanent loss” as our definition of risk.

Limiting Permanent Loss of Capital

Permanent loss of capital typically arises from one of three sources. The first is paying too much for a security. The fastest way permanently to lose 50% of your capital is to pay \$100 for a security worth only \$50. The moment of purchase is the time of maximum risk if by risk you mean a permanent loss. Protection against overpaying begins with a sensible search strategy. Purchases of overpriced securities occur most often when markets are experiencing extreme overvaluation, which, after the fact, are called “bubbles.” They are characterized by outsized ratios of market prices to underlying measures of economic value, such as price-to-sustainable earnings ratios above 30 or price-to-book ratios of 8 or more for typical stocks. These valuations invariably appear as a consequence of rising securities prices over an extended period. This was the case of Japanese stocks in the late 1980s; tech, Internet, and telecom stocks in the late 1990s; and housing and associated financing institutions in the run-up to the financial crisis of 2008. Avoiding these situations ought to be an essential characteristic of a well-conceived search strategy. If there is any significant probability that a group of securities are in bubble territory, those stock should not be purchased. At best, they will fall into the too-tough-to-call category. Sooner or later there are always more attractive alternative opportunities.

More generally, value-based protection against permanent loss of capital at purchase depends on a margin of safety between the price paid and the estimated value of an investment. For a disciplined Graham and Dodd investor, an adequate margin of safety is the most important measure of investment safety. This approach requires a valuation process that produces an accurate assessment of intrinsic worth, whether a discounted cash flow for near-term fixed return opportunities, an asset value or earnings power value for non-franchise businesses, or assets and cash plus prospective growth returns for franchise businesses. For some companies, notably new, rapidly growing franchise businesses, it may be impossible to arrive at a sufficiently accurate valuation on which to base an investment. These companies, however attractive they appear, should be filed away in the too-tough-to-call category and, for risk management reasons, only purchased when more accurate estimates are possible.

A second source of permanent loss is excessive leverage. Levels of indebtedness at the company level that lead to bankruptcy or even extended impairment of the company's competitive position will convert temporary losses into permanent ones. Ideally debt at the company level should be low enough so that even in reasonable worst

case situations, say three consecutive bad years, operating earnings and capital recoveries are adequate not only to cover any debt service requirements but also to avoid any doubts about future solvency that might materially impair the firm's standing with customers, suppliers, employees, regulators, or necessary sources of future funding. At the investment portfolio level, margin debt that leads to forced sales in the face of short-term market declines will also convert temporary losses into permanent ones. Leverage of this sort should also be avoided. Investing in time-limited options that enhance upside returns is a kind of leverage that is equally risky. If a temporary downturn persists past the date of option expiration, then the options will expire worthless and the loss will become permanent. The second principle of permanent loss-focused risk management is to avoid excessive leverage, which can be lethal. If you are uncertain when leverage becomes excessive, it is wise to avoid it altogether.

Diversification

If securities are bought with an adequate margin of safety and excessive leverage is avoided, then permanent losses will generally arise for only two reasons. The first is overestimation of the security's intrinsic worth at the time of purchase—a simple mistake in valuation. The second is a major unanticipated adverse development affecting the financial position of the issuing company. For careful disciplined investors both these events should be essentially random occurrences. For an investor who carefully monitors his or her past performance and learns to avoid repeated systematic errors, such as persistent optimism with respect to value-creating business growth or sustainable earnings valuations that fail to incorporate the possibility of recession years, overvaluations will only occur in the face of new developments that are not sufficiently anticipated. By their very nature, these mistakes should be unsystematic. Unanticipated events that permanently impair a company's position will generally be either company or industry specific. At the company level, its new product initiative or geographic expansion turns out to be a disastrous failure, a new competitor makes unexpected inroads enabled by a company management's current inattention, or a new management focuses on ill-advised expansion at the expense of operating efficiency. At the industry level, such events should be specific to particular types of businesses and geographic areas. These might include new technologies that undermine an industry's current economic position or the value of its

assets, government regulatory changes that whittle down industry profitability, the adverse impact of globalization on markets, or the collapse of cooperative industry norms under the weight of new competitors. Developments like these will rarely have economy-wide implications.

Another kind of idiosyncratic risk arises in the area of event-driven investing. Arbitrage positions, investments in companies being taken over or in newly spun-off divisions, investments in securities of companies in Chapter 11, investments in companies in the process of liquidation are all investments where the size of the pay-off may be well defined, but the timing is uncertain. It is usually independent of the timing of other such events and the general economic conditions that determine the returns of other securities. The arrival of a pay-off is an unsystematic event causing unsystematic risk.

Unsystematic risks of this sort can generally be diversified away. Consider the case of a small oil company that drills a single well with a 20% chance of success. For an investor in only that company, a dry hole, which occurs 80% of the time, represents a total permanent loss of capital. The small oil company investment appears to be extremely risky. Now suppose there are 1000 other small oil companies with similar prospects for drilling in different areas whose 20% probabilities of striking oil are completely unrelated to each other.⁴ Overall, 200 will be successful. The likely range of successful wells run from 150 to 250.⁵ If an investor spreads his or her investment over the entire 1000 wells, placing 0.1% of the total in each one, the maximum loss is not a 100% loss 80% of the time but a 25% loss (50/200) that occurs only about 5% of the time. In fact the situation is better than this. If a successful well returns $5\frac{1}{2}$ times its cost, which is necessary to provide a reasonable 10% average return, then the 150 worst-case successful wells will produce a return of 825 times the amount invested or a loss of only 17.5% $((1000 - 825) \div 1000)$, which again occurs only about 5% of the time. Diversification reduces the potential loss from 100% in 80% of the instances to 17.5% less than 5% of the time. Risk in the diversified portfolio of small oil companies does not disappear entirely,

⁴ The level of 1000 small companies may sound extreme, but the oil well example is particularly risky and therefore even small oil companies will drill many wells, so that much of the diversification across projects takes place and should take place within companies. Active risk reduction from diversification will be less than this extreme example, but it still will be 90% or more.

⁵ This example assumes a normal distribution of results.

but it is greatly reduced. The expected maximum loss falls by more than 98%.⁶

The industry-wide risk of a decline in oil prices that affects all the small oil companies, and large ones as well, is not reduced by diversifying among them. However, diversification across industries will reduce even this risk. If our model investor has roughly equal holdings across 15 industries, only one of which is heavily dependent on oil prices (the oil companies themselves), then only about 7% (1/15) of the effect of the lower oil prices will appear as a reduction in the value of the overall portfolio. More realistically, although some industries will suffer from the decline in oil prices, those for whom petroleum is an input are likely to benefit more than enough to offset those direct losses, so that the overall exposure is likely to be below 7%. In both these cases, diversification reduces the worst-case permanent losses by 90% or more. This means that the risks of idiosyncratic errors and unanticipated developments can be largely eliminated by holding a properly diversified portfolio.

Much of this risk reduction can be achieved by holding as few as 30 securities, appropriately diversified across industries and geographies. Complete diversification, holding a portfolio in which all securities are proportionally represented, is not necessary. A value investor can be adequately diversified while still holding a relatively concentrated portfolio of bargains. However, since permanent impairment of capital tends to arise from idiosyncratic events at the company and industry levels, a sensible level of diversification (i.e., 30 or more stocks from a range of industries and geographies) is even more important from a permanent loss perspective than it is for a variance-based measure of risk. In his writing and practice at Graham-Newman, Benjamin Graham explicitly endorsed the importance of diversification.⁷

For a diversified portfolio of stocks purchased with an adequate margin of safety and using appropriately low levels of leverage, the remaining risks are those that affect economies and financial markets as a whole. Permanent impairment of capital at this level typically is due to either markets at unsustainable price levels (Japan in the late 1980s) or national economies in terminal meltdown (Venezuela in the middle 2000s). As the recovery of securities markets in the wake of the 2008–10 global economic crisis has shown, economies in

⁶ The expected maximum loss from the single company investment is 80% (100% loss 80% of the time). For the diversified portfolio it is around 17.5% about 5% of the time or 0.875%.

⁷ Graham, Benjamin *The Intelligent Investor*, Fourth Revised Edition (New York: Harper and Row, 1973), p. 54.

well-functioning national environments are extraordinarily resilient. This may be due in part to government commitments to do whatever is required to sustain macroeconomic stability. There were wide variations in the degree and nature of government responses to the 2008–10 crisis, and yet financial markets, with few exceptions, have recovered. This experience suggests that for the long run, comprehensive economic and financial impairment will be rare. Despite this sanguine view, there will always be areas and assets at risk of permanent damage. Unanticipated inflation of any size and duration is rarely, if ever, followed by equivalent price declines. Under these circumstances, fixed-income assets with long maturities suffer significant and permanent losses, as institutions that relied on long-term bond income in the late 1950s found to their enduring cost. Equity markets can also experience more than decade-long periods of falling or flat valuations. Although the resilience of the underlying economies and businesses has usually ensured an adequate long-run return, low interim results can inflict serious pain on investors. It is prudent, therefore, to secure some protection against long-lived but less than permanent macroeconomic impairment.

Global portfolio diversification should provide protection against the possibility of national economic difficulties. All investors, especially those in relatively unstable emerging societies, need some degree of global diversification. But as the 2008 crisis demonstrated, global markets are increasingly integrated, and national market returns have become more highly correlated. This convergence has meant that even global diversification will not provide complete protection against national macroeconomic dislocations.

Managing Global Macroeconomic Risks

For a disciplined value investor who is well diversified and selects individual investments with an adequate margin of safety, management of macroeconomic risks will generally be a matter of significant but secondary importance. This position is in stark contrast to the practice of adherents of Modern Portfolio Theory, for whom managing macroeconomic risk is a primary concern. This MPT approach involves detailed attention to allocations among narrowly defined asset classes and to portfolios selected to have tailored exposures to intricately, and usually statistically, defined macro “risk factors” (so-called factor investing). Most value investors think of macroeconomic risk as falling into one of two categories. First are inflationary environments, periods of relative

prosperity for business but with high and unpredictable price inflation. Second are deflationary periods in which prices are stable or even falling, but overall business conditions and growth rates are depressed. Inflation and deflation can actually co-exist. Stagflation, weak growth and high inflation, characterized many developed economies in the 1970s and 1980s. Still, it is more useful to think of managing these two risks separately.

Asset classes can be organized into four broad categories. First, there are assets with fixed nominal returns: bonds, notes, insurance policies, and other long-lived debt instruments both public and private. These tend to do well in deflationary environments and poorly in inflationary ones. Second, there are real assets: natural resources, real estate, and non-franchise business. These do well in inflationary environments where their values over the long run rise with price levels and their returns are enhanced by prosperity.⁸ They tend to do poorly during deflationary periods. Third, there are franchise businesses, such as Coca-Cola, with pricing power and protection from the effects of intensified competition that typically comes with deflationary environments. They tend to be relatively immune to the adverse effects of both inflation and deflation. And finally, there are low risk and very low return alternatives such as cash and cash equivalents.

For a value investor, bottom-up analysis will determine the investment choices within these three non-cash categories. If fixed-income assets are cheap, by which we mean bonds with high yields when interest rates are likely to decline, then a portfolio will contain more of them. The same applies to real and franchise assets. This process by itself provides some degree of overall protection against macroeconomic risks as investments are concentrated in cheaper, presumably less vulnerable asset classes. But a portfolio will inevitably be subject to residual macro-risks unless by chance it is composed entirely of franchise businesses. A portfolio dominated by fixed-income holdings will require protection against inflation. A portfolio heavily weighted toward real assets will require insurance against deflation.

It is only these residual portfolio level risks that should be insured against, and the insurance has to be managed centrally. There are three basic types of insurance. The first is to hold assets such as cash or gold that will maintain or increase their value as other asset prices fall. Often this

⁸ This relationship applies only in environments of low to modest and generally stable inflation. High and unstable inflationary environments typically lead to impaired business operations and, in turn, low real asset returns. These are stagflationary situations.

allocation occurs naturally. If the search and valuation process reveals a scarcity of opportunities in all three non-cash asset classes, then by default cash holdings will be high, and an investor may choose to hold some fraction of them in an asset such as gold. Gold prices have historically had a negative correlation with broader financial market prices and have provided long-term protection against inflation.⁹

The second form of insurance is selective short sales. This is the risk mitigation tactic that hedge funds attempt to supply on a decentralized basis. If done well, it provides useful insurance. However, there are three disadvantages to short sales. First, for taxable investors, short sale gains tend to get taxed as ordinary income while long-term gains are generally taxed at preferential capital gains rates. This asymmetry makes short sales relatively expensive. Second, if the proceeds of short sales are held in cash, the average return on cash is typically well below the average returns on the shorted asset classes. For equities, the difference is usually somewhere between 4 and 8%. Therefore, short sales involve a significant average cost most of the time. Third, short sales have unfortunate risk characteristics in themselves. If a short goes wrong, the prices of the underlying security rises and so too does the size of the short position. Larger positions mean greater risks to the portfolio. For short positions, miscalculations compound into significantly greater risks. On the long side, in contrast, miscalculations lead to lower prices, shrinking positions, and less risk. To adjust for this asymmetry, investors who hedge with short positions tend to hold many more of them than they do long ones. But this means that each position is less well analyzed, which also has the unfortunate effect of increasing risk.

The third insurance tactic is to use options. One way to employ them is to sell securities and purchase call options, which preserve upside returns and limit losses to the cost of the options. Another approach is to buy put options that enable an investor to sell securities at a set price no matter how low the market price goes. But because options are time-limited, in order to provide continuous protection an investor has to roll over a sequence of future dated options so that the options do not all expire at the same time. If they were to expire simultaneously, the degree of protection would fall to zero as that date approached. Also options can be expensive, and the tax treatment of put options is usually the same as for short sales. However there is a special advantage

⁹ Cash typically has a near-zero correlation with nominal financial market returns, and interest rates on cash have generally risen with rates of inflation, although not perfectly.

to option strategies if done well. Options are cheapest when markets are least volatile, which is when investors are most optimistic about the absence of disruptive future events. Historically, this level of confidence leads to high asset prices and greater risk. As a consequence, option strategies are most valuable when they are cheapest. Carefully implemented option strategies means buying options protection when the conventional wisdom is that they are unnecessary. For a disciplined value investor concerned with aggregate portfolio risk, this is probably the best form of insurance. But the insurance should cover only the long-term net vulnerability of the portfolio as a whole. Insuring cash or stable franchise businesses would be an unnecessary expense. Insuring fixed-income and real asset portfolios separately rather than overall net exposures to inflationary and deflationary environments is equally inefficient.

Behavioral Risks

What we have discussed so far are risks that arise from the uncertainties of financial markets and the businesses whose unpredictable activities underlie the values of the securities involved.¹⁰ Another source of risk is the behavior of investors themselves, which is just as important as economic risks. For value managers, a danger arises when there are virtually no securities being sold at prices that offer adequate margins of safety. In these times, the manager is either selling without reinvesting the proceeds or, if still reasonably satisfied with the existing portfolio, doing nothing at all. But watching cash accumulate or doing nothing does not satisfy the urge for action that motivates most hard-working investment managers. The natural temptation is to force the issue and buy something, which under these conditions is generally a mistake. The way to deal with this temptation, like most temptations, is to develop a plan in advance and then to stick to that plan, especially when the temptation to abandon it is strong.

When there are no identifiable attractive buying opportunities, a value investor is ironically in a position similar to that which all investors are assumed to face under the Efficient Market Hypothesis. Knowing nothing worth doing is close to knowing nothing advantageous at all, which is the unmistakable implication of the EMH. Thus, following

¹⁰ The operations of households and governments, which often underwrite debt securities, can also be erratic.

the prescriptions of academic finance theory, the best course is a mix of a broadly diversified portfolio and a risk-free asset. The specific mix depends on the degree of risk aversion of the investor in question. For value investors, who place a low value on market returns under these circumstances, the allocation will be heavily weighted toward cash. The most risk-averse value investors will hold all cash. On the other hand, equity fund managers who are judged against a market index, such as the S&P 500, will minimize their career risk with an all S&P 500 portfolio.

There is an issue of how best to define a broad market portfolio. Both history and insights from behavioral finance suggest that in markets with few if any value opportunities, a statistical portfolio of cheap stocks (low market-to-book, low price-to-earnings, or low by other valuation measures) will still outperform a market portfolio. Also stock portfolios that weight component stock holdings equally have historically performed better than portfolios, such as the S&P 500, that weight stock holdings according to their market capitalizations. In low opportunity environments, then, a value investor might want to identify an equal-weighted portfolio, with less invested in mega market capitalization stocks, as the broad portfolio option in a cash-portfolio mix. It is still important to remember that nothing works all the time, and if the manager is evaluated relative to the cap weighted S&P 500, that is the safest place to hide until genuine opportunities appear.

It is important to have thought ahead of time about the appropriate cash-portfolio mix and the right security portfolio so that this default strategy can be automatically executed in a no-opportunity environment. The alternative is to search energetically for opportunities that are not there, relax margin of safety standards, and make poor individual investment decisions. The temptation to do something is powerful. Warren Buffett's notion that investing is a game with no called strikes so that you can always wait for a fat pitch is only part of the story. The problem is that in a world where manager performance is judged against average investor returns, which necessarily equal average asset returns, they run up the score whether you swing or not. A good default strategy is essential to being comfortable with this reality and acting rationally in this world.

For non-professional investors, the costs of bad behavior have historically been severe. In periods where market returns have been between 7 and 11%, average individual investor returns are typically

1 to 5%, a difference of 600 basis points.¹¹ In addition, the average money-weighted returns for investment funds have typically trailed the time-weighted returns by the same 600 basis points.¹² In the first case, individual investors are getting into securities when prices are high and out of them when prices are low, exactly the wrong times. In the second case, investors are doing the same thing for individual investment funds, getting in and out at the wrong times. To put these losses in perspective, a truly outstanding investor will typically outperform the market over long periods by 3–5% per year. This is measurably less than the 6% loss attributable to the herd-like behavior of individual investors. On a positive note, someone has to be on the wrong side of that trade.

The chances that multitudes of individuals will modify their behavior and stop buying high and selling low are slim. If the behavioral psychologists and their economic associates are right, these kinds of reactions are hard-wired in our brains, the result of evolution over many generations. But for at least some investment managers, there is hope. They can either try to select investors who don't behave this way, an option more available to hedge funds and other investment partnerships than to typical mutual funds, or they can try to educate their clients about the folly of chasing performance. The extent to which investment advisors can persuade their clients to stay the course in good times and bad will have a greater beneficial impact on long-term investor returns than all but the very best investment decision-making.

A simple rebalancing of portfolios is one way to accomplish this goal. When a particular class of investments does well relative to other classes, it will represent a larger share of the overall portfolio value. Trimming under these conditions to return to the preset asset mix while adding to those classes that have lagged counteracts the natural tendency to buy when prices are high and rising and sell when they are low and falling. Systematic rebalancing has in practice significantly enhanced portfolio returns. It is a valuable tool for managing behavioral risk.

¹¹ Terrance Odean has written voluminously on the behavior and performance of individual investors. See his website <http://faculty.haas.berkeley.edu/odean/>.

¹² The money-weighted returns take into account the amount of assets in the fund in that year so that years in which the fund is big count more than years in which it has fewer assets. Time-weighted returns look only at the net return, counting each year the same. The consulting firm Dalbar has measured investor returns compared to fund returns since 1964. See <https://www.dalbar.com/QAIB/Index>.

Structures for Risk Management

The optimal structure for risk management is different from the optimal structure for security selection. Security selection can be decentralized and, because specialization improves investment choices, benefits from decentralization. Risk management, as we pointed out at the beginning of this chapter, must be centralized. It is possible that the two functions may reside in the same organization, so that specialized industry- and geography-focused analysts recommend position sizes based on their levels of conviction, and portfolio managers have control over the overall portfolio and are therefore responsible for risk management. Such a structure makes sense when, as in the case of the Baupost Group, an organization is fully informed about all of its clients' holdings. When wealth owners parcel out their assets among a number of independent asset managers, there must be some central authority that is responsible for overall portfolio structure and risk management.

Sophisticated wealth owners with the time necessary to supervise their holdings may perform this centralized risk-management task themselves. But more often wealth owners will lack the capacity, the time, or the inclination. Thus, the recent evolution of wealth management has been in the direction of dedicated family offices, sometimes with a multi-family client base, and institutional asset management groups. These offices parcel out funds to investment managers who will presumably invest shared pools of funds differently from one another. The investment managers must communicate the level of risk they are taking on and the broad nature of their investments so that wealth managers can control risks by (1) properly allocating funds among investment managers, (2) specifying position limits for these managers (e.g., if a client wealth owner has a major illiquid interest in the family automotive business, portfolio investments in autos should be strictly limited), and (3) selectively hedging the risks of the aggregate portfolio, including the value of any ongoing business operations of the wealth-owning institution or family.

As supervisory offices increasingly become responsible for portfolio construction and diversification, individual investment managers are likely to become more specialized and concentrated in a smaller number of securities. Having a portfolio of 30 or more stocks diversified across regions and industries may make sense for a total portfolio. But if the wealth manager has funds with 10 or more such managers, that overall diversification will involve either substantial redundancy or diversification over roughly 300 stocks, an unnecessarily large degree of

diversification in most instances. Allocating funds into only specialized regional or industry managers eliminates this redundancy. But having 10 such managers, each with three specialties, holding three to five positions in each specialty will lead to a portfolio of more than 100 stocks. Thus, the portfolios at the bottom of the pyramid, each run by a specialized manager, may have to consist of only five to seven major positions.

This degree of concentration will inevitably create problems for the primary investment managers, meaning the stock pickers. Concentrated portfolios will have relatively high levels of unsystematic risk that, while not diversified for the investment managers, is diversified away for the client wealth owners. Specialized managers, even those with as many as five specialties, may face extended periods where there are no attractive value investments in their target areas. Under these circumstances, investment managers should ideally communicate the lack of opportunity to clients. But this candor will be difficult if the investment manager must thereby put itself out of business for extended periods. Contracts between investment and risk managers will have to be designed in ways that protect investment managers without significantly reducing incentives to invest well. This will not be easy. Despite prior agreement on the specific roles of investors and risk managers, the urge to ask, "What have you done for me lately?" will be difficult to contain.

Investor Profiles

In the first edition of this book, we thought that it would be helpful to look beyond the basic principles of value investing and to describe in some detail what accomplished value investors actually do when they are at work. Benjamin Graham wrote two books—*Security Analysis*, with David Dodd, and *The Intelligent Investor*—that laid out his theory of investing and described his own practices. Investing was his primary profession; he taught as an adjunct, even though his course was famous and virtually mandatory for all serious Wall Street professionals. While we had some investment experience, it did not come close to Graham's level of practical knowledge. To compensate, we included a number of investor profiles in the first edition of this book. The chapters described how successful value-oriented investors did their jobs, and it included real examples from their experience. In alphabetical order, the investors included Warren Buffett, Mario Gabelli, Glenn Greenberg, Robert Heilbrunner, Seth Klarman, Michael Price, Walter and Edwin Schloss, and Paul Sonkin. The profiles were based on interviews, presentations made to classes in the Value Investing course at Columbia Business School (CBS), and extensive written records. Our investors ranged in age from 35 to over 90. Three had worked for or with Graham himself. Two others had worked for Max Heine, another legend in the field who first met Graham as his broker and maintained a close relationship with him. The remaining three attended Columbia Business School, where they took courses with Graham's successors Roger Murray and Bruce Greenwald.

These investors practiced a broad range of value investing styles. Some concentrated on superior businesses purchased at reasonable

prices; they intended to hold these firms for decades if not longer. Others looked for damaged companies and assets offered for sale at distressed prices. Some ran portfolios with six or eight stocks; others owned hundreds of assorted investments at a time. Some of them bought the bonds of companies headed for or actually in bankruptcy, calculating that the bonds would earn them an attractive return either in cash, new debt, or equity in a post-bankruptcy reorganized company. Others sought to buy the shares of stable companies that they felt would command a premium above the public market price from a strategic investor purchasing the whole firm. Some sought to avoid the crowd by concentrating on small, obscure companies overlooked by investors with large amounts of capital to deploy. Others, who themselves had charge of mountains of investment funds, concentrated on large, well-established firms. Some of our investors bought only marketable securities. Others purchased entire companies or real estate. All the profiled investors specialized in one or some combination of these alternatives. Their approaches covered a full spectrum of investment possibilities.

In the nearly 20 years since the first edition was written, some of the original investors no longer actively manage portfolios. We have included the profiles of Walter and Edwin Schloss and Robert Heilbrunner virtually unchanged. Walter Schloss, one of the most decent people in a sometimes nasty business, died in 2012. Warren Buffett has been very active and has produced 20 new annual reports with a wealth of investing wisdom. We have rewritten his profile to incorporate these insights; as before, we have relied almost entirely on Buffett's own writing, which is as skillful as his investing. In the case of the remaining investors, we have taken advantage of improvements in technology. Mario Gabelli, Glenn Greenberg, Seth Klarman, and Michael Price have all spoken to the Value Investing course at CBS almost every year for the last 20 years. These talks have been recorded and edited versions are available on the web. The necessary information to access them is at the end of this chapter. To supplement the videos, we have incorporated the narratives from the first edition for these investors in a single chapter and made that chapter also available online. We refer interested readers to both the videos and the web versions of the first edition profiles.

Since the first edition of this book appeared in 2001, value investors have faced major challenges. When we began writing in 1999, value investors were assumed to be a dying if not dead breed. By the time the book hit the stands, in June 2001, the dot-com bubble had crashed, and value investors benefitted from both their actual returns and the relative

gap between them and the major indices, which took a dive when sock puppets hit the ground. But for the last ten years, since the recovery from the Great Recession, value portfolios made up of stocks trading at lower than average P/E or market-to-book ratios have produced returns that trail both broad market indices and portfolios of growth stocks. In these unprecedented circumstances, many previously successful value investors have performed poorly. Several economic factors appear to be responsible. First, as economic activity has migrated from manufacturing to services, standard accounting measures have become less relevant as indicators of intrinsic value. To a large extent, the assets of service businesses are intangibles—product portfolios, brand images, customer relationships, trained workers, and organizational capital—which are treated as current expenses and do not appear on traditional balance sheets. To get a useful estimate of a firm's intrinsic value based on its assets, value investors need to find creative but realistic ways of measuring and incorporating the value of intangible assets. At the same time, both investments in and depreciation of intangible assets are not well identified in traditional income statements. Expenditures on sales activity, advertising, training, and especially research and development that may have a substantial investment component, meaning that they generate revenue and income beyond the current year, are generally treated as current costs. The accounting treatment of depreciation and amortization typically does a poor job of measuring the true depreciation of intangible assets, if it measures it at all. Again, successful value investors must learn to adjust their earnings measures to account for these factors.

The increasing importance of service businesses has a second consequence for investors, whether value oriented or not. Services are generally locally produced and consumed. Service markets tend to be small and local rather than large and global. Small local markets, by contrast with large and global ones, are more easily dominated by single firms who enjoy economies-of-scale advantages. At the same time, service businesses involve more frequent and more important face-to-face contacts with customers than do firms that produce goods. Customer captivity grows out of these interactions. For both reasons, barriers to entry, what Warren Buffett and others refer to as “moats,” are likely to play a larger part in services than in manufacturing. The ability to value franchise businesses, including the value of growth, has become more important as services constitute a larger fraction of economic activity. To succeed, value investors must adjust their intrinsic value estimates to take this new reality into account.

As information technology moves toward the cloud, hardware costs decline and software becomes more important. This move reinforces the consequences of the trend toward services. Artificial intelligence tends to be highly specialized. Popular game-playing programs are game specific: chess, Jeopardy, and Go programs, not general game-playing programs. In medicine, AI programs are specific to a medical specialty and disease. The cost structure of software products invariably means high fixed and low marginal costs, a structure that generates significant economies of scale. Specialized markets in product space, like geographically local markets in services, are more easily dominated than more general markets for microchips and other computer hardware. More than hardware, software use entails more continuous, direct customer interaction, which is a pathway to customer captivity. So technology, like services, tends to increase the importance of franchises, making growth a more important part of intrinsic value.

Global investing has been an important consequence of the larger globalization trend of the last several decades. Successful value investors must now deal with global opportunity sets. The number and complexity of financial instruments available to investors has increased exponentially, presenting another challenge to value investors. The striking success of Warren Buffett and other value investors, including those we have profiled, attracted growing numbers of investors to the Graham-Dodd-Buffett camp. This trend was accelerated by comprehensive statistical studies that clearly demonstrated the superiority of a value approach to investing. Today there are far more value investors with instantaneous access to comprehensive accounting and business information on the Internet assessing value opportunities than there were in 2000, in the wake of the tech and telecom mania. More skilled investors make it more difficult to achieve superior returns.

To address these developments, we have expanded our set of profiles of successful value practitioners. We have added Thomas Russo, who specializes in global consumer product companies, especially those that are family run; Andrew Weiss, who focuses on complex financial instruments, often with intricate administrative requirement; Paul Hilal, a serial specialist who undertakes active interventions to push management to improve the operations of a small number of companies; and Jan Hummel, who invests in small-cap businesses with dominant service components restricted to Germany, the Nordic countries, and the UK. We have provided brief introductory profiles for each. Edited videos for all of them are available online (<http://www.wiley.com/go/greenwald/valueinvesting2e>).

Warren Buffett

Investing Is Allocating Capital

Few will dispute the claim that Warren Buffett is the most illustrious investor ever. The recognition he has earned rests both on the exceptional returns he has provided for his partners and shareholders over more than six decades and on his singular ability to explain the intricacies of his craft in a manner as long on clarity, modesty, and humor as it is short on pretense or self-exaltation. In the years he ran his limited partnerships, from 1956 through 1969, and beginning again in 1977 as chairman of Berkshire Hathaway, Buffett has written annual letters that report the major choices he has made during the year and, of more permanent significance, the investment philosophy that has guided his actions. To say that these letters are a rich source of investment wisdom is to understate their true worth, as all the authors who collect royalties on books mined from this correspondence will attest.¹

We have selected passages from the letters during the Berkshire Hathaway years and organized them to reveal what we consider the most significant elements of Buffett's approach to investing. The full text of these letters is available online and may be read and downloaded from the company's web site. All the material is copyrighted by Warren Buffett, and it is quoted here with his permission.

¹ In this section we have inserted a few comments, which are set in Garamond font to distinguish them from Buffet's text, set in Times New Roman, the same font as the rest of the book.

In addition to the selection and organization of these excerpts, we will confine our contribution to a few remarks about Buffett as an investor. First, as he himself says frequently in the letters, his association with his Berkshire partner Charlie Munger helped to move him away from an orthodox Benjamin Graham preference for buying assets at a deep discount to their value, no matter how miserable the company in which they were found, to buying good or excellent businesses at a reasonable price. While Buffett in the Berkshire years still speaks with reverence about Graham, he looks for companies that have strong franchises and outstanding management, even though they may sell for multiples of their book value.

Second, while many investors say that they think of themselves as buying businesses rather than stocks or bonds, Buffett really means it. Berkshire Hathaway itself is a hybrid; it owns some businesses outright, including a number of insurance companies, and it has large investments in companies whose shares are still publicly traded. As Berkshire has increased in size, the number of market-based opportunities capable of absorbing significant amounts of its ocean of investible capital has diminished dramatically. As a result, Buffett has devoted an ever greater fraction of that capital to the acquisition and development of whole businesses. But this evolution has not required any readjustment of his basic investment philosophy. Carol Loomis, an observer and friend of Buffett for many years, wrote that

The key point about the two Buffetts, the investor and the businessman, is that they look at the ownership of businesses in exactly the same way. The investor sees the chance to buy *portions* of a business in the stock market at a price below intrinsic value—that is, below what a rational buyer would pay to own the entire establishment. The manager sees the chance to buy the *whole* business at no more than intrinsic value.

The kind of merchandise that Buffet wants is simple described also: “good businesses.” To him that essentially means operations with strong franchises, above-average returns on equity, a relatively small need for capital investment, and the capacity therefore to throw off cash. That list may sound like motherhood and apple pie. But finding and buying such businesses isn’t easy; Buffett likens the hunt to bagging “rare and fast-moving elephants.”²

Or, to say the same thing in somewhat different terms, no matter what is being bought, investing is allocating capital.

² Loomis, Carol. (2013). *Tap Dancing to Work: Warren Buffett on Practically Everything, 1966–2012: A Fortune Magazine Book*. New York: Portfolio, p. 65.

Finally, value investors share the conviction that they are most successful when they stay within their circle of competence. Buffett speaks repeatedly of looking for businesses that he can understand, and his avoidance of companies whose fortunes depend on their technological excellence is well chronicled. Resting within the confines of businesses that he definitely does understand, which include insurance, media companies, and consumer goods firms, he has earned a lot of money for himself and for those who have invested with him over the years. It may seem odd to claim that the greatest investor, ever, has made fortunes by following the humble precept of sticking to his knitting, and surely his superiority stems from more than adhering to this rule. Still, the gallery of investors is replete with geniuses who stumbled when they wandered away from their zones of excellence. Just as Coca-Cola discovered it was better at making and selling soft drinks than in producing films, Buffett has had the sense to recognize the boundaries of his own franchise.

General Investment Principles

It's been an easy matter for Berkshire and other owners of American equities to prosper over the years. Between December 31, 1899, and December 31, 1999, to give a really long-term example, the Dow rose from 66 to 11,497. This huge rise came about for a simple reason: Over the century American businesses did extraordinarily well and investors rode the wave of their prosperity. Businesses continue to do well. But now shareholders, through a series of self-inflicted wounds, are in a major way cutting the returns they will realize from their investments.

The explanation of how this is happening begins with a fundamental truth: With unimportant exceptions, such as bankruptcies in which some of a company's losses are borne by creditors, the most that owners in aggregate can earn between now and Judgment Day is what their businesses in aggregate earn. True, by buying and selling that is clever or lucky, investor A may take more than his share of the pie at the expense of investor B. And, yes, all investors feel richer when stocks soar. But an owner can exit only by having someone take his place. If one investor sells high, another must buy high. For owners as a whole, there is simply no magic—no shower of money from outer space—that will enable them to extract wealth from their companies beyond that created by the companies themselves. (2005)

So that was my argument—and now let me put it into a simple equation. If Group A (active investors) and Group B (do-nothing

investors) comprise the total investing universe, and B is destined to achieve average results before costs, so, too, must A. Whichever group has the lower costs will win. (The academic in me requires me to mention that there is a very minor point—not worth detailing—that slightly modifies this formulation.) And if Group A has exorbitant costs, its shortfall will be substantial.

There are, of course, some skilled individuals who are highly likely to out-perform the S&P over long stretches. In my lifetime, though, I've identified—early on—only ten or so professionals that I expected would accomplish this feat.

There are no doubt many hundreds of people—perhaps thousands—whom I have never met and whose abilities would equal those of the people I've identified. The job, after all, is not impossible. The problem simply is that the great majority of managers who attempt to over-perform will fail. The probability is also very high that the person soliciting your funds will not be the exception who does well. Bill Ruane—a truly wonderful human being and a man whom I identified 60 years ago as almost certain to deliver superior investment returns over the long haul—said it well: “In investment management, the progression is from the innovators to the imitators to the swarming incompetents.”

Further complicating the search for the rare high-fee manager who is worth his or her pay is the fact that some investment professionals, just as some amateurs, will be lucky over short periods. If 1,000 managers make a market prediction at the beginning of a year, it's very likely that the calls of at least one will be correct for nine consecutive years. Of course, 1,000 monkeys would be just as likely to produce a seemingly all-wise prophet. But there would remain a difference: The lucky monkey would not find people standing in line to invest with him.

Finally, there are three connected realities that cause investing success to breed failure. First, a good record quickly attracts a torrent of money. Second, huge sums invariably act as an anchor on investment performance: What is easy with millions, struggles with billions (sob!). Third, most managers will nevertheless seek new money because of their *personal* equation—namely, the more funds they have under management, the more their fees. (2016)

Objective of Investment for Berkshire Hathaway

Our long-term economic goal (subject to some qualifications mentioned later) is to maximize Berkshire's average annual rate of gain in intrinsic

business value on a per-share basis. We do not measure the economic significance or performance of Berkshire by its size; we measure by per-share progress. We are certain that the rate of per-share progress will diminish in the future—a greatly enlarged capital base will see to that. But we will be disappointed if our rate does not exceed that of the average large American corporation. (From “Owner Related Business Principles”; many years.)

Because Berkshire Hathaway is committed to issue new shares only when the intrinsic value received in exchange exceeds the value of the issued shares, and to repurchase shares only when they are available at prices below Berkshire’s intrinsic value, maximizing the growth of intrinsic value per share is equivalent to maximizing Berkshire’s intrinsic value. The definition and estimation of intrinsic value is at the heart of Buffett and Munger’s investment philosophy.

Performance Measurement

From the start, Charlie and I have believed in having a rational and unbending standard for measuring what we have—or have not—accomplished. That keeps us from the temptation of seeing where the arrow of performance lands and then painting the bull’s eye around it.

Selecting the S&P 500 as our bogey was an easy choice because our shareholders, at virtually no cost, can match its performance by holding an index fund. Why should they pay us for merely duplicating that result? (2009)

Realistic Return Expectations

Let’s revisit some data I mentioned two years ago: During the 20th Century, the Dow advanced from 66 to 11,497. This gain, though it appears huge, shrinks to 5.3% when compounded annually. An investor who owned the Dow throughout the century would also have received generous dividends for much of the period, but only about 2% or so in the final years. It was a wonderful century.

Think now about this century. For investors to merely match that 5.3% market-value gain, the Dow—recently below 13,000—would need to close at about 2,000,000 on December 31, 2099. We are now eight years into this century, and we have racked up less than 2,000 of the 1,988,000 Dow points the market needed to travel in this hundred years to equal the 5.3% of the last.

Dividends continue to run about 2%. Even if stocks were to average the 5.3% annual appreciation of the 1900s, the equity portion of plan assets—allowing for expenses of .5%—would produce no more than 7% or so. And .5% may well understate costs, given the presence of layers of consultants and high-priced managers (“helpers”).

I should mention that people who expect to earn 10% annually from equities during this century—envisioning that 2% of that will come from dividends and 8% from price appreciation—are implicitly forecasting a level of about 24,000,000 on the Dow by 2100. If your adviser talks to you about double-digit returns from equities, explain this math to him—not that it will faze him. Many helpers are apparently direct descendants of the queen in Alice in Wonderland, who said: “Why, sometimes I’ve believed as many as six impossible things before breakfast.” Beware the glib helper who fills your head with fantasies while he fills his pockets with fees. (2007)

Measuring Intrinsic Value

Our equity-investing strategy remains little changed from what it was fifteen years ago, when we said in the 1977 annual report: “We select our marketable equity securities in much the way we would evaluate a business for acquisition in its entirety. We want the business to be one (a) that we can understand; (b) with favorable long-term prospects; (c) operated by honest and competent people; and (d) available at a very attractive price.” We have seen cause to make only one change in this creed: Because of both market conditions and our size, we now substitute “an attractive price” for “a very attractive price.”

But how, you will ask, does one decide what’s “attractive”? In answering this question, most analysts feel they must choose between two approaches customarily thought to be in opposition: “value” and “growth.” Indeed, many investment professionals see any mixing of the two terms as a form of intellectual cross-dressing.

We view that as fuzzy thinking (in which, it must be confessed, I myself engaged some years ago). In our opinion, the two approaches are joined at the hip: Growth is always a component in the calculation of value, constituting a variable whose importance can range from negligible to enormous and whose impact can be negative as well as positive.

In addition, we think the very term “value investing” is redundant. What is “investing” if it is not the act of seeking value at least sufficient to justify the amount paid? Consciously paying more for a stock than its

calculated value—in the hope that it can soon be sold for a still-higher price—should be labeled speculation (which is neither illegal, immoral nor—in our view—financially fattening).

Whether appropriate or not, the term “value investing” is widely used. Typically, it connotes the purchase of stocks having attributes such as a low ratio of price to book value, a low price-earnings ratio, or a high dividend yield. Unfortunately, such characteristics, even if they appear in combination, are far from determinative as to whether an investor is indeed buying something for what it is worth and is therefore truly operating on the principle of obtaining value in his investments. Correspondingly, opposite characteristics—a high ratio of price to book value, a high price-earnings ratio, and a low dividend yield—are in no way inconsistent with a “value” purchase.

Similarly, business growth, per se, tells us little about value. It's true that growth often has a positive impact on value, sometimes one of spectacular proportions. But such an effect is far from certain. For example, investors have regularly poured money into the domestic airline business to finance profitless (or worse) growth. For these investors, it would have been far better if Orville had failed to get off the ground at Kitty Hawk: The more the industry has grown, the worse the disaster for owners.

Growth benefits investors only when the business in point can invest at incremental returns that are enticing—in other words, only when each dollar used to finance the growth creates over a dollar of long-term market value. In the case of a low-return business requiring incremental funds, growth hurts the investor.

In *The Theory of Investment Value*, written over 50 years ago, John Burr Williams set forth the equation for value, which we condense here: The value of any stock, bond or business today is determined by the cash inflows and outflows—discounted at an appropriate interest rate—that can be expected to occur during the remaining life of the asset. Note that the formula is the same for stocks as for bonds. Even so, there is an important, and difficult to deal with, difference between the two: A bond has a coupon and maturity date that define future cash flows; but in the case of equities, the investment analyst must himself estimate the future “coupons.” Furthermore, the quality of management affects the bond coupon only rarely—chiefly when management is so inept or dishonest that payment of interest is suspended. In contrast, the ability of management can dramatically affect the equity “coupons.”

The investment shown by the discounted-flows-of-cash calculation to be the cheapest is the one that the investor should purchase—irrespective of whether the business grows or doesn't, displays volatility

or smoothness in its earnings, or carries a high price or low in relation to its current earnings and book value. Moreover, though the value equation has usually shown equities to be cheaper than bonds, that result is not inevitable: When bonds are calculated to be the more attractive investment, they should be bought.

Leaving the question of price aside, the best business to own is one that over an extended period can employ large amounts of incremental capital at very high rates of return. The worst business to own is one that must, or will, do the opposite—that is, consistently employ ever-greater amounts of capital at very low rates of return. Unfortunately, the first type of business is very hard to find: Most high-return businesses need relatively little capital. Shareholders of such a business usually will benefit if it pays out most of its earnings in dividends or makes significant stock repurchases.

Though the mathematical calculations required to evaluate equities are not difficult, an analyst—even one who is experienced and intelligent—can easily go wrong in estimating future “coupons.” At Berkshire, we attempt to deal with this problem in two ways. First, we try to stick to businesses we believe we understand. That means they must be relatively simple and stable in character. If a business is complex or subject to constant change, we’re not smart enough to predict future cash flows. Incidentally, that shortcoming doesn’t bother us. What counts for most people in investing is not how much they know, but rather how realistically they define what they don’t know. An investor needs to do very few things right as long as he or she avoids big mistakes.

Second, and equally important, we insist on a margin of safety in our purchase price. If we calculate the value of a common stock to be only slightly higher than its price, we’re not interested in buying. We believe this margin-of-safety principle, so strongly emphasized by Ben Graham, to be the cornerstone of investment success. (1992)

Increasing Wealth Is Not the Same as Increasing Size

When Charlie and I buy stocks—which we think of as small portions of businesses—our analysis is very similar to that which we use in buying entire businesses. We first have to decide whether we can sensibly estimate an earnings range for five years out, or more. If the answer is yes, we will buy the stock (or business) if it sells at a reasonable price in relation to the bottom boundary of our estimate. If, however, we lack the ability

to estimate future earnings—which is usually the case—we simply move on to other prospects. In the 54 years we have worked together, we have never foregone an attractive purchase because of the macro or political environment, or the views of other people. In fact, these subjects never come up when we make decisions.

It's vital, however, that we recognize the perimeter of our "circle of competence" and stay well inside of it. Even then, we will make some mistakes, both with stocks and businesses. But they will not be the disasters that occur, for example, when a long-rising market induces purchases that are based on anticipated price behavior and a desire to be where the action is. (2013)

Measuring Berkshire's Intrinsic Value

The first component of value is our investments: stocks, bonds and cash equivalents. At year-end these totaled \$158 billion at market value.

Insurance float—money we temporarily hold in our insurance operations that does not belong to us—funds \$66 billion of our investments. This float is "free" as long as insurance underwriting breaks even, meaning that the premiums we receive equal the losses and expenses we incur. Of course, underwriting results are volatile, swinging erratically between profits and losses. Over our entire history, though, we've been significantly profitable, and I also expect us to average breakeven results or better in the future. If we do that, all of our investments—those funded both by float and by retained earnings—can be viewed as an element of value for Berkshire shareholders.

Berkshire's second component of value is earnings that come from sources other than investments and insurance underwriting. These earnings are delivered by our 68 non-insurance companies. In Berkshire's early years, we focused on the investment side. During the past two decades, however, we've increasingly emphasized the development of earnings from non-insurance businesses, a practice that will continue.

For the forty years, our compounded annual gain in pre-tax, non-insurance earnings per share is 21.0%. During the same period, Berkshire's stock price increased at a rate of 22.1% annually. Over time, you can expect our stock price to move in rough tandem with Berkshire's investments and earnings. Market price and intrinsic value often follow very different paths—sometimes for extended periods—but eventually they meet.

There is a third, more subjective, element to an intrinsic value calculation that can be either positive or negative: the efficacy with which retained earnings will be deployed in the future. We, as well as many other businesses, are likely to retain earnings over the next decade that will equal, or even exceed, the capital we presently employ. Some companies will turn these retained dollars into fifty-cent pieces, others into two-dollar bills.

This “what-will-they-do-with-the-money” factor must always be evaluated along with the “what-do-we-have-now” calculation in order for us, or anybody, to arrive at a sensible estimate of a company’s intrinsic value. That’s because an outside investor stands by helplessly as management reinvests his share of the company’s earnings. If a CEO can be expected to do this job well, the reinvestment prospects add to the company’s current value; if the CEO’s talents or motives are suspect, today’s value must be discounted. The difference in outcome can be huge. A dollar of then-value in the hands of Sears Roebuck’s or Montgomery Ward’s CEOs in the late 1960s had a far different destiny than did a dollar entrusted to Sam Walton. (2010)

A Final Word on Measuring Intrinsic Value and Investment versus Speculation

Leaving aside tax factors, the formula we use for evaluating stocks and businesses is identical. Indeed, the formula for valuing all assets that are purchased for financial gain has been unchanged since it was first laid out by a very smart man in about 600 B.C. (though he wasn’t smart enough to know it was 600 B.C.).

The oracle was Aesop and his enduring, though somewhat incomplete, investment insight was “a bird in the hand is worth two in the bush.” To flesh out this principle, you must answer only three questions. How certain are you that there are indeed birds in the bush? When will they emerge and how many will there be? What is the risk-free interest rate (which we consider to be the yield on long-term U.S. bonds)? If you can answer these three questions, you will know the maximum value of the bush—and the maximum number of the birds you now possess that should be offered for it. And, of course, don’t literally think birds. Think dollars.

Aesop’s investment axiom, thus expanded and converted into dollars, is immutable. It applies to outlays for farms, oil royalties, bonds, stocks, lottery tickets, and manufacturing plants. And neither the advent of the

steam engine, the harnessing of electricity nor the creation of the automobile changed the formula one iota—nor will the Internet. Just insert the correct numbers, and you can rank the attractiveness of all possible uses of capital throughout the universe.

Common yardsticks such as dividend yield, the ratio of price to earnings or to book value, and even growth rates have nothing to do with valuation except to the extent they provide clues to the amount and timing of cash flows into and from the business. Indeed, growth can destroy value if it requires cash inputs in the early years of a project or enterprise that exceed the discounted value of the cash that those assets will generate in later years. Market commentators and investment managers who glibly refer to “growth” and “value” styles as contrasting approaches to investment are displaying their ignorance, not their sophistication. Growth is simply a component—usually a plus, sometimes a minus—in the value equation.

Alas, though Aesop’s proposition and the third variable—that is, interest rates—are simple, plugging in numbers for the other two variables is a difficult task. Using precise numbers is, in fact, foolish; working with a range of possibilities is the better approach.

Usually, the range must be so wide that no useful conclusion can be reached. Occasionally, though, even very conservative estimates about the future emergence of birds reveal that the price quoted is startlingly low in relation to value. (Let’s call this phenomenon the IBT—Inefficient Bush Theory.) To be sure, an investor needs some general understanding of business economics as well as the ability to think independently to reach a well-founded positive conclusion. But the investor does not need brilliance nor blinding insights.

At the other extreme, there are many times when the most brilliant of investors can’t muster a conviction about the birds to emerge, not even when a very broad range of estimates is employed. This kind of uncertainty frequently occurs when new businesses and rapidly changing industries are under examination. In cases of this sort, any capital commitment must be labeled speculative.

Now, speculation—in which the focus is not on what an asset will produce but rather on what the next fellow will pay for it—is neither illegal, immoral nor un-American. But it is not a game in which Charlie and I wish to play. We bring nothing to the party, so why should we expect to take anything home?

The line separating investment and speculation, which is never bright and clear, becomes blurred still further when most market participants have recently enjoyed triumphs. Nothing sedates rationality

like large doses of effortless money. After a heady experience of that kind, normally sensible people drift into behavior akin to that of Cinderella at the ball. They know that overstaying the festivities—that is, continuing to speculate in companies that have gigantic valuations relative to the cash they are likely to generate in the future—will eventually bring on pumpkins and mice. But they nevertheless hate to miss a single minute of what is one helluva party. Therefore, the giddy participants all plan to leave just seconds before midnight. There's a problem, though: They are dancing in a room in which the clocks have no hands. (2000)

Risk Management—Who's Responsible

In my view a board of directors of a huge financial institution is *derelict* if it does not insist that its CEO bear full responsibility for risk control. If he's incapable of handling that job, he should look for other employment. And if he fails at it—with the government thereupon required to step in with funds or guarantees—the financial consequences for him and his board should be severe. (2009)

Diversification and Concentration; Risk and Return

The strategy we've adopted precludes our following standard diversification dogma. Many pundits would therefore say the strategy must be riskier than that employed by more conventional investors. We disagree. We believe that a policy of portfolio concentration may well decrease risk if it raises, as it should, both the intensity with which an investor thinks about a business and the comfort-level he must feel with its economic characteristics before buying into it. In stating this opinion, we define risk, using dictionary terms, as "the possibility of loss or injury."

Academics, however, like to define investment "risk" differently, averring that it is the relative volatility of a stock or portfolio of stocks—that is, their volatility as compared to that of a large universe of stocks. Employing data bases and statistical skills, these academics compute with precision the "beta" of a stock—its relative volatility in the past—and then build arcane investment and capital-allocation theories around this calculation. In their hunger for a single statistic to measure risk, however, they forget a fundamental principle: It is better to be approximately right than precisely wrong.

For owners of a business—and that's the way we think of shareholders—the academics' definition of risk is far off the mark, so much so that it produces absurdities. For example, under beta-based theory, a stock that has dropped very sharply compared to the market—as had Washington Post when we bought it in 1973—becomes “riskier” at the lower price than it was at the higher price. Would that description have then made any sense to someone who was offered the entire company at a vastly-reduced price?

In fact, the true investor welcomes volatility. Ben Graham explained why in Chapter 8 of *The Intelligent Investor*. There he introduced “Mr. Market,” an obliging fellow who shows up every day to either buy from you or sell to you, whichever you wish. The more manic-depressive this chap is, the greater the opportunities available to the investor. That's true because a wildly fluctuating market means that irrationally low prices will periodically be attached to solid businesses. It is impossible to see how the availability of such prices can be thought of as increasing the hazards for an investor who is totally free to either ignore the market or exploit its folly.

In assessing risk, a beta purist will disdain examining what a company produces, what its competitors are doing, or how much borrowed money the business employs. He may even prefer not to know the company's name. What he treasures is the price history of its stock. In contrast, we'll happily forgo knowing the price history and instead will seek whatever information will further our understanding of the company's business. After we buy a stock, consequently, we would not be disturbed if markets closed for a year or two. We don't need a daily quote on our 100% position in See's or H. H. Brown to validate our well-being. Why, then, should we need a quote on our 7% interest in Coke?

In our opinion, the real risk that an investor must assess is whether his aggregate after-tax receipts from an investment (including those he receives on sale) will, over his prospective holding period, give him at least as much purchasing power as he had to begin with, plus a modest rate of interest on that initial stake. Though this risk cannot be calculated with engineering precision, it can in some cases be judged with a degree of accuracy that is useful. The primary factors bearing upon this evaluation are:

1. The certainty with which the long-term economic characteristics of the business can be evaluated;
2. The certainty with which management can be evaluated, both as to its ability to realize the full potential of the business and to wisely employ its cash flows;

3. The certainty with which management can be counted on to channel the rewards from the business to the shareholders rather than to itself;
4. The purchase price of the business;
5. The levels of taxation and inflation that will be experienced and that will determine the degree by which an investor's purchasing-power return is reduced from his gross return.

These factors will probably strike many analysts as unbearably fuzzy, since they cannot be extracted from a data base of any kind. But the difficulty of precisely quantifying these matters does not negate their importance nor is it insuperable. Just as Justice Stewart found it impossible to formulate a test for obscenity but nevertheless asserted, "I know it when I see it," so also can investors—in an inexact but useful way—"see" the risks inherent in certain investments without reference to complex equations or price histories. (1978)

Good Management, from the Berkshire Perspective

The most important factors determining a business' intrinsic value are the quality of management and the underlying economic quality of the business itself. Buffett has described at length what both good managements and good businesses look like.

In 1983, I set down 13 owner-related business principles that I thought would help new shareholders understand our managerial approach.

- *Although our form is corporate, our attitude is partnership. Charlie Munger and I think of our shareholders as owner partners, and of ourselves as managing partners. (Because of the size of our shareholdings we are also, for better or worse, controlling partners.) We do not view the company itself as the ultimate owner of our business assets but instead view the company as a conduit through which our shareholders own the assets.*
- *In line with Berkshire's owner-orientation, most of our directors have a significant portion of their net worth invested in the company. We eat our own cooking.*
- *Accounting consequences do not influence our operating or capital-allocation decisions. When acquisition costs are similar, we much prefer to purchase \$2 of earnings that is not reportable by us under standard accounting principles than to purchase \$1 of earnings that is reportable. This is precisely the choice that often faces us since entire businesses (whose earnings will be fully reportable)*

frequently sell for double the pro-rata price of small portions (whose earnings will be largely unreportable). In aggregate and over time, we expect the unreported earnings to be fully reflected in our intrinsic business value through capital gains.

- *We use debt sparingly. We will reject interesting opportunities rather than over-leverage our balance sheet. This conservatism has penalized our results but it is the only behavior that leaves us comfortable, considering our fiduciary obligations to policyholders, lenders and the many equity holders who have committed unusually large portions of their net worth to our care. (As one of the Indianapolis “500” winners said: “To finish first, you must first finish.”)*
- *A managerial “wish list” will not be filled at shareholder expense. We will not diversify by purchasing entire businesses at control prices that ignore long-term economic consequences to our shareholders. We will only do with your money what we would do with our own, weighing fully the values you can obtain by diversifying your own portfolios through direct purchases in the stock market.*
- *We feel noble intentions should be checked periodically against results. We test the wisdom of retaining earnings by assessing whether retention, over time, delivers shareholders at least \$1 of market value for each \$1 retained. To date, this test has been met. We will continue to apply it on a five-year rolling basis. As our net worth grows, it is more difficult to use retained earnings wisely.*
- *We will issue common stock only when we receive as much in business value as we give. This rule applies to all forms of issuance—not only mergers or public stock offerings, but stock-for-debt swaps, stock options, and convertible securities as well. We will not sell small portions of your company—and that is what the issuance of shares amounts to—on a basis inconsistent with the value of the entire enterprise. (Owner’s Manual, Updated Version)*

Every day, in countless ways, the competitive position of each of our businesses grows either weaker or stronger. If we are delighting customers, eliminating unnecessary costs and improving our products and services, we gain strength. But if we treat customers with indifference or tolerate bloat, our businesses will wither. On a daily basis, the effects of our actions are imperceptible; cumulatively, though, their consequences are enormous.

When our long-term competitive position improves as a result of these almost unnoticeable actions, we describe the phenomenon as “widening the moat.” And doing that is essential if we are to have the kind of business we want a decade or two from now. We always, of course, hope to earn more money in the short-term. But when short-term and long-term conflict, widening the moat must take

precedence. If a management makes bad decisions in order to hit short-term earnings targets, and consequently gets behind the eight-ball in terms of costs, customer satisfaction or brand strength, no amount of subsequent brilliance will overcome the damage that has been inflicted. Take a look at the dilemmas of managers in the auto and airline industries today as they struggle with the huge problems handed them by their predecessors. Charlie is fond of quoting Ben Franklin's "An ounce of prevention is worth a pound of cure." But sometimes no amount of cure will overcome the mistakes of the past.

Our managers focus on moat-widening—and are brilliant at it. Quite simply, they are passionate about their businesses. Usually, they were running those long before we came along; our only function since has been to stay out of the way. If you see these heroes—and our four heroines as well—at the annual meeting, thank them for the job they do for you.

It's hard to overemphasize the importance of who is CEO of a company. Before Jim Kilts arrived at Gillette in 2001, the company was struggling, having particularly suffered from capital-allocation blunders. In the major example, Gillette's acquisition of Duracell cost Gillette shareholders billions of dollars, a loss never made visible by conventional accounting. Quite simply, what Gillette received in business value in this acquisition was not equivalent to what it gave up. (Amazingly, this most fundamental of yardsticks is almost always ignored by both managements and their investment bankers when acquisitions are under discussion.) (2005)

General Acquisition Behavior

As our history indicates, we are comfortable both with total ownership of businesses and with marketable securities representing small portions of businesses. We continually look for ways to employ large sums in each area. (But we try to avoid small commitments—"If something's not worth doing at all, it's not worth doing well".) Indeed, the liquidity requirements of our insurance and trading stamp businesses mandate major investments in marketable securities.

Our acquisition decisions will be aimed at maximizing real economic benefits, not at maximizing either managerial domain or reported numbers for accounting purposes. (In the long run, managements stressing

accounting appearance over economic substance usually achieve little of either.)

Regardless of the impact upon immediately reportable earnings, we would rather buy 10% of Wonderful Business T at X per share than 100% of T at 2X per share. Most corporate managers prefer just the reverse, and have no shortage of stated rationales for their behavior.

However, we suspect three motivations—usually unspoken—to be, singly or in combination, the important ones in most high-premium takeovers:

1. Leaders, business or otherwise, seldom are deficient in animal spirits and often relish increased activity and challenge. At Berkshire, the corporate pulse never beats faster than when an acquisition is in prospect.
2. Most organizations, business or otherwise, measure themselves, are measured by others, and compensate their managers far more by the yardstick of size than by any other yardstick. (Ask a Fortune 500 manager where his corporation stands on that famous list and, invariably, the number responded will be from the list ranked by size of sales; he may well not even know where his corporation places on the list Fortune just as faithfully compiles ranking the same 500 corporations by profitability.)
3. Many managements apparently were overexposed in impressionable childhood years to the story in which the imprisoned handsome prince is released from a toad's body by a kiss from a beautiful princess. Consequently, they are certain their managerial kiss will do wonders for the profitability of Company T(target).

Such optimism is essential. Absent that rosy view, why else should the shareholders of Company A(cquisitor) want to own an interest in T at the 2X takeover cost rather than at the X market price they would pay if they made direct purchases on their own?

In other words, investors can always buy toads at the going price for toads. If investors instead bankroll princesses who wish to pay double for the right to kiss the toad, those kisses had better pack some real dynamite. We've observed many kisses but very few miracles. Nevertheless, many managerial princesses remain serenely confident about the future potency of their kisses—even after their corporate backyards are knee-deep in unresponsive toads . . . (1981)

Businesses—the Great, the Good and the Gruesome

Let's take a look at what kind of businesses turn us on. And while we're at it, let's also discuss what we wish to avoid.

Charlie and I look for companies that have (a) a business we understand; (b) favorable long-term economics; (c) able and trustworthy management; and (d) a sensible price tag. We like to buy the whole business or, if management is our partner, at least 80%. When control-type purchases of quality aren't available, though, we are also happy to simply buy small portions of great businesses by way of stock market purchases. It's better to have a part interest in the Hope Diamond than to own all of a rhinestone.

A truly great business must have an enduring "moat" that protects excellent returns on invested capital. The dynamics of capitalism guarantee that competitors will repeatedly assault any business "castle" that is earning high returns. Therefore a formidable barrier such as a company's being the low cost producer (GEICO, Costco) or possessing a powerful world-wide brand (Coca-Cola, Gillette, American Express) is essential for sustained success. Business history is filled with "Roman Candles," companies whose moats proved illusory and were soon crossed.

Our criterion of "enduring" causes us to rule out companies in industries prone to rapid and continuous change. Though capitalism's "creative destruction" is highly beneficial for society, it precludes investment certainty. A moat that must be continuously rebuilt will eventually be no moat at all.

Additionally, this criterion eliminates the business whose success depends on having a great manager. Of course, a terrific CEO is a huge asset for any enterprise, and at Berkshire we have an abundance of these managers. Their abilities have created billions of dollars of value that would never have materialized if typical CEOs had been running their businesses.

But if a business requires a superstar to produce great results, the business itself cannot be deemed great. A medical partnership led by your area's premier brain surgeon may enjoy outsized and growing earnings, but that tells little about its future. The partnership's moat will go when the surgeon goes. You can count, though, on the moat of the Mayo Clinic to endure, even though you can't name its CEO.

Long-term competitive advantage in a stable industry is what we seek in a business. If that comes with rapid organic growth, great. But even without organic growth, such a business is rewarding. We will simply take the lush earnings of the business and use them to buy similar

businesses elsewhere. There's no rule that you have to invest money where you've earned it. Indeed, it's often a mistake to do so: Truly great businesses, earning huge returns on tangible assets, can't for any extended period reinvest a large portion of their earnings internally at high rates of return.

Let's look at the prototype of a dream business, our own See's Candy. The boxed-chocolates industry in which it operates is unexciting: Per-capita consumption in the U.S. is extremely low and doesn't grow. Many once-important brands have disappeared, and only three companies have earned more than token profits over the last forty years. Indeed, I believe that See's, though it obtains the bulk of its revenues from only a few states, accounts for nearly half of the entire industry's earnings.

At See's, annual sales were 16 million pounds of candy when Blue Chip Stamps purchased the company in 1972. (Charlie and I controlled Blue Chip at the time and later merged it into Berkshire.) Last year See's sold 31 million pounds, a growth rate of only 2% annually. Yet its durable competitive advantage, built by the See's family over a 50-year period, and strengthened subsequently by Chuck Huggins and Brad Kinstler, has produced extraordinary results for Berkshire.

We bought See's for \$25 million when its sales were \$30 million and pre-tax earnings were less than \$5 million. The capital then required to conduct the business was \$8 million. (Modest seasonal debt was also needed for a few months each year.) Consequently, the company was earning 60% pre-tax on invested capital. Two factors helped to minimize the funds required for operations. First, the product was sold for cash, and that eliminated accounts receivable. Second, the production and distribution cycle was short, which minimized inventories.

Last year See's sales were \$383 million, and pre-tax profits were \$82 million. The capital now required to run the business is \$40 million. This means we have had to reinvest only \$32 million since 1972 to handle the modest physical growth—and somewhat immodest financial growth—of the business. In the meantime pre-tax earnings have totaled \$1.35 billion. All of that, except for the \$32 million, has been sent to Berkshire (or, in the early years, to Blue Chip). After paying corporate taxes on the profits, we have used the rest to buy other attractive businesses. Just as Adam and Eve kick-started an activity that led to six billion humans, See's has given birth to multiple new streams of cash for us. (The biblical command to “be fruitful and multiply” is one we take seriously at Berkshire.)

There aren't many See's in Corporate America. Typically, companies that increase their earnings from \$5 million to \$82 million require, say, \$400 million or so of capital investment to finance their growth. That's

because growing businesses have both working capital needs that increase in proportion to sales growth and significant requirements for fixed asset investments.

A company that needs large increases in capital to engender its growth may well prove to be a satisfactory investment. There is, to follow through on our example, nothing shabby about earning \$82 million pre-tax on \$400 million of net tangible assets. But that equation for the owner is vastly different from the See's situation. It's far better to have an ever-increasing stream of earnings with virtually no major capital requirements. Ask Microsoft or Google.

One example of good, but far from sensational, business economics is our own FlightSafety. This company delivers benefits to its customers that are the equal of those delivered by any business that I know of. It also possesses a durable competitive advantage: Going to any other flight-training provider than the best is like taking the low bid on a surgical procedure.

Nevertheless, this business requires a significant reinvestment of earnings if it is to grow. When we purchased FlightSafety in 1996, its pre-tax operating earnings were \$111 million, and its net investment in fixed assets was \$570 million. Since our purchase, depreciation charges have totaled \$923 million. But capital expenditures have totaled \$1.635 billion, most of that for simulators to match the new airplane models that are constantly being introduced. (A simulator can cost us more than \$12 million, and we have 273 of them.) Our fixed assets, after depreciation, now amount to \$1.079 billion. Pre-tax operating earnings in 2007 were \$270 million, a gain of \$159 million since 1996. That gain gave us a good, but far from See's-like, return on our incremental investment of \$509 million.

Consequently, if measured only by economic returns, FlightSafety is an excellent but not extraordinary business. Its put-up-more-to-earn-more experience is that faced by most corporations. For example, our large investment in regulated utilities falls squarely in this category. We will earn considerably more money in this business ten years from now, but we will invest many billions to make it.

Now let's move to the gruesome. The worst sort of business is one that grows rapidly, requires significant capital to engender the growth, and then earns little or no money. Think airlines. Here a durable competitive advantage has proven elusive ever since the days of the Wright Brothers. Indeed, if a farsighted capitalist had been present at Kitty Hawk, he would have done his successors a huge favor by shooting Orville down.

The airline industry's demand for capital ever since that first flight has been insatiable. Investors have poured money into a bottomless pit, attracted by growth when they should have been repelled by it. And I, to my shame, participated in this foolishness when I had Berkshire buy U.S. Air preferred stock in 1989. As the ink was drying on our check, the company went into a tailspin, and before long our preferred dividend was no longer being paid. But we then got very lucky. In one of the recurrent, but always misguided, bursts of optimism for airlines, we were actually able to sell our shares in 1998 for a hefty gain. In the decade following our sale, the company went bankrupt. Twice. (2007)

Even Homer Nods, or Why Good Businesses Are Better Than Poor Ones

Mistakes of the First Twenty-five Years (A Condensed Version)

To quote Robert Benchley, "Having a dog teaches a boy fidelity, perseverance, and to turn around three times before lying down." Such are the shortcomings of experience. Nevertheless, it's a good idea to review past mistakes before committing new ones. So let's take a quick look at the last 25 years.

- My first mistake, of course, was in buying control of Berkshire. Though I knew its business—textile manufacturing—to be unpromising, I was enticed to buy because the price looked cheap. Stock purchases of that kind had proved reasonably rewarding in my early years, though by the time Berkshire came along in 1965 I was becoming aware that the strategy was not ideal.

If you buy a stock at a sufficiently low price, there will usually be some hiccup in the fortunes of the business that gives you a chance to unload at a decent profit, even though the long-term performance of the business may be terrible. I call this the "cigar butt" approach to investing. A cigar butt found on the street that has only one puff left in it may not offer much of a smoke, but the "bargain purchase" will make that puff all profit.

Unless you are a liquidator, that kind of approach to buying businesses is foolish. First, the original "bargain" price probably will not turn out to be such a steal after all. In a difficult business, no sooner is one problem solved than another surfaces—never is there just one cockroach in the kitchen. Second, any initial advantage you secure will be quickly

eroded by the low return that the business earns. For example, if you buy a business for \$8 million that can be sold or liquidated for \$10 million and promptly take either course, you can realize a high return. But the investment will disappoint if the business is sold for \$10 million in ten years and in the interim has annually earned and distributed only a few percent on cost. Time is the friend of the wonderful business, the enemy of the mediocre.

You might think this principle is obvious, but I had to learn it the hard way—in fact, I had to learn it several times over. Shortly after purchasing Berkshire, I acquired a Baltimore department store, Hochschild Kohn, buying through a company called Diversified Retailing that later merged with Berkshire. I bought at a substantial discount from book value, the people were first-class, and the deal included some extras—unrecorded real estate values and a significant LIFO inventory cushion. How could I miss? — three years later I was lucky to sell the business for about what I had paid. After ending our corporate marriage to Hochschild Kohn, I had memories like those of the husband in the country song, “My Wife Ran Away With My Best Friend and I Still Miss Him a Lot.”

I could give you other personal examples of “bargain-purchase” folly but I’m sure you get the picture: It’s far better to buy a wonderful company at a fair price than a fair company at a wonderful price. Charlie understood this early; I was a slow learner. But now, when buying companies or common stocks, we look for first-class businesses accompanied by first-class managements.

- That leads right into a related lesson: Good jockeys will do well on good horses, but not on broken-down nags. Both Berkshire’s textile business and Hochschild, Kohn had able and honest people running them. The same managers employed in a business with good economic characteristics would have achieved fine records. But they were never going to make any progress while running in quicksand.

I’ve said many times that when a management with a reputation for brilliance tackles a business with a reputation for bad economics, it is the reputation of the business that remains intact. I just wish I hadn’t been so energetic in creating examples. My behavior has matched that admitted by Mae West: “I was Snow White, but I drifted.”

- A further related lesson: Easy does it. After 25 years of buying and supervising a great variety of businesses, Charlie and I have not

learned how to solve difficult business problems. What we have learned is to avoid them. To the extent we have been successful, it is because we concentrated on identifying one-foot hurdles that we could step over rather than because we acquired any ability to clear seven-footers

- My most surprising discovery: the overwhelming importance in business of an unseen force that we might call “the institutional imperative.” In business school, I was given no hint of the imperative’s existence and I did not intuitively understand it when I entered the business world. I thought then that decent, intelligent, and experienced managers would automatically make rational business decisions. But I learned over time that isn’t so. Instead, rationality frequently wilts when the institutional imperative comes into play.

For example: (1) As if governed by Newton’s First Law of Motion, an institution will resist any change in its current direction; (2) Just as work expands to fill available time, corporate projects or acquisitions will materialize to soak up available funds; (3) Any business craving of the leader, however foolish, will be quickly supported by detailed rate-of-return and strategic studies prepared by his troops; and (4) The behavior of peer companies, whether they are expanding, acquiring, setting executive compensation or whatever, will be mindlessly imitated.

Institutional dynamics, not venality or stupidity, set businesses on these courses, which are too often misguided. After making some expensive mistakes because I ignored the power of the imperative, I have tried to organize and manage Berkshire in ways that minimize its influence. Furthermore, Charlie and I have attempted to concentrate our investments in companies that appear alert to the problem

- Our consistently-conservative financial policies may appear to have been a mistake, but in my view were not. In retrospect, it is clear that significantly higher, though still conventional, leverage ratios at Berkshire would have produced considerably better returns on equity than the 23.8% we have actually averaged. Even in 1965, perhaps we could have judged there to be a 99% probability that higher leverage would lead to nothing but good. Correspondingly, we might have seen only a 1% chance that some shock factor, external or internal, would cause a conventional debt ratio to produce a result falling somewhere between temporary anguish and default.

We wouldn’t have liked those 99:1 odds—and never will. A small chance of distress or disgrace cannot, in our view, be offset by a large

chance of extra returns. If your actions are sensible, you are certain to get good results; in most such cases, leverage just moves things along faster. Charlie and I have never been in a big hurry: We enjoy the process far more than the proceeds—though we have learned to live with those also. (1989)

A Final Word on Good Businesses and Good Management

My conclusion from my own experiences and from much observation of other businesses is that a good managerial record (measured by economic returns) is far more a function of what business boat you get into than it is of how effectively you row (though intelligence and effort help considerably, of course, in any business, good or bad). Some years ago I wrote: “When a management with a reputation for brilliance tackles a business with a reputation for poor fundamental economics, it is the reputation of the business that remains intact.” Nothing has since changed my point of view on that matter. Should you find yourself in a chronically leaking boat, energy devoted to changing vessels is likely to be more productive than energy devoted to patching leaks. (1985)

Concluding Thoughts on Investing

Investing is often described as the process of laying out money now in the expectation of receiving more money in the future. At Berkshire we take a more demanding approach, defining investing as the transfer to others of purchasing power now with the reasoned expectation of receiving more purchasing power—after taxes have been paid on nominal gains—in the future. More succinctly, investing is forgoing consumption now in order to have the ability to consume more at a later date.

From our definition there flows an important corollary: The riskiness of an investment is not measured by beta (a Wall Street term encompassing volatility and often used in measuring risk) but rather by the probability—the reasoned probability—of that investment causing its owner a loss of purchasing-power over his contemplated holding period. Assets can fluctuate greatly in price and not be risky as long as they are reasonably certain to deliver increased purchasing power over their holding period. And as we will see, a non-fluctuating asset can be laden with risk.

Investment possibilities are both many and varied. There are three major categories, however, and it's important to understand the characteristics of each. So let's survey the field.

- Investments that are denominated in a given currency include money-market funds, bonds, mortgages, bank deposits, and other instruments. Most of these currency-based investments are thought of as "safe." In truth they are among the most dangerous of assets. Their beta may be zero, but their risk is huge.

Over the past century these instruments have destroyed the purchasing power of investors in many countries, even as the holders continued to receive timely payments of interest and principal. This ugly result, moreover, will forever recur. Governments determine the ultimate value of money, and systemic forces will sometimes cause them to gravitate to policies that produce inflation. From time to time such policies spin out of control.

Even in the U.S., where the wish for a stable currency is strong, the dollar has fallen a staggering 86% in value since 1965, when I took over management of Berkshire. It takes no less than \$7 today to buy what \$1 did at that time. Consequently, a tax-free institution would have needed 4.3% interest annually from bond investments over that period to simply maintain its purchasing power. Its managers would have been kidding themselves if they thought of any portion of that interest as "income."

Beyond the requirements that liquidity and regulators impose on us, we will purchase currency-related securities only if they offer the possibility of unusual gain—either because a particular credit is mispriced, as can occur in periodic junk-bond debacles, or because rates rise to a level that offers the possibility of realizing substantial capital gains on high-grade bonds when rates fall. Though we've exploited both opportunities in the past—and may do so again—we are now 180 degrees removed from such prospects. Today, a wry comment that Wall Streeter Shelby Cullom Davis made long ago seems apt: "Bonds promoted as offering risk-free returns are now priced to deliver return-free risk."

- The second major category of investments involves assets that will never produce anything, but that are purchased in the buyer's hope that someone else—who also knows that the assets will be forever unproductive—will pay more for them in the future. Tulips, of all things, briefly became a favorite of such buyers in the 17th century.

This type of investment requires an expanding pool of buyers, who, in turn, are enticed because they believe the buying pool will expand still further. Owners are not inspired by what the asset itself can produce—it will remain lifeless forever—but rather by the belief that others will desire it even more avidly in the future.

The major asset in this category is gold, currently a huge favorite of investors who fear almost all other assets, especially paper money (of whose value, as noted, they are right to be fearful).

Today the world's gold stock is about 170,000 metric tons. If all of this gold were melded together, it would form a cube of about 68 feet per side. (Picture it fitting comfortably within a baseball infield.) At \$1,750 per ounce—gold's price as I write this—its value would be \$9.6 trillion. Call this cube pile A.

Let's now create a pile B costing an equal amount. For that, we could buy all U.S. cropland (400 million acres with output of about \$200 billion annually), plus 16 Exxon Mobils (the world's most profitable company, one earning more than \$40 billion annually). After these purchases, we would have about \$1 trillion left over for walking-around money (no sense feeling strapped after this buying binge). Can you imagine an investor with \$9.6 trillion selecting pile A over pile B?

A century from now the 400 million acres of farmland will have produced staggering amounts of corn, wheat, cotton, and other crops—and will continue to produce that valuable bounty, whatever the currency may be. Exxon Mobil will probably have delivered trillions of dollars in dividends to its owners and will also hold assets worth many more trillions (and, remember, you get 16 Exxons). The 170,000 tons of gold will be unchanged in size and still incapable of producing anything. You can fondle the cube, but it will not respond.

- Our first two categories enjoy maximum popularity at peaks of fear: Terror over economic collapse drives individuals to currency-based assets, most particularly U.S. obligations, and fear of currency collapse fosters movement to sterile assets such as gold. We heard “cash is king” in late 2008, just when cash should have been deployed rather than held. Similarly, we heard “cash is trash” in the early 1980s just when fixed-dollar investments were at their most attractive level in memory. On those occasions, investors who required a supportive crowd paid dearly for that comfort.

My own preference—and you knew this was coming—is our third category: investment in productive assets, whether businesses, farms, or

real estate. Ideally, these assets should have the ability in inflationary times to deliver output that will retain its purchasing-power value while requiring a minimum of new capital investment. Farms, real estate, and many businesses such as Coca-Cola, IBM and our own See's Candy meet that double-barreled test. Certain other companies—think of our regulated utilities, for example—fail it because inflation places heavy capital requirements on them. To earn more, their owners must invest more. Even so, these investments will remain superior to nonproductive or currency-based assets. (2011)

Bonds as Businesses

Washington Public Power Supply System

From October, 1983 through June, 1984 Berkshire's insurance subsidiaries continuously purchased large quantities of bonds of Projects 1, 2, and 3 of Washington Public Power Supply System ("WPPSS"). This is the same entity that, on July 1, 1983, defaulted on \$2.2 billion of bonds issued to finance partial construction of the now-abandoned Projects 4 and 5. While there are material differences in the obligors, promises, and properties underlying the two categories of bonds, the problems of Projects 4 and 5 have cast a major cloud over Projects 1, 2, and 3, and might possibly cause serious problems for the latter issues. In addition, there have been a multitude of problems related directly to Projects 1, 2, and 3 that could weaken or destroy an otherwise strong credit position arising from guarantees by Bonneville Power Administration.

Despite these important negatives, Charlie and I judged the risks at the time we purchased the bonds and at the prices Berkshire paid (much lower than present prices) to be considerably more than compensated for by prospects of profit.

As you know, we buy marketable stocks for our insurance companies based upon the criteria we would apply in the purchase of an entire business. This business-valuation approach is not widespread among professional money managers and is scorned by many academics. Nevertheless, it has served its followers well (to which the academics seem to say, "Well, it may be all right in practice, but it will never work in theory.") Simply put, we feel that if we can buy small pieces of businesses with satisfactory underlying economics at a fraction of the per-share value of the entire business, something good is likely to happen to us—particularly if we own a group of such securities.

We extend this business-valuation approach even to bond purchases such as WPPSS. We compare the \$139 million cost of our year end investment in WPPSS to a similar \$139 million investment in an operating business. In the case of WPPSS, the “business” contractually earns \$22.7 million after tax (via the interest paid on the bonds), and those earnings are available to us currently in cash. We are unable to buy operating businesses with economics close to these. Only a relatively few businesses earn the 16.3% after tax on unleveraged capital that our WPPSS investment does and those businesses, when available for purchase, sell at large premiums to that capital. In the average negotiated business transaction, unleveraged corporate earnings of \$22.7 million after-tax (equivalent to about \$45 million pre-tax) might command a price of \$250–\$300 million (or sometimes far more). For a business we understand well and strongly like, we will gladly pay that much. But it is double the price we paid to realize the same earnings from WPPSS bonds.

However, in the case of WPPSS, there is what we view to be a very slight risk that the “business” could be worth nothing within a year or two. There also is the risk that interest payments might be interrupted for a considerable period of time. Furthermore, the most that the “business” could be worth is about the \$205 million face value of the bonds that we own, an amount only 48% higher than the price we paid.

This ceiling on upside potential is an important minus. It should be realized, however, that the great majority of operating businesses have a limited upside potential also unless more capital is continuously invested in them. That is so because most businesses are unable to significantly improve their average returns on equity—even under inflationary conditions, though these were once thought to automatically raise returns.

(Let’s push our bond-as-a-business example one notch further: if you elect to “retain” the annual earnings of a 12% bond by using the proceeds from coupons to buy more bonds, earnings of that bond “business” will grow at a rate comparable to that of most operating businesses that similarly reinvest all earnings. In the first instance, a 30-year, zero-coupon, 12% bond purchased today for \$10 million will be worth \$300 million in 2015. In the second, a \$10 million business that regularly earns 12% on equity and retains all earnings to grow, will also end up with \$300 million of capital in 2015. Both the business and the bond will earn over \$32 million in the final year.)

Our approach to bond investment—treating it as an unusual sort of “business” with special advantages and disadvantages—may strike you

as a bit quirky. However, we believe that many staggering errors by investors could have been avoided if they had viewed bond investment with a businessman's perspective. For example, in 1946, 20-year AAA tax-exempt bonds traded at slightly below a 1% yield. In effect, the buyer of those bonds at that time bought a "business" that earned about 1% on "book value" (and that, moreover, could never earn a dime more than 1% on book), and paid 100 cents on the dollar for that abominable business.

If an investor had been business-minded enough to think in those terms—and that was the precise reality of the bargain struck—he would have laughed at the proposition and walked away. For, at the same time, businesses with excellent future prospects could have been bought at, or close to, book value while earning 10%, 12%, or 15% after tax on book. Probably no business in America changed hands in 1946 at book value that the buyer believed lacked the ability to earn more than 1% on book. But investors with bond-buying habits eagerly made economic commitments throughout the year on just that basis. Similar, although less extreme, conditions prevailed for the next two decades as bond investors happily signed up for twenty or thirty years on terms outrageously inadequate by business standards. (In what I think is by far the best book on investing ever written—*The Intelligent Investor*, by Ben Graham—the last section of the last chapter begins with, "Investment is most intelligent when it is most businesslike." This section is called "A Final Word," and it is appropriately titled.)

We will emphasize again that there is unquestionably some risk in the WPPSS commitment. It is also the sort of risk that is difficult to evaluate. Were Charlie and I to deal with 50 similar evaluations over a lifetime, we would expect our judgment to prove reasonably satisfactory. But we do not get the chance to make 50 or even 5 such decisions in a single year. Even though our long-term results may turn out fine, in any given year we run a risk that we will look extraordinarily foolish. (That's why all of these sentences say "Charlie and I," or "we.")

Most managers have very little incentive to make the intelligent-but-with-some-chance-of-looking-like-an-idiot decision. Their personal gain/loss ratio is all too obvious: if an unconventional decision works out well, they get a pat on the back and, if it works out poorly, they get a pink slip. (Failing conventionally is the route to go; as a group, lemmings may have a rotten image, but no individual lemming has ever received bad press.) (1984)

Some Thoughts About Investing

Investment is most intelligent when it is most businesslike. — The Intelligent Investor by Benjamin Graham

It is fitting to have a Ben Graham quote open this discussion because I owe so much of what I know about investing to him. I will talk more about Ben a bit later, and I will even sooner talk about common stocks. But let me first tell you about two small non-stock investments that I made long ago. Though neither changed my net worth by much, they are instructive.

This tale begins in Nebraska. From 1973 to 1981, the Midwest experienced an explosion in farm prices, caused by a widespread belief that runaway inflation was coming and fueled by the lending policies of small rural banks. Then the bubble burst, bringing price declines of 50% or more that devastated both leveraged farmers and their lenders. Five times as many Iowa and Nebraska banks failed in that bubble's aftermath than in our recent Great Recession.

In 1986, I purchased a 400-acre farm, located 50 miles north of Omaha, from the FDIC. It cost me \$280,000, considerably less than what a failed bank had lent against the farm a few years earlier. I knew nothing about operating a farm. But I have a son who loves farming and I learned from him both how many bushels of corn and soybeans the farm would produce and what the operating expenses would be. From these estimates, I calculated the normalized return from the farm to then be about 10%. I also thought it was likely that productivity would improve over time and that crop prices would move higher as well. Both expectations proved out.

I needed no unusual knowledge or intelligence to conclude that the investment had no downside and potentially had substantial upside. There would, of course, be the occasional bad crop and prices would sometimes disappoint. But so what? There would be some unusually good years as well, and I would never be under any pressure to sell the property. Now, 28 years later, the farm has tripled its earnings and is worth five times or more what I paid. I still know nothing about farming and recently made just my second visit to the farm.

In 1993, I made another small investment. Larry Silverstein, Salomon's landlord when I was the company's CEO, told me about a New York retail property adjacent to NYU that the Resolution Trust Corp. was selling. Again, a bubble had popped—this one involving commercial real estate—and the RTC had been created to dispose of the assets of failed savings institutions whose optimistic lending practices had fueled the folly.

Here, too, the analysis was simple. As had been the case with the farm, the unleveraged current yield from the property was about 10%. But the property had been under managed by the RTC, and its income would increase when several vacant stores were leased. Even more important, the largest tenant—who occupied around 20% of the project's space—was paying rent of about \$5 per foot, whereas other tenants averaged \$70. The expiration of this bargain lease in nine years was certain to provide a major boost to earnings. The property's location was also superb: NYU wasn't going anywhere.

I joined a small group, including Larry and my friend Fred Rose, that purchased the parcel. Fred was an experienced, high-grade real estate investor who, with his family, would manage the property. And manage it they did. As old leases expired, earnings tripled. Annual distributions now exceed 35% of our original equity investment. Moreover, our original mortgage was refinanced in 1996 and again in 1999, moves that allowed several special distributions totaling more than 150% of what we had invested. I've yet to view the property.

Income from both the farm and the NYU real estate will probably increase in the decades to come. Though the gains won't be dramatic, the two investments will be solid and satisfactory holdings for my lifetime and, subsequently, for my children and grandchildren.

I tell these tales to illustrate certain fundamentals of investing:

- You don't need to be an expert in order to achieve satisfactory investment returns. But if you aren't, you must recognize your limitations and follow a course certain to work reasonably well.³ Keep things simple and don't swing for the fences. When promised quick profits, respond with a quick "no."
- Focus on the future productivity of the asset you are considering. If you don't feel comfortable making a rough estimate of the asset's future earnings, just forget it and move on. No one has the ability to evaluate every investment possibility. But omniscience isn't necessary; you only need to understand the actions you undertake.

Forming macro opinions or listening to the macro or market predictions of others is a waste of time. Indeed, it is dangerous because it may blur your vision of the facts that are truly important. (When I hear

³ It should be noted that in both these cases Buffett relied on trustworthy expert advice (from his son on Nebraska farmland and a successful New York investor on his New York real estate.) Also the other sides of these trades—a bank disposing of distressed assets and a US government agency doing the same thing—were hardly well informed, calculating investors.

TV commentators glibly opine on what the market will do next, I am reminded of Mickey Mantle's scathing comment: "You don't know how easy this game is until you get into that broadcasting booth.") (2013)

A Final Note on Buffett Related Investing

In the first edition of this book, we sought to identify what investing well within a circle of competence looked like by quoting Buffett on investments in areas—insurance, consumer non-durables, and newspapers—with which he was intimately familiar. We think a better approach is simply to refer readers to his annual letters. In particular, reading his successive annual commentaries on his insurance businesses—National Indemnity (early on), Berkshire Re, General Re, GEICO, and other specialty lines—should provide a better idea of what a competent business understanding looks like. Readers should then consider how well they would fare in an insurance business transaction if someone like Warren Buffett were on the other side of the trade.

Robert H. Heilbrunn

Investing in Investors

In 1929, shortly after Robert Heilbrunn had enrolled at the Wharton School, his father died. Heilbrunn left school to take over the management of the family leather trade business. That would have been a large enough challenge for anyone his age, but the onset of the Depression made things even more difficult. In addition to the business, his father had also left an investment portfolio of stocks and bonds. This too became Heilbrunn's responsibility. It was not one of the joyful moments in investment history. The markets were already down a sickening amount in that year alone. And Heilbrunn, though he had worked in the leather business in summers and on vacation, had no experience in managing money.

To gain some understanding, he enrolled in courses given by the stock exchange and by New York University. With none of them providing him with the kind of practical information he needed, he remembered that his father had told him about an investment advisor he knew and trusted, a man named Ben Graham. Heilbrunn looked up Graham in the telephone book and called him. Graham did remember Heilbrunn's father, and the two made an appointment. As Graham later told Heilbrunn, given the circumstances of the time he thought Robert was coming to ask for a loan. In fact Heilbrunn wanted more; he wanted Graham to become his investment advisor and help run the portfolio. Graham agreed but told Heilbrunn there would be a fee of \$25 a month. Heilbrunn, recognizing a bargain when he saw one, accepted.

They started to examine the holdings in the portfolio. Heilbrunn's father had taken a large position in high-grade utility bonds. Though some utility firms had become holding company pyramids and collapsed in the crash, Heilbrunn's bonds held up well; they were paying interest and selling at close to par. Graham's advice took Heilbrunn by surprise. He said they should sell the bonds. Why, Heilbrunn asked; they are good securities. Exactly, Graham told him, and they will never be worth more than they are today. And what would they buy with the proceeds from the sale, Heilbrunn wanted to know. The bonds of Fisk Tire and Rubber, Graham responded. He told Heilbrunn that although the company was in bankruptcy, and the bonds were selling at \$.30 on the dollar, Graham was confident that Fisk would reorganize and that holders of the bonds would receive \$700 in new securities for each \$1,000 bond they held. This was Heilbrunn's introduction to value investing.

Heilbrunn decided to follow his advice—after all, he was paying \$25 per month for it—and called his broker with instructions to sell the utility bonds and buy Fisk. An hour or so after he placed the order, the brokerage firm called him back. They would not buy the Fisk bonds, they told him. They were a high class firm, and they thought their reputation might be tarnished if word got out that they were dealing in bankrupt paper. When Heilbrunn recounted this response to Graham, Graham told him the brokerage firm was simply wrong, that the bonds would prove a successful investment. Heilbrunn moved his business to another broker, who was Graham's brother, and stayed with him for many years. The Fisk bonds did come through as Graham predicted, and Heilbrunn was convinced that Graham was indeed a brilliant investor. He sent other members of his family to see Graham, and he himself began to advise them using the knowledge and insights he was picking up from Graham.

In 1934 Graham and Dodd published their book *Security Analysis*. Heilbrunn was so taken with it that he enrolled in Graham's course at Columbia, taught then as an extension course at night and thus available to people working during the day. Heilbrunn found him a superb teacher, and the fact that Graham dissected the financial statements of companies that he was buying for Heilbrunn's portfolio did nothing to diminish Heilbrunn's interest. By the time the course was over, Heilbrunn realized that he had had enough of the leather business. He wanted to work for Ben Graham, a desire shared by many students fortunate enough to take the course. But Graham at that time had no need of another employee. Instead, he suggested that Heilbrunn

become an independent investigator. He and Graham would discuss certain investment ideas, and Heilbrunn would do the leg work: call the companies, visit them, and find less obvious ways to get more information about them. Though the Securities Act of 1933 and the Securities Exchange Act of 1934 had recently been passed, it took time for companies to make public all the information we now expect. There was no Internet, no EDGAR database, no Free Edgar or any of the other wonderful tools investors can now summon at the click of a mouse. So much the better for energetic and clever researchers.

Around 1937, Benjamin Graham bought a large block of stock in a gas pipeline company for Graham-Newman, the investment partnership he had formed with Jerome Newman. To get the information he needed, Graham went to the state public utility commission, where all public utilities were required to file detailed documents describing their operations. Heilbrunn adapted the approach to investigating Government Employees Insurance Company, an insurance company in Texas, and found what he was looking for in the state insurance office. Graham-Newman owned a large position, and Heilbrunn was able to enrich their knowledge and understanding of the company. But Heilbrunn also discovered that it was a violation of the law for an investment company to own a controlling interest in an insurance company. Graham-Newman solved this problem by distributing the shares directly to their limited partners, who then owned it directly. Government Employees Insurance Company, now known as GEICO, has been the object of value investors' interest for decades. After Graham-Newman distributed the shares, the company attracted the attention of Warren Buffett, who made a well-chronicled trip to Washington to learn what he could about the company and spent a Sunday talking with the chairman. After some ups and downs—the company nearly went bankrupt—it eventually was bought out entirely by Berkshire Hathaway.

Heilbrunn worked on other Graham-Newman Depression-era investments that also paid off handsomely. The real estate market in New York was pummeled by the Depression, the damage made more severe by overbuilding in the 1920s. When developers and owners defaulted on their mortgages, title companies would take a number of them, packaged them together, and sell them as bonds. (Securitization of loans has a longer history than many of us realize.) Bonds backed by mortgages in default sold at deep discounts to their face value, and Heilbrunn, along with Graham-Newman, bought a substantial amount.

They anticipated that at some point the demand for New York City real estate would return. Another series of bonds had been issued to finance the building of the Waldorf-Astoria Hotel. They came out in 1929, paying 6% interest. Even the city's most prestigious hotel could not fill its rooms in the early 1930s, and interest payments were suspended. With the bonds in default, prices dropped to \$.30 on the dollar, or \$300 for a \$1,000 bond. At that price they looked attractive, and since the Chase Bank was willing to lend buyers \$250 on each bond, the cost to the investor was only \$50. Some years later the bonds were repaid in full, including all the accrued interest. Buying damaged goods paid off handsomely.

Heilbrunn continued to invest with Graham-Newman, to do research, which he shared with them, and to do some investing on his own. He sold his leather business to concentrate on investing. His approach was to apply what he had learned from Graham—find the bargains. In that period, when *Barron's* published annually a list of low-priced stocks, Heilbrunn and Graham would examine the companies on the list and buy the 10 best as a basket. Some might disappear, but the ones that worked out more than made up for the losers. And he followed Graham's practice of comparing two companies in the same industry, like Bethlehem and Crucible Steel, to see which was cheaper on an intrinsic value basis. Their focus was the balance sheet, not the income statement. They were able to discuss these ideas and pick up other suggestions from a community of value investors that had formed around Graham.

One lasting interest of Benjamin Graham and this circle was the search for quantitative trading formulas that could be used to direct market investment strategies in a disciplined way. Heilbrunn contributed to the development of these kinds of rules in an article published in 1958. The method prefigured many of the formulas used by quantitatively oriented value investors today. Heilbrunn examined the price, earnings, and dividend histories of specific companies to establish the ranges of the P/E multiple and the dividend yield within which the securities had traded. The investment strategy based on this information is to buy stocks when they trade in the lower portion of their historical P/E multiple range, within the higher portion of their dividend yield range, or both. By establishing the ranges with precision, this approach provides a check on the emotions that can distort investment judgment, both the exuberance engendered by a rising market and the despair occasioned by a falling one. It applies a discipline for buying stocks—when they are cheap—and, usually more valuable, a discipline for selling them—when

they are dear. In a paragraph perhaps more timely today than when he initially wrote it, Heilbrunn warned that:

A feeling of over-optimism in bull markets is one which must be very carefully guarded against by the professional investor as well as by the amateur, since it is generally acknowledged that ... (both are) ... influenced by the tremendous quantity of bullish sentiment ... in newspaper and magazine article, speeches, reports, analyses ... which emanate from the financial district. This statement is in no way to be construed as a criticism of the security analyst, but being human, and this is probably a disadvantage in this profession, he is subject to the same psychological pressures as everyone else.

Heilbrunn's innovation was to focus on the variability of a single stock as it traded within its historic ranges, identifying its highs and lows as compared with itself. The more customary value approach has been to search for stocks with low P/Es, high dividend yields, or low price-to-book multiples as measured simultaneously against other stocks in the universe. Modern quantitative techniques developed by major value-oriented institutions such as Sanford Bernstein have combined the two approaches. They look at where stocks are trading relative to their historic valuation ranges, and then compare stocks with one another based on these results. They identify those at the lower end of their own ranges, and then test each stock against certain other criteria. If the price of the stock no longer falls when additional bad news is announced, that is a good sign. If insiders and other knowledgeable investors are buying, that is another positive sign. The initial quantitative screen as confirmed by the subsequent stock-by-stock examination produces a disciplined overall evaluation.

In his own practice, Heilbrunn embodied one of the core principles of value investing. The circle of professionals that formed around Ben Graham included two men who actually worked for Graham-Newman, Walter Schloss and, in the mid-1950s, Warren Buffett. Buffett, as all his followers know, had come to Columbia Business School to study with Graham, after having read *The Intelligent Investor*. After a while Heilbrunn began to think that the best investments he could make were in Graham, Schloss, and Buffett. He put money into Buffett's partnership a year or so after it started, and he also entrusted some funds to Walter Schloss after initially deciding against it. In later years he added other prominent value investors to his portfolio of managers. These decisions paid off handsomely, and Heilbrunn was able to retire, more or less, from direct active investing.

By entrusting his assets to other managers, Heilbrunn embodied a version of one of the enduring precepts of value investing: know what you know and stay within your circle of competence. Investments records are not definitive, but it does seem clear that a small number of professional investors, disproportionately of the value persuasion, have been able to earn above market returns over the long term. The likelihood of mere good fortune as an explanation is small. When that performance is tied to a carefully designed investment approach and expertise in particular industries, and when the abilities involved are available to others at a reasonable price—Graham charged Heilbrunn \$25 per month, but then that was in 1929—then investing through these individuals or institutions makes a great deal of sense. Knowing when other full-time investors are likely to outperform your own part-time efforts may be the most fundamental of all value insights.

Several years ago, when Wells Fargo Bank looked like a promising investment to the value firm Tweedy, Browne, the company was about to assign an analyst to study Wells Fargo in detail. When they discovered that Berkshire Hathaway had acquired a large position in the bank, Tweedy, Browne reassigned the analyst and simply bought the stock. They felt that their own research was unlikely to be superior to Warren Buffett's. Over the years, Robert Heilbrunn exercised a similar degree of judgment, enriched by his own experiences as a value investor, in evaluating candidates to manage his money. He made wise choices.

Walter and Edwin Schloss

Keep It Simple, and Cheap

Walter Schloss started his limited partnership in the middle of 1955. He tracks his performance from January 1, 1956, a date sufficiently historic to give him one of the longest uninterrupted records—same manager, same organization—in investment history. He also has one of the best. Over the entire 45-year period from 1956 through 2000, Schloss and his son Edwin, who joined him in 1973, have returned for their investors a compounded return of 15.3% per year. During the same period, the S&P Industrial Index¹ had comparable total returns of 11.5%. Every dollar a fortunate investor entrusted with Schloss at the start of 1956 had grown to \$662 by the end of 2000, including all charges for management. A dollar invested in the S&P index would have been worth \$118. The Schlosses's accomplishment is even better than this initial comparison suggests. Over that entire 45-year period, their portfolio has had 7 years in which it lost money; the S&P Index has had 11. The average loss in the Schloss partnership

¹ Walter Schloss began using the S&P Industrial Index in 1955 because, without utilities or transportation companies, it more accurately matched the investments in his portfolio. He has kept that index as his benchmark comparison even as the S&P 500 has become the proxy of choice. Comparing the two over the last two decades, we found that they tracked each other very closely and that the Industrial had a slightly higher return than the 500.

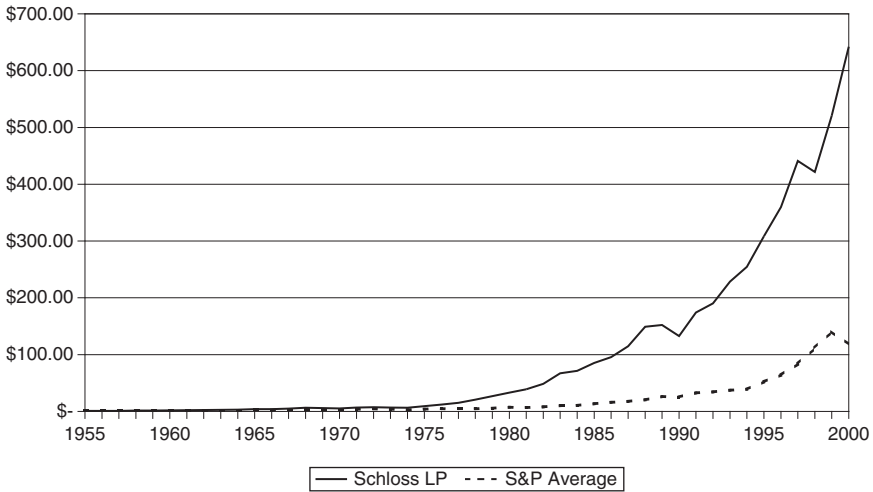


Figure Schloss #1 Returns of the partnership versus the S&P, 1956–2000

was 7.6%; in the S&P, 10.6%. Modern investment theory argues that return is compensation for risk, that higher returns are achieved only by increasing the volatility of the portfolio. The investment success of the Schlosses does not confirm the theory.

Walter and Edwin Schloss are minimalists. Their office—Castle Schloss has one room—is spare, they don't visit companies, they rarely speak to management, they don't speak to analysts, they don't use the Internet. Not wanting to be swayed to do something they shouldn't, they limit their conversations. There is an abundance of articulate and intelligent people in the investment world, most of whom can cite persuasive reasons for buying this stock or that bond. The Schlosses would rather trust their own analysis and their long-standing commitment to buying cheap stocks. This approach leads them to focus almost exclusively on the published financial statements that public firms must produce each quarter. They start by looking at the balance sheet. Can they buy the company for less than the value of the assets, net of all debt? If so, the stock is a candidate for purchase.

This may sound familiar. If Walter Schloss was not present at the creation of value investing, he showed up shortly thereafter. He started on Wall Street in 1934, at eighteen, in the midst of the Depression. During the late 1930s, Schloss took courses from Benjamin Graham at the New York Stock Exchange Institute. He was in good company; his fellow students included Gus Levy, head of the arbitrage department at Goldman Sachs, Cy Winters of Abraham, at one time president of the New York Society of Security Analysts, and other Wall Street heavyweights. At the

time Schloss was working at Carl M. Loeb & Company, Ben's brother Leon was a customer's man at the firm, and Ben kept his account at Loeb Rhodes, allowing Schloss to confirm that Graham did indeed practice what he preached in class. And he preached value, the advantage of paying less for stocks than for the value of the current assets after deducting all liabilities.

A favorite teaching strategy of Graham was to analyze two companies side by side, even if they were in different industries, and compare the balance sheets. He would take Coca-Cola and Colgate, related to one another only by alphabetical proximity, and ask which stock was more of a bargain relative to the net asset values. Graham's primary concern was the margin of safety, a focus which prevented him from recognizing the great growth potential in Coke. Not all of Graham's tactics worked out. He would buy a leading company in an industry, such as the Illinois Central Railroad, and sell short a secondary one, such as Missouri Kansas Texas, as a hedge. As it turned out, the two securities were not correlated, and the hedge did not work. Another type of hedge, which Graham used repeatedly, was to buy a convertible preferred stock and short the common. If the common rose, he was protected by the convertible feature. If it fell, he made money on the short. In either case, he collected the dividend. This approach has become a standard practice in the industry even though it no longer has the tax advantages it once did. Schloss sees Graham as a legitimate genius, someone whose thinking was original and often contrary to established wisdom. Graham's motivation, Schloss thinks, was primarily intellectual. He was more interested in the ideas than in the money, although that too had its rewards.

Looking for Cheap Stocks

Ask either Schloss about their investment strategy, and you will get the same succinct response: We buy cheap stocks. Identifying "cheap" means comparing price with value. What generally brings a stock to the Schlosses' attention is that the price has fallen. They scrutinize the new lows list to find stocks that have come down in price. If they find that the stock is at a two- or three-year low, so much the better. Some brokers they have done business with over the years call them with suggestions. These securities tend to be at the opposite end of the spectrum from the momentum stocks that most brokers are peddling. The Schlosses are especially attracted to stocks that have gapped down in price, stocks where the price decline has been precipitous.

This taste for fiasco is very contrarian. Stock prices sink when investors have been disappointed, either by a recent event like an earnings announcement below expectations, or by continued unsatisfactory performance that ultimately induces even patient investors to throw in the towel. Over the many years the Schlosses have managed money, they have found themselves investing in different industries, in large, medium, and small companies, in companies with shares that have plummeted in price and in those that have slid downward gradually but persistently. The unifying theme is that the stuff they buy is on sale.

The other term in their strategy is equally important. They buy stocks. They don't buy derivatives, indices, or commodities. They don't short stocks; they have in the past, and made some money, but the experience was uncomfortable. They don't try to time the market, although they do let the market tell them which stocks are cheap. At some points in their careers, the Schlosses did invest in bankrupt bonds, and if the situation presented itself to them, they might again. But that field has become more crowded over the years, and like most value investors, they don't want too much company. As for ordinary fixed income investments, they steer clear. The potential returns are limited, and they can be negative if interest rates rise. Their business is making money for their partners by investing in cheap stocks.

When they find a cheap stock, they may start to buy even before they have completed their research. They have at least a rudimentary knowledge of thousands of companies, and they can consult Value Line or the S&P stock guide for a quick check into the company's financial position. Both believe that the only way to really know a security is to own it, so they sometimes stake out their initial position and then send for the financial statements. The market today is so fast moving that they are almost forced to move quickly.

What Is It Worth? Valuing Assets, Earnings, and Companies

For the nine and half years that Walter Schloss worked for Ben Graham and for some years after he left to run his own partnership, he was able to find stocks selling for less than two-thirds of working capital. But sometime after 1960, as the Depression became a distant memory, those opportunities generally disappeared. Today, companies that meet that requirement are either so burdened by liabilities or losing so much money that their future is in jeopardy. Instead of a margin of safety, there is an aura of doubt.

Nevertheless, Walter has retained his preference for valuation based on assets. A company's assets are more stable than its earnings. If a company has a tangible book value of \$15 per share, even if it is not earning money at the moment, the chances are good that the value of the assets will not drop precipitously. An investor paying \$10 or even \$12 per share has some comfort in knowing that the assets are there to back up the shares. And in Schloss's long experience, companies whose shares can be bought for less than the value of the assets will, more often than not, either return to profitability or be taken over by another firm. All of this may take time; their average holding period for a stock is around four years. He has the patience to hold on. The underlying bet he is making is that overreaction by the market has offered him a bargain, and that given enough time, he will be rewarded. "Something good will happen," he likes to say. And in the interim, the asset value provides some protection against another steep drop in the price of the shares. Though he tends to make his initial purchase before the stock has bottomed, and likes the opportunity to add to his position at lower prices, he also sleeps better at night knowing that if there is a cliff out there, his shares have already fallen over it.

Edwin Schloss also pays attention to asset values, but he is more willing to look at a company's earnings power. He also wants some asset protection. If he finds a cheap stock based on normalized earnings power, he generally will not consider it if he has to pay more than three times book value. There are some durable companies in industries such as food, defense, and even plain old manufacturing that sell for more than book value even when their share prices are depressed. Depending on their estimate of what the companies can earn, Edwin may still find the stock cheap enough to buy.

When they begin to take a hard look at a new company, the Schlosses make sure to read the annual reports thoroughly. The financial statements are important, no doubt, but so are the footnotes. They want to be certain that there are no significant off-balance sheet liabilities. They look at the history of capital spending to see what condition the fixed assets are in. A company that has a fully depreciated plant may be reporting higher earnings than a rival that has just completed a new factory, but if it has spent its money carefully, it is likely to have a more modern and more efficient operation. Ten years of advertising expenses don't show up on the balance sheet but do create some value for a brand, provided the company knows how to exploit it. The Schlosses are looking for recovery potential. The stocks they buy have become cheap for a reason, and their success lies in their ability to form a sufficiently accurate estimate of whether or not the market has overreacted. They do not try

to get inside the business, to know the details of the operations better than management itself. They don't claim or want that expertise. Instead, they limit their exposure to any single company and use their broad and deep investment experience to guide their judgment.

Because the Schlosses have been in the business so long, they have been forced to adjust their criteria as market conditions have changed. When markets are very expensive, their definition of cheap has to be somewhat more flexible and relative. As certain strategies, such as investing in bankrupt bonds, became popular, they moved to other areas. Like many great athletes and some other value investors, they let the game come to them. They have core principles that do not change. They buy cheap stocks, and they like to hold them until they have recovered. But otherwise, they are willing to take what the market offers them on the grounds that if they have bought correctly—the stock was sufficiently cheap—the chances are that something good will happen.

Keeping Track

The Schlosses joke that they will go to corporate annual meetings that are held within a 20-block radius of their office. Since they work in mid-Manhattan (New York, not Kansas), that is a less severe restriction than might initially appear. When they do show up, they like to be lonely, not surrounded by analysts and investment managers. Once they owned the shares of Asarco, a copper mining and smelting company; they went to the meeting and found the room full. On closer inspection, the other attendees turned out to be wives of directors, employees, and people working for the company's investment relations firm. Needless to say, the cheap stock had not yet been discovered. In this case, the company did recover from its price decline and was ultimately bought out by Grupo Mexico.

Because the Schlosses hold their positions on average for four or five years, they have time to become more familiar with the company. They continue to look at each quarterly report. But they do not obsess about day-to-day price swings or two cents per share earnings disappointments or positive surprises. Their approach, as we said, is minimalist. If a company announces an acquisition they regard as foolish, that would be cause for concern, and they might decide to sell. Since everything about their approach orients them toward companies that are not in rapidly changing industries where technological innovation may undermine value in weeks if not days, they can afford to sit back and wait.

They are not entirely passive. Having started with a bottom-up approach to finding a cheap stock, now that they own it they look laterally to analyze other firms in the industry. Are these also cheap, and for the same reasons? They may decide that one of these other companies is a better investment than their initial purchase. Perhaps it is a higher quality company, with better profit margins or lower debt levels. If so, they may trade up in quality, provided they can still take advantage of the depressed status of the industry.

When to Buy, When to Sell

The notion that an investor can buy a stock that has reached the bottom of its fall is a fantasy. No one can accurately predict tops, bottoms, or anything in between. More often than not, value investors will start to buy a stock on the way down. The disappointments or reduced expectations that have made it cheap are not going away anytime soon, and there will still be owners of the stock who haven't yet given up when the value investor makes an initial purchase. If it is toward the end of the year, then selling to take advantage of tax losses can drive the price down even more. Because they are aware that they are, to use an industry cliché, catching a falling knife, value investors are likely to try to scale into a position, buying it in stages. For some, like Warren Buffett, that may not be so easy. Once the word is out that Berkshire Hathaway is a buyer, the stock shoots up in price. Graham himself, Walter Schloss recounts, confronted this problem. He divulged a name to a fellow investor over lunch; by the time he was back in the office, the price had risen so much that he could not buy more and still maintain his value discipline. This is one of the reasons why the Schlosses limit their conversations.

Still, when asked to name the mistake he makes most frequently, Edwin Schloss confesses to buying too much of the stock on the initial purchase and not leaving himself enough room to buy more when the price goes down. If it doesn't drop after his first purchase, then he has made the right decision. But the chances are against him. He often does get the opportunity to average down, meaning to buy additional shares at a lower price. The Schlosses have been in the business too long to think that the stock will now oblige them and only rise in price. Investing is a humbling profession, but when decades of positive results confirm the wisdom of the strategy, humility is tempered by confidence.

Value investors buy too soon and sell too soon, and the Schlosses are no exceptions. The cheap stocks generally get cheaper. When they

recover and start to improve, they reach a point at which they are no longer bargains. The Schlosses start to sell them to investors who are delighted that the prices have gone up. In many instances, they will continue to rise, sometimes dramatically, while the value investor is searching for new bargains. The Schlosses bought the investment bank Lehman Brothers a few years ago at \$15 a share, below book value. When it reached \$35 they sold out. A few years later it had passed \$130. Obviously that last \$100 did not end up in the pockets of value investors. Over the years, they have had similar experiences with Longines-Wittnauer, Clark Oil, and other stocks that moved from undervalued through fairly valued to overvalued without blinking. The money left on the table, to cite yet another investment cliché, makes for a good night's sleep.

The decision to sell a stock that has not recovered requires more judgment than selling a winner. At some point, everyone throws in the towel. For value investors like the Schlosses, the trigger will generally be a deterioration in the assets or the earnings power beyond what they had initially anticipated. The stock may still be cheap, but the prospects of recovery have now started to fade. Even the patience of the most tolerant investor can ultimately be exhausted. There are always other places to invest the money. Also, a realized loss has at least some tax benefits for the partners, whereas the depressed stock is just a reminder of a mistake.

The Portfolio: Diversification with Leeway

In the minds of some money managers, diversification is a defense against ignorance. The thoroughly informed investor, knowledgeable about the industry, the company, and even the economy, can take fewer and larger positions in situations in which he or she is fully informed. Value investors come down on both sides of the question of diversification, although all of them think there is an important role for active stock selection. The Schlosses run a diversified portfolio, but they do it without prescribed limits as to the size of a position they will take. Though they may own 100 names, it is typical for the largest 20 positions to account for around 60% of the portfolio. They have occasionally had up to 20% of their fund in a single security, but that degree of concentration is a rarity. They are buying cheap stocks, we must remember, not great companies with golden futures. Though history has shown that most of their investments work out, there are always some that don't. The difficult task is to tell which will be which

ahead of time. Diversification is a safeguard against uncertainty and an essential feature of the Schlosses's successful strategy.

Here as with other aspects of their approach, they rely on judgment rather than fixed rules. Although they are not going to end up with the portfolio invested in one or two industries, they will overweight their holdings when they find cheap stocks clustered together in out of favor sectors. At times like these, they can pick the better companies within these discarded securities. If the price of a commodity like copper has plummeted, then copper-related stocks will be on sale. Unless copper disappears permanently from use as an industrial and communications material, the supply and demand cycles have a way of righting themselves. Companies with low costs and not overburdened by debt are safe bets at these times, primarily because nobody wants to own them. A cheap price can make up for a multitude of cyclical, operational, and even managerial shortcomings.

Take Care of the Clients

When Walter Schloss had been in business for twenty years, including several with Edwin, Warren Buffett sent a letter to some friends describing the Schloss partnership which had then reached its twentieth anniversary. Schloss left Graham-Newman, Buffett told his readers,

in 1955. And Graham-Newman closed up in 1956. I would prefer not to dwell on the implications of this sequence.

In any event, armed only with a monthly stock guide, a sophisticated style acquired largely from association with me, a sub-lease on a portion of a closet at Tweedy, Browne and a group of partners whose names were straight from a roll call at Ellis Island, Walter strode forth to do battle with the S&P.

We have seen the results of that contest.

Other than an additional 25 years of superior performance, little else has changed. The Schlosses still sublet from Tweedy, Browne, although they have moved into a full-sized room. They supplement the monthly stock guide with Value Line and a quotation machine. And the nature of their clients has persisted, something that does distinguish the Schlosses' partnership from most similarly structured funds. There are partners in the fund whose parents were partners; some are even third-generation clients. As a group, they are not wealthy by the standards of limited partnerships. The money invested with the Schlosses is important to them,

which is one reason why the Schlosses are determined not to lose it. It may also explain why the Schlosses do not disclose to their partners the names of the companies whose shares they own. In the main, they invest in unpresentable securities, stocks no one wants to brag about at cocktail parties or anywhere else. Through painful experience—agony both for the limited partners and themselves—they have found that letting the dogs out of the bag does not add to their clients' comfort level. Quite the reverse; some people have left the fund out of fear that the beaten down shares in the portfolio were too risky. Despite all their experience with value investing and how the Schlosses practice it, these former clients were unable to incorporate the idea that at the right price—very low—the shares of a troubled company make a good investment.

The Schlosses are very attentive to the taxes their partners will have to pay. They do not like to sell shares in which they have a profit while the sale would constitute a short-term capital gain. This occasionally may put an investment at some risk; tax laws currently in place make it difficult if not impossible to protect the gain by a hedge until it goes long term. Given the different tax rate between short- and long-term gains, that is a risk the Schlosses are willing to take.

There are two policies the Schlosses follow in their partnership that set them apart from most money managers running similarly structured investment funds. First, they assume that they will distribute all the realized gains to their partners each year. If a partner asks that the money be kept in the fund, they will oblige, naturally. Most partnerships require an affirmative request, made many weeks in advance, for the withdrawal of any money whether realized gains or not. The Schlosses do not regard the money entrusted to them as captive, requiring a rescue by the partner to pry it free. This policy also helps them keep a rough cap on the size of the fund. They don't want to manage billions of dollars; returning a large portion of the gains each year is like pruning back a shrub to the desired height. Second, the Schlosses get paid for their work by taking a portion of the investment returns. This is the typical practice for limited investment partnerships. Where they depart from the standard is that they also assume the same portion of the losses. If the fund has dropped in value over the year, the accounts of the limited partners decline less than the fund itself. The Schlosses are charged their proportional share of the loss. Also, they do not take a management fee from the fund; most of their peers do. They only get paid for performance. As their long history shows, in seven years out of 45, they ended up worse off than they began. This arrangement is another incentive for them not to lose money.

Example 1: Asarco: Cheap Assets Find a Buyer

In 1999, Asarco was a copper company with a past more glorious than its current situation might suggest. Once a member of the Dow Jones Industrial Average, it lost \$1.70 per share in 1998, down from \$5.65 in 1995. When the stock dropped below \$15 per share, its market cap fell to under \$600 million, less than the \$885 million in long-term debt on its books. Yet despite all these troubles, Asarco had a book value of around \$40 per share. Its assets included its 50% ownership of the Southern Peru Copper Company, the equivalent of one share per each share of Asarco. Southern Peru traded between \$10 and \$14 for most of the year. A purchase of Asarco at anywhere from \$15 to \$20 would leave the investor with a margin of safety of at least 50% based on book value; he or she would be paying between \$5 and \$10 for the potential earnings of the company, after deducting the value of the Southern Peru share.

Walter and Edwin had actually been buying and then selling Asarco since 1993. They picked up some at around \$20 and sold it out above \$30 the following year. When the stock fell in 1999, they moved back in. This turned out well. The company agreed to a merger of equals with Cyprus Amax Minerals. The larger copper firm Phelps Dodge then bid for both companies, trying to buy them before the merger. They turned Phelps down, even though its bid for Asarco was worth around \$22. Now that Asarco was in play, it was just a matter of time before a higher offer appeared. The Schlosses finally sold their shares to Grupo Mexico for almost \$30 in cash. If the assets are there, as Walter likes to say, something good will happen.

Example 2: J.M. Smucker Co.: Selling Sugar to Americans

Warren Buffett didn't say it, to our knowledge, but some of his investments—Coca-Cola, See's Candies, Dairy Queen—suggest a faith in the notion that no one ever went broke selling sugar to Americans. That is the business the Smucker family has been in for years, packaging the sugar in the form of jams, jellies, and other sweet treats. But though the company has earned money consistently, their outside shareholders have not fared so well. The shares hit a high of \$39 in 1992, but from then through 1999, they seldom sold for as much as \$30. Earnings varied little over this period, from \$1.27 per share in 1993 to \$1.26 in 1999; the book value increased from \$7.55 to \$11. The price-to-book

ratio never fell below 1.5 to 1, and the price-to-earnings ratio only dropped below 15 on a bad day in 1999. On neither an asset nor earnings-based valuation did Smucker qualify as a value investment.

Then, in 2000, the price dropped below \$15 per share. Food stocks in general were a depressed group, and Smucker fell along with the others. The Schlosses bought some. At that price, the company was selling for 10 times earnings, well below its historic range. Though it was not a company with a franchise, it did have an established brand and a share of supermarket shelves. On an earnings power basis, it was cheap enough to take a position.

Two events made this a sweet investment. First, when Best Foods was bought out by Unilever, the prices of other food stocks rose in sympathy. Then, the Smucker family, which were controlling shareholders of the firm, decided to simplify the share structure and do away with a class of super voting stock. This reorganization made the stock more liquid and also introduced the possibility that the company could be taken over, now that the super voting shares were eliminated. Thanks to Unilever and the family, the stock rose from \$15 to \$25 within seven or eight months. The Schlosses now had to choose between taking a short-term capital gain, not something they like, or continuing to hold a stock after it had ceased to be cheap and may have become overvalued. They took the rational course and decided that paying the tax would be less painful than seeing the shares decline to \$15. Naturally the stock rose to \$27 before the end of the year, but now, at this higher price level, it is someone else's worry.

Mario Gabelli

Mario Gabelli graduated from Columbia Business School in 1967. He had been a student of Roger Murray, who kept the value flame alive after Ben Graham and David Dodd retired. For the next ten years, Gabelli was a sell-side analyst, first in the automotive industry and then in entertainment. In 1977, he formed Gabelli Asset Management (GAMCO) and began his career in active investing. He took the company public in 1999. After a long record of superior returns, the firm managed \$36.5 billion at the end of 2019. From the start, he has been a bottom-up investor, carefully researching individual companies. He estimated a company's value as the sum of the values of the individual divisions for which financial information was available. At the company level, he then considered how this sum-of-the-parts value would be affected by the quality of corporate management, especially with regard to capital allocation. Gabelli did not adjust his investments based on forecasts of macroeconomic variables such as interest rates or overall financial market conditions.

Having entered the business in the late 1960s, at a time of high valuations, Gabelli recognized, as did Warren Buffett at about the same time, that successful value investing could no longer rely on finding Graham-type "net-net" opportunities. Few if any companies were selling at less than two-thirds of net working capital less all other liabilities. He was forced to look for better businesses with favorable long-run prospects, whose value often depended on cash flows in the distant future. Unfortunately for Gabelli, he understood that commonly used discounted cash flow estimate of intrinsic value were highly sensitive to

assumptions about future growth rates and costs of capital. He had to look elsewhere to find reliable intrinsic value estimates.

Gabelli's innovative response to this challenge was to look at what he referred to as a company's "private market value" (or PMV). He defined this as "the price an informed industrialist would be willing to pay to purchase assets with similar characteristics." These prices could be deduced from transactions in which whole companies, divisions of companies, or individual assets such as cable television systems changed hands. Prices could be scaled against both accounting measures, such as dollars paid per dollar of average EBITDA, and direct operating statistics, such as dollars paid per cable customer acquired. Gabelli developed systematic databases of relevant transactions. The advantages of this approach were two-fold. Instead of relying on the market price, and by extension projected earnings and cash flows, it had a grounded estimate of what a knowledgeable buyer might pay. And the industrial buyer incorporated in his or her price the value of a control premium due to improved management and any other hidden potential in the underlying business, such as local synergies with existing businesses.

At the same time, Gabelli recognized that private market values could be relatively unstable, fluctuating with overall market conditions. The sooner the overall market value was embodied in the market price of a stock, the lower the risk that it might be undermined by adverse market developments. So Gabelli looked for catalysts that would lead public market prices to converge with private market values. Catalysts could be company-specific events—takeover bids, management replacements, corporate restructurings including spin-offs, and share repurchases—or developments affecting whole industries, such as regulatory changes or shifts in demand. These catalysts would draw positive attention to the investment in question and lead either directly, as in the case of takeovers, or indirectly to increases in the public market price.

Private market values plus catalysts have become a defining characteristic of Gabelli's investment strategy.

Glenn Greenberg

Glenn Greenberg's successful career in investing is a tribute to the value of general ability and education rather than either technical training or early experience. At Yale he studied English literature. His first job after graduation was teaching, first in elementary school and then in high school, where he also served as a principal. Since his greatest satisfaction came from managing the school, he took his boss's advice that he develop his managerial talent by going to business school. At Columbia Business School, the two projects that appealed to him most were, first, the analysis of a company from the perspective of a potential investor (he picked TWA and advised against investing in it) and second, a detailed and critical study of his family's business Gimbel Brothers, which also owned Saks Fifth Avenue. When he graduated in 1973, he considered positions in consulting and investment banking but decided to join the asset management department of J.P. Morgan.

His experience at Morgan was very useful. As he has said, he learned how not to manage money. Managers assembled their portfolios from an approved list of supposedly impregnable companies about which they were largely ignorant. From the beginning of 1973 to the end of 1974, portfolios of these stocks fell by 40 percent or more, no better than the S&P 500. In 1978, Greenberg moved to a smaller firm whose typical portfolio held ten or fewer stocks, each of which had been exhaustively researched. There he learned to be skeptical of the prevailing wisdom and to make sure that he personally understood the reason for investing in any stock owned.

In 1984 Greenberg and John Shapiro established their own firm, Chieftain Capital Management, based on these investment principles. They would not initiate an investment in a stock unless they were confident that their knowledge of the business justified an investment of at least 5 percent of their assets. They also intended to hold these positions for extended periods. They made only a few investment decisions in any year, allowing them to do extensive research on each firm they selected and on those on which they passed. All of the firms' four professionals were fully informed about each decision. They attended company meetings, individually scrutinized all financial filings, and tracked industry developments. They had to unanimously agree on all purchase decisions. They continued to devote the same scrutiny to companies already in their portfolio. The consequence of this focus was that Chieftain was better informed than most industry experts and seldom relied on outside opinions, which they generally found mistaken.

In selecting investments to target, Greenberg looked at a universe confined to hundreds of large, stable companies with strong competitive positions, high returns on capital, and the ability to distribute cash to investors even as they invested for rapid growth. These opportunities were characterized by three features. First, Greenberg looked for good businesses protected from relentless competition. They were often duopolies with histories of successful coexistence, or local monopolies, such as cable systems, whose profitability was not threatened by government intervention. He also sought out companies that could grow by taking market share, such as Airbus, Progressive, or HCA. Second, he looked for managements devoted to shareholder interest. That meant a focus on efficient operations, value-enhancing capital allocation, and the distribution of excess funds either as dividends or share buybacks. Finally, Greenberg sought a price for these purchases well below their values conservatively measured.

Greenberg was not interested in speculating on potential improvements either in management or the industry. He was not attracted to turnarounds. Rather, he looked for good businesses whose market prices were depressed by misperceptions of current or anticipated difficulties. For example, he was an aggressive purchaser of cable stocks where there was widespread concern, which he thought mistaken, that satellite competition and "cord-cutting" would limit cable's growth. Covid-19 presented many opportunities as investors confused a short-term event with a permanent reduction of corporate value.

In 2010, Greenberg and Shapiro separated, and Greenberg established Brave Warrior Advisors. He continues to place all his eggs in a small number of baskets and to watch those baskets very closely. The basket now includes technology stocks as his partners are of a younger age and able to help him figure out which of these faster-growing companies embody qualities he has always valued.

Paul Hilal

Paul Hilal was raised in a distinguished academic and medical family in New York City. After completing his undergraduate (B.A. Biochemistry, Harvard 1988) and graduate studies (J.D. and M.B.A., Columbia 1992), he spent five years doing M&A advisory work at Broadview Associates, the leading advisor to the tech sector at the time. After five intense years at Broadview, he took a sabbatical to travel around the world.

In January 1998, he joined his elder brother Peter's value-oriented investment management firm, Hilal Capital Management, where he created and ran the firm's information technology investment program. While there, he uncovered untapped potential in publicly traded Worldtalk Communication, a struggling software vendor. He led the fund's significant investment in the company and took charge as board chairman and acting CEO from 1999 to 2000. The reorganization, restructuring, and sale of the firm were highly profitable. The experience forever changed Hilal's thoughts on investing.

In 2002, Hilal opened his own investment management firm, Caliber Capital Management, to pursue a small number of concentrated investments, primarily focused on IT companies. In December 2005, his college roommate and close friend Bill Ackman asked him to join Pershing Square Capital, where Hilal could increase the scope and scale of his activities and yet continue to practice his focused investment style including active engagement with management.

The six investments on which Hilal worked over the subsequent decade, including three high profile projects, expanded Pershing Square's reach into a number of new industries. His first venture involved the

transformation of an IT company like WorldTalk; in this case it was industry laggard Ceridian. Pershing Square secured seats on Ceridian's board for Hilal and three other nominees. Almost before the ink was dry, Ceridian received an offer for the whole company so attractive that Pershing accepted and abandoned its longer-term restructuring plans. It was a bittersweet outcome for Hilal, who had looked forward to improving Ceridian's performance.

Hilal's next public engagement was far removed from IT. Long study had convinced him that railroads represented a fertile ground for his concentrated, activist approach. Canadian Pacific (CP), one of two Canadian transcontinental railroads, was North America's worst performer. Hilal believed its operations could be quickly and materially improved despite management's claims that structural challenges specific to CP limited its performance. Based on his now extensive industry knowledge, Hilal identified and recruited a replacement CEO capable of transforming CP. Hunter Harrison was a respected but controversial railroad operator who had been forcibly retired by the board of industry leader Canadian National despite having greatly improved its performance. Hilal initially sought to reach an agreement with CP's board for a cooperative restructuring. When that did not work, Pershing Square initiated a proxy contest and won support of 90% of the shareholders and board seats for all seven of its candidates, including Hilal and Ackman. Harrison was installed and began a successful transformation of CP's operations.

Hilal's next project again involved months of study of a new industry, this time industrial gases. As a fixed facility distribution business, it bore some similarity to railroads. He again identified a specific target for restructuring, Air Products and Chemicals. Of the four major industrial gas providers, Air Products was the worst performer. Benchmarking it against Praxair, the industry leader, revealed that Air Products had room for major improvement. As he had done with CP, Hilal found and recruited an exceptional manager to transform the company. Seifi Ghasemi was a well-respected industry veteran. Pershing Square negotiated a settlement with the existing management that led to board representation and the installation of Ghasemi as CEO. Restructuring, including a spin-off, and dramatic operational improvement followed, sharply increasing the value of the company.

These three projects along with three smaller investments were responsible for a substantial fraction of Pershing Square's returns during Hilal's tenure. He left in January 2016, with Ackman's support, to form his own firm, Mantle Ridge. He defined Mantle Ridge's mission as

creating value through the focused engagement with companies, which he had practiced successfully at Pershing Square.

Hilal's first target at Mantle Ridge was CSX, another underperforming railroad. He again selected Hunter Harrison as his change agent. In order to install Harrison at CSX, Hilal had not only to gain the agreement of CSX's board but also to induce CP to release Harrison from his current management obligations and a non-compete contract. This part of the task took \$100 million; an agreement with CSX soon followed. Hilal and four others from his team joined the CSX board, and Harrison became CEO. Though Harrison died only nine months after this move, the transformation was still completed and the railroad's value more than doubled.

Hilal's guiding principle at Mantle Ridge is that "professional investors assume stewardship of their partners' precious capital. Boards and managements assume stewardship of industry and the interests of the shareholders and other corporate stakeholders. Engaged professional investors do both." His success in applying this principle depends on "constructive and cooperative engagement between boards, management teams, and engaged shareholders." The key to making this approach work is comprehensive industry knowledge on the part of the engaged shareholders, which is acquired with an industry focused, if serial, specialization of the kind that Paul Hilal has practiced throughout his career. It also requires the discipline needed to pass on all but a few extraordinary opportunities to which the necessary unremitting attention can be applied.

Jan Hummel

Jan Hummel, perhaps unusual for someone growing up in the countryside in Sweden, bought his first share, of a Swedish IPO, at the age of 16. He graduated with an MS from the Stockholm School of Economics and was awarded a fellowship to study financial economics and mathematics at Stanford. He was then recruited to Booz, Allen, Hamilton in London. Subsequently, Hummel received his MBA from Harvard Business School where Bruce Greenwald was one of his professors. His first position after business school was as an investment banker with Merrill Lynch. Attracted more to investing than investment banking, he left Merrill to pursue turnaround and restructuring opportunities in a private equity setting. After several years, he recognized that much of the benefit of private equity investment was dissipated in the often frenetic bidding for deals, raising the costs and lowering subsequent returns. As a result, he turned his attention to investing in publicly traded equities and started Paradigm Capital in 2001 as a private investment company. His focused strategy and strong track record attracted Mario Gabelli's participation. In July 2007, he founded the Paradigm Capital Value Fund, one of Europe's first value investing funds, with his own capital and Gabelli's group as an anchor investor.

Hummel focuses on smaller- and medium-sized businesses. He typically invests in companies with market capitalizations between 200 million and 5 billion Euros. This is an area of the market generally ignored by the large institutions—major banks, insurance companies, and pension funds—that dominate the investment scene in Europe. With many billions of Euros under management, it is rarely worthwhile

for them to look at companies of this size. In Paradigm Capital's universe, shareholders tend to be relatively conservative family business owners, individuals, and smaller institutions.

Hummel invests exclusively in companies located in Austria, Germany, Ireland, the Nordics (Denmark, Norway, Sweden, and Finland), and the UK, countries that benefit from good corporate governance and well-regulated securities markets. Only a limited number of companies fit Hummel's size, geographic, and business quality criteria, allowing him and his team of analysts to closely study the firms and managements within this universe.

Hummel looks for franchise firms earning high returns on invested capital whose operations are protected by strong barriers to entry. These franchises, given the size of his target companies, are necessarily local, dominating their markets either geographically or, if they sell globally, focused on narrow product niches. They are often service businesses with local economies of scale in purchasing, advertising, sales, service infrastructure, and management. Examples include local distribution businesses for both people (e.g., airlines) and goods (e.g., logistics, retail and wholesale companies), local contractors (e.g., plumbing and electrical, real estate, and education), and financial service providers (e.g., specialty insurers, locally focused banks). One particularly promising category is manufacturers for whom production costs are low relative to local sales and service costs. They are often evaluated as small manufacturers rather than locally dominant service providers. Hummel seeks to acquire companies below or at fair value, often when an opportunity for value enhancement is available to an engaged investor, such as growth typically in an adjacent market, capital structure optimization, changes in top management, or possibly a complementary acquisition.

Hummel looks for high-quality managements along several dimensions. First, he wants a relentless focus on efficient day-to-day operations. Second, he seeks a growth strategy that creates value by investing only when the returns on growth exceed the cost of capital. In practice, this means expanding in markets to which existing economies of scale carry over. Finally, Paradigm Capital looks for management that is attentive to its shareholders, distributing any funds that cannot be profitably invested in the business and financing the business in the most efficient way possible by using appropriate amounts of debt and not holding excess cash. Executive compensation should typically be tied to long-term returns on capital employed.

The limited geographic footprint of its investments allows Paradigm Capital continuous access to management. Hummel is not shy about

conveying his views to them if he feels they need to improve their operations. Attractive buying opportunities often emerge from managerial underperformance, even in companies in which Paradigm Capital is already a shareholder. His preference is to establish a cooperative relationship with a company; to be a “suggestivist” rather than a traditional activist who is apt to use a more forceful approach. If necessary, Paradigm has taken control of companies and installed new people with proven managerial talent. Its successes in these instances have depended on Paradigm’s ability to identify capable, locally available managers, all made possible by its geographic focus.

Hummel considers a concentrated portfolio to be a critical part of Paradigm Capital’s success. The portfolio has around 75 % invested in a set of around ten positions, each accounting for about 6–12 % of the fund. These major positions are typically held for extended periods. Before investing, Hummel must be thoroughly familiar with their business operations and their managements. The remaining 25 % consist of promising exploratory positions, of which there are up to ten, some event-driven holdings that are cash-like, and cash itself. After initial purchase, these exploratory investments are subjected to continuous attention, including developing a relationship with and an evaluation of management. Only a few become full positions, a decision very much driven by the valuation and thus the expected return.

The process by which Paradigm Capital selects these exploratory positions begins with about 50 to 70 ideas a year, of which around 20 survive a preliminary screening. Each of Paradigm’s five analysts investigates about four to six of these 20; they are assigned based on industry or other specialization. The list is further winnowed until only about a handful are left, one per analyst. These become exploratory investments, each followed intently by the analyst who recommended it. Ultimately, they are either dropped or elevated to one of the major positions in the portfolio after a discussion by all investment professionals. Companies that do not make the final cut are added to a library of target companies so that Paradigm can move quickly if prices fall sufficiently to make the investment attractive.

Risk management plays an important part in Hummel’s investment philosophy. It is handled by requiring a margin of safety before any money is invested. Diversification also reduces risks. In vulnerable market periods, when both prices and investor confidence are high, Hummel reduces market risk further with various derivative positions that in these circumstances are often available at relatively low cost. In these periods, Paradigm Capital also holds cash or cash-like arbitrage

and other event-driven positions intended to produce higher than pure cash returns. Paradigm Capital considers itself to have no edge in determining the future of currency movements and therefore hedges all non-Euro positions, and for overseas US investors, it hedges out the overall currency risk between the Euro and the dollar. As a consequence, Paradigm Capital's value investment strategy coupled with strict risk discipline has produced returns with very low correlation to the market.

Paradigm Capital creates its edge through highly disciplined, deep fundamental research. Its overall strategy is to look within its intentionally limited universe for (1) good businesses with (2) high-quality management, (3) good corporate governance, and (4) competitive advantages or (5) assets at advantageous prices. Through this focused approach, Hummel has made investments that earned well above the market returns without any financial leverage, showing that traditional value investing principles of sticking with quality and buying what one knows are still as relevant in Europe as ever. Discipline, focus, specialization, and hard work coupled with careful risk management have done well for Paradigm Capital and its investors.

Seth Klarman

Seth Klarman began investing as a teenager and was introduced to value investing during his college years. He worked with Max Heine and Michael Price for a summer while in college and then for 18 months before he entered Harvard Business School. When he graduated in 1982, two faculty members hired him to help manage their wealth and that of two other families. After an extensive search of outside investment advisors, they decided that Klarman himself could probably do a better job than any of the available alternatives. That decision must rank as among the very best in the history of money management. They established The Baupost Group with Klarman as portfolio manager and began investing in early 1983 with \$27 million in assets under management. By the end of 1999, assets under management were approaching \$2 billion without having raised any significant amount of outside funds. That figure had grown to more than \$28 billion by the end of 2019.

Klarman has been in an unusual position at Baupost. Most institutional investment managers control only a portion of their clients' wealth. As a result, they are not primarily risk managers, although risk controls affect their investment choices. Baupost continues to manage significant assets for its founding families as well as of the assets of many of Klarman's employees who have made a point of eating their own cooking and collectively make up the firm's largest client. It is no surprise that Klarman's much treasured book on investment should be titled *Margin of Safety: Risk Averse Investing Strategies for the Thoughtful Investor*. Baupost's goal is to achieve attractive risk-adjusted returns over an extended period of time.

In implementing his approach, Klarman begins to look at all investments by asking, "What is my edge?" He realizes that every time he purchases an asset, thinking it will be a good investment, someone else is selling it, usually with a less positive view on its value. Without an edge, Klarman has no reason to think that over time, he will end up on the right side of the transaction more often than not. Fortunately, Baupost's structure and its long-term relationship to its clients means that Klarman has patient money, does not have to respond to daily market pressures, and can weather any short-term adverse developments without having to sell his investments at inopportune times. Klarman focuses on markets and securities with motivated or unreflective sellers. These include distressed fixed-income securities being disposed of by managers who are constrained to hold only investment grade assets; spin-offs from larger companies the stocks of which are almost automatically sold by money managers for materiality reasons; distressed or non-performing assets being sold by financial institutions; assets sold by government agencies, such as the Resolution Trust Company in the early 1990s, forced to dispose of assets from failed savings and loan banks and other institutions that had acquired them for reasons of economic stability; stocks removed from a market index that must be sold by index funds; and less than prime real estate, such as half-empty buildings or other hard-to-sell assets.

Another attractive characteristic of these kinds of markets is that Klarman faces much less competition from other well-informed and liquid buyers. Over time, however, as investors discovered such opportunities, other buyers arrived and bid up prices. Klarman was forced repeatedly to find new fields in which to prospect. For example, Klarman developed a network of local real estate operating partners for whom he was the primary or exclusive source of sophisticated, ready to act external capital.

In selecting particular investments, Klarman concentrates first on the downside, which he defines as the likelihood and magnitude of permanent loss. This approach enables him to acquire assets with volatile prices but generally predictable long-run value. A case in point was his acquisition of senior Texaco bonds in the 1980s. In response to an adverse legal judgment, Texaco was forced to protect itself by declaring bankruptcy and halting interest payments on all its debt. Interest due continued to accumulate during the process of reorganization. Texaco owned assets that fully covered the outstanding principal and many years of accumulated interest on the senior debt, even after an adverse legal judgment. The investment uncertainty stemmed largely from the timing of this

return, not its eventual payment. Unless resolution of the bankruptcy took an unusually long time, the bonds were selling at a price that was likely to produce an annualized return of at least 15 percent. But the prices of the bonds might fluctuate in the interim, in response to developments in the bankruptcy process. Having a long-term perspective and defining risk as permanent loss of capital, Klarman was immune to these concerns. As things worked out, he earned an annualized return of well over 20 percent over a few years with little risk.

Another example of a distressed credit with limited downside risk and sizeable upside was in the debt of Enron. Enron bonds were selling at pennies on the dollar after a major fraud was exposed. While the exact value of the assets was uncertain, the market price of the bonds seemed to incorporate an excessive discount for the risks.

Klarman has other advantages. He combines a high intelligence, not a rarity in the profession, with an exceptional discipline that helps keep him immune to the pull or push of market trends or other enthusiasm. He and his firm do meticulous research, and because of his stellar reputation, he has the ability to attract and retain first-rate people. However, the extraordinary results he has achieved might not have been possible with less patient investors and more focus on “What have you done for me lately.” He has also built a cohesive firm with a strong culture of collaboration and a drive for excellence.

Michael Price

Michael Price's long and successful career as a value investor began when he became a research analyst for Max Heine, a legend in the field, shortly after graduating from college in 1973. When Price started at Heine Securities, its fund Mutual Shares series had roughly \$5 million under management. When Heine died in 1988, Price succeeded to control of the firm. He sold it in 1996 to Franklin Resources but continued to manage the funds until 1998. By that time, assets under management exceeded \$15 billion without any substantial advertising, promotion, or sales commissions. The growth came from outstanding investment performance that did not deteriorate as the funds grew in size. The record provides testimony to Price's investing abilities and the strength of the value approach he practices.

Price embraced an eclectic approach to investing. Most funds were invested in value equities, with the intention of holding them for extended periods. Price also invested in event-driven return opportunities, including bankruptcies and reorganizations—which were a specialty of Max Heine—merger arbitrage, and activist intervention. In constructing his portfolio, Price aimed for a 15% return with limited downside risk, all based on acquiring assets and businesses at a price that afforded a large margin of safety.

Long-term value equity investments typically made up two-thirds of the portfolio. These investments were widely diversified and spread among dozens of stocks. Price looked for a discount between price and intrinsic value of 30–40 % in the stocks he purchased. To find these companies, he had to avoid areas in which manic investor enthusiasms, as in the dot.com bubble, or the promotion activities of Wall Street firms, as

in hyped IPOs, created inflated prices. He was patient and disciplined, sometimes waiting years for a stock to meet his price requirements. The remaining one-third of the portfolio consisted of event-driven investments that Price regarded as having cash-like values independent of overall market fluctuations. Carefully selected merger-arbitrage opportunities would often yield annualized returns of 15% or more. The risk that these deals would collapse and produce negative returns was not correlated with whatever the overall market was doing. By investing in 5–10 carefully selected deals from the hundreds available, the risk of a deal breaking could be minimized and largely diversified away. At that same time, Price always held at least 5% of the portfolio in cash. His balanced portfolio had a record of stability, with major declines 40% less severe than those of the stock market as a whole.

Price's approach to estimating intrinsic values relied heavily on private market valuations. He began by examining individual operating businesses and assets using available financial data. For each, he looked at the historical record of what "a knowledgeable buyer was willing to pay" for a similar asset. Most of this information comes from merger and arbitrage transactions, which are accompanied by detailed investment banking analysis of each revenue stream and the non-operating assets, such as holdings of publicly traded securities. Systematic collection and analysis of the information for relevant, recent transactions reveal what knowledgeable buyers have been willing to pay for similar operations. Summing the values of the individual pieces that make up a company will yield something like a maximum value. If the current market price of the company under examination is at or above this value, then the company's stock is a poor bargain and to be avoided. (In the first edition of this book, Price did such a calculation for General Electric, at that time one of the largest companies in the world, which showed that the stock was significantly overvalued.) The much touted positive synergies tend to be minimal or non-existent. On the other hand, poor overall company management may destroy a significant part of whatever potential synergies exist. So the last step in Price's analysis was to examine carefully the historical performance of the firm's management. If this showed major value destruction, Price was prepared to intervene, as he did successfully with Chase Manhattan Bank in 1995.

After selling Mutual Shares in 1995, Price has continued to manage money for himself, some friends, and a few institutions. His investment style has not changed.

He is focused and efficient in looking for potential buying opportunities. If an intrinsic value calculation of a promising firm shows that the price is not cheap enough to buy, he does not simply discard the work and move on. He continues to follow the firm and increase his understanding so that if it trades at a price which provides a sufficient margin of safety, he is ready to act quickly. After he has sold a firm because it became fully priced, he keeps track of it in case the price retreats to make it a bargain again. He applies this discipline to bankruptcies. He follows a firm through every stage of the process, from pre-filing distress to the filing itself, through the various reorganization plans, and finally emergence from bankruptcy. At each stage, he is prepared for any attractive investment opportunity that appears.

Price begins the search for new opportunities with a careful reading of the business press. He is not interested in general market information. He looks for company news, especially bad news, and industry developments, such as consolidation through mergers, that are likely to alter market prices and intrinsic values and may thus create new opportunities. He then has his team begin a comprehensive analysis of any affected company, including its particular position in the industry and its management. Price's underlying strategy is to be patient in waiting for opportunities and well prepared to act quickly when they appear.

Thomas Russo

After graduating from Dartmouth in 1977, Thomas Russo enrolled in Stanford in 1980 for a joint JD/MBA degree. In the business school he took a course from Professor Jack McDonald, who—in the face of a finance department deeply committed to the efficient market theory—taught value investing in the Graham and Dodd tradition. A guest appearance during the course by Warren Buffett cemented Russo’s attachment to value principles. He graduated in 1984 with degrees in law and business and was hired by the investment firm Ruane, Cunniff to work in their Sequoia Fund, one of the funds that Buffett profiled in his famous “Superinvestors of Graham and Doddsville” article. After five years he left for Gardner, Russo & Gardner, where he has managed the Semper Vic family of funds ever since.

Russo’s investment approach is rooted in three fundamental principles. First, he looks for “fifty-cent dollars,” stocks selling at 50 % or less of his estimate of intrinsic value. Second, he focuses on companies with both “great brands” and significant opportunities to reinvest in the extension of these brands at attractive returns. Third, he looks for companies and managements that have a “capacity to suffer,” by which he means the willingness to make long-term investments at the expense of current reported earnings. Short-term investors often respond to a decline in earnings by selling the stock and driving down the price until it meets Russo’s “fifty-cent dollar” criterion. For the “great brands” companies in which Russo invests, those that can earn more than the cost of capital on their investments for growth, the long-run benefits ultimately become apparent and the stock price rises.

One example of this process was the global expansion of Brown-Foreman. The company's whiskey brands, most notably Jack Daniel's, were popular in the US but had only a small presence internationally. Since domestic sales were stagnant, the most obvious path to growth was overseas. At the time, foreign sales accounted for less than 10% of total company revenue. The campaign to increase foreign sales was costly. It involved funds for advertising, distribution, and sales that increased operating expenses well in advance of any corresponding increase in revenue or income. The result was declining reported earnings for a company that continued to show stagnant overall sales. The stock price declined enough to meet Russo's "fifty-cent dollar" requirement. Ultimately, overseas sales came to represent half of Brown-Foreman's revenue, and profit grew accordingly. Russo saw a similar pattern in the development of the House and Garden Cable Television (HGTV) channel. House and Garden was part of E. W. Scripps, a family-controlled business that owned newspapers, magazines, and local television stations. Creating programming and arranging distribution increased expenses well ahead of any visible benefits. But with patience, Scripps and Russo saw the investment pay off. Ultimately, HGTV added more than \$1 billion to Scripps' value.

The success of these investments depended on two factors. First, the opportunity to earn a return above the cost of capital by extending a brand's geographic reach, in the case of Brown-Foreman, or to expand existing content to new channels of distribution, in the case of Scripps, must actually be there. Russo had to be able to distinguish between "strong brands" with profitable reinvestment prospects, and weak ones, with limited ability to move into new markets. Second, management must be capable of accurately identifying the markets and be willing to accept short-term pain while still executing the expansion strategy effectively. Identifying managers with these skills requires detailed familiarity with managers across an industry. Russo has developed this expertise by focusing on a few industries, chiefly consumer brand businesses, and on the managers who work in them.

Over time, he has recognized that family-controlled businesses are a particularly fruitful area on which to concentrate. Family-controlled firms tend to take a much longer-term view than companies with managers who need to respond to the market's reaction to quarterly results. Also in a world in which most companies are led by professional managers with negligible ownership stakes, there are often irrational

prejudices against family-owned firms, which tend to reduce their share prices. Sometimes these judgments are justified. Within his target industries, Russo is especially attentive to how effective family management, often overseas, can be distinguished from dysfunctional ones.

To make the most of his specialized knowledge, Russo has had to invest globally. He was an early leader among US value managers in looking abroad. He has also run a concentrated portfolio. At the end of 2019, Russo's top five positions accounted for roughly half of his roughly \$13 billion portfolio. Four of these were widely recognized global consumer brands, including MasterCard, and the other was his longtime holding in Berkshire Hathaway. His next ten investments accounted for another 40% of the portfolio. They included six global consumer brand companies and three companies that, like Berkshire, he had followed for his whole career. One of these was family controlled. The remaining 10% of the portfolio was made up of around 30 small positions, some exploratory and some in the process of liquidation. Many of these were also global consumer brands.

With a concentrated portfolio and a long holding period corresponding to his long-term investment horizon, Russo makes only a few major investment decisions each year, allowing him to run with a small research staff. He is very much in the "put your eggs in one basket and watch that basket carefully" school. Among the concentrated value investors who were the subjects of a careful, comprehensive study by Carla Knobloch, it was those with specialized expertise who performed best. Russo was first among them. But his long-term success has required the right kind of investors. As he acknowledged, "I have been very fortunate in having been provided with the capacity to take my time in earning returns."

Andrew Weiss

Andrew Weiss began his professional career as an academic economist. He majored in political economy at Williams College and after spending a few years as a public school teacher, he earned a PhD in economics from Stanford University in 1977. He then taught economics at Columbia while also working as a member of the economics group at Bell Labs, then one of the world's preeminent research institutions. In 1986 he moved to Boston University, where he remained until retirement. The papers he published, widely cited, earned him a distinguished reputation. During this period, he was a highly disciplined and successful investor, first of his own and some family money and then establishing a fund in 1991. When he tired of the politics and administrative demands of faculty life in 2004, he became an *emeritus* in order to focus on investment management.

As an investor, Weiss has always been attracted to overlooked investments that are undervalued because they are obscure and complex, rather than individual stocks. In the late 1960s, when US investors were almost exclusively focused on US securities, he looked overseas. He became interested in Japanese industrials at a time when Japan, before its days as an economic powerhouse, was known chiefly as a producer of simple, cheap, and low quality consumer goods. He identified a US closed-end fund that invested in Japanese manufacturing stocks. A closed-end fund, in contrast to the much more common open-end fund, sells a finite number of shares to the public in an IPO and invests the proceeds. Those shares then trade on a stock exchange. Although the holdings of the fund are publicly known and the net asset value of the fund can be easily calculated, the fund often trades at a discount to this value and less frequently

at a premium. Weiss saw that while the companies in the Japan fund were trading at very low earnings multiples, sometimes as low as five; his own research had found strong growth prospects for the Japanese economy as a whole. The shares of fund itself traded at a significant discount to the market value of the stocks in the portfolio, and there was a possibility of the fund liquidating in the future, at which point the underlying stocks would be sold and the proceeds distributed. All of this, though obvious to Weiss, was either overlooked or ignored by investors who had disdain for the Japanese companies. As Japan gained recognition as an industrial powerhouse, the underlying shares appreciated in value. By the early 1980s the Japanese stock market was fully valued and Weiss sold his position. Weiss's return benefitted both from the repricing of these shares and the narrowing of the discount at which he had originally bought them. The complexity and obscurity created a double margin of safety—cheap stocks selling at a discount to their market value.

He also began to focus on complex investments in more established markets that leveraged his economics expertise to find opportunities that markets were mispricing. For example, in the manically optimistic atmosphere associated with German unification in the early 1990s, not only did German stocks appreciate into bubble territory, but the Germany Fund, a closed-end fund investing in German securities, traded at a price that was well above the value of the securities in its portfolio. The stocks themselves had run up considerably in response to a macroeconomic change that according to Weiss's analysis was not going to improve their future earnings (subsidies to East Germany paid for by West Germany was not going to help the after-tax earnings of West German companies).

Weiss began managing money professionally in 1991, and he continued to apply the two principles evident in his Japanese investment. He looked at opportunities globally, often in overlooked, undesirable markets, and he sought out complex securities that he could buy with a margin of safety by investing in the highly diversified portfolios of closed-end funds. In the 1990s and early 2000s, those opportunities often lay in the post-communist countries of Eastern Europe and former Soviet States such as Kazakhstan. At the time, governments were privatizing the state-owned enterprises by distributing the stock among their workers. The workers, in turn, having no aptitude for or interest in stock ownership simply dumped the share on newly established local stock markets. At the same time, rampant corruption and a tradition of dishonest reporting deterred most overseas and many local purchasers. As a result, many of these companies sold at a tiny fraction of their asset values and as low as one-half their annual earnings. Several closed-end

funds, organized to take advantage of these bargains, were selling at significant discounts after the promised high returns did not materialize immediately. Weiss, who is naturally risk averse, ultimately found opportunities to invest in funds at prices that more than compensated for the risk and that were relatively well managed. In some cases he was buying shares for less than the value of the entitlement to cash held by the fund; he was getting an interest in operating companies for less than zero.

Another opportunity arose in the mid-1990s from geographic differences in opinion about the prospects for Brazil. A Brazilian closed-end fund traded in Europe at a significant premium. At the same time, a fund with similar holdings was trading in Argentina at a large discount. Buying the Argentinian fund while shorting the European one provided a high expected return at low risk.

Financial disaster created other opportunities. Just as US mortgage securities in the early 2000s were sliced and diced to create a spectrum of “safe” to “risky” opportunities, in the 1990s British asset managers had created sliced and diced securities based on ordinary debt or equity portfolios. The constituent securities were referred to as “splits.” An extra level of complexity was added by the fact that there were several different constituent securities, each of which had different governance rights as well as different priority of claims in the event of payment suspensions. When the values of the underlying securities collapsed in the spring and summer of 2002 there were widespread suspensions of payoffs to holders of a share class who thought that all the dividend income would automatically be distributed to them. Investors whose supposedly “safe” securities proved very unsafe dumped them without performing the complicated calculations necessary to determine the true value. Weiss and his analysts developed careful methods to value the underlying securities and also the administrative and legal capacity to enforce the legal entitlements and to exercise associated governance rights, which often required physical presence at meetings in the Channel Islands or Scotland.

Finally, there are often systematic local anomalies that create attractive opportunities. Korean companies have always issued both ordinary and preference shares. These preference shares are quite different from the preferred stock issued in developed financial markets. They have the same fractional claim on a company’s earnings as the ordinary share, but they pay slightly higher dividends. On the downside, preference shares have no votes in corporate elections and other matters that don’t directly affect the preference shares. Since most Korean managements operate without any shareholder input, this is a minimal sacrifice.

Yet the preference shares trade at sometimes substantial—often greater than 50%—discounts to the ordinaries. The problem is that these discounts vary markedly across time and companies. Weiss and his associates studied the historical variations in discounts and developed a portfolio of preference shares that they could hedge with the corresponding ordinary shares.

Given the seemingly endless global capacity of investment banks to create such complex instruments, opportunities of this sort seem to be continuously available. Weiss has built a firm with the analytical and administrative capacity to exploit them effectively as well as to be creative in the forms of financing they can offer to firms whose structure and needs made them a poor fit for the terms and conditions of the customary commercial or investment bank contracts. This highly focused organization, with about \$2 billion under management, has produced healthy returns from 1991 through the end of 2019 with no down years.

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